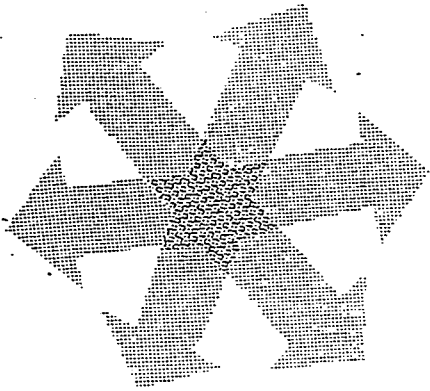


SOLUTIONS MANUAL



to accompany

CRANDALL & DAHL

An introduction to the mechanics of solids

SECOND EDITION

Prepared by

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서

문

본 책자는 Crandall & Dahl의 "An Introduction to the mechanics of Solids"의 연습문제를 풀이한 것이다.

위 책자는 역학을 공부하는 이들이 필히 익혀야 할 책으로서 PANGAL, N. NAYAK과 RANGNATH NAYAK에 의해서 제 7 판에 대한 영자판 해답집이 이미 시중에 나온 바 있다.

제 2 판이 출판된데 즈음하여 이에 대한 한글판 해답집을 엮어내기에 이르렀다.

위 책자는 역학을 공부하는데 있어 하나의 플이에르서 도움을 줄 수도 있지만, 무의미하게 이용한다면 엄청난 손해를 야기할 수도 있다는 것을 말씀드린다. 다시 말해서 위 플이들이 완전하다고 생각하고 비판없이 받아 들인다면, 스스로 생각하고 문제를 해결하는 능력을 마비시킬 수도 있다는 것을 꼭 명심하기 바란다.

따라서, 펴내는 이의 바램은 본 해답집이 보시는 이들로부터 Crandall 책의 문제를 풀이한 후에 심심플이로 문제 해를 비교해 보시는 정도로만 이용되었으면 한다.

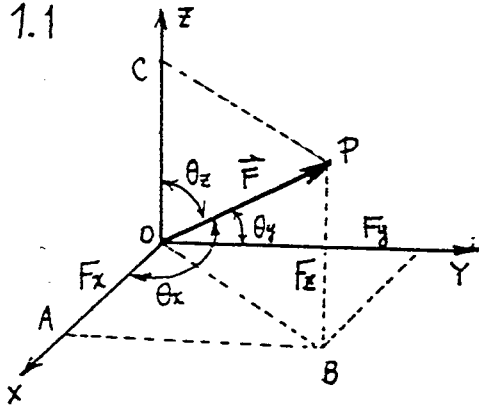
아무쪼록 이 플이집이 보시는 이에게 조금이나마 도움이 되길 바랍니다.

CONTENTS

Chapter 1. Fundamental principals of mechanics	1.
Chapter 2. Introduction to mechanics of deformable bodies	24.
Chapter 3. Forces and moments transmitted by slender members	59.
Chapter 4. Stress and strain	85.
Chapter 5. Stress-strain-temperature relations	105
Chapter 6. Tortions	141
Chapter 7. Stresses due to bending	164
Chapter 8. Deflections due to bending	214
Chapter 9. Stability of equilibrium : buckling	259

CHAPTER 1.

1.1



$CP \parallel OB$, $OB \perp OC \quad \therefore CP \perp OC$

따라서

$$\cos \theta_z = \frac{F_z}{|\vec{F}|}$$

마찬가지로

$$\cos \theta_y = \frac{F_y}{|\vec{F}|}$$

$$\cos \theta_x = \frac{F_x}{|\vec{F}|}$$

피타고라스 정리에 의하면

$$F_x^2 + F_y^2 + F_z^2 = |\vec{F}|^2$$

$$|\vec{F}|^2 \cos^2 \theta_x + |\vec{F}|^2 \cos^2 \theta_y + |\vec{F}|^2 \cos^2 \theta_z = |\vec{F}|^2$$

$$\cos^2 \theta_x + \cos^2 \theta_y + \cos^2 \theta_z = 1$$

1.2

그림에서

$$\vec{i} = \vec{a} \cos \theta - \vec{b} \sin \theta$$

$$\vec{j} = \vec{a} \sin \theta + \vec{b} \cos \theta$$

또한 $\vec{i} \cdot \vec{a} = |\vec{i}| |\vec{a}| \cos \theta = \cos \theta$

$$\vec{i} \cdot \vec{b} = -\sin \theta$$

$$\vec{j} \cdot \vec{a} = \cos(\frac{\pi}{2} - \theta) = \sin \theta$$

$$\vec{j} \cdot \vec{b} = \cos \theta$$

일반적으로 $\vec{i} = (\vec{i} \cdot \vec{a}) \vec{a} + (\vec{i} \cdot \vec{b}) \vec{b}$

$$\vec{j} = (\vec{j} \cdot \vec{a}) \vec{a} + (\vec{j} \cdot \vec{b}) \vec{b}$$

$$\vec{F} = F_x \vec{i} + F_y \vec{j} = F_x (\vec{a} \cos \theta - \vec{b} \sin \theta) + F_y (\vec{a} \sin \theta + \vec{b} \cos \theta)$$

$$= \vec{a} (F_x \cos \theta + F_y \sin \theta) + \vec{b} (F_y \cos \theta - F_x \sin \theta)$$

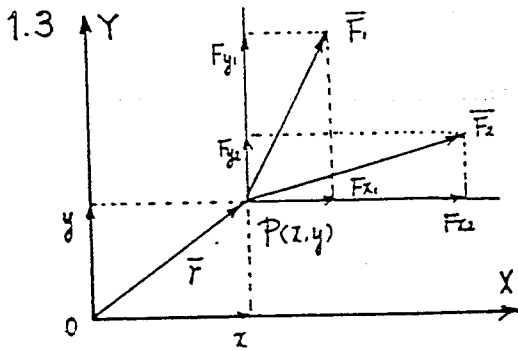
$$\therefore F_a = F_x \cos \theta + F_y \sin \theta \quad \dots \textcircled{1}$$

$$F_b = -F_x \sin \theta + F_y \cos \theta \quad \dots \textcircled{2}$$

①과 ②로부터

$$F_x = F_a \cos \theta - F_b \sin \theta$$

$$F_y = F_a \sin \theta + F_b \cos \theta$$



$$\vec{r} = \vec{i}x + \vec{j}y$$

$$\vec{F}_1 = \vec{i}F_{x1} + \vec{j}F_{y1}$$

$$\vec{F}_2 = \vec{i}F_{x2} + \vec{j}F_{y2}$$

$$\vec{M}_0 = \vec{r} \times (\vec{F}_1 + \vec{F}_2)$$

$$= (\vec{i}x + \vec{j}y) \times [\vec{i}(F_{x1} + F_{x2}) + \vec{j}(F_{y1} + F_{y2})]$$

$$= \{x(F_{y1} + F_{y2}) - y(F_{x1} + F_{x2})\} \vec{k}$$

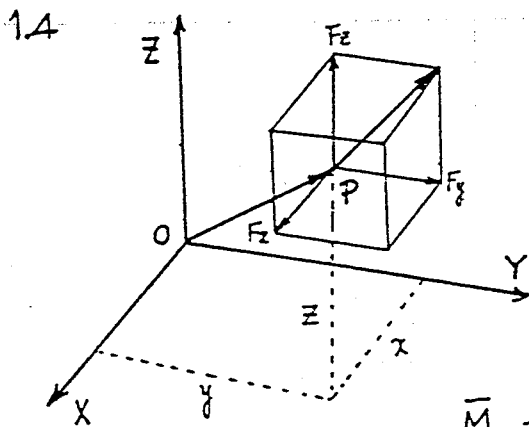
0점에 대한 $\vec{F}_1$ 과 $\vec{F}_2$ 의 Moment를 각각 생각하면

$$\vec{M}_{01} = \vec{r} \times \vec{F}_1 = (xF_{y1} - yF_{x1}) \vec{k}$$

$$\vec{M}_{02} = \vec{r} \times \vec{F}_2 = (xF_{y2} - yF_{x2}) \vec{k}$$

$$\sum_{i=1}^2 M_{0i} = \{x(F_{y1} + F_{y2}) - y(F_{x1} + F_{x2})\} \vec{k} = \vec{M}_0$$

$$\therefore \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2 = \vec{r} \times (\vec{F}_1 + \vec{F}_2)$$



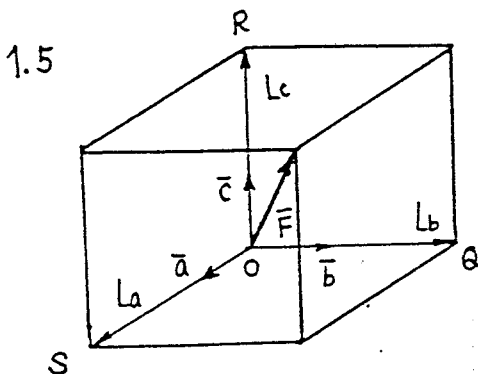
$$\vec{M}_0 \text{ (0점에 관한 Moment)} = \vec{r} \times \vec{F}$$

$$\vec{M}_0 = (x\vec{i} + y\vec{j} + z\vec{k}) \times (F_x\vec{i} + F_y\vec{j} + F_z\vec{k})$$

$$= \vec{i}(yF_z - zF_y) + \vec{j}(zF_x - xF_z) + \vec{k}(xF_y - yF_x)$$

따라서 $\vec{M}_0$ 는 다음과 같이 쓸수 있다.

$$\vec{M}_0 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$



$$(\vec{F} \times \vec{b}) \cdot \vec{c} = \{(L_a \vec{a} + L_b \vec{b} + L_c \vec{c}) \times \vec{b}\} \cdot \vec{c}$$

$$= L_a(\vec{a} \times \vec{b}) \cdot \vec{c} + L_b(\vec{b} \times \vec{b}) \cdot \vec{c} + L_c(\vec{c} \times \vec{b}) \cdot \vec{c}$$

$$(\vec{c} \times \vec{b}) \cdot \vec{c} = 0 \quad (\because \vec{c} \times \vec{b} \perp \vec{c})$$

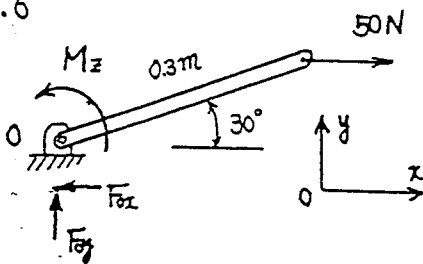
$$\vec{b} \times \vec{b} = 0$$

$$\therefore (\vec{F} \times \vec{b}) \cdot \vec{c} = L_a(\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$\therefore L_a = \frac{(\vec{F} \times \vec{b}) \cdot \vec{c}}{(\vec{a} \times \vec{b}) \cdot \vec{c}}$$

같은 방법으로 $L_b = \frac{(\bar{F} \times \bar{c}) \cdot \bar{a}}{(\bar{a} \times \bar{b}) \cdot \bar{c}}, \quad L_c = \frac{(\bar{F} \times \bar{a}) \cdot \bar{b}}{(\bar{a} \times \bar{b}) \cdot \bar{c}}$

1.6



평형조건으로부터

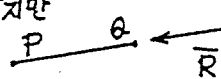
$$\begin{aligned} \Sigma F_x &= 0 & F_{0x} &= 50 \text{ N} \\ \Sigma F_y &= 0 & F_{0y} &= 0 \\ \Sigma M_{0z} &= 0 & M_z &= 50 \times 0.3 \sin 30^\circ \\ & & &= 75 \text{ (N}\cdot\text{m)} \end{aligned}$$

Ans. $F_{0x} = 50 \text{ N} \quad M_z = 75 \text{ (N}\cdot\text{m)}$

1.7 (a) $\Sigma F_a = 0, \Sigma F_b = 0$ 이면 $\Sigma \bar{F} = 0$ 를 만족하나 이 조건만으로는 x-y 면에서 Couple (짝힘)이 생길 수 있다.

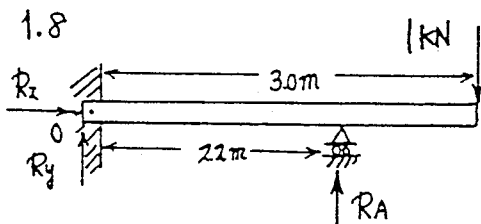
임의의 점에 대하여 $\Sigma M_b = 0$ 가 되어야 Couple이 방지될 수 있다.

(b) $\Sigma M_p = 0, \Sigma M_a = 0$ 이면 Couple은 발생하지 않지만

$\bar{P}$ 방향의 resultant force $\bar{R}$ 이 발생할 수 있다. 

추가로 $\Sigma \bar{F}_p = 0$ 의 조건이 가해지면 resultant force가 없어진다

(c) $\Sigma M_o = 0, \Sigma M_p = 0, \Sigma M_a = 0$ 라면 Couple과 resultant force가 각각 발생하지 않는다



$$\begin{aligned} \Sigma F_x &= 0 & \therefore R_x &= 0 \\ \Sigma F_y &= 0 & R_y + R_A &= 1000 \text{ --- ①} \\ \Sigma M_o &= 0 & 2.2 \times R_A &= 3 \times 1000 \end{aligned}$$

Ans) $\begin{cases} R_A = 1363.6 \text{ N} \\ R_y = -363.6 \text{ N} \end{cases}$ ①에서부터

1.9

i) 위 실린더의 평형 조건

$$\Sigma F_x = 0 \quad F_p \cos 45^\circ = F_c$$

$$\Sigma F_y = 0 \quad F_p \sin 45^\circ = 900 \text{ N}$$

$$\therefore F_p = 1272.8 \text{ N}, \quad F_c = 900 \text{ N}$$

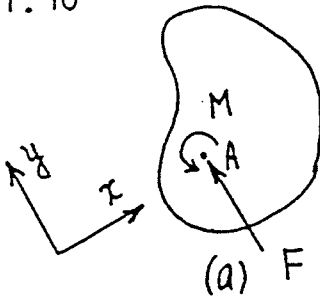
ii) 밑 받침대의 평형 조건

$$\sum F_x = 0 \quad F_A = F \cos 45^\circ = 900 \text{ N}$$

$$\sum F_y = 0 \quad F_B = 900 + F \sin 45^\circ = 1800 \text{ N}$$

$$\text{Ans.} \quad \begin{cases} F_A = 900 \text{ N} \\ F_B = 1800 \text{ N} \\ F_C = 900 \text{ N} \end{cases}$$

1.10



i) 그림 (a)에서의 평형 조건

$$\sum F_x = 0, \quad \sum F_y = F, \quad \sum M_A = M$$

ii) 그림 (b)에서의 평형 조건

$$\sum F_x = 0, \quad \sum F_y = F$$

$$\sum M_A = a \cdot F = \frac{M}{F} \cdot F = M$$

$\therefore$ 그림 (a)와 (b)로부터 힘의 평형식이 같

A점에 관한 moment 크기도 같다

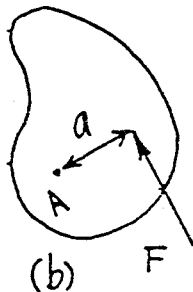
그러므로 그림 (a)와 같이 힘계에 힘과

모멘트가 작용하는 경우는 그림 (b)와 같이

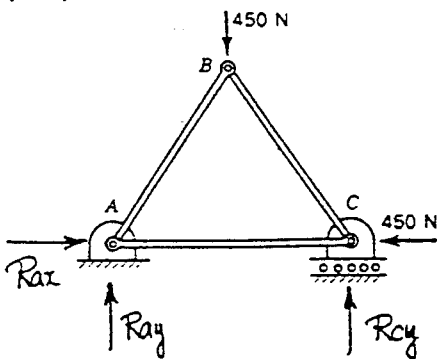
거리 $a = M/F$ 만큼 이동한 점에서

힘만을 가한 경우와 대등하다

$$a = \frac{|M|}{|F|}$$



1.11



i) A점에 관하여 ;

$$\sum M_A = 0 ; -450 \times \frac{l}{2} + R_{cy} \cdot l = 0$$

$$\sum F_x = 0 ; R_{ax} = 450 \text{ N}$$

$$\sum F_y = 0 ; R_{ay} = 225 \text{ N} \quad \left. \begin{array}{l} R_{ax} = 450 \text{ N} \\ R_{ay} = 225 \text{ N} \end{array} \right\} R_{cy} = 225 \text{ N}$$

ii) B점에 관하여 ;

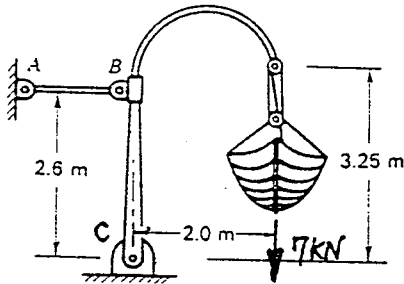
$$\sum F_x = 0 ; \therefore F_{AB} = F_{BC} \quad F_{AB} \quad 60^\circ \quad F_{BC}$$

iii) C점에 관하여 ; $\sum F_y = 0 ; F_{BC} \sin 60^\circ = 225$

$$\sum F_x = 0 ; F_{AC} - 450 + F_{BC} \cos 60^\circ = 0 \therefore F_{AC} = 320 \text{ N}, F_{BC} = 259.8 \text{ N}$$

$$\text{Ans.)} \quad F_{AB} = F_{BC} = 259.8 \text{ N}, \quad F_{AC} = 320 \text{ N}$$

1.12

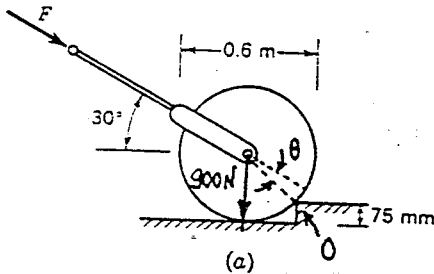
 $\Sigma M_C = 0$ 로부터

$$F_{AB} \times 2.6 - 7000 \times 2 = 0$$

$$\therefore F_{AB} = 5384.6 \text{ N}$$

$$\text{Ans.) } F_{AB} = 5384.6 \text{ N}$$

1.13

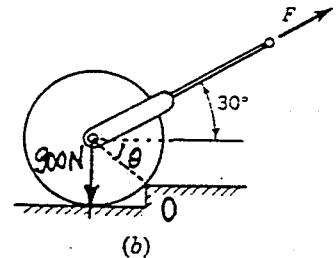


$$(a) \quad \theta = 60^\circ - \cos^{-1} \frac{30 - 7.5}{30} = 18.59^\circ$$

$$\Sigma M_O = 0 ; 900 \text{ N} \sqrt{0.3^2 - (0.3 - 0.075)^2}$$

$$- F \times 0.3 \sin 18.59^\circ = 0$$

$$\therefore F = 1867 \text{ N}$$



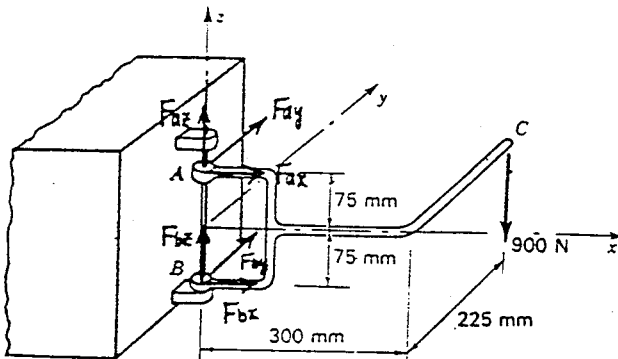
$$(b) \quad \theta = 30^\circ + \sin^{-1} \frac{30 - 7.5}{30} = 78.59^\circ$$

$$\Sigma M_O = 0 ; 900 \text{ N} \sqrt{0.3^2 - (0.3 - 0.075)^2}$$

$$- F \times 0.3 \sin 78.59^\circ = 0$$

$$\therefore F = 607 \text{ N}$$

1.14



연 구조물의 평형 조건으로부터

$$\Sigma F_x = F_{Ax} + F_{Bx} = 0$$

$$\Sigma F_y = F_{Ay} + F_{By} = 0$$

$$\Sigma F_z = F_{Az} + F_{Bz} - 900 = 0$$

$$\Sigma M_x = 0 ; F_{By} \times 0.15 - 900 \times 0.225 = 0$$

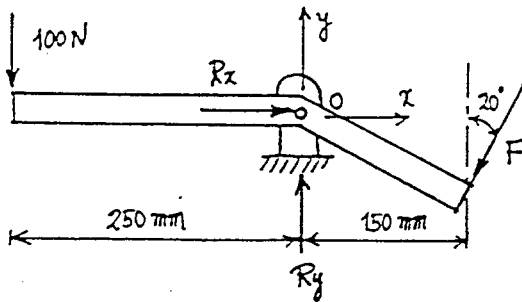
$$\therefore F_{By} = -F_{Ay} = 1350 \text{ N}$$

$$\Sigma M_y = 0 ; F_{Ax} \times 0.15 + 900 \times 0.3 = 0$$

$$\therefore F_{Ax} = -F_{Bx} = -1800 \text{ N} \quad , \quad F_{Bz} = 900 \text{ N} (\because F_{Az} = 0)$$

$$\text{Ans.) } F_{Ax} = -1800 \text{ N}, F_{Ay} = -1350 \text{ N}, F_{Az} = 0, F_{Bx} = 1800 \text{ N}, F_{By} = 1350 \text{ N}, F_{Bz} = 900 \text{ N}$$

1.15



$$\sum F_x = 0; -F \sin 20^\circ + R_x = 0$$

$$\therefore R_x = 53.6 \text{ N}$$

$$\sum F_y = 0; R_y - F \cos 20^\circ - 100 = 0$$

$$\therefore R_y = 247 \text{ N}$$

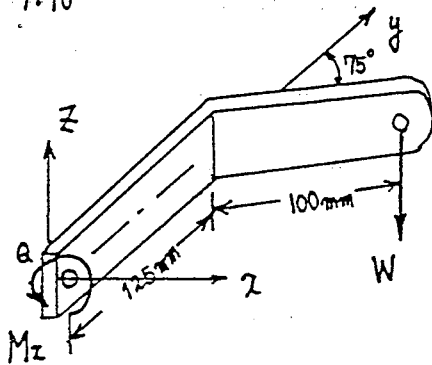
$$\sum M_o = 0;$$

$$100 \times 0.25 - F \times 0.15 \times \frac{1}{\cos 20^\circ} = 0$$

$$\therefore F = 156.6 \text{ N}$$

Ans.) $R_x = 53.6 \text{ N}$, $R_y = 247 \text{ N}$, $F = 156.6 \text{ N}$

1.16



$$T_{\max} = 100 \text{ (N}\cdot\text{m)}$$

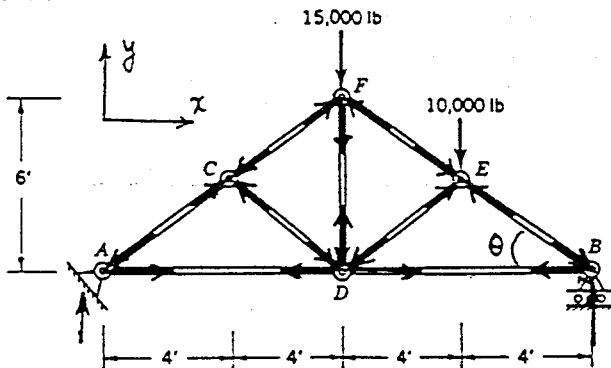
$$\sum M_{\max} = 0;$$

$$M_x - W(0.125 + 0.1 \cos 75^\circ) = 0$$

$$\therefore W_{\max} = \frac{100}{0.125 + 0.1 \cos 75^\circ} = 682.8 \text{ N}$$

Ans.) $W_{\max} = 682.8 \text{ N}$

1.17



Example 1-4로부터

$$F_{AC} = 16670 \text{ (comp.)}$$

$$F_{AD} = 13330 \text{ (tens.)}$$

$$F_B = 15000 \text{ (upward)}$$

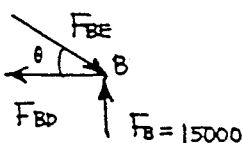
$$\sin \theta = \frac{3}{5}$$

$$\cos \theta = \frac{4}{5}$$

i) B점

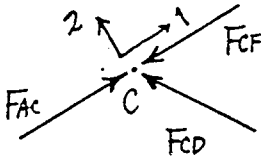
$$\sum F_y = 0; F_{BE} \sin \theta = 15000 \therefore F_{BE} = 25000$$

$$\sum F_x = 0; F_{BD} = F_{BE} \cos \theta \therefore F_{BD} = 20000$$



(뒷장 계류)

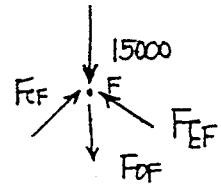
ii) C점



$$\sum F_x = 0; \therefore F_{CD} = 0$$

$$\sum F_y = 0; F_{AC} = F_{CF} = 16670$$

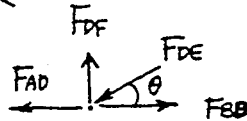
iii) F점



$$\sum F_x = 0;$$

$$F_{EF} = F_{CF} = 16670$$

iv) D점



$$\sum F_x = 0;$$

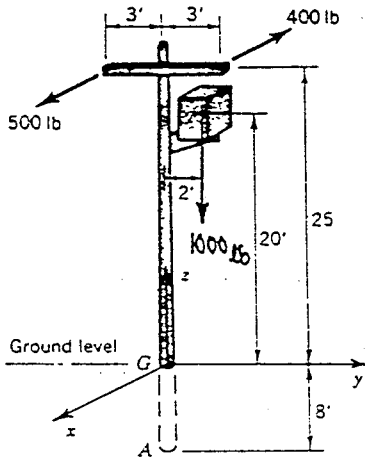
$$F_{DE} \cos \theta = F_{BD} - F_{AD} \therefore F_{DE} = \frac{5}{4} (20000 - 13330) = 8337.5$$

$$\sum F_y = 0; F_{DF} = F_{DE} \sin \theta = 8337.5 \cdot \frac{3}{5} = 5002.5$$

Ans.) $\left[F_{BD} = 20000 \text{ lb (ten.)} \quad F_{EB} = 25000 \text{ lb (comp.)} \right]$

$F_{CD} = 0, F_{DE} = 8337.5 \text{ lb (comp.)}, F_{CF} = F_{EF} = 16670 \text{ lb (comp.)}$

1.18



G에 작용하는 힘과 모멘트는 전선과 변압기에 의한 양으로 나눌 수 있다

i) 전선에 의한 힘과 모멘트 성분

$$F_{Gx} = 100 \vec{i}$$

$$M_{Gy} = (500 \times 25 - 400 \times 25) \vec{j} = 2500 \vec{j}$$

$$M_{Gz} = (3 \times 500 + 3 \times 400) \vec{k} = 2700 \vec{k}$$

ii) 변압기에 의한 성분

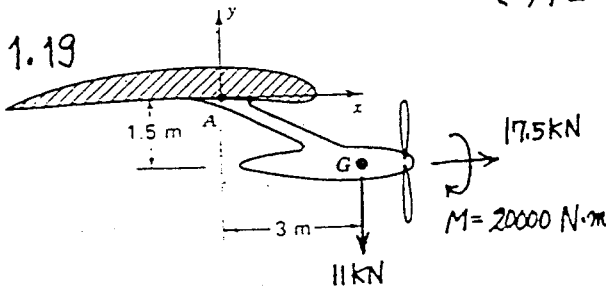
$$M_{Gz} = -1000 \vec{k}$$

$$M_{Gz} = -2 \times 1000 \vec{i} = -2000 \vec{i}$$

Ans.) G에서 작용하는 힘과 모멘트 $\left[\vec{F} = 100 \vec{i} - 1000 \vec{k} \text{ (lb)} \right]$

$\vec{M} = -2000 \vec{i} + 2500 \vec{j} + 2700 \vec{k} \text{ (ft-lb)}$

1.19



$$F_{Ax} = 17.5 \text{ kN}, F_{Ay} = -11 \text{ kN}$$

$$M_z = 20000 \text{ N}\cdot\text{m}$$

$$M_z = 17.5 \times 1.5 - 11 \times 3 = -6.75 \text{ (kN}\cdot\text{m)}$$

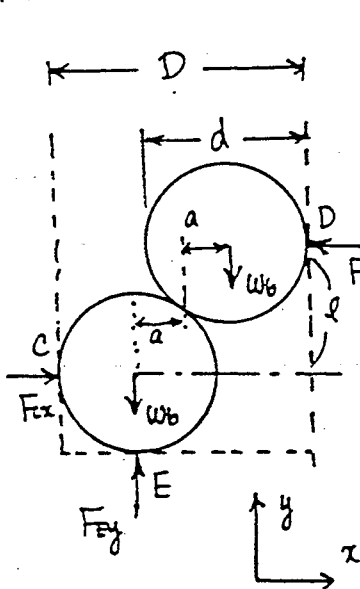
(뒷면이 계속)

$$\vec{F} = 17500 \vec{i} - 11000 \vec{j} \text{ (N)} \quad \vec{M} = 20000 \vec{i} - 6750 \vec{k} \text{ (N.m)}$$

Ans.) $\vec{F} = 17500 \vec{i} - 11000 \vec{j} \text{ (N)}, \vec{M} = 20000 \vec{i} - 6750 \vec{k} \text{ (N.m)}$

1.20

* 우선 구의 표면에 마찰력이 없다고 가정하면.



$$a = \frac{D-d}{2}, \quad l = 2\sqrt{\left(\frac{d}{2}\right)^2 - a^2} = \sqrt{D(2d-D)}$$

평형 조건식 ; $\sum F_y = 0 \quad F_{Ey} = 2W_b$

$$\sum F_x = 0 \quad F_{Cx} = F_{Dx}$$

$$\sum M_{Cz} = 0 ; (F_{Dx})l + (F_{Ey})\frac{d}{2} = W_b\frac{d}{2} + W_b\left(\frac{d}{2} + 2a\right)$$

또는 $F_{Cx} = F_{Dx} = \frac{W_b(D-d)}{l} = \frac{W_b(D-d)}{\sqrt{D(2d-D)}}$

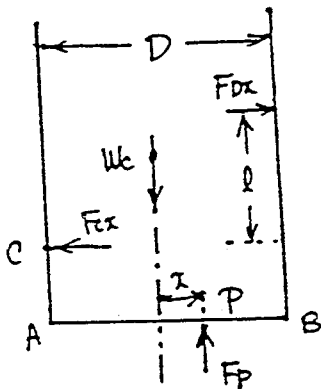
* AB 중심선으로부터 2만큼 떨어진 점에 작용하는

회전의 반력을 F_p 라 하면 $\sum M_{Bz} = 0 ;$

$$W_b \cdot \frac{D}{2} + F_{Cx} \cdot \frac{d}{2} = F_{Dx} \left(\frac{d}{2} + l\right) + F_p \left(\frac{D}{2} - x\right)$$

$$\sum F_y = 0 ; F_p = W_b$$

위 두식으로부터 $x = \frac{W_b}{W_c} (D-d)$



i) Orange juice can에 대하여 $D=50, d=45$

$$W_b = 2.0 \quad W_c = 0.57$$

$$x = \frac{2.0}{0.57} (50-45) = 17.5$$

$\therefore x < \frac{D}{2}$ 이므로 AB선상 내에 반력이 존재하며

따라서 "안정" 상태

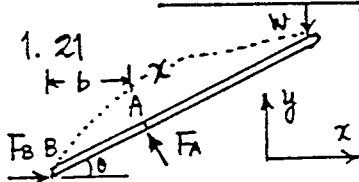
ii) Beer can $D=70, d=45 \quad W_b=2.0 \quad W_c=1.0$

$$x = \frac{2.0}{1.0} (70-45) = 50. \quad \therefore x > \frac{D}{2}$$

(Reaction) 큰 반력이 AB선상 밖에 존재하므로 평형을 이룰수 없다

따라서 "불안정"

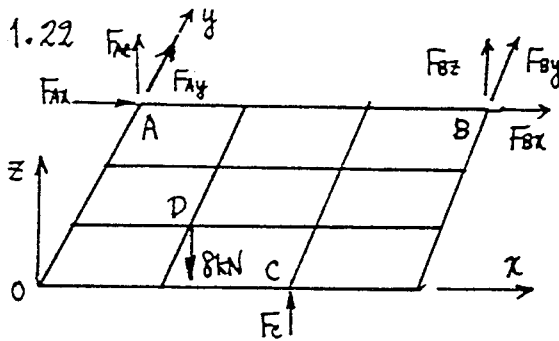
Ans.) Orange juice can ; 안정 , Beer can ; 불안정



$$\sum F_x = 0 ; F_B = F_A \sin \theta$$

$$\sum F_y = 0 ; F_A \cos \theta = W$$

$$\sum M_{Bz} = 0 ; F_A b / \cos \theta = W x \cos \theta$$



가정

- a) No friction at hinges
- b) prop is rigid
- c) No friction at prop

$$\sum F_z = 0 ; F_{Ax} + F_{Bx} = 0$$

부등방성문제에서 주어진 가정을 취하면

$$F_{Ax} = F_{Bx} = 0$$

$$\sum F_y = 0 ; F_{Ay} + F_{By} = 0 \quad , \quad \sum F_z = 0 ; F_{Az} + F_{Bz} + F_c = 8000$$

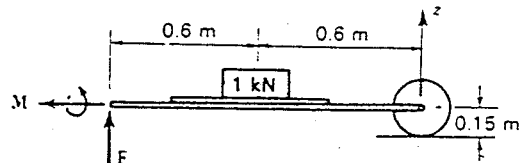
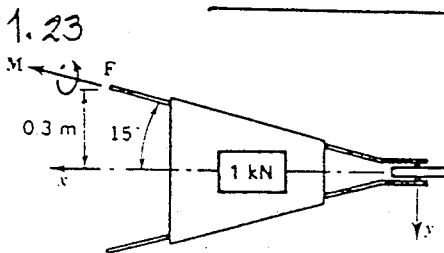
$$\sum M_{Bz} = 0 ; \therefore F_{Ay} = 0 \quad , \quad F_{By} = 0$$

$$\sum M_{Ay} = 0 ; F_{Bz} \times 3.9 + F_c \times 2.6 = 8000 \times 1.3$$

$$\sum M_{Az} = 0 ; 8000 \times 2 = F_c \times 3$$

위치를 각각 풀면 F를 구할 수 있다.

Ans.) $\left[\begin{array}{l} F_{Bz} = -888.7 \text{ N} , F_{Az} = 3555.7 \text{ N} , F_c = 5333 \text{ N} \\ F_{Ax} = F_{Bx} = F_{Ay} = F_{By} = 0 \end{array} \right.$

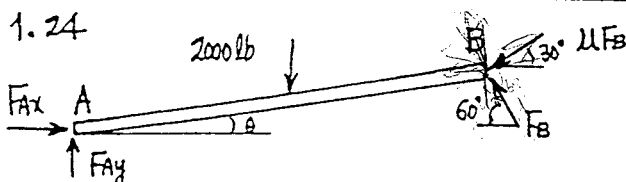


$$\sum M_y = 0 ; 0.6 \times 1000 = 1.2 F + M \sin 15^\circ$$

$$\sum M_z = 0 ; 0.3 F = M \cos 15^\circ \quad \therefore M (4 \cos 15^\circ + \sin 15^\circ) = 600$$

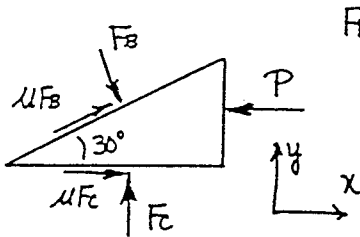
$$\therefore M = 145.54 \text{ (N}\cdot\text{m)}, \quad F = 468.6 \text{ (N)}$$

Ans.) $M = 145.54 \text{ N}\cdot\text{m} , F = 468.6 \text{ N}$



a) $\sum M_A = 0 ;$

$$2000 \times 2 = F_B \times 4 \sin 60^\circ + \frac{1}{2} F_B \times 4 \sin 30^\circ$$



$$F_B = 1396.6 \text{ N}$$

쇠끼의 평형 상하로부터

$$\sum F_y = 0; F_C - F_B \cos 30^\circ + \mu F_B \sin 30^\circ = 0$$

$$\sum F_x = 0; -P + \mu F_C + (\mu \cos 30^\circ + \sin 30^\circ) F_B = 0$$

구해면 F_B 로부터

$$F_C = 1000 \text{ lb}$$

$$P = 1000 \left[\mu + \frac{0.866\mu + 0.5}{0.866 - 0.5\mu} \right], \mu = 0.3 \text{ 이라면 } P = 1364 \text{ lb}$$

b) 쇠끼가 미끄러져 내려가는 것을 방지하려면 $P=0$ 인 외력이 없
상하에서 마찰력의 작용 방향은 (a)의 경우와 반대가 되므로, 등식을 세우면

$$0 = 1000 \left[-\mu + \frac{0.5 - 0.866\mu}{0.866 + 0.5\mu} \right] \text{ 가 된다.}$$

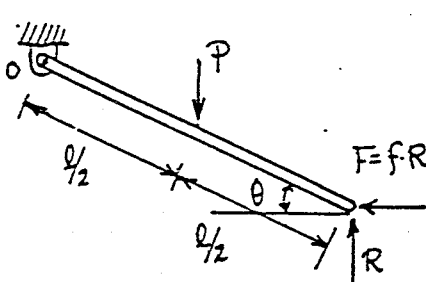
이식을 풀면 $\mu = 0.27$

Ans.) a) $P = 1364 \text{ lb}$

b) $\mu = 0.27$

1.25

a)



$$\sum M_0 = 0;$$

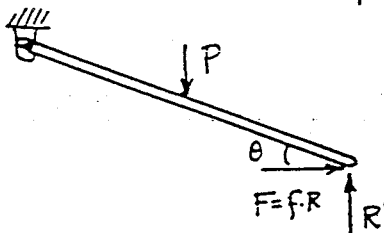
$$P \cdot \frac{l}{2} \cos \theta + F \cdot l \sin \theta = R \cdot l \cos \theta$$

$$F = fR \text{ 이므로}$$

$$\frac{P}{2} \cos \theta = \frac{F}{f} \cos \theta - F \sin \theta$$

$$\therefore F = \frac{Pf \cos \theta}{2(\cos \theta - f \sin \theta)} \quad \dots \textcircled{1}$$

b)



$$\sum M_0 = 0;$$

$$P \cdot \frac{l}{2} \cos \theta = R \cdot l \cos \theta + fR \cdot l \sin \theta$$

$$\frac{P}{2} \cos \theta = \frac{F}{f} (\cos \theta + f \sin \theta) \quad (\because R = \frac{F}{f})$$

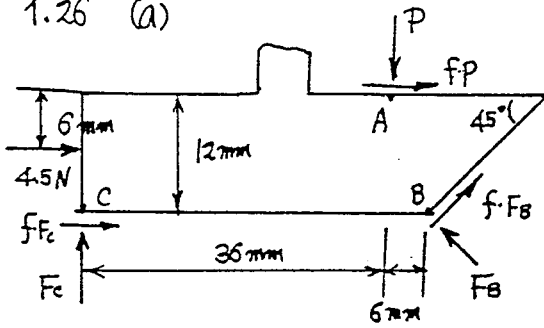
$$\therefore F = \frac{Pf \cos \theta}{2(\cos \theta + f \sin \theta)} \quad \dots \textcircled{2}$$

(a)의 경우와 마찰력이 (b)의 경우 보다 크며 (a)의 경우에 friction lock가 생길 수 있다.

Ans.) a) $F = \frac{Pf \cos \theta}{2(\cos \theta - f \sin \theta)}$

b) $F = \frac{Pf \cos \theta}{2(\cos \theta + f \sin \theta)}$

1.26 (a)



$$\Sigma M_B = 0 ;$$

$$6P = 36F_c + 12f \cdot P + 4.5 \times 6$$

$$\Sigma F_x = 0 ;$$

$$4.5 + fP + fF_c + fF_B \cos 45^\circ = F_B \cos 45^\circ$$

$$\Sigma F_y = 0 ;$$

$$F_c + F_B \sin 45^\circ + fF_B \cos 45^\circ = P$$

$$\therefore P - 1.3 \cos 45^\circ F_B - F_c = 0$$

연립하여 풀면 $P = 21.2 \text{ N}$, $F_B = 22.34 \text{ N}$, $F_c = 0.6633 \text{ N}$
 만약 문의 hinge 까지 거리를 L이라 하고 hinge로부터 모우먼트를
 생각하면 $\Sigma M_{\text{hinge}} = 0$;

$$PL = F_B \sin 45^\circ (L+6) + fF_B \sin 45^\circ (L+6)$$

$$= F_B (1+f) (L+6) \sin 45^\circ$$

$$L+6 \cong L \quad (\because L \gg 6 \text{ mm})$$

$$\therefore P = F_B (1+f) \sin 45^\circ = 22.34 \times 1.3 \times \sin 45^\circ = 20.54 \text{ (N)}$$

(b) $f=0$ 인 경우에 대하여 같은 수를 유도하면

$$P = 6F_c + 4.5$$

$$4.5 = F_B \cos 45^\circ$$

$$P = F_c + F_B \sin 45^\circ$$

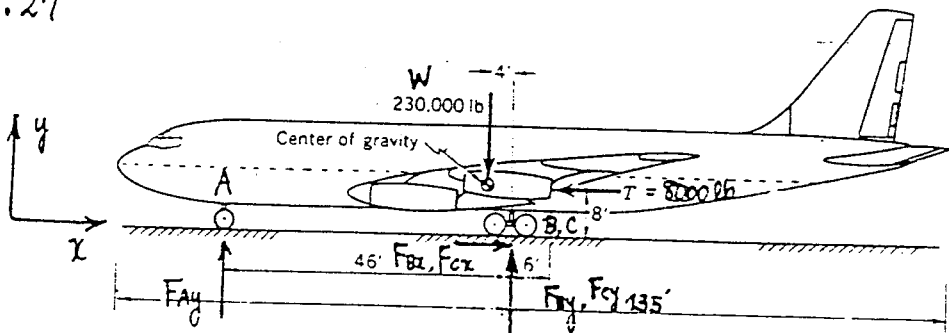
$$\therefore F_B = 6.364 \text{ (N)}$$

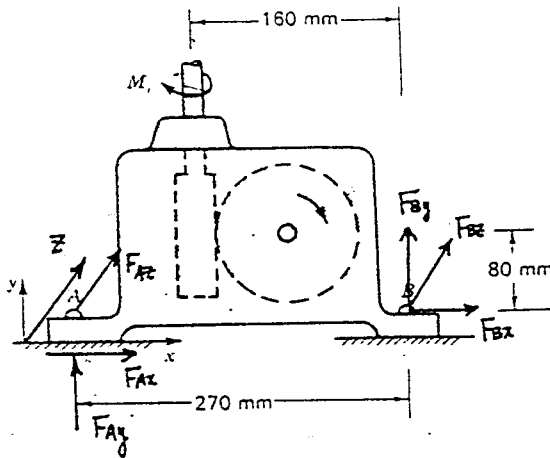
$$\text{또한 } P = F_B \sin 45^\circ \text{ 이므로}$$

$$P = 6.364 \times \sin 45^\circ = 4.5 \text{ (N)}$$

$$\text{Ans.) (a) } P = 20.54 \text{ N} \quad (b) \quad P = 4.5 \text{ N}$$

1.27





$$\Sigma F_y = 0; F_{Ay} + F_{By} = 0$$

$$\Sigma F_z = 0; F_{Az} + F_{Bz} = 0$$

$$\therefore F_{Az} = -F_{Bz} = \frac{M_i}{0.27} = \frac{15}{0.27}$$

$$= 55.56 \text{ (N)}$$

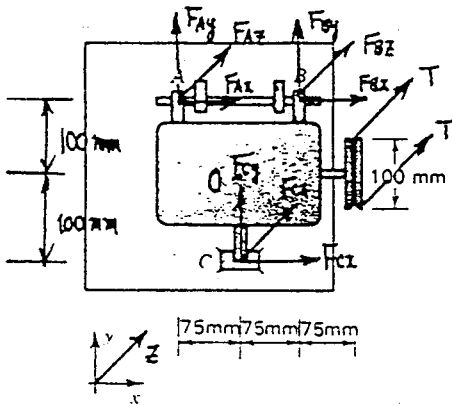
$$F_{By} = -F_{Ay} = \frac{M_o}{0.27} = \frac{50 \times 15}{0.27}$$

$$= 2777.8 \text{ (N)}$$

$$(\because M_i = 15 \text{ N}\cdot\text{m}, M_o = 50 M_i)$$

Ans.) $F_{Az} = -F_{Bz} = 55.56 \text{ N}, F_{By} = -F_{Ay} = 2777.8 \text{ N}$

1.29



$$W = 80 \text{ N}, T = 125 \text{ N}$$

$$\Sigma F_x = 0; F_{Ax} + F_{Bx} + F_{Cx} = 0$$

$$\Sigma F_y = 0; F_{Ay} + F_{By} + F_{Cy} = 0$$

$$\Sigma F_z = 0; F_{Az} + F_{Bz} + F_{Cz} = 2T$$

$$\Sigma M_{xO} = 0;$$

$$(F_{Az} + F_{Bz}) \times 0.1 + W \times 0.1 = F_{Cz} \times 0.1 + 2T$$

$$\Sigma M_{yB} = 0;$$

$$2T \times 0.075 + F_{Cz} \times 0.075 + F_{Az} \times 0.15 = 0$$

$$\Sigma M_{zO} = 0; F_{Ay} \times 0.075 = F_{By} \times 0.075$$

$$F_{Ax} = F_{Bx} = F_{Cx} = 0 \text{ (By assumption, } F_{Ax}, F_{Bx}, F_{Cx} \text{; indeterminate)}$$

$$F_{Cy} = 0 \text{ (}\because \text{C point is fixed in the y-direction, so it is not possible)}$$

$$\Sigma F_y = 0 \text{ 와 } \Sigma M_{zO} = 0 \text{ 에서}$$

$$F_{Ay} + F_{By} = W, F_{Ay} = F_{By} \therefore F_{Ay} = F_{By} = \frac{W}{2} = 40 \text{ N}$$

$$\Sigma F_z = 0, \Sigma M_{xO} = 0, \Sigma M_{yB} = 0 \text{ 에서}$$

$$F_{Az} + F_{Bz} + F_{Cz} = 2T \quad \dots \textcircled{1}$$

$$F_{Az} + F_{Cz} = -2T \quad \dots \textcircled{2}$$

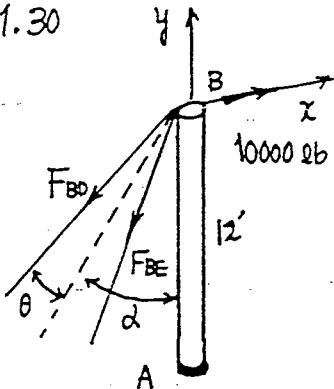
$$F_{Az} + F_{Bz} - F_{Cz} = 2T - W \quad \dots \textcircled{3}$$

①②③으로 부터 $F_{Cz} = 151.5 \text{ N}$, $F_{Bz} = 299.25 \text{ N}$, $F_{Az} = -200.75 \text{ N}$

$\therefore \vec{F}_A = 40\vec{j} - 200.75\vec{k}$, $\vec{F}_B = 40\vec{j} + 299.25\vec{k}$, $\vec{F}_C = 151.5\vec{k}$

Ans.) $\left[\begin{array}{l} \vec{F}_A = 40\vec{j} - 200.75\vec{k} \text{ (N)} \\ \vec{F}_B = 40\vec{j} + 299.25\vec{k} \text{ (N)}, \vec{F}_C = 151.5\vec{k} \text{ (N)} \end{array} \right]$

1.30



$\Sigma M_{Az} = 0$;

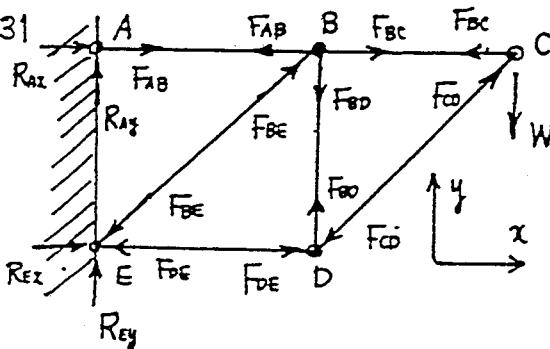
$2 F_{BD} \cos \theta \cdot \sin \alpha \times 12 = 10000 \times 12$

$\sin \alpha = \frac{10}{\sqrt{10^2 + 12^2}} \quad \cos \theta = \frac{\sqrt{10^2 + 12^2}}{\sqrt{8^2 + 10^2 + 12^2}}$

$\therefore F_{BD} = F_{BE} = 8775 \text{ N}$

Ans.) $F_{BD} = F_{BE} = 8775 \text{ N}$

1.31



$\Sigma M_E = 0$; $2W = -R_{Ax}$

$\Sigma F_x = 0$; $R_{Ex} = -R_{Ax} = 2W$

$\Sigma F_y = 0$; $R_{Ay} + R_{Ey} = W$

CE; $\Sigma F_y = 0$;

$F_{CD} \cos 45^\circ = W$

$\therefore F_{CD} = 1.41 W$ (압축력)

$\Sigma F_x = 0$; $F_{BC} = F_{CD} \cos 45^\circ = W$ (인장력)

D점, $\Sigma F_y = 0$; $F_{BD} = F_{CD} \sin 45^\circ = W$ (인장력)

$\Sigma F_x = 0$; $F_{ED} = F_{CD} \cos 45^\circ = W$ (압축력)

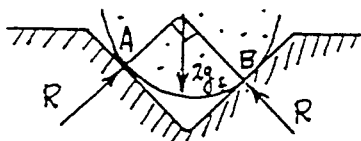
B점, $\Sigma F_y = 0$; $F_{BE} \cos 45^\circ = F_{BD}$ $\therefore F_{BE} = 1.41 W$ (압축력)

$\Sigma F_x = 0$; $F_{BA} = F_{BC} + F_{BE} \sin 45^\circ$ $\therefore F_{AB} = 2W$ (인장력)

Ans.) $F_{AB} = 2W$, $F_{BD} = F_{BC} = W$ (인장력), $F_{BE} = F_{CD} = 1.41 W$, $F_{DE} = W$

(압축력)

1.32



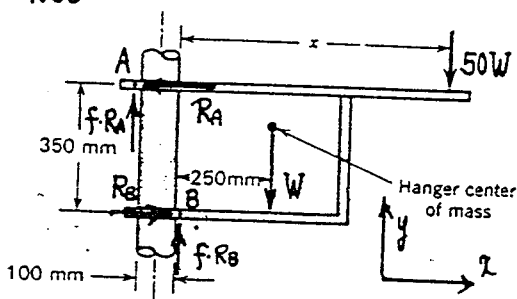
A, B점에서의 반력

$R = 2 \times 980 \times \cos 45^\circ = 1386 \text{ dyn}$

AB에서의 마찰력 ; $f = 0.2 \times R = 0.2 \times 1386 = 277 \text{ (dyn)}$

Ans.) $R = 1386 \text{ dyn}$, $f = 277 \text{ dyn}$

1.33



$\Sigma M_B = 0 ;$

$250W + 50Wx + 100fR_A = 350R_A$

$\Sigma F_x = 0 ; R_A = R_B$

$\Sigma F_y = 0 ; f(R_A + R_B) = 51W$

두개를 연결하면

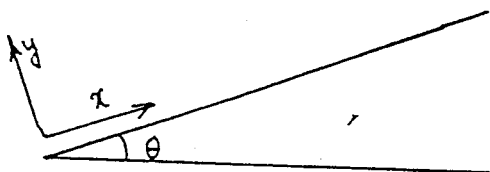
$R_A = R_B = \frac{51W}{2 \times 0.3} = 85W$

$(250 + 50x)W = (350 - 100 \times 0.3)R_A = 27200W$

$\therefore x = 539 \text{ (mm)}$

Ans.) 539 mm

1.34



무엇, 차의 마찰이 충분히 커서

차가 오를 수 있는 경사각은 마찰력이
좌우되다고 가정 하면,

$\Sigma M_A = 0 ;$

$LN_2 = 0.4L \cdot W \cos \theta - hW \sin \theta$

$\Sigma F_x = 0 ; fN_1 = W \sin \theta$

($\because$ 뒷바퀴에만 동력이 걸림)

$\Sigma F_y = 0 ; N_1 + N_2 = W \cos \theta$

$\Sigma M_B = 0 ; N_1 = (0.6 \cos \theta + \frac{h}{L} \sin \theta) W$

$\Sigma M_A = 0 ; N_2 = (0.4 \cos \theta - \frac{h}{L} \sin \theta) W$

$\therefore f(0.6 \cos \theta + \frac{h}{L} \sin \theta) = \sin \theta$

$\tan \theta = \frac{0.6L \cdot f}{L - f \cdot h} = \frac{0.6 \times 100 \times 0.7}{100 - 0.7 \times 2 \times 12} = 0.504$

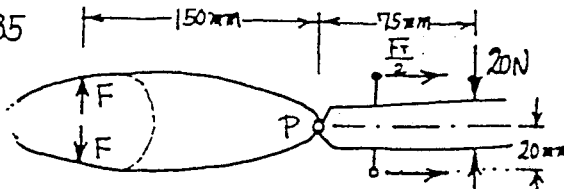
$\therefore \theta = 26.8$

만약 마찰이 충분히 못하다면 오를 수 있는 경사각은 다른
결정된다.

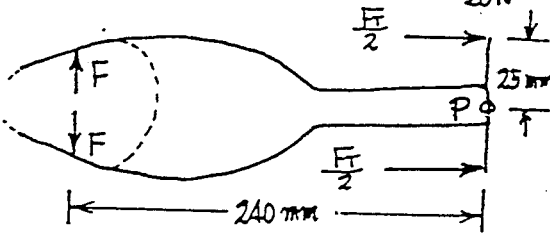
Ans.) 26

1.35

a)



b)



a) $\sum M_P = 0$;

$$150 \times F = \frac{120}{2} \times 20 + 75 \times 20$$

$$\therefore F = 18 \text{ (N)}$$

b) $\sum M_P = 0$

$$-\frac{F}{2} \times 25 + F \times 240 = 0$$

$$\therefore F = -6.25 \text{ (N)}$$

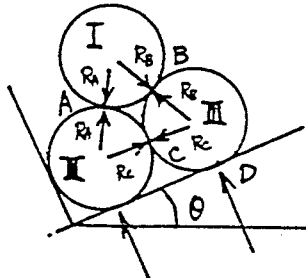
(b)의 경우 forcep blade가

테아의 머리를 누르는 힘이

없으며, birth canal이 테아에 작용하는 힘과 서로 상쇄되는 결과가
되므로 테아에 좋은 forcep은 (b)의 Divergent-lever forcep이다.

Ans.) { a) 18N b) -6.25N
Divergent-lever forcep이 좋다.

1.36



i) I통나무에 대해서

$$\sum F_x = 0 ; R_A \sin(30-\theta) = R_B \sin(30+\theta)$$

$$\sum F_y = 0 ; W = R_A \cos(30-\theta) + R_B \cos(30+\theta)$$

$$\therefore R_A = \frac{W \sin(30+\theta)}{\sin 60^\circ}$$

$$\therefore R_B = \frac{W \sin(30-\theta)}{\sin 60^\circ}$$

ii) III통나무에 대해서

$$\sum F_x = 0 ; R_C + R_B \sin 30 = W \sin \theta$$

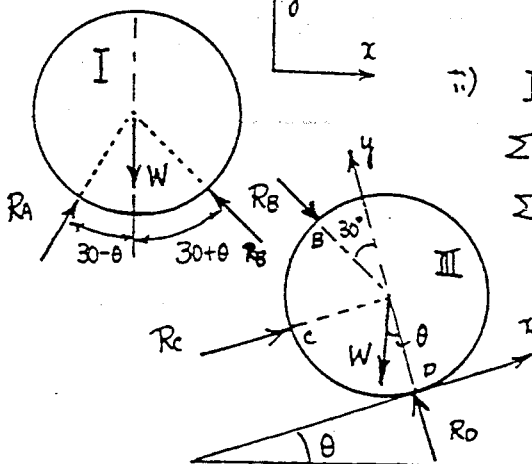
$$\sum F_y = 0 ; R_D = W \cos \theta + R_B \cos 30^\circ$$

$R_C \geq 0$ 이라야 원상태 유지

$R_C = 0$ 이면

$$R_B \sin 30^\circ = W \sin \theta$$

$$\frac{\sin(30-\theta)}{\sin 60^\circ} = \frac{\sin \theta}{\sin 30^\circ}$$



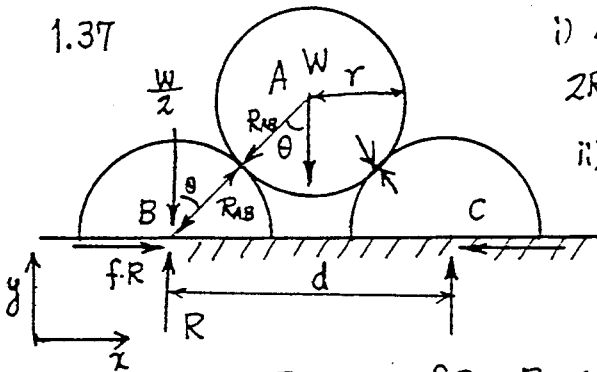
$$\tan \theta = \left(\frac{1}{2} - \frac{\sqrt{3}}{2} \tan \theta \right) \frac{1}{\sqrt{3}}$$

$$\theta = 10.89^\circ$$

$$\therefore \theta = \tan^{-1} \frac{1}{3\sqrt{3}}$$

Ans.) 10.89°

1.37



1) A7에 대하여

 $2R_{AB}$

ii) B7에 대하여

$$\Sigma F_y = 0; \quad R = \frac{W}{2} + R_{AB} \cos \theta$$

$$R_{AB} = \frac{W}{2 \cos \theta} \quad (\because R=W)$$

$$\Sigma F_z = 0; \quad f \cdot R = R_{AB} \sin \theta = \frac{W}{2} \tan \theta$$

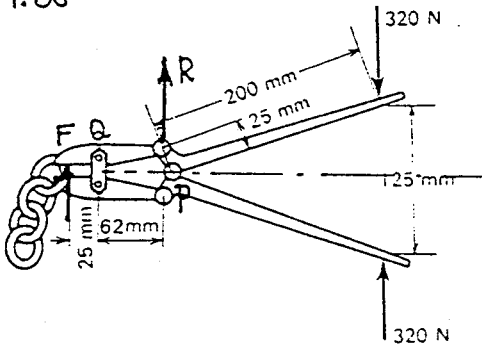
$$\therefore f = \frac{1}{2} \tan \theta = \frac{\frac{d}{2}}{\sqrt{(2r)^2 - \left(\frac{d}{2}\right)^2}} = \frac{d}{2\sqrt{4r^2 - d^2}}$$

 d 를 f 와 g 로 표현하면

$$d = \frac{8fr}{\sqrt{1+4f^2}}$$

Ans.) $d_{\max} = \frac{2.71}{\sqrt{1 + 4f^2}}$

1.38



Q점에 대한 모우멘트를 생각하면

$F \times 25 = R \times 62$
 $R = \frac{125}{2} \text{ mm}$

$$R \times 25 \sin \theta = \sqrt{200^2 - \left(\frac{125}{2}\right)^2} \times 320$$

$$\sin \theta = \frac{\frac{125}{2}}{200}$$

$$\therefore R = 7781.7 \text{ N}$$

$$F = \frac{62}{25} \times R = 19298.7 \text{ N}$$

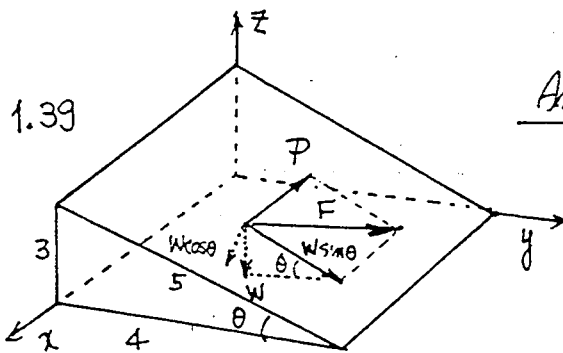
Ans.) 19298.7 N

$$F = \sqrt{p^2 + (W \sin \theta)^2}$$

$$= \sqrt{(0.4W)^2 + (0.6W)^2}$$

$$= 0.72 \text{ W}$$

1.39



블록이 움직이지 위해서는 F_f 마찰력 f 보다 커야 한다

$$R_f = f \cdot W \cos \theta$$

$$0.72W \geq f \cdot W \cos \theta$$

$$f \leq 0.72 \times \frac{5}{4}$$

$$\therefore f \leq 0.9$$

Ans.) $f \leq 0.9$

1.40 가정; Post는 pin-joint이므로 급힘 모멘트를 받지 못한다

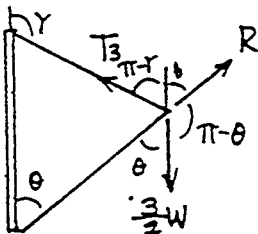
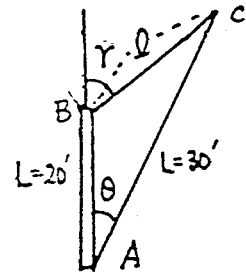
i) γ 와 θ 의 관계식

$$l \sin \gamma = 30 \sin \theta \quad \dots ①$$

$$-l \cos \gamma + 30 \cos \theta = 20 \quad \dots ②$$

$$① \rightarrow ② ; -\cot \gamma \sin \theta + \cos \theta = \frac{2}{3} \quad \dots ③$$

ii) $\triangle ABC$ 를 포괄하는 직면에 대하여



$\overline{DB}$, $\overline{BE}$, $\overline{BC}$ 에 걸리는 장력을 T_1 , T_2 , T_3 라 한다

$$\sum F_x = 0; T_3 \sin(\pi - \gamma) = R \sin \theta \quad \dots ④$$

$$\sum F_y = 0; T_3 \cos(\pi - \gamma) + R \cos \theta = \frac{3}{2}W \quad \dots ⑤$$

$$④⑤에서 T_3 = \frac{1500}{(\cot \theta \sin \gamma - \cos \gamma)} \leq 1200 \quad \dots ⑥$$

$$\therefore \cot \theta \sin \gamma - \cos \gamma \geq \frac{5}{4} \quad \dots ⑦$$

iii) B점을 포괄하며 직면 평행한 평면에 대하여

(참고; 2의 범위는 T_1, T_2 가 마찰이므로 양의 값으로 존재해야 함) (다음 표까지 계속)

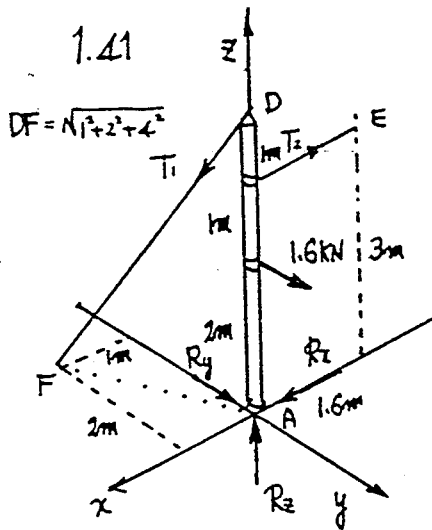
$\theta (^{\circ})$	$\frac{\cos \theta - \frac{5}{4}}{\sin \theta}$	$\gamma (^{\circ})$	$\cot \theta \sin \gamma - \cos \gamma$	α (도)	2의 범위
5	3.781	14.8	1.953	임의값	$0 < \alpha < 60^{\circ}$
10	1.832	28.63	1.839	"	$0 < \alpha < 60^{\circ}$
20	0.798	51.41	1.523	"	"
24	0.607	58.74	1.401	"	"
25	0.567	60.45	1.372	56.83	$3.17 < \alpha < 56.83$
27	0.494	63.70	1.316	51.18	$8.82 < \alpha < 51.18$
29	0.429	66.78	1.264	46.88	$13.12 < \alpha < 46.88$
30	0.399	68.25	1.238	$\Rightarrow T_3$ 가 한계 하중을 넘는다.	

$$\frac{|T_2 \cos 40^\circ|}{\sin(\pi - \alpha)} = \frac{|T_3 \sin \theta|}{\sin 60^\circ} \quad (\because \text{라미의 정리에 의해})$$

$$T_2 = \frac{\sin \alpha \sin \theta}{\cos 40^\circ \sin 60^\circ} T_3 \leq 1200 \dots \textcircled{8}$$

③, ⑦, ⑧로부터 θ, r, α 의 허용치를 구할 수 있다.

즉, $\theta = 30^\circ$ 를 보면 T_3 가 한계 하중을 초과하여 부정확하다.



$$\sum M_x = 0; 1600 \times 2 = \frac{\sqrt{5}}{\sqrt{21}} \times \frac{2}{\sqrt{5}} \times T_1 \times 4$$

$$\sum M_y = 0; T_2 \times 3 = \frac{\sqrt{5}}{\sqrt{21}} \times \frac{1}{\sqrt{5}} \times T_1 \times 4$$

$$\text{위 두 식을 연립하여 } T_1 = 1833 \text{ N}$$

$$T_2 = 533 \text{ N}$$

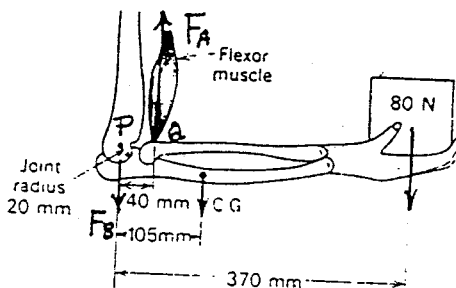
$$\sum F_x = 0; R_x = T_2 - \frac{1}{\sqrt{21}} T_1 = 133 \text{ N}$$

$$\sum F_y = 0; R_y = \frac{2}{\sqrt{21}} T_1 - R_z = -800 \text{ N}$$

$$\sum F_z = 0; R_z = T_1 \cdot \frac{4}{\sqrt{21}} = 1600 \text{ N}$$

Ans.) $\left[\begin{array}{l} T_1 = 1833 \text{ N}, T_2 = 533 \text{ N} \\ R_x = 133 \text{ N}, R_y = -800 \text{ N}, R_z = 1600 \text{ N} \end{array} \right]$

1.42



i) 만약 팔꿈치에 마찰이 없다면

$$\sum M_P = 0;$$

$$F_A \times 40 = 16 \times 105 + 80 \times 370$$

$$\therefore F_A = 782 \text{ N}$$

$$\sum M_B = 0;$$

$$F_B \times 40 = 16 \times 65 + 80 \times 330$$

$$\therefore F_B = 686 \text{ N}$$

ii) 마찰 계수 $f = 0.015$ 이라면

(a) 하중을 들어 올릴 때

$$\sum M_P = 0; F_A \times 40 = 16 \times 105 + 80 \times 370 + 0.015 \times F_B \times 20$$

$$\sum F_y = 0; F_A = F_B + 16 + 80 \quad \therefore F_B = F_A - 96$$

두부를 연결하면 $F_A \times 40 = 16 \times 105 + 80 \times 370 + 0.015 \times (F_A - 96) \times 20$
 $F_A = 787 \text{ N}$ $\therefore F_B = 787 - 96 = 691 \text{ N}$

(b) 하중을 단순히 지지하고 있을 때

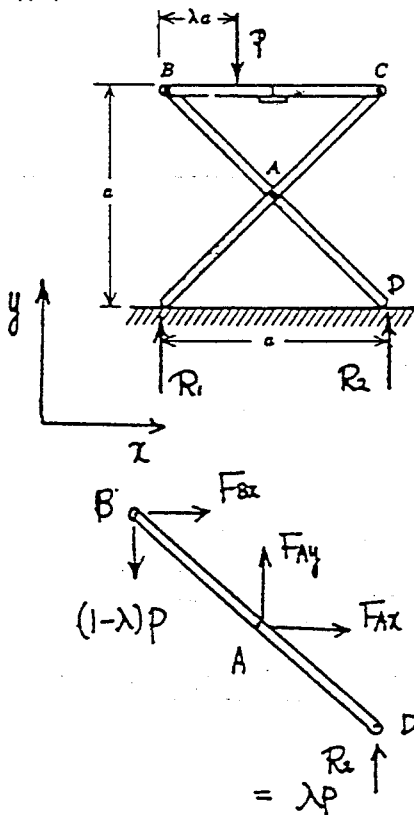
지지하는 상하부에는 마찰력이 작용하지 않으므로 $F_{\text{friction}} = 0$

$\therefore F_A = 782 \text{ N}$

Ans.) i) $F_A = 782 \text{ N}$, $F_B = 686 \text{ N}$

ii) (a) $F_A = 787 \text{ N}$, $F_B = 691 \text{ N}$, (b) $F_A = 782 \text{ N}$, $F_B = 686 \text{ N}$

1.43



a)

$\sum F_y = 0$; $R_1 + R_2 = P$

$\sum M_B = 0$; $a R_2 = \lambda a P$

$\therefore R_2 = \lambda P$, $R_1 = (1 - \lambda)P$

* Bar BD의 평형 조건

$\sum F_x = 0$; $F_{Ax} + F_{Bx} = 0$

$\sum F_y = 0$; $F_{Ay} = (1 - \lambda)P - \lambda P$
 $= (1 - 2\lambda)P$

$\sum M_A = 0$; $F_{Bx} \cdot \frac{a}{2} - (1 - \lambda)P \cdot \frac{a}{2} - \lambda P \cdot \frac{a}{2} = 0$ $\therefore F_{Bx} = P$

$\therefore \begin{cases} F_{Ax} = -P \\ F_{Ay} = (1 - 2\lambda)P \end{cases}$

$F_A = \sqrt{F_{Ax}^2 + F_{Ay}^2} = P \sqrt{2 - 4\lambda + 4\lambda^2}$

b) λ 에 대해서 최대인 값을 구하려면

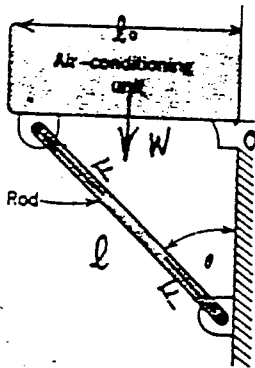
$\frac{dF_A}{d\lambda} = P \frac{8\lambda - 4}{2\sqrt{2 - 4\lambda + 4\lambda^2}} = 0$ 이므로

$\lambda = \frac{1}{2}$ 이면 F_A 는 최소, 최대는 $\lambda = 0$, $\lambda = 1$ 인 경계에서

$F_{\text{max}} = \sqrt{2}P$

Ans.) a) $F_A = \sqrt{2 - 4\lambda + 4\lambda^2} P$, b) $F_{\text{max}} = \sqrt{2}P$ ($\lambda = 0, \lambda = 1$)

1.44



Rod에 걸리는 압축력을 F 라 하면,

$$\sum M_O = 0; F \cos \theta = \frac{W}{2} \therefore F = \frac{W}{2 \cos \theta}$$

바람이 가장 뜨게 되기 위해서는 재료가

뜨게 소모 되야 하므로 $\frac{dV}{d\theta} = 0$ 인

θ 값을 찾으면 된다. ($l_0 = l \sin \theta$)

$$V = \pi r^2 l = \pi r^2 \frac{l_0}{\sin \theta} \dots \textcircled{1}$$

$$F = A \cdot \sigma_r = \pi r^2 \sigma_r = \frac{W}{2 \cos \theta} \dots \textcircled{2}$$

① ② 식으로부터 V 와 θ 에 대해서 정리하면

$$V \sin \theta \cdot 2 \cos \theta = \frac{W l_0}{\sigma_r} = C \dots \textcircled{3} (\because W, l_0 \text{은 주어진 값, } \sigma_r \text{은 압축 응력})$$

③ 식을 변형 V 와 θ 에 대해서 미분하면 $V \sin 2\theta = C$ 로 부터

$$dV \sin 2\theta + 2V \cos 2\theta d\theta = 0 \quad \frac{dV}{d\theta} = -2V \frac{\cos 2\theta}{\sin 2\theta} = 0 \quad (0 < \theta < \frac{\pi}{2})$$

$$\therefore \cos 2\theta = 0$$

$$\theta = \frac{\pi}{4}$$

Ans.) $\theta = \frac{\pi}{4}$

1.45 i) 트렁크를 밀어 올리기 시작 할때 뜨는 힘 ($\theta = 30^\circ$)

$$F = W \sin 30^\circ + f W \cos 30^\circ \dots \textcircled{1}$$

ii) 트렁크가 밀려 내려오지 못하게 버티는 힘 ($\theta = 60^\circ$)

$$F = W \sin 60^\circ - f W \cos 60^\circ \dots \textcircled{2}$$

① = ② 에서

$$f(\cos 30^\circ + \cos 60^\circ) = \sin 60^\circ - \sin 30^\circ$$

$$\therefore f = 0.268$$

Ans.) $f = 0.268$

$$\sum F_x = 0; F \sin \theta = f \cdot F \cos \theta$$

$$\therefore f = \tan \theta$$

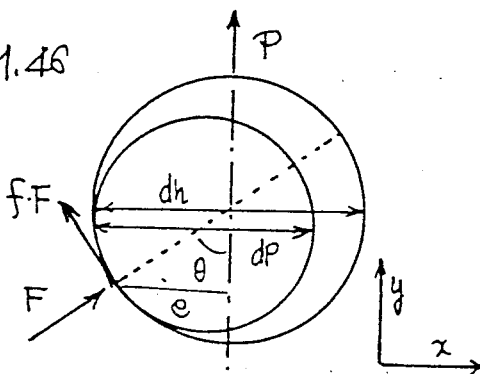
$$\sum F_y = 0; F \cos \theta + f \cdot F \sin \theta = P$$

$$\therefore F = \frac{P \cos \theta}{\cos \theta + f \sin \theta} \quad f = \tan \theta = 0.15 \quad \therefore \theta = 8.53^\circ$$

$$e = \frac{dh}{2} \sin \theta = \frac{0.255}{2} \sin 8.53^\circ$$

$$= 0.0189 \text{ (in)} \quad \text{Ans.) } e = 0.0189 \text{ (in)}$$

1.46



The diagram illustrates a tapered wing aircraft in a coordinate system with x and y axes. The wing is shown in plan view, with the leading edge at the origin. Key dimensions and parameters include:

- Wing Dimensions:** The total span is $130'$. The root chord is $15'$ and the tip chord is $5'$. The taper ratio is indicated as $95'$ for the spanwise distance from the root to the tip.
- Aerodynamic Parameters:**
 - D_c : Drag coefficient at the root.
 - D_w : Drag coefficient at the wing.
 - D_t : Drag coefficient at the tail.
 - L_c : Lift coefficient at the root.
 - L_w : Lift coefficient at the wing.
 - L_t : Lift coefficient at the tail.
 - w : Downward velocity vector at the wing.
 - ϵ : Angle of attack at the tail.
 - T : Thrust vector at the tail.
- Centroids and Axes:**
 - AC_c : Aerodynamic center at the root.
 - CG : Center of gravity.
 - AC_w : Aerodynamic center at the wing.
 - AC_t : Aerodynamic center at the tail.
- Angles:**
 - 15° : Angle of attack at the root.
 - 20° : Angle of attack at the tail.

가체의 힘의 평형 상자를 생각하면.

$$\Sigma F_x = 0; \quad D_c + D_w + D_t = T \cos 5^\circ \quad \dots \textcircled{1}$$

$$\Sigma F_y = 0; \quad L_w + L_t + T \sin 15^\circ = W \quad \dots \textcircled{2}$$

$$\Sigma M_{ACW} = 0 ; \quad D_c \times 135 \sin 15^\circ - W_5 \cos 15^\circ - L_1 \times 95 \cos 15^\circ - D_1 \times 95 \sin 15^\circ + T \cdot 20 = 0 \quad \dots \textcircled{3} \quad \text{or } \uparrow$$

또한 주어진 값은 $(L/p)_t = 1.2$, $T = 160,000 \text{ lb}$

$W = 400,000 \text{ lb}$, $D_c = 100 \text{ lb}$ 이므로 위식에 대입.

$$D_w + D_t = 154448 \quad \dots \textcircled{1}'$$

$$L_w + L_t = 358589 \dots \textcircled{2}'$$

$$L \cos 15^\circ + D \sin 15^\circ = 13385.7 \dots (3)$$

③'에 $L_t = 1.2 D_t$ 를 대입하면

$$(1.2 \cos 15^\circ + \sin 15^\circ) D_4 = 13385.7$$

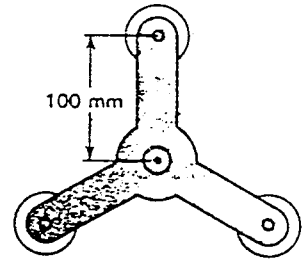
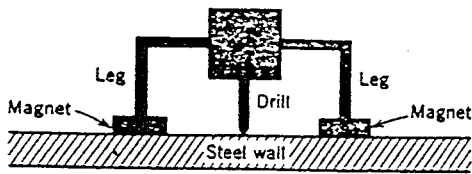
$$\therefore D_A = 9440.3 \text{ lb}, \quad D_W = 144507.7 \text{ lb}$$

$$L_t = 11328.4 \text{ lb} \quad , \quad L_w = 347260.6 \text{ lb}$$

구하고자 하는 lift-to-drag ratio $(L/D)_w$ 는

$$(L/D)_w = \frac{347260.6}{144507.7} \approx 2.4$$

Ans.) $(L/D)_w = 2.4$



$$T = 15 \text{ N}\cdot\text{m}$$

$$\text{수직력} = 50 \text{ N}$$

$$f = 0.4$$

우선 드릴의 회전력에 의한

$$F_f = f \cdot R_N$$

반력 R_T 에 대한 평형식

$$T = 3 R_T \times 0.1 = 15 \text{ N}\cdot\text{m}$$

$$\therefore R_T = 50 \text{ N}$$

$$\text{마찰력 } F_f = f \cdot R_N$$

(R_N ; 벽에 작용되는 힘)

그런데 드릴에 가해지는

수직력 50N으로 인한

leg에서의 반력 R_z 는

$$R_z = \frac{50}{3} \text{ N}$$

$$\therefore R_N = F_M - R_z$$

(단, F_M ; 자중에 의해 작용되는 힘)

결국 $F_f > R_T$ 이어야 하므로

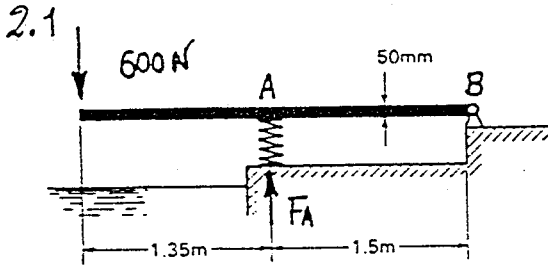
$$f \times (F_M - R_z) \geq R_T \quad \therefore F_M \geq \frac{50}{0.4} + \frac{50}{3}$$

$$F_M \geq 141.67 \text{ (N)}$$

따라서, 자중이 141.67 N 이상이어야 드릴이 가능하다

Ans.) 141.67 N

CHAPTER. 2.



[Given $k = 35000 \text{ (N/m)}$]

$$\sum M_B = 0;$$

$$(1.35 + 1.5) \times 600 = 1.5 F_A$$

$$\therefore F_A = 1140 \text{ N}$$

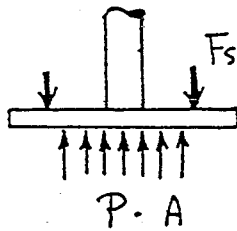
$$F_A = k \delta_A$$

$$\therefore \delta_A = \frac{F_A}{k} = \frac{1140}{35000} = 0.03257 \text{ (m)}$$

별판의 강도가 강한 것과 약한 것을 비교하여 보면
강도가 약한 것은 굽힘에 의해서 moment-arm이 작아지기 때문에
실제로 처짐은 강한 것보다 작아지게 되나 그 영향은
driving board의 경우 무시 할 수 있다.

Ans.) $\delta = 3.257 \text{ cm}$

2.2



압축된 스프링의 압축력

$$F_s = 120,000 \times \frac{(250 - 207)}{1000}$$

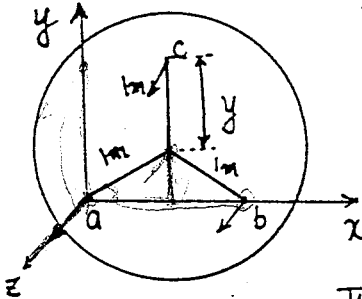
용기내 압력이 별판을 마는 힘

$$F_s = P \cdot A = P \cdot \frac{\pi}{4} (0.05)^2$$

$$\therefore P = \frac{120,000 \times \frac{43}{1000}}{\frac{\pi}{4} (0.05)^2} = 2627966 \text{ (N/m}^2\text{)}$$

Ans.) $P = 2.628 \text{ MN/m}^2$

2.3



$$R_a = R_b = 14 \text{ kN/m}$$

$$R_c = 16 \text{ kN/m} \quad W = 1.1 \text{ kN}$$

$$\sum F_z = 0; \quad W = 2R_a \sin 30^\circ + R_c \cos 30^\circ$$

$$\sum M_z = 0;$$

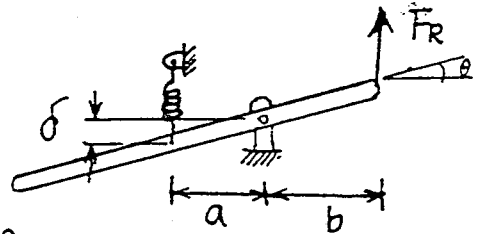
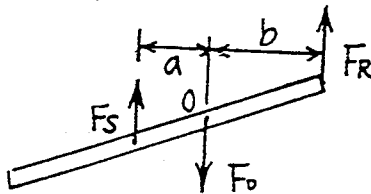
$$W \sin 30^\circ = R_c \sin 30^\circ + R_a \cos 30^\circ$$

$$\delta_a = \delta_c \text{ 이므로 } W \sin 30^\circ = \frac{k_c W}{2k_a + k_c} (\sin 30^\circ + y)$$

$$\therefore y = \frac{2k_a}{k_c} \sin 30^\circ = \frac{2 \times 14}{16} \times \frac{1}{2} = 0.825 \text{ (m)}$$

Ans.) 0.825 m

2.4 만약 θ 가 충분히 작다면 $\delta \approx a\theta$ ($\because \theta \ll 1$)



힘-변형 관계식은 $F_s = k\delta$

모우멘트에 관한 평형을 생각하면 $\sum M_o = 0$, 곧 $aF_s = bF_R$

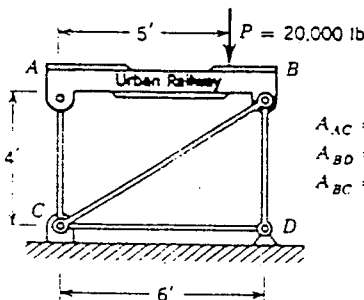
위식에서 bF_R 이 상수이므로 aF_s 도 역시 일정한 값을 알 수 있다. 따라서 가해지는 힘의 크기는

스프링의 탄성 계수나 그 위치에 따라 변하는 것이 아니며 작동자가 힘을 주게 들어려면 레버의 끝을 잡게 하여 점으로부터 힘이 작용 위치를 옮겨야 한다.

이때 가해지는 힘이 감소되어 적은 하는 일이 감소되는 것을 의미하는 것이 아니고 다만 작동에 효율을 떨어뜨릴 수 있을 뿐이다.

2.5

탄성 계수 $E = 30 \times 10^6 \text{ psi}$ 라 하자.



$$\begin{aligned} A_{AC} &= 15 \text{ in.}^2 \\ A_{BD} &= 10 \text{ in.}^2 \\ A_{BC} &= 3 \text{ in.}^2 \end{aligned}$$

$$\begin{cases} F_{AC} = \frac{1}{6} P \\ F_{BD} = \frac{5}{6} P \\ F_{BC} = 0 \end{cases}$$

$$\delta = \frac{FL}{EA} \text{ 로 부러}$$

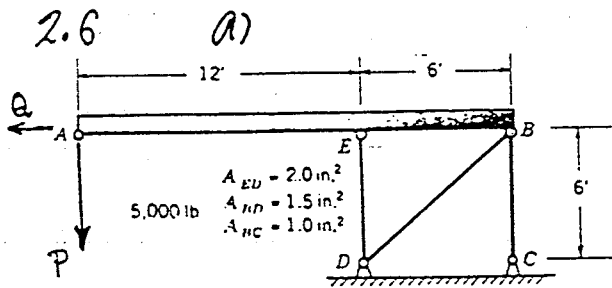
δ_{AC} 와 δ_{BD} 를 구하면

$$\delta_{AC} = \left(\frac{FL}{EA} \right)_{AC} = \frac{\frac{1}{6} PL}{15E} = \frac{20,000 \times 4 \times 12}{90 \times 30 \times 10^6} = 3.56 \times 10^{-4} \text{ (in)}$$

$$\delta_{BD} = \left(\frac{FL}{EA} \right)_{BD} = \frac{PL}{12E} = \frac{20,000 \times 4 \times 12}{12 \times 30 \times 10^6} = 2.67 \times 10^{-3} \text{ (in)}$$

$$\theta = \frac{\delta_{BD} - \delta_{AC}}{6 \times 12} = \frac{(2.67 - 3.56) \times 10^{-4}}{6 \times 12} = 3.2 \times 10^{-5} \text{ (rad)}$$

Ans.) $\theta = 3.2 \times 10^{-5} \text{ rad.}$



$$\sum M_B = 0 ;$$

$$6 F_{DE} = 18 \times 5000$$

$$\therefore F_{DE} = 15,000 \text{ (lb)}$$

(압축력)

$$\sum F_y = 0 ;$$

$$\therefore F_{BC} = -10,000 \text{ lb}$$

(인장력)

$$5000 = 15000 + F_{BC}$$

$$\sum F_x = 0 ; \therefore F_{BD} = 0$$

($\because$ 2방향의 수평 성분 힘이 존재하지 않으므로)

b) A점에서의 수직 및 수평 변위를 얻기 위해서는 각 부재의 변형에 의한 A점의 변위를 고려하여야 함으로

$$\delta = \sum \frac{F L}{EA} \frac{\partial F}{\partial P} \text{ 로 표현 할 수 있다.}$$

(단, F 는 영향력을 받는 부재에 가해지는 외력,

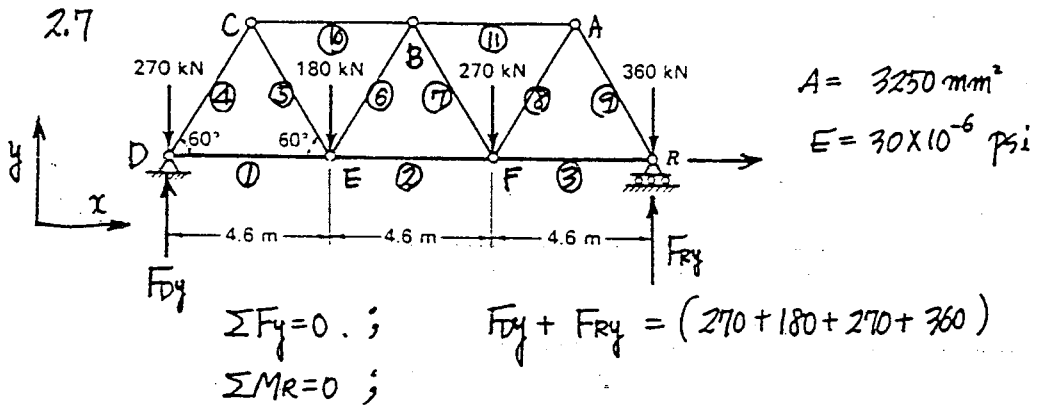
P 는 구하고자 하는 변위 방향의 힘 성분)

AB 부재가 충분히 강하므로 굽힘에 의한 처짐을 무시하면

A점의 수직 및 수평 처짐은 다음 표와 같다.

	F	L/EA	$\frac{\partial F}{\partial P}$	$\frac{\partial F}{\partial \theta}$	$u = \frac{FL}{EA} \frac{\partial F}{\partial P}$	$v = \frac{FL}{EA} \frac{\partial F}{\partial \theta}$
ED	-15000	1×10^{-7}	-3	0	4.5×10^{-3}	0
BC	10000	2×10^{-7}	2	1	4.0×10^{-3}	2×10^{-3}
BD	0	1.39×10^{-7}	0	$-\sqrt{2}$	0	0
$E = 30 \times 10^6 \text{ psi}$					8.5×10^{-3}	2.0×10^{-3}

따라서, 수평 변위 $\delta_v = u = 8.5 \times 10^{-3} \text{ ft} = 0.102 \text{ in } (\downarrow)$
 수평 변위 $\delta_H = v = 2 \times 10^{-3} \text{ ft} = 0.024 \text{ in } (\leftarrow)$ at
Ans.) $\delta_v = 0.102 \text{ in}$, $\delta_H = 0.024 \text{ in}$

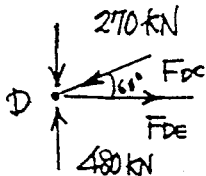


$$F_{Dy} \times (3 \times 4.6) = 270 \times (4.6 \times 3) + 180 \times (2 \times 4.6) + 270 \times 4.6$$

$$\therefore F_{Dy} = 480 \text{ kN}, \quad F_{Ry} = 600 \text{ kN}$$

변형 후의 각 점을 '로 표기하자. (D는 D')

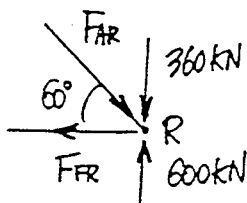
$$\delta_R = R'R \cong DR' - DR = (DE' - DE) + (EF' - EF) + (FR' - FR) \\ = \delta_{DE} + \delta_{EF} + \delta_{FR}$$



$$\sum F_y = 0; \quad 480 = 270 + F_{DE} \sin 60^\circ$$

$$\sum F_x = 0; \quad F_{DE} \cos 60^\circ = F_{Dx}$$

$$\therefore F_{DE} = \frac{210}{\sin 60^\circ} \cos 60^\circ = 121.24 \text{ kN}$$



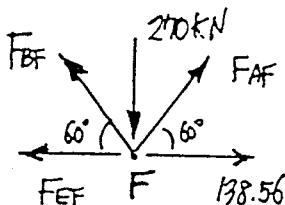
$$\sum F_y = 0; \quad 600 = 360 + F_{AR} \sin 60^\circ$$

$$\sum F_x = 0; \quad F_{AR} \cos 60^\circ = F_{FR}$$

$$\therefore F_{AR} = 277 \text{ (kN)}, \quad F_{FR} = 138.56 \text{ kN}$$

또한 F_{DE}와 F_{FR}의 합이 평형을 이루려면

$$F_{AF} = F_{AR} = 277 \text{ kN} \text{ 이므로}$$



$$\sum F_y = 0; \quad 277 \sin 60^\circ + F_{BF} \sin 60^\circ = 270$$

$$\sum F_x = 0; \quad F_{FE} = 138.56 + 277 \sin 60^\circ - F_{BF} \sin 60^\circ$$

위 두식을 연립하면

$$F_{BF} = 34.64 \text{ kN}, \quad F_{EF} = 259.81 \text{ kN} \text{ 이다.}$$

$$\delta = \frac{FL}{EA} \text{ 이므로 수평 변위 } \delta_H \text{ 는}$$

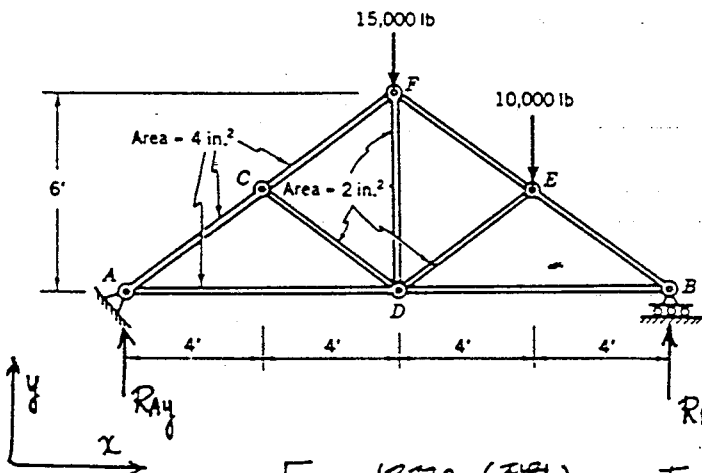
$$\delta_H = 6.9 \times 10^{-6} (121.24 + 138.56 + 259.81)$$

$$= 3.588 \times 10^{-3} \text{ (m)}$$

$$\therefore \delta_H = 3.588 \text{ mm (오른쪽으로)}$$

Ans.) 오른쪽으로 3.588 mm 이동

2.8



$$\sum F_y = 0;$$

$$R_{Ay} + R_{By} = 25000$$

$$\sum M_A = 0;$$

$$15000 \times 8 + 12 \times 10000$$

$$= 16 R_{By}$$

$$\therefore R_{By} = 15000$$

$$R_{Ay} = 10000$$

여기서 풀이를 이용하면

$$F_{AD} = 13330 \text{ (장력)}$$

$$F_{BD} = 20000 \text{ (장력)}$$

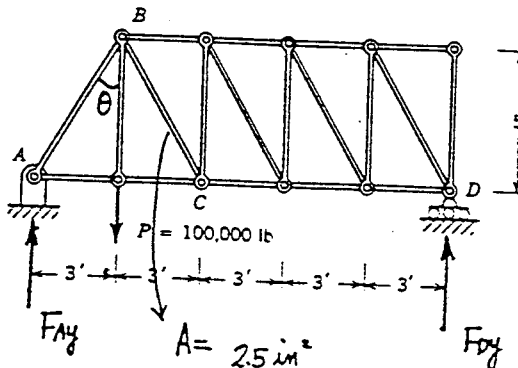
$$\delta_H = \delta_{AD} + \delta_{BD}$$

$$= \frac{L}{EA} (F_{AD} + F_{BD}) = \frac{8}{2 \times 10 \times 10^6} \times (13330 + 20000)$$

$$= 6.666 \times 10^{-3} \text{ (ft)} = 0.079 \text{ in}$$

$$\text{Ans.) } \delta_H = 0.08 \text{ in}$$

2.9



$$\sum M_D = 0; F_{Ay} \times 15$$

$$= P \times 12$$

$$\therefore F_{Ay} = 80,000 \text{ lb}$$

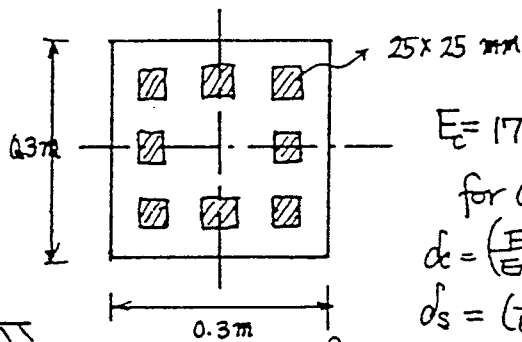
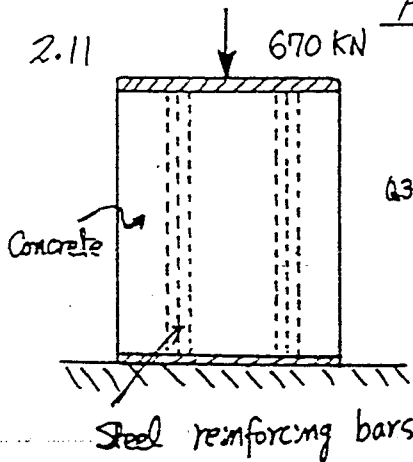
$$\sum F_y = 0;$$

$$F_{Ay} + F_{Dy} = 100,000$$

$$\therefore F_{Dy} = 20,000 \text{ lb}$$

$$\delta_v = \sum_{i=1}^n \frac{\pi}{E A_i} \frac{F_i L_i}{2W} \frac{\partial F_i}{\partial W} = 12.657 \frac{WL}{EA}$$

Ans.) $\delta_v = 12.657 \frac{WL}{EA}$



$E_c = 17 \text{ GN/m}^2$
for Concrete

$$\delta_c = \left(\frac{E}{EA} \right)_c L$$

$$\delta_s = \left(\frac{E}{EA} \right)_s L$$

$$\delta_c = \delta_s \quad (\because \text{H로 변형되어야 함})$$

$$A_c = 0.085 \text{ m}^2, \quad A_s = \frac{0.005}{8} \text{ m}^2$$

$$F_c + 8 F_s = 670 \text{ kN} \quad \dots \textcircled{1}$$

$$\frac{F_c \times L}{17 \times 10^9 \times 0.085} = \frac{F_s \times L}{20 \times 10^9 \times \frac{0.005}{8}} \quad \therefore F_c = 11.56 F_s$$

$$\textcircled{1} \text{ 식으로 부터 } (11.56 + 8) F_s = 670 \text{ kN} \quad \therefore F_s = 34.25 \text{ kN}$$

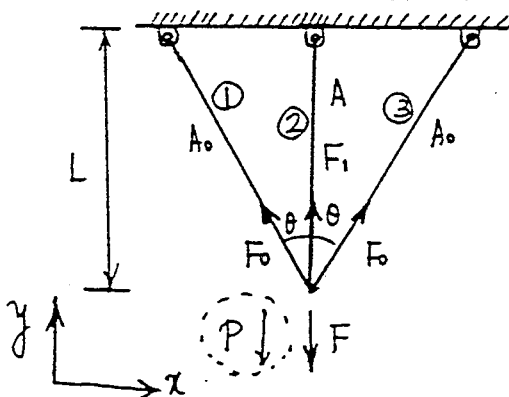
$$\textcircled{2} \text{ 에 대입하면 } F_c = 396.0 \text{ kN}$$

$$\sigma_s = F_s / A_s = 34.25 / \frac{0.005}{8} = 54800 \text{ kN/m}^2$$

$$\sigma_c = F_c / A_c = 396.0 / 0.085 = 4658.8 \text{ kN/m}^2$$

Ans.) $\sigma_s = 54.8 \text{ MN/m}^2, \quad \sigma_c = 4.66 \text{ MN/m}^2$

2.12



$$\sum F_y = 0;$$

$$F_1 + 2 F_0 \cos \theta = F \quad \dots \textcircled{1}$$

bar ①, ③ 판이 F를 받는다

$$\delta_1 = \frac{F_1 \times L}{EA \cos \theta} \quad \frac{1}{2 \cos \theta} \times 2$$

$$(\because \delta = \frac{FL}{EA} \frac{\partial F}{\partial P})$$

bar ②가 작용강도로 이에 작용하는 힘 F_1 과 대응하는 처짐 δ_2 를 구하여 보면

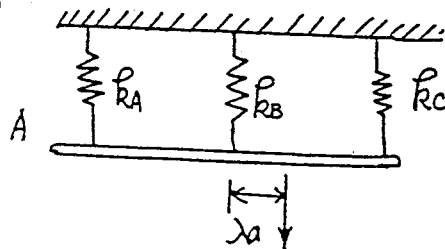
$$\delta_1 - \delta_2 = \frac{F_1 \cdot L}{EA} \Rightarrow \frac{F_1 \cdot L}{2EA_0 \cos^3 \theta} = \frac{F_1 \times L}{EA}$$

$$\therefore AF = (A + 2A_0 \cos^3 \theta) F_1 \dots \dots \textcircled{2}$$

①, ②에서 $F_0 = \frac{A \cdot F}{A + 2A_0 \cos^3 \theta}$, $F_1 = \frac{FA_0 \cos^3 \theta}{A + 2A_0 \cos^3 \theta}$

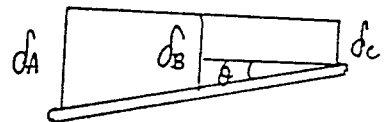
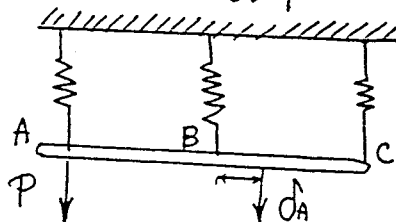
Ans.) $F_0 = \frac{AF}{A + 2A_0 \cos^3 \theta}$, $F_1 = \frac{FA_0 \cos^3 \theta}{A + 2A_0 \cos^3 \theta}$

2.13 a)



$$\delta_A = \frac{P[2k_C - \lambda(k_B + 2k_C)]}{k_A k_B + 4k_A k_C + k_B k_C}$$

b)



$$k_A \delta_A + k_B \delta_B + k_C \delta_C = P, \quad \delta_A + \delta_C = 2\delta_B$$

$$\sum M_C = 0; \quad 2k_A \delta_A + k_B \delta_B = 2P$$

$$\begin{bmatrix} k_A & k_B & k_C \\ 1 & -2 & 1 \\ 2k_A & k_B & 0 \end{bmatrix} \begin{bmatrix} \delta_A \\ \delta_B \\ \delta_C \end{bmatrix} = \begin{bmatrix} P \\ 0 \\ 2P \end{bmatrix}$$

$$\det. A = \begin{vmatrix} k_A & k_B & k_C \\ 1 & -2 & 1 \\ 2k_A & k_B & 0 \end{vmatrix} = k_A k_B + k_B k_C + 4k_A k_C$$

$$\delta_A = \frac{1}{\det A} \begin{vmatrix} P & k_B & k_C \\ 0 & -2 & 1 \\ 2P & k_B & 0 \end{vmatrix} = \frac{1}{\det A} (2k_B P + 4k_C P - k_B P)$$

$$\therefore \delta_A = \frac{2k_B P + 4k_C P - k_B P}{k_A k_B + k_B k_C + 4k_A k_C} = \frac{(k_B + 4k_C) P}{k_A k_B + k_B k_C + 4k_A k_C}$$

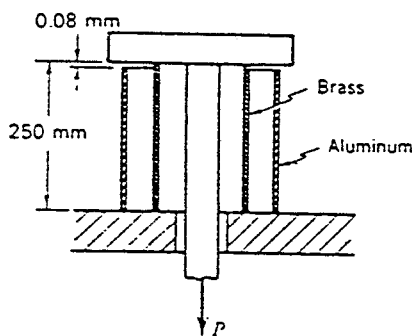
$$\begin{aligned} \delta_C &= \frac{1}{\det A} \begin{vmatrix} k_A & k_B & P \\ 1 & -2 & 0 \\ 2k_A & k_B & 2P \end{vmatrix} = \frac{1}{\det A} (-4k_A P + k_B P + 4k_A P - 2k_B P) \\ &= \frac{-k_B P}{k_A k_B + k_B k_C + 4k_A k_C} \end{aligned}$$

$$\theta = \frac{\delta_A - \delta_C}{2a} \quad \therefore \delta = \delta_C + (1-\lambda) a \theta$$

$$\begin{aligned} \therefore \delta &= \frac{1-\lambda}{2} \delta_A + \frac{1+\lambda}{2} \delta_C \\ &= \frac{P}{k_A k_B + k_B k_C + 4k_A k_C} \left[-k_B + \frac{1-\lambda}{2} (4k_C + 2k_B) \right] \\ &= \frac{P}{k_A k_B + k_B k_C + 4k_A k_C} \left[2k_C - \lambda (k_B + 2k_C) \right] \end{aligned}$$

— Q.E.D —

2.14



$$A = \pi r^2 - \pi (r-t)^2$$

$$= \pi (2r-t) t$$

$$= \pi (D-t) t$$

r : 반경, D : 직경, t : 두께.

$$\begin{aligned} A_{Al} &= \pi (0.25 - 0.00625) \times 0.00625 \\ &= 479 \times 10^{-3} \text{ (m}^2\text{)} \end{aligned}$$

$$A_{Br} = \pi (0.15 - 0.00625) \times 0.00625 = 2.82 \times 10^{-3} \text{ (m}^2\text{)}$$

$$E_{Al} = 69 \times 10^9 \text{ N/m}^2, \quad E_{Br} = 103 \times 10^9 \text{ N/m}^2 \text{ 라고}$$

놓으면 하중-처짐 곡선은 다음과 같이 구하여진다.

$$i) \quad 0 \leq \delta \leq 8 \times 10^{-5} (m)$$

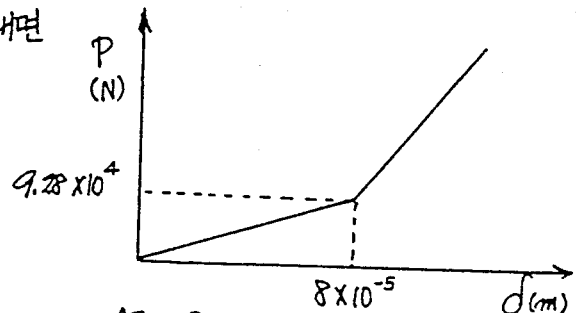
$$\delta = \frac{PL_B}{E_B A_B} = \frac{P \times 0.25}{(103 \times 10^9) \times (2.82 \times 10^{-3})} = 8.61 \times 10^{-10} P$$

$$\therefore P = 1.16 \times 10^9 \delta$$

$$ii) \quad \delta \geq 8 \times 10^{-5} (m)$$

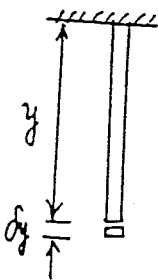
$$\begin{aligned} \phi &= \frac{E_B A_B}{L_B} \delta + \frac{E_A A_A}{L_A} (\delta - 8 \times 10^{-5}) \\ &= \frac{(103 \times 10^9)(2.82 \times 10^{-3})}{0.25} \delta + \frac{(69 \times 10^9)(4.79 \times 10^{-3})}{(0.25 - 8 \times 10^{-5})} (\delta - 8 \times 10^{-5}) \\ &= 2.48 \times 10^9 \delta - 105797.1 \end{aligned}$$

그래프를 그려 나타내면
오른쪽 그림과 같다.



Ans.)

2.15



$$\delta = \frac{FL}{AE}$$

$$\therefore R = \frac{AE}{L}$$

$$F = \frac{AE}{L} \delta$$

$$\left[\begin{aligned} R &= 10^6 \text{ lb/in} \\ W &= 1.6 \text{ lb/ft} \end{aligned} \right.$$

$$d\delta = \frac{W y}{EA} dy \text{ 이므로}$$

$$\begin{aligned} \delta &= \int_0^L \frac{W y}{EA} dy \\ &= \frac{WL^2}{2EA} \end{aligned}$$

$$AE = kL \text{ 이므로}$$

$$\therefore \delta = \frac{WL}{2R} \text{ 이다.}$$

여기서 R과 같은 5980 ft rope spring constant

$$R = \frac{10^6}{5980} (\text{lb/in}) \quad \therefore \delta = \frac{(1.6 \times \frac{1}{12}) \times (5980 \times 12)}{2 \times (10^6/5980)}$$

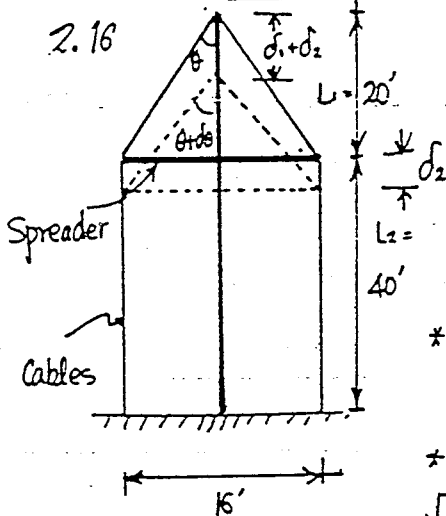
$$= 28.6 \text{ in} = 2.384 \text{ ft}$$

$$6000 - (5980 + 2.384) = 17.616 \text{ ft.}$$

6'인 사람이 매달린다면 약 12ft 거양이 모자란다 이를 rope 끝에 W의 무게를 달아 줄을 늘어뜨린다면

$$W = R \cdot d' = \frac{10^6}{5980} \times 12 \times 12 = 24080 \text{ (lb)}$$

Ans.) 24080 lb 무게의 추를 로프 끝에 매달아 놓는다.



* 가정

- i) Spreader는 견고히 mast에 부착된다
- ii) Cable과 spreader 사이는 마찰이 없다.
- iii) 감판은 견고하다.
- iv) turnbuckle의 길이는 무시할 수 있다.

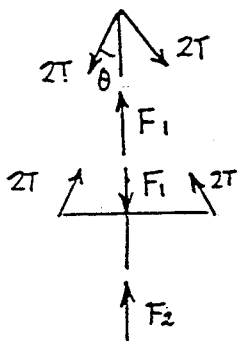
* 변형 전

$$\sqrt{L_1^2 + 8^2} + L_2 = L_3 \dots \textcircled{1}$$

* 변형 후

$$\sqrt{(L - d_1)^2 + 8^2} + (L_2 - d_2) = L_3 - d_3 \dots \textcircled{2}$$

또한, $d_3 = d_3' - d_3''$ (d_3' ; turnbuckle의 조임에 의한 길이 감소
이다. d_3'' ; 장력에 의한 길이 증가



Mast 위에서 평형식

$$F_1 = 2T \cos \theta \times 2 = 4T \cos \theta \dots \textcircled{3}$$

Mast 아래에서의 평형식

$$\sum F_y = 0; 4T \cos \theta - 4T - F_1 + F_2 = 0 \dots \textcircled{4}$$

$$\textcircled{3}, \textcircled{4} \text{로부터 } T = F_2/4, F_1 = F_2 \cos \theta$$

$$(1 + \epsilon)^n \approx 1 + n\epsilon \quad (\text{단, } \epsilon \ll 1) \text{를 이용하여}$$

③식을 정리하면

$$\sqrt{L_1^2 + 8^2} - \frac{d_1 L_1}{\sqrt{L_1^2 + 8^2}} + L_2 + d_2 = L_3 - d_3$$

①식을 사용하면

$$-\frac{d_1 L_1}{\sqrt{L_1^2 + 8^2}} = d_2 - d_3 \quad \text{or} \quad -0.93 d_1 = d_2 - d_3 \dots \textcircled{5}$$

$$d_1 = \frac{F L_1}{A_1 E_1} = \frac{F_2 \cos \theta \times 20}{20 \times 1.6 \times 10^6} = 5.8 \times 10^{-7} F_2 \text{ (ft)}$$

마찬가지로 $\delta_2 = 12.5 \times 10^{-7} \text{ F}_2 \text{ (ft)}$

$\delta_3' = \frac{15}{20} \times \frac{1}{12} = 0.0625 \text{ ft.}$

$\delta_3'' = \frac{P_3}{12 \times 10,000} = 3.67 \times 10^{-5} \text{ T (ft)} = 9.18 \times 10^{-6} \text{ F}_2 \text{ (ft)}$

$\delta_3 = \delta_3' - \delta_3'' = 0.0625 - 9.18 \times 10^{-6} \text{ F}_2$

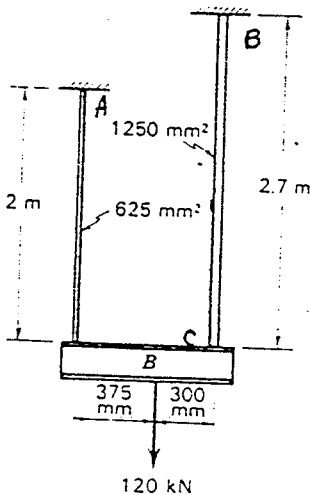
④이제 $\delta_1, \delta_2, \delta_3$ 를 각각 대입하면

$-(0.93)(5.8 \times 10^{-7}) \text{ F}_2 = 12.5 \times 10^{-7} \text{ F}_2 - 0.0625 + 9.18 \times 10^{-6} \text{ F}_2$

$\therefore \text{F}_2 = 5700 \text{ lb}$

Ans.) $\text{F}_2 = 5700 \text{ lb}$

2.17



$E_s = 200 \text{ GN/m}^2$

$\sum M_c = 0 ; \text{F}_A \times 675 = 120,000 \times 300$

$\therefore \text{F}_A = 53333 \text{ N}, \text{F}_B = 66667 \text{ N}$

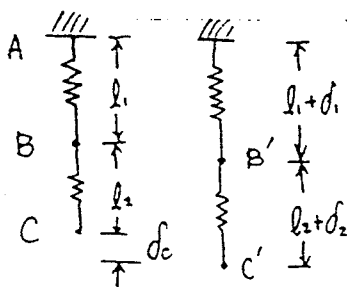
$\delta_A = \frac{53333 \times 2}{(200 \times 10^9)(1250 \times 10^{-6})} = 8.122 \times 10^{-4} \text{ (m)}$

$\delta_c = \frac{66667 \times 2.7}{(200 \times 10^9)(625 \times 10^{-6})} = 6.853 \times 10^{-4} \text{ (m)}$

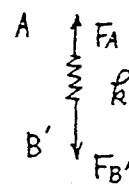
$\therefore \delta_B = \delta_c + \frac{\delta_A - \delta_c}{L_{AC}} L_{BC} = 6.853 \times 10^{-4} + \frac{300}{675} (8.122 - 6.853) \times 10^{-4} = 7.417 \times 10^{-4} \text{ (m)}$

Ans.) $\delta_B = 7.417 \times 10^{-4} \text{ m}$

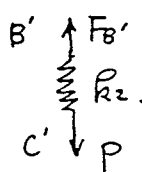
2.18.



AB의 스프링.



BC의 스프링.



정리하면

$\text{B'C'} ; \text{F}_{B'} = P, \text{AB'} ; \text{F}_A = \text{F}_{B'} = P$

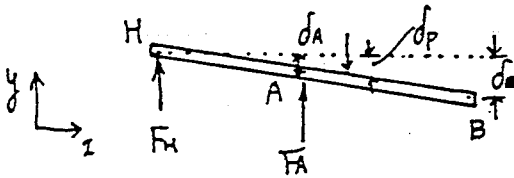
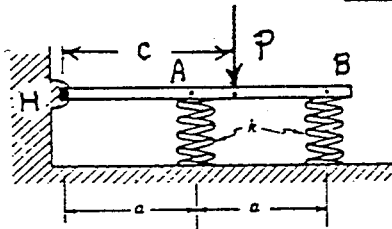
합-처짐 $\delta_1 = \frac{P}{k_1}$, $\delta_2 = \frac{P}{k_2}$

가하력 작용 후 좌합조건식 ; C의 변위 $= (l_1 + \delta_1) + (l_2 + \delta_2) - (l_1 + l_2)$
 $= \delta_1 + \delta_2 = P \left(\frac{1}{k_1} + \frac{1}{k_2} \right)$

전체 스프링 Constant $= \frac{P}{\delta} = \frac{k_1 k_2}{k_1 + k_2}$

Ans.) $\frac{1}{K} = \frac{1}{k_1} + \frac{1}{k_2}$

2.19



가하력 작용 조건으로부터

$\delta_A = \frac{1}{2} \delta_B$, $\delta_P = \frac{c}{2a} \delta_B$

합-변형 관계 ;

$F_A = k \delta_A = \frac{k}{2} \delta_B$

$F_B = k \delta_B$

$\sum M_{Az} = 0$;

$P \cdot c = F_A \cdot a + F_B \cdot 2a = \frac{5}{2} k \cdot a \delta_B$

$\therefore P = \frac{5}{2} k \delta_B \frac{a}{c}$

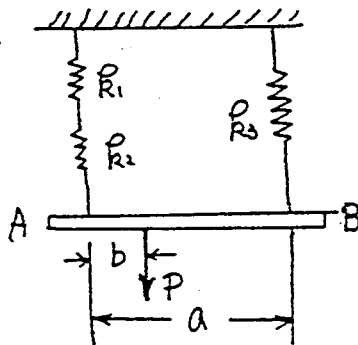
$\frac{P}{\delta_P} = \frac{\frac{5}{2} k \delta_B \frac{a}{c}}{\frac{c}{2a} \delta_B} = 5 \frac{a^2}{c^2} k$, $\frac{P}{\delta_P} = \frac{20}{9} k$ 이라면

$5 \frac{a^2}{c^2} k = \frac{20}{9} k$

$\therefore c = 1.5a$

Ans.) $c = 1.5a$

2.20.



문제(18)로부터 k_1, k_2 의

합성 스프링상수 $k_4 = \frac{k_1 k_2}{k_1 + k_2}$

A, B에 주어진 힘을

각각 F_A, F_B 라 하면

$F_A = k_4 \delta \dots \dots \textcircled{1}$

$F_B = k_3 \delta \dots \dots \textcircled{2}$

평형식 ; $F_A + F_B = P$, $F_A \cdot a = P(a-b)$

$\therefore F_A = P \left(1 - \frac{b}{a} \right)$, $F_B = P \frac{b}{a}$

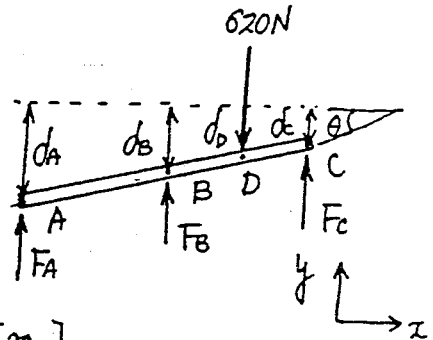
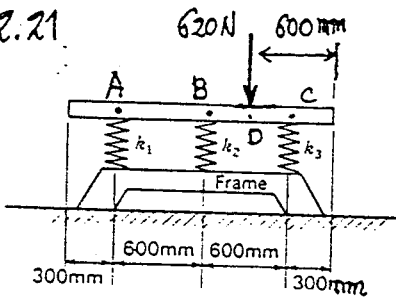
①, ②로부터 $\frac{F_A}{k_1} = \frac{F_B}{k_3}$ or $\frac{(b-a)}{ak_1} = \frac{b}{ak_3}$

또, $\frac{b}{a} = \frac{k_3}{k_1+k_2} = \frac{k_3(k_1+k_2)}{k_1k_2+k_3(k_1+k_2)}$ 이며

$$\delta = \frac{P(k_1+k_2)}{k_1k_2+k_3k_1+k_3k_2}$$

Ans.) $b = \frac{(k_1+k_2)k_3}{k_1k_2+k_3k_1+k_3k_2} a$, $\delta = \frac{(k_1+k_2)P}{k_1k_2+k_3k_1+k_3k_2}$

2.21



$d_B = d_A - 0.6\theta$ $\left\{ \begin{array}{l} \delta: [m] \\ \theta: [radian] \end{array} \right.$

$d_D = d_A - 0.9\theta$

$d_C = d_A - 1.2\theta$

$k_1 = k_3 = 17 \text{ kN/m}$, $k_2 = 21 \text{ kN/m}$

힘-처짐 관계식 $F_A = k_1 d_A$, $F_B = k_2 d_B$, $F_C = k_3 d_C$ 을

위식에 대입 $F_A = 17 \times 10^3 d_A$, ... ①

$F_B = 21 \times 10^3 (d_A - 0.6\theta)$... ②

$F_C = 17 \times 10^3 (d_A - 1.2\theta)$... ③

$\sum F_y = 0$; $F_A + F_B + F_C = 620 \text{ N}$

위식에 ①, ②, ③을 대입 $55 d_A - 33\theta = 0.62$... ④

$\sum M_{Dz} = 0$; $0.9 F_A + 0.3 F_B = 0.3 F_C$

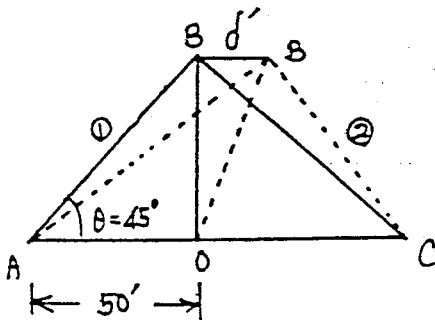
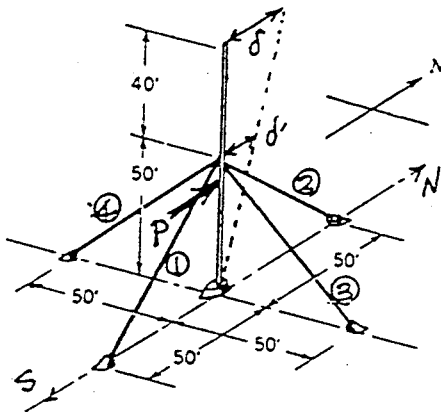
or $55 d_A + 7.8\theta = 0$, ... ⑤

$40.8\theta = -0.62$ $\therefore \theta = -0.0152$, $d_A = 2.155 \times 10^{-3}$

$F_A = 36.59 \text{ N}$ $F_B = 236.78 \text{ N}$, $F_C = 346.72 \text{ N}$

Ans.) $\theta = 0.0152 \text{ rad}$ (시계 방향)

2.22



- 가정 ;
 - i) Ball joint는 마찰이 없고
모두언로를 전달 뿐이 못한다.
 - ii) 깃대 는 견고하다.
 - iii) 깃대의 처짐은 S-N plane
내에 있다.
 - iv) 처짐이 상당히 작아 이에 따른
각 변하는 무시 할 수 있다.

로프 ①과 ②의 변형을 생각하면

$$\begin{aligned} AB' &\approx AB + BB' \cos(\angle BB'A) \\ &\approx AB + BB' \cos 45^\circ = AB + d' \cos 45^\circ \\ \therefore d &= \text{elongation of rope} \text{ ①} \\ &= d' \cos 45^\circ \\ &= \frac{5}{2} \times 0.707 d' \end{aligned}$$

② 로오프의 응력은 압축력이 될 수 없으므로 ($\because$ wire는 압축력을 받을 수 없다) ③에 걸리는 힘은 zero, 또한 d' 의 발생은 SN plane에서 일어나므로 ③, ④ 로오프의 신장도 무시할 수 있다.

힘-변위 관계식 $T_1 = A_1 E_1 \left(\frac{d_1}{L} \right) = \frac{\pi}{4} \left(\frac{1}{4} \right)^2 \times 30 \times 10^6 \times \frac{\frac{5}{9} \times 0.707 d}{1.414 \times 50}$

$$\therefore T_1 = 8230^\circ$$

기대의 평형식 ; $T_1 \left(\frac{1}{\sqrt{2}} \right) \times 50 = 900 \times 45$

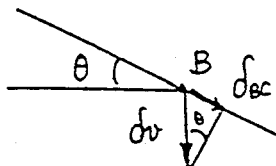
$$\therefore T_1 = 1140 \text{ lb}$$

$$8230 \sigma = 1140$$

$$\therefore \delta = 0.139 \text{ ft} = 1.68 \text{ in}$$

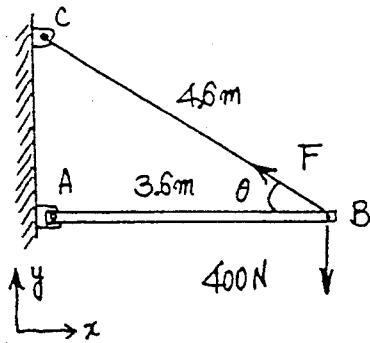
Ans.) $\delta = 1.68 \text{ in}$

2.23



$$\delta = \frac{WL}{EA} = \frac{400 \times 4.6}{E \cdot A} = 23 \times 10^{-3}$$

$$\therefore E \cdot A = 80,000 \text{ V}$$



$$\Sigma F_y = 0; F \sin \theta = 400$$

$$\therefore F = \frac{4.6}{\sqrt{4.6^2 - 3.6^2}} \times 400$$

$$= 642.6 \text{ N}$$

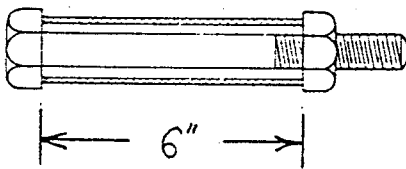
$$\delta_{BC} = \frac{FL}{EA} = \frac{642.6 \times 4.6}{80,000}$$

$$= 0.0369 \text{ (in)}$$

$$\therefore \delta_v = \frac{\delta_{BC}}{\sin \theta} = 0.05935 \text{ m}$$

Ans.) $\delta_v = 59.35 \text{ mm}$

2.24. Bolt area = 1.00 in^2 , Sleeve area = 0.60 in^2



$$\delta_b = \delta_b' - \delta_b''$$

$$\left\{ \begin{array}{l} \delta_b'; 90^\circ \text{ 회전시 의한 길이변화} \\ \delta_b''; \text{인장에 의한 } \end{array} \right.$$

$$\delta_b' = \frac{1}{6} \times \frac{1}{2} \times \frac{1}{4} = \frac{1}{768} \text{ (ft)}, \quad \delta_b'' = \frac{T \cdot L}{EA_b}$$

볼트의 인장력과 Sleeve의 압축력이 같고 $\delta_b = \delta_s$ 이므로

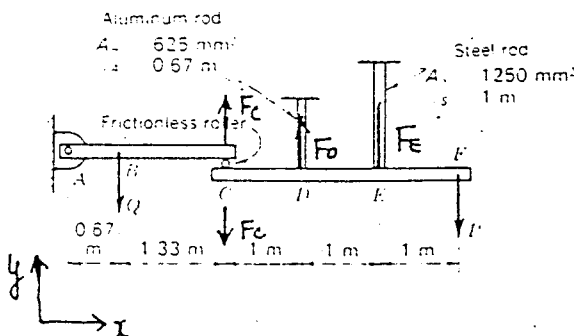
$$\delta_s = \frac{TL}{EA_s} = \delta_b = \frac{1}{768} - \frac{TL}{EA_b}$$

$$\frac{TL}{E} \left(\frac{1}{A_s} + \frac{1}{A_b} \right) = \frac{1}{768} \quad \therefore T = \frac{E}{768L} \frac{A_s A_b}{A_s + A_b}$$

$$\therefore T = 2441.406 \text{ lb}$$

Ans.) $T = 2441.406 \text{ lb}$

2.25



* 알루미늄 rod AC에서

$$\Sigma M_A = 0; 0.67Q = 2F_c$$

$$\therefore F_c = 0.335Q$$

* Steel rod에서

$$\Sigma F_y = 0; F_c + P = F_d + F_e$$

$$\therefore 0.335Q = F_d + F_e - P \dots \textcircled{1}$$

$$3P = F_d + 2F_e \dots \textcircled{2}$$

$$\delta_D = \frac{F_D L_A}{E_A A_A}, \quad \delta_E = \frac{F_E L_E}{E_S A_S} \quad \delta_D = \delta_E \quad \dots \quad (3)$$

$$\therefore \frac{F_D \times 0.67}{69 \times 10^9 \times 625} = \frac{F_E \times 1.0}{200 \times 10^9 \times 1250}$$

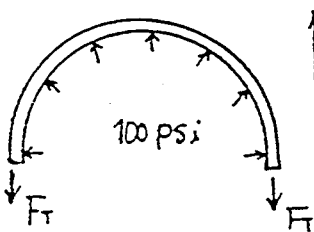
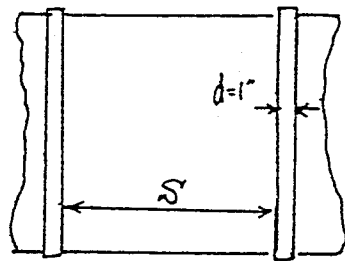
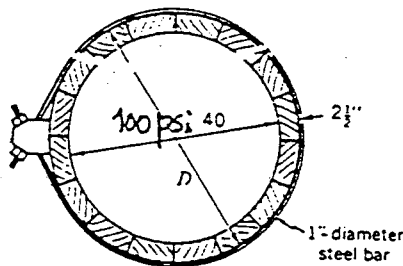
$$F_E = 3.884 F_D \quad \dots \quad (4)$$

②④로부터 $F_D = 0.342 P, \quad F_E = 1.329 P,$

①로부터 $Q = \frac{(0.342 + 1.329 - 1) P}{0.335} = 2.003 P$

Ans.) $Q = 2.003 P$

2.26



$$\sum F_y = 0; \quad F_T \times 2 = 40 \times l \times 100$$

$$\therefore F_T = 2000 S' \text{ (lb)}$$

$$\text{Strain (tangential)} = \frac{2000 S'}{\frac{\pi}{4} d^2 E}$$

$$= \frac{2000 S'}{\frac{\pi}{4} (1)^2 \times 30 \times 10^6} = \frac{dL}{L}$$

$$L = 2\pi R \quad \therefore dL = 2\pi dR$$

$$\frac{dL}{L} = \frac{dR}{R} = \frac{2000 S'}{\frac{\pi}{4} 30 \times 10^6}$$

$$dR \leq \left(\frac{0.03}{2}\right)''$$

$$\text{한계 값 } dR = \left(\frac{0.03}{2}\right)''$$

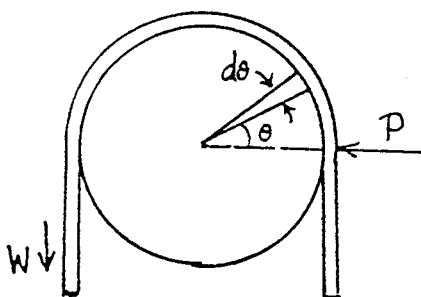
$$\therefore l = \frac{\pi}{4} \frac{30 \times 10^6}{2000} \times \frac{0.03/2}{22.5} = 7.65''$$

Ans.) 7.65 in

2.27. 예제 2.7로부터 $\frac{dT}{d\theta} = f \cdot T$

구분하면 $T = C e^{f\theta} \quad (C = \text{const})$

$\theta = \pi$ 이면, $T = W = C e^{\pi f} \quad \therefore C = W e^{-\pi f}$



또한, $T = W e^{f(\theta - \pi)}$ 이며
 $\theta = 0$ 일때 $T_0 = W e^{-f\pi}$

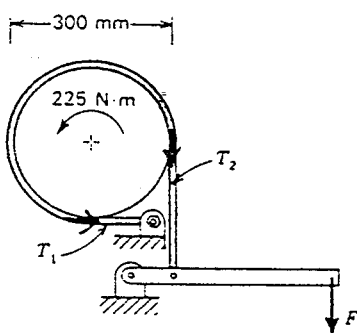
이러한 경우가 가능하려면

$$f \cdot P = T_0$$

$$\therefore P = \frac{W}{f} e^{-f\pi}$$

Ans.) $P = \frac{W}{f} e^{-f\pi}$

2.28



$$(T_2 - T_1) \times 0.15 = 225$$

$$T_2 = T_1 e^{3\pi f}$$

$$\therefore T_2 = T_1 + 1500$$

$$T_2 = T_1 e^{0.6\pi}$$

위값을 구하면 $T_1 = 268.5 \text{ lb}$

$$T_2 = 1768.5 \text{ lb}$$

Ans.) $T_1 = 268.5 \text{ lb}$ $T_2 = 1768.5 \text{ lb}$

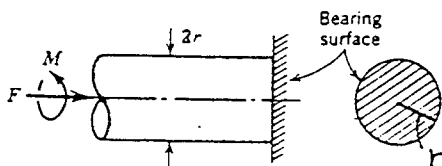
2.29.

$$T = T_0 e^{f\theta} \quad (\theta = 8\pi, f = 0.3, T_0 = 200 \text{ N})$$

$$\therefore T = 200 e^{8\pi \times 0.3} = 376300 \text{ (N)}$$

Ans.) $T = 376.3 \text{ kN}$

2.30



$$\sigma = \frac{F}{A} = \frac{F}{\pi r^2}$$

$$\tau = f \cdot \sigma = \frac{F \cdot f}{\pi r^2}$$

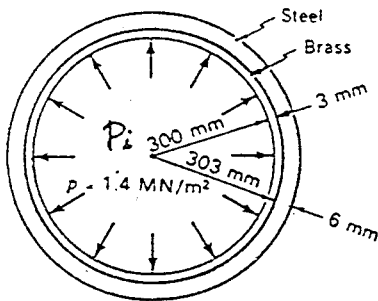
(단위 면적당 작용하는 마찰력)

$$\therefore dM = \tau \cdot R \cdot 2\pi R \cdot dR = \frac{2Ff}{r^2} R^2 dR$$

$$M = \frac{2Ff}{r^2} \int_0^r R^2 dR = \frac{2f \cdot r \cdot F}{3}$$

Ans.) $M = \frac{2f r F}{3}$

2.31



$$E_B = 103 \text{ GN/m}^2$$

$$E_S = 205 \text{ GN/m}^2$$

$$\Delta D_S = \frac{\delta_S}{\pi} = \frac{F_S \times 0.612}{(205 \times 10^9)(0.006 \times 0.006)} = 1.659 \times 10^{-7} \frac{F_S}{2}$$

$$\Delta D_B = \Delta D_S ; \quad F_B = 0.255 F_S \dots \dots \textcircled{2}$$

$$\textcircled{1} \textcircled{2} \Rightarrow F_S = 2008 \text{ N}, \quad F_B = 512 \text{ N}$$

$$\text{Ans.) } F_S = 2008 \text{ N}, \quad F_B = 512 \text{ N}$$

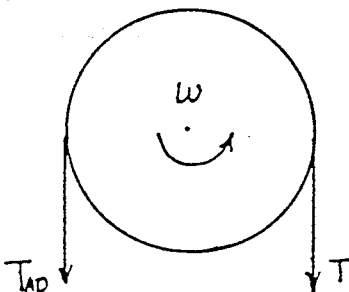
$$2.32 \quad \Delta D_B = 6.505 \times 10^{-7} \frac{F_B}{2} = 3.33 \times 10^{-4} \times \frac{1}{2} (\text{m})$$

$$= 0.1665 (\text{mm})$$

$$\therefore \Delta R_B = \frac{\Delta D_B}{2} = 0.08326 \text{ mm}$$

$$\text{Ans.) } \Delta R_B = 0.08326 \text{ mm}$$

2.33



$$\Delta N - T \Delta \theta = 0.$$

$$\Delta T - f \Delta N = 0$$

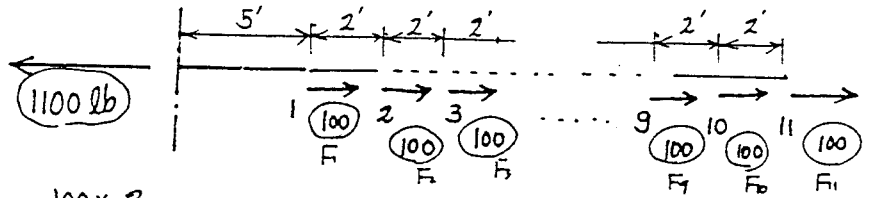
$$\therefore T = T_{A0} e^{f\theta}$$

$$\frac{dT}{T} = \frac{f R d\theta}{R} = \frac{f}{R} R d\theta = \frac{f R d\theta}{R}$$

$$\therefore \frac{dT}{T} = \frac{f}{R} R d\theta = \frac{f R d\theta}{R}$$

- A.E.D -

2.34



$$\delta_{11} = \frac{100 \times 2}{EA}$$

$$\delta_{10} = \frac{200 \times 2}{EA}$$

$$\delta_2 = \frac{1000 \times 2}{EA}$$

$$\delta_1 = \frac{1100 \times 5}{EA}$$

$$EA \text{를 구하기 위해서 } R = \frac{EA}{L}$$

$$EA = RL = 29400 \text{ lb/in} \times 1 \text{ ft}$$

$$= 29400 \times 12 \text{ lb}$$

$$\therefore EA = 352800 \text{ lb}$$

$$\delta = \frac{1}{EA} [1100 \times 5 + 200(1+2+\dots+10)]$$

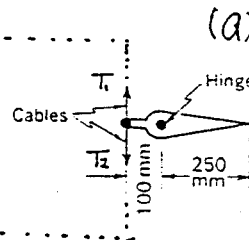
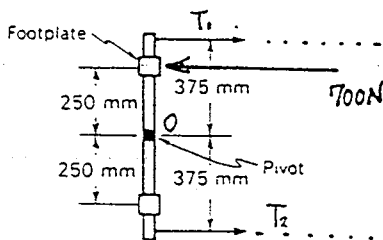
$$= 0.0468 \text{ [ft]}$$

$$\delta_{\text{total}} = 2\delta = 0.0936 \text{ (ft)}$$

$$L = 50 + 2 \times 4 + 0.0936 = 58.094 \text{ ft}$$

$$\text{Ans.) } L = 58.094 \text{ ft}$$

2.35



$$(a) \quad \sum M_0 = 0;$$

$$375T_1 = 375T_2$$

$$+ 250 \times 700$$

$$\therefore T_1 - T_2 = \frac{2}{3} \times 700$$

$$T_0 = 1.4 \text{ kN}, \quad EA = RL = 60 \times 10^6 \times 0.025 \text{ (N)}$$

$$\therefore \delta_1 = \frac{(T_1 - T_0) \times 6}{EA}, \quad \delta_2 = \frac{(T_0 - T_2) \times 6}{EA} \quad (\text{단, } T_2 > 0)$$

$$T_2 > 0 \text{ 면 } \delta_1 = \delta_2 \text{ 이어야 한다}$$

$$\therefore T_1 - T_0 = T_0 - T_2$$

$$\therefore T_1 + T_2 = 2T_0 = 2800$$

위 두식을 연립하면

$$T_1 = 1633.6 \text{ N}$$

$$T_2 = 1166.7 \text{ N}$$

$$\delta = \frac{(1633.3 - 1400) \times 6}{60 \times 10^6 \times 0.025} = 9.33 \times 10^{-4} \text{ (m)}$$

$$\theta = \frac{\delta}{0.1} = 0.00933 \text{ (rad)} = 9.33 \times 10^{-3} \text{ (rad)} = 0.535^\circ$$

(b) $T_2 = 0$ 라면

$$375 T_1 = 250 \times 700$$

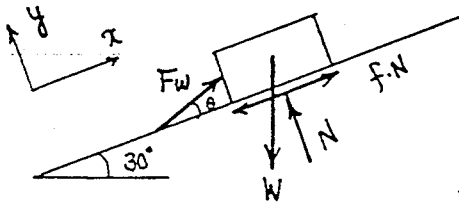
$$\therefore T_1 = \frac{2}{3} \times 700$$

$$\delta = \frac{\frac{2}{3} \times 700 \times 6}{60 \times 10^6 \times 0.025} = 1.867 \times 10^{-3} \text{ (m)}$$

$$\theta_1 = \frac{\delta_1}{0.1} = 1.867 \times 10^{-2} \text{ (rad)} = 1.07^\circ$$

Ans.) (a) 0.535° (b) 1.07°

2.36



$$W = 10 \text{ tons} \quad f = 0.2 \quad \tan \theta = \frac{3}{4}$$

i) 미끄러져 내려올 때

$$\sum F_x = 0; \quad W \sin 30^\circ = f \cdot N + F_w \cos \theta$$

$$\sum F_y = 0; \quad W \cos 30^\circ = N + F_w \sin \theta$$

위 두식을 연립하면

$$F_w = \frac{W(\sin 30^\circ - f \cos 30^\circ)}{\cos \theta - f \sin \theta} = 7651.5 \text{ (lb)}$$

$$\therefore \delta_w = \frac{F_w L}{EA} = \frac{7651.5 \times \sqrt{2^2 + 5^2}}{(1.6 \times 10^6) \times (2 \times 2)} = 6.438 \times 10^{-3} \text{ (ft)} = 0.07726 \text{ (in)}$$

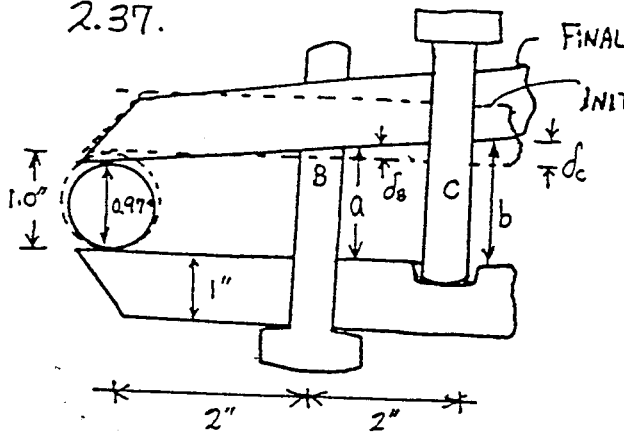
ii) 끌어 올릴 때

$$F_w = \frac{W(\sin 30^\circ + f \cos 30^\circ)}{\cos \theta + f \sin \theta} = 13427.1 \text{ (lb)}$$

$$\therefore \delta_w = \frac{F_w L}{AE} = \frac{13427.1 \times \sqrt{2^2 + 5^2}}{(1.6 \times 10^6) \times (2 \times 2)} = 0.011298 \text{ (ft)} = 0.1356 \text{ (in)}$$

Ans.) $\left[\begin{array}{l} \text{i) 미끄러져 내려올 때 : } \delta_w = 0.07726 \text{ in} \\ \text{ii) 끌어 올릴 때 : } \delta_w = 0.1356 \text{ in} \end{array} \right.$

2.37.



C볼트를 N번
회전시켰을때

δ_B = 볼트 B의 신장량

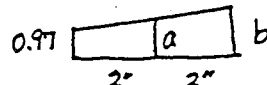
δ_C = 볼트 C의 수축량
라 하자.

$$a = 1 + \delta_B \quad \dots \textcircled{1}$$

$$b = 1 + \frac{N}{20} - \delta_C \quad \dots \textcircled{2}$$

Jo가 절고하다면 변형 후 기하학적 조건으로 부터

$$2a = 0.97 + b \quad \dots \textcircled{3}$$



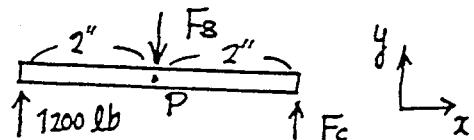
①, ②를 ③에 대입

$$2\delta_B + \delta_C = \frac{N}{20} - 0.03 \quad \dots \textcircled{4}$$

Equilibrium

$$\sum F_y = 0; 1200 + F_C = F_B$$

$$\sum M_P = 0; 1200 \times 2 = F_C \times 2$$



$$\therefore F_B = 2400 \text{ lb}, \quad F_C = 1200 \text{ lb}$$

$$\text{힘-변형 관계식: } \delta_B = \frac{F_B L_B}{A_B E_B} = \frac{2400 \times 1}{0.1 \times 30 \times 10^6} = 0.0008 \text{ in}$$

$$\delta_C = \frac{F_C L_C}{A_C E_C} = \frac{1200 \times 1}{0.1 \times 30 \times 10^6} = 0.0004 \text{ in}$$

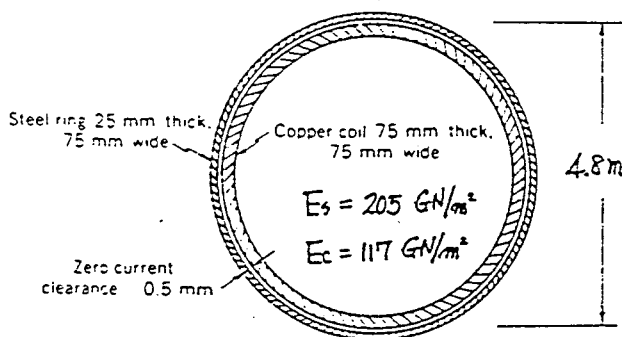
$$N = 0.64 \quad (\because \textcircled{4} \text{로부터})$$

$$\underline{\text{Ans.) } N = 0.64 \text{ (회전)}}$$

$$2.38 \quad \delta_{RS} = \delta_{RC} - 0.5 \quad \dots \textcircled{1}$$

결과 구리/나이에 작용하는 압력 = P_i

$$P_0 = \frac{70,000}{0.075} = 933333.3 \text{ N/m}^2$$



$$\delta_R = \frac{\Delta P \cdot r^2}{Et}$$

$$\delta_{RS} = \frac{P_i r^2}{E_s t_s}$$

$$\delta_{RC} = \frac{(P_o - P_i) r^2}{E_c t_c}$$

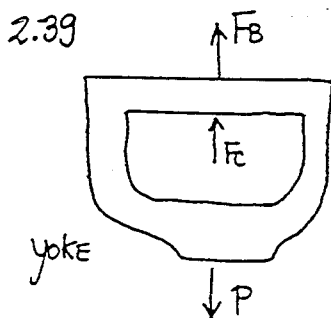
$$\delta_{RS} = \delta_{RC} - 0.5 \text{ 이므로}$$

$$\frac{P_i (2.4)^2 \times 1000}{(205 \times 10^9) \times 0.025} = \frac{(P_o - P_i) \times (2.4)^2 \times 1000}{(117 \times 10^9) \times 0.075} - 0.5$$

$$\therefore P_i = 63275 \text{ (N/m}^2\text{)}, \quad F_c = (P_o - P_i) r \text{ b로 부터}$$

$$F_c = (933333.3 - 63275) \times 2.4 \times 0.075 = 156610.50 \text{ (N)}$$

$$\text{Ans.) } F_c = 156610.5 \text{ N}$$



yoke의 평형 조건

$$P = F_B + F_C \quad \dots \text{ ①}$$

$$(P = 325 \text{ lb})$$

$$\frac{\delta_B}{L_B} = f_B \left(\frac{F_B}{A_B} \right), \quad \frac{\delta_C}{L_C} = f_C \left(\frac{F_C}{A_C} \right)$$

f_B, f_C 는 응력-변형도 곡선의 기울기에 해당.

$$\delta_B = \delta_C = \delta \text{ 를 이용, } L_B f_B \left(\frac{F_B}{A_B} \right) = L_C f_C \left(\frac{F_C}{A_C} \right) \quad \dots \text{ ②}$$

$$\langle \text{플리과정} \rangle \quad \left(\frac{F_B}{A_B} \right) A_B + \left(\frac{F_C}{A_C} \right) A_C = \left(\frac{P}{A} \right)_B \times 0.01 + \left(\frac{P}{A} \right)_C \times 0.02 = P$$

$$\text{또한, } \left(\frac{\delta}{L} \right)_B L_B = \left(\frac{\delta}{L} \right)_C L_C \quad \therefore \left(\frac{\delta}{L} \right)_C / \left(\frac{\delta}{L} \right)_B = \frac{L_B}{L_C} = \frac{3}{2}$$

* 도표

TRIAL	$\left(\frac{\delta}{L} \right)_B$	$\left(\frac{\delta}{L} \right)_C$	$\left(\frac{P}{A} \right)_B$	$\left(\frac{P}{A} \right)_C$	F_B	F_C	P
1	.003	.0045	7,500	13,400	75	268	343
2	.0028	.0042	7,000	13,900	70	260	330
3	.0026	.0039	6,500	12,800	65	256	321

< 앞면으로 부터 계측 >

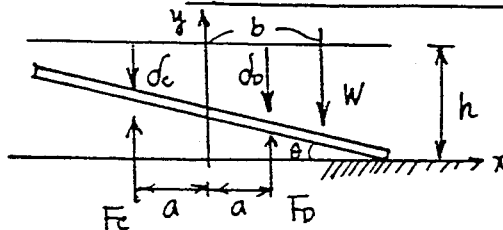
4	.0027	.0041	6750	12963	67.5	259.0	326
---	-------	-------	------	-------	------	-------	-----

* 극한 해 $\Rightarrow P_B = 67 \text{ (lb)}, P_C = 258 \text{ (lb)}$

이때 $(\frac{d}{L})_B = 0.0027 \quad \therefore d = 1.8 \times 0.0027 = 4.86 \times 10^{-3} \text{ in}$

Ans.) $d = 4.86 \times 10^{-3} \text{ in}$

2.40.



$$W = F_c + F_D$$

$$F_D = \frac{a+b}{2a} W$$

$$F_c = \frac{-(b-a)}{2a} W$$

$$\theta = \frac{d_b - d_c}{2a} = \frac{h - \frac{1}{2}(d_c + d_b)}{L} \dots \dots \dots \textcircled{1}$$

$$F_D = \frac{a+b}{2a} W = R_D d_b, \quad F_c = \frac{a-b}{2a} W = R_c d_c \dots \textcircled{2}$$

$$\textcircled{1} \text{을 정리} \quad L(d_b - d_c) = 2ah - a(d_c + d_b) \dots \textcircled{1'}$$

$\textcircled{2} \rightarrow \textcircled{1'}$

$$L \left(\frac{a+b}{2a} \frac{W}{R_D} - \frac{a-b}{2a} \frac{W}{R_c} \right) + a \left(\frac{a-b}{2a} \frac{W}{R_c} + \frac{a+b}{2a} \frac{W}{R_D} \right) = 2ah$$

$$R_c/R_D = \nu, \quad a/L = s \text{ 라 하면}$$

$$\left[\nu \left(s + \frac{b}{L} \right) - \left(s - \frac{b}{L} \right) \right] + s \left[\left(s - \frac{b}{L} \right) + \nu \left(s + \frac{b}{L} \right) \right] = \frac{4s^2 R_c}{W} h$$

$$\therefore (\nu + 1 - s + \nu s) \frac{b}{L} = \frac{4s^2 R_c h}{W} + s - \nu s - s^2 - \nu s^2$$

$$= s^2 \left(\frac{4R_c h}{W} + \frac{1}{s} - \frac{\nu}{s} - 1 - \nu \right)$$

$$\text{그런데} \quad \nu + 1 - s + \nu s = 2 + \nu - 1 + \nu s - s$$

$$= 2 + (\nu - 1)(s + 1) = 2 - (1 - \nu)(1 + s)$$

$$\frac{1}{s} - \frac{\nu}{s} - 1 - \nu = -2 + \frac{1}{s} - \frac{\nu}{s} + 1 - \nu$$

$$= -2 + (1 - \nu) \left(1 + \frac{1}{s} \right) \quad \text{오직}$$

b/L 를 정리하면

$$\therefore \frac{b}{L} = \frac{S^2}{2 - (1-\nu)(1+S)} \left[\frac{4Rch}{W} - 2 + (1-\nu)\left(1 + \frac{1}{3}\right) \right]$$

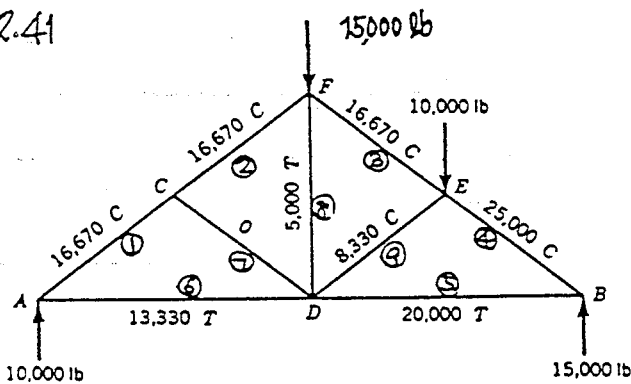
$$= \frac{S^2}{2 - \frac{1}{2}(1-\nu)(1+S)} \left[\frac{2Rch}{W} - 1 + \frac{1}{2}(1-\nu)\left(1 + \frac{1}{3}\right) \right]$$

(단, $\nu = R_c/R_o$, $S = a/L$)

$$\therefore, \frac{b}{L} = \frac{(a/L)^2}{1 - \frac{1}{2}(1-\nu)(1 + a/L)} \left[\frac{2Rch}{W} - 1 + \frac{1}{2}(1-\nu)\left(1 + \frac{L}{a}\right) \right]$$

- Q.E.D -

2.41



에너지 방법을

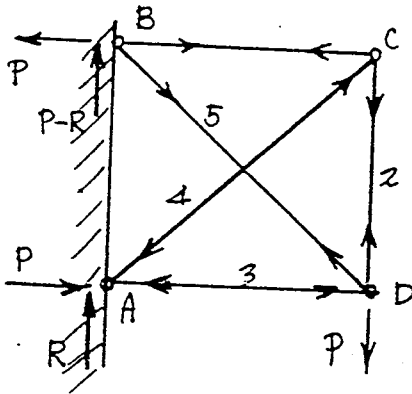
이용한다

$$\delta_P = \sum_{i=1}^n F_i \frac{L_i}{A_i E_i} \frac{\partial F_i}{\partial P}$$

	F	$\frac{L}{EA}$	$\frac{\partial F}{\partial P}$	$\frac{FL}{EA} \cdot \frac{\partial F}{\partial P}$
1	-16670	$\frac{5}{4 \times (10 \times 10^6)}$	$-\frac{2.5}{3}$	$208375 \frac{1}{10 \times 10^6 \times 12}$
2	-16670	$\frac{5}{4 \times (10 \times 10^6)}$	$-\frac{2.5}{3}$	"
3	-16670	"	$-\frac{2.5}{3}$	"
4	-25,000	"	$-\frac{2.5}{3}$	$312500 \frac{1}{10 \times 10^6 \times 12}$
5	20,000	"	$\frac{2}{3}$	$200000 \frac{1}{10 \times 10^6 \times 12}$
6	13,330	"	$\frac{2}{3}$	$133000 \frac{1}{10 \times 10^6 \times 12}$
7	0	$\frac{5}{2 \times (10 \times 10^6)}$	0	0
8	5000	$\frac{6}{2 \times (10 \times 10^6)}$	0	0
9	-8330	$\frac{5}{2 \times (10 \times 10^6)}$	0	0
Σ				0.011 (ft)

Ans.) $\delta_{vH} = 0.011 \text{ ft}$

2.42



A점; $F_1 = \sqrt{2}R$

$F_3 = P - F_4 \cos 45^\circ = P - R$

B점; $F_5 = \sqrt{2}(P - R)$

$F_1 = P - F_5 \cos 45^\circ = R$

C점; $F_2 = F_4 \cos 45^\circ = R$

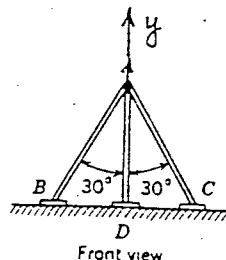
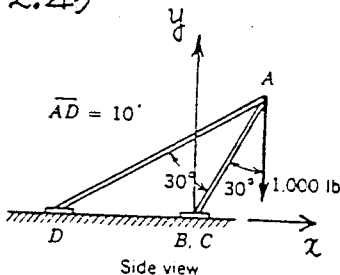
	F	$\frac{L}{EA}$	$\frac{\partial F}{\partial P}$	$\frac{FL}{EA} \cdot \frac{\partial F}{\partial P}$
1	R	$\frac{L}{EA}$	1	$\frac{RL}{EA}$
2	R	"	1	"
3	$-(P - R)$	"	1	$\frac{(R - P)L}{EA}$
4	$-\sqrt{2}R$	$\sqrt{2} \frac{L}{EA}$	$-\sqrt{2}$	$\frac{2\sqrt{2}RL}{EA}$
5	$\sqrt{2}(P - R)$	"	$-\sqrt{2}$	$\frac{2\sqrt{2}(R - P)L}{EA}$
Σ				$(3 + 4\sqrt{2})R - (1 + 2\sqrt{2})P / \frac{EA}$

$\delta_{AV} = 0$ 일때 $(3 + 4\sqrt{2})R = (1 + 2\sqrt{2})P$

$\therefore R = \frac{1 + 2\sqrt{2}}{3 + 4\sqrt{2}} P = 0.442 P$ $\therefore F_{CD} = 0.442 P$ (tension)

Ans.) $F_{CD} = 0.442 P$ (tension)

2.43



$AD = 10'$, $A_{AD} = 1 \text{ in}^2$

$A_{AB} = A_{AC} = 2.1 \text{ in}^2$

$E_{AD} = E_{AC} = 30 \times 10^6 \text{ psi}$

* 가정; 부중량인

구조물을 경량 구조물로

보기 위하여 base와 member 사이의 bolt joint를 pin joint로 가정하고 마찰이 없다고 본다.

$$\overline{AB} = \overline{AC} = \sqrt{(AD \sin 30^\circ \tan 30^\circ)^2 + (AD \sin 30^\circ \frac{1}{\cos 30^\circ})^2} = 6.455 \text{ (ft)}$$

$$\sum M_z = 0 ; \quad F_{AD} (5 \tan 30^\circ) = 1000 \times (5 \tan 30^\circ)$$

$$\therefore F_{AD} = 1000 \text{ lb}$$

$$\vec{F}_{AB} = F_{AB} \frac{1}{6.455} (5 \tan 30^\circ \vec{i} + 5 \vec{j} - 5 \tan 30^\circ \vec{k})$$

$$\sum F_x = 0 ; \quad 2 \left(\frac{F_{AB} 5 \tan 30^\circ}{6.455} \right) = 1000 \cos 30^\circ \quad (\because A_{AB} = A_{AC})$$

$$\therefore F_{AB} = 968.25 \text{ (lb)}$$

① 수평 이동

	F (lb)	$\frac{L}{EA}$	$\frac{\partial F}{\partial P}$	$\frac{FL}{EA} \cdot \frac{\partial F}{\partial P}$
AD	1000	10^{-6}	1	10^{-3}
AB	-968.25	1.0246×10^{-7}	-0.96825	9.6056×10^{-5}
AC	"	"	"	"
Σ				$1.192 \times 10^{-3} \text{ (ft)}$

$$\therefore \delta_{AV} = 1.192 \times 10^{-3} \text{ (ft)} = 0.0143 \text{ (in)} \quad (\text{아래방향으로})$$

② 수평 이동

$$\sum M_z = 0 ; \quad 5 = F_{AD}' 5 \tan 30^\circ \quad \therefore F_{AD}' = 1.732$$

$$F_{AB}' = F_{AB} \frac{1}{6.455} (5 \tan 30^\circ \vec{i} + 5 \vec{j} - 5 \tan 30^\circ \vec{k})$$

$$\sum F_x = 0 ; \quad F_{AB}' \frac{1}{6.455} 5 \tan 30^\circ = F_{AD}' - 1$$

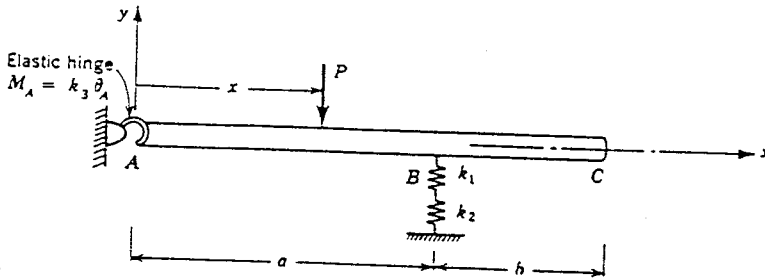
$$\therefore F_{AB}' = 1.559$$

	F (lb)	$\frac{L}{EA}$	$\frac{\partial F}{\partial P}$	$\frac{FL}{EA} \cdot \frac{\partial F}{\partial P}$
AD	1000	10^{-6}	1.732	1.732×10^{-3}
AB	-968.25	1.0246×10^{-7}	-0.559	5.546×10^{-5}
AC	"	"	"	"
Σ				$1.84 \times 10^{-3} \text{ (ft)}$

$$\therefore \delta_{AH} = 1.84 \times 10^{-3} \text{ ft} = 0.0221 \text{ in (right)}$$

Ans.) $\delta_{AH} = 0.0221 \text{ in}, \delta_{AV} = 0.0143 \text{ in}$

2.44



$$(a) \sum M_{kA} = 0 ; P x = k_3 \theta_A + k a \theta_A \cdot a \quad \dots \dots \textcircled{1}$$

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} \rightarrow k = \frac{k_1 k_2}{k_1 + k_2} \quad \dots \dots \textcircled{2}$$

$$F_B = k a \theta_A \quad \therefore \theta_A = F_B / k a \quad \dots \dots \textcircled{3}$$

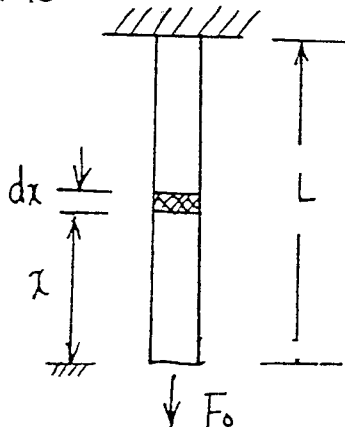
$$\textcircled{3} \text{을 } \textcircled{1} \text{에 대입} \quad P x = \frac{k_3 F_B}{k a} + \frac{k a^2 F_B}{k a}$$

$$\therefore F_B = \frac{k a P x}{k_3 + k a^2} = \frac{k_1 k_2 a P x}{(k_1 + k_2) k_3 + k_1 k_2 a^2}$$

$$b) \theta_A = \frac{F_B}{k a} = \frac{(k_1 + k_2) P x}{(k_1 + k_2) k_3 + k_1 k_2 a^2}$$

$$\text{Ans.)} \left[\begin{array}{l} F_B = \frac{k_1 k_2 a P x}{(k_1 + k_2) k_3 + k_1 k_2 a^2} \\ \theta_A = \frac{(k_1 + k_2) P x}{(k_1 + k_2) k_3 + k_1 k_2 a^2} \end{array} \right]$$

2.45



(a) 막대기의 자중을 생각하면

$$F x = w x + F_0, \quad dU = \frac{F x^2}{2EA} dx$$

$$\therefore U = \int_0^L \frac{L(w x + F_0)^2}{2EA} dx$$

$$= \frac{1}{2EA} \left[\frac{w^2}{3} x^3 + w F_0 x^2 + F_0^2 x \right]_0^L$$

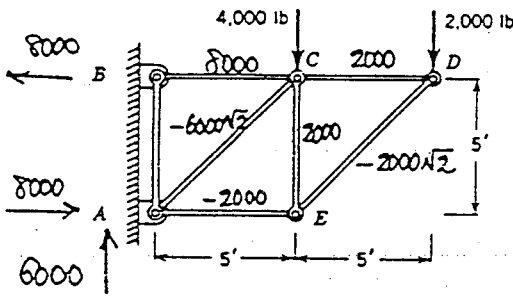
$$= \frac{1}{6EA} \left(\frac{w_0}{L^2} + 3 w F_0 L + 3 F_0^2 \right)$$

(b) 막대기 자중 없이 생각하면

$$U = \frac{F_0^2 L}{2EA}$$

$$\text{Ans.} \left\{ \begin{array}{l} (a) \quad U = \frac{L}{6EA} (W_0 L^2 + 3W_0 L + 3F^2) \\ (b) \quad U = \frac{F^2 L}{2EA} \end{array} \right.$$

2.46



$$(a) \text{ 경우 } (P_c)_{\max} = 6000\sqrt{2} = 8485 \text{ lb}$$

$$(P_t)_{\max} = 8000 \text{ lb}$$

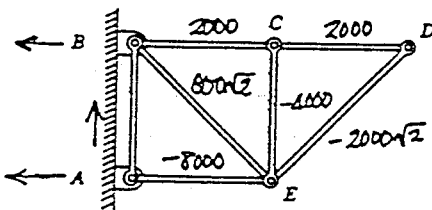
$$(1) \quad (\sigma_{\max})_t = \frac{8000}{2.17} = 3686.6 \text{ psi} < 20000 \text{ psi}$$

$$\frac{10^8 I}{L^2} = \frac{10^8 \times 0.59}{(5\sqrt{2} \times 12)^2} = 8194.41 \text{ (lb)} > 2000\sqrt{2} \text{ (o)}$$

그러나 $8194.4 < (P_c)_{\max} = 8485 \text{ lb}$ 이므로

(1)의 재료를 사용 할 수 없다. (2) 재료를 4층에 생각.

(b) 경우



$$(P_t)_{\max} = 6000\sqrt{2} \text{ lb}$$

$$(P_c)_{\max} = 8000 \text{ lb}$$

$$(1) \quad (\sigma_{\max})_t = \frac{6000\sqrt{2}}{2.17}$$

$$= 3910.3 \text{ psi} < 20000 \text{ (o)}$$

$$\frac{10^8 I}{L^2} = \frac{10^8 \times 0.59}{(5 \times 12)^2} = 16388.9 \text{ lb} > 8000 \text{ lb (o)}$$

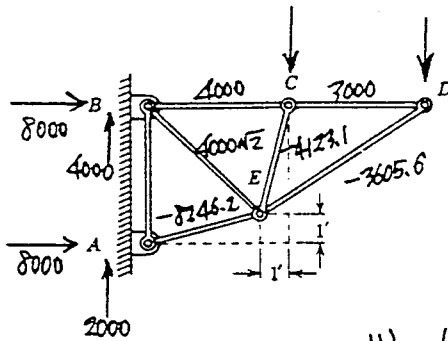
$$\text{총 중량} = 293.566 \text{ lb}$$

(1)의 재료가 합당하므로 (2)의 경우는 고려하지 않아도 된다.

(c) 경우

$$(P_t)_{\max} = 4000\sqrt{2} \text{ lb}$$

$$(P_c)_{\max} = 8246.2 \text{ lb}$$



$$(1) (\sigma_{max})_t = \frac{4000\sqrt{2}}{2.17} = 2606.8 \text{ psi} < 20,000 \text{ psi} \quad (0)$$

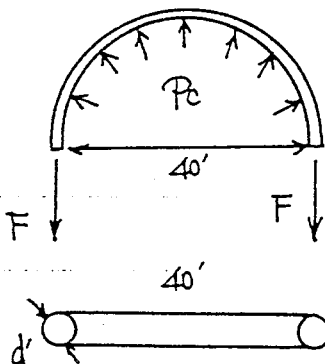
$$i) \frac{10^8 I}{L^2} = \frac{10^8 \times 0.59}{(\sqrt{17} \times 2)^2} = 24101.3 > 8246 \quad (0)$$

$$ii) \frac{10^8 I}{L^2} = \frac{10^8 \times 0.59}{(\sqrt{36+16} \times 12)^2} = 7879.3 > 3605.6 \quad (0)$$

$$\text{중량} = (5 \times 3 + 4\sqrt{2} + 2\sqrt{17} + \sqrt{52}) \times 7.5 = 270.9 \text{ lb} \quad (2b)$$

∴ (C)의 경우로 설계하고 (1)의 재료를 사용하면 주어진
중량에 최소 무게 (270.9 lb)로 설계할 수 있다.

2.47



$$A = 170 \times \frac{\pi}{4} \left(\frac{1}{4}\right)^2$$

$$d' = \sqrt{\frac{4}{\pi} A} = \frac{1}{4} \sqrt{170}$$

$$2F = P_c \cdot d' \times (40 \times 12)$$

$$= P_c \frac{1}{4} \sqrt{170}$$

$$\therefore F = 782.3 P_c \text{ lb}$$

(단, P_c 단위는 psi)

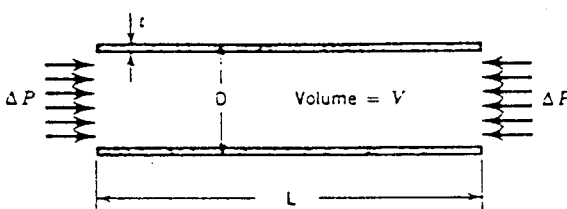
$$P_i \times (31 \times 12) \left(\frac{75 \times 12}{310} \right)$$

$$= 2FA = 2 \times 782.3 P_c \left\{ 170 \frac{\pi}{4} \left(\frac{1}{4}\right)^2 \right\}$$

$$\therefore P_i = 12.1 P_c$$

Ans.) $F = 782.3 P_c \text{ lb}$, $P_i = 12.1 P_c \text{ psi}$

2.48



$$\sigma_R = \frac{\Delta P r^2}{E t} \left(1 + \frac{t}{2r} \right)$$

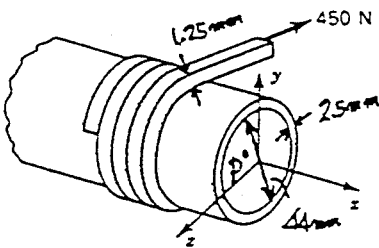
$$V = \frac{\pi D^2}{4} L$$

$$\begin{aligned}
 \Delta V &= \pi \left(\frac{D}{2} + d_R \right)^2 L - \pi D^2 L = \pi L d_R (D + d_R) \\
 R &= \frac{\Delta V}{V} \frac{1}{\Delta P} = \frac{1}{V} \cdot \pi L d_R (D + d_R) \frac{1}{\Delta P} \\
 &= \frac{1}{V} \pi L D d_R \frac{1}{\Delta P} \quad (\because d_R^2 \ll 1) \\
 &= \frac{1}{V} \pi L D \frac{\Delta P r^2}{E t} \left(1 + \frac{t}{2r} \right) \frac{1}{\Delta P} = \frac{\pi L r^2}{V} \frac{1}{E t} (D + t) \\
 &= \frac{1}{E t} (D + t) = \frac{11.25 \times 10^{-3}}{(205 \times 10^9) (1.25 \times 10^{-3})} \\
 &= 4.39 \times 10^{-11} \text{ (m/N)} = 4.39 \times 10^{-5} \text{ (mm/N)} \\
 \text{Ans.) } R &= 4.39 \times 10^{-5} \text{ (mm/N)}
 \end{aligned}$$

$$\begin{aligned}
 2.49. \quad F &= \pi r^2 p = \pi \times (0.05)^2 \times \left(\frac{60}{760} \times 101325 \right) \\
 &= 62.8 \text{ (N)}
 \end{aligned}$$

$$\text{Ans.) } F = 62.8 \text{ N}$$

2.50



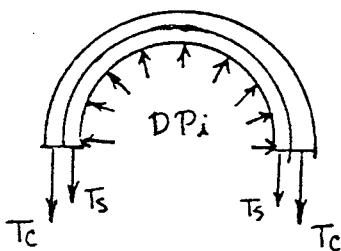
(a)

$$\begin{aligned}
 2 \times 450 &= P_0 \times (49 \times 10^{-3}) (1.25 \times 10^{-3}) \\
 \therefore P_0 &= 146938 \text{ (MN/m}^2\text{)} \\
 d_R &= \frac{P r^2}{E t} \left(1 - \frac{t}{2r} \right) \\
 &= \frac{146938 \times 10^6 \times (24.5 \times 10^{-3})^2}{(140 \times 10^9) (2.5 \times 10^{-3}) \left(1 - \frac{25}{49} \right)} \\
 &= 2.391 \times 10^{-5} \text{ (m)}
 \end{aligned}$$

$$\therefore d_R = 0.02391 \text{ (mm)}$$

$$(b) \quad P_i D_b = 2 (T_s + T_c) \quad \dots \dots \dots \textcircled{1}$$

$$d_{RS} = \frac{T_s L_s}{2\pi E_s A_s}, \quad d_{RC} = \frac{T_c L_c}{2\pi E_c A_c}$$



$$\delta_{RS} = \delta_{RC}$$

$$\frac{T_c \cdot \pi (44 + 2.5) \times 10^{-3}}{(140 \times 10^9) (2.5 \times 10^{-3}) \times (1.25 \times 10^{-3})} = \frac{T_s \pi (49 + 1.25) \times 10^{-3}}{(205 \times 10^9) (1.25 \times 10^{-3})^2}$$

$$\therefore T_c = 1.476 T_s \quad \dots \textcircled{2}$$

① ② 식으로 풀기

$$P_i (44 \times 10^{-3}) (1.25 \times 10^{-3}) = 2 (1.476 + 1) T_s$$

$$\therefore P_i = 90036.478 T_s = 61000.194 T_c$$

$$(T_c)_{\max} = (210 \times 10^6) \times (2.5 \times 10^{-3}) \times (1.25 \times 10^{-3}) = 656.25 \text{ N}$$

$$(T_s)_{\max} = (1.0 \times 10^9) \times (1.25 \times 10^{-3})^2 = 1562.5 \text{ N}$$

또한 외부에서 450 N의 작용하므로

$$(T_s)'_{\max} = (T_s)_{\max} - 450 = 1112.5 \text{ (N)}$$

$$(T_c)'_{\max} = (T_c)_{\max} + 450 = 1106.25 \text{ (N)} \quad \dots \textcircled{3}$$

②, ③ 식으로 풀기 T_c 가 먼저 극한값에 도달 한다는 것을 알 수 있다.

$$\therefore P_i \leq 61000.194 \times 1106.25 \text{ (N/m}^2\text{)}$$

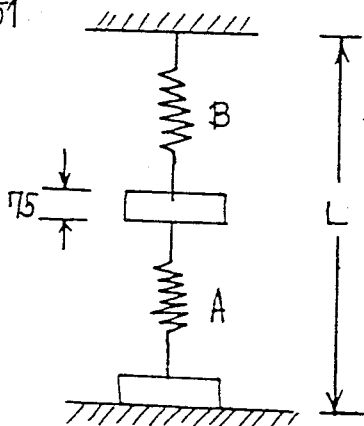
$$\leq 67.481 \text{ (MN/m}^2\text{)}$$

Ans.)

$$(a) \delta_r = 2.391 \times 10^{-5} \text{ m}$$

$$(b) P_i \leq 67.481 \text{ MN/m}^2$$

2.51



Spring free length Spring Constant

A 250 mm 14 MN/m

B 200 mm 0.875 MN/m

$$\delta_A + \delta_B = L - (600 - 75) \quad \dots \textcircled{1}$$

$$k_A \delta_A = k_B \delta_B \quad \dots \textcircled{2}$$

① ②로부터 $\delta_A = \frac{R_B}{R_A + R_B} (L - 525)$, $\delta_B = \frac{R_A}{R_A + R_B} (L - 525)$

하중 W로 δ 만큼 처짐이 생길때 코iled 늘어난 Spring 길이

$$\delta_B' = \delta_B + \delta, \quad \delta_A' = \delta_A - \delta$$

$$\therefore F_B = R_B (\delta_B + \delta), \quad F_A = R_A (\delta_A - \delta)$$

$$\Sigma F_y = 0, \quad W = F_B - F_A$$

$$W = (R_A + R_B) \delta + R_B \delta_B - R_A \delta_A$$

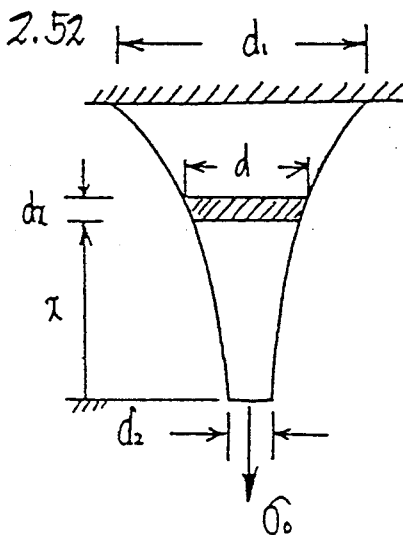
$$= (R_A + R_B) \delta + \frac{R_A R_B}{R_A + R_B} (L - 525) - \frac{R_A R_B}{R_A + R_B} (L - 525)$$

$$\therefore \frac{W}{\delta} = R_A + R_B$$

$$\text{Sensitivity} = \frac{\delta}{W} = \frac{1}{R_A + R_B} = \frac{1}{1.4 + 0.875}$$

$$= 0.43956 \text{ mm/kN}$$

[결론] Sensitivity는 R_A 및 R_B 에만 관계될 뿐
길이 L에는 무관하다.



$$\frac{\pi d^2}{4} \sigma_0 = \int_0^x r \frac{\pi d^2}{4} dr + \frac{\pi d^2}{4} \sigma_0$$

양변을 x에 대해 미분

$$\frac{\pi}{4} \sigma_0 \frac{d}{dx} (d^2) = r \frac{\pi}{2} d^2$$

$$\frac{r}{\sigma_0} dx = \frac{d(d^2)}{d^2} \quad \text{양변 적분}$$

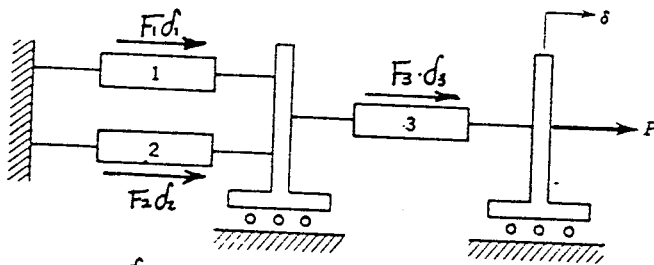
$$\frac{r}{\sigma_0} x = \ln(d^2) + C$$

$$\therefore d^2 = A e^{\frac{r}{\sigma_0} x}$$

$$d^2 = d_2^2 e^{\frac{r}{\sigma_0} x} \quad \therefore d = d_2 e^{\frac{r}{2\sigma_0} x}$$

Ans.) $d = d_2 e^{\frac{r}{2\sigma_0} x}$

2.53



$$\int_0^\delta p d\delta = \sum_{i=1}^3 \int_0^{e_i} F_i de_i$$

$$U = \int_0^{e_1} F_1 de_1 + \int_0^{e_2} F_2 de_2 + \int_0^{e_3} F_3 de_3$$

$d_1 = d_2$ 이며 $F_1 + F_2 = F_3 = P$ 이므로

$$U = \int_0^{e_1} (F_1 + F_2) de_1 + \int_0^{e_3} F_3 de_3$$

$$= \int_0^{e_1} P de_1 + \int_0^{e_3} F_3 de_3 \quad (\because F_1 + F_2 = F_3 = P)$$

또한, $d_1 + d_3 = e_1 + e_3 = dP = \text{const}$ 이므로

$$U = \int_0^{e_1+e_3} P d(e_1+e_3) = \int_0^\delta P d\delta$$

$$\therefore U = \int_0^\delta P d\delta = \sum_{i=1}^3 \int_0^{e_i} F_i de_i$$

2.54. $U^* = \int_0^{F_1} e_1 dF_1 + \int_0^{F_2} e_2 dF_2 + \int_0^{F_3} e_3 dF_3$ — A.E.D —

$$= \int_0^{F_1+F_2} e_1 d(F_1+F_2) + \int_0^{F_3} e_3 d(F_3)$$

$(\because e_1 = e_2, e_1 = \text{const})$

$$= \int_0^P e_1 dP + \int_0^P e_3 dP \quad (\because F_1+F_2 = F_3 = P)$$

$$= \int_0^P \delta dP \quad (\because e_1+e_3 = \delta)$$

$$\therefore U^* = \int_0^P \delta dP = \sum_{i=1}^3 \int_0^{F_i} e_i dF_i$$

— A.E.D —

2.55 $\sum_i \int_{S_i} \vec{P}_i \cdot d\vec{S}_i = U$

S_i 위의 다른 변위는 없다고 본 상태에서

S_i 에 대해 대변히 작은 ΔS_i 가 발생했다고 가정

변위는 평행을 유지하며 서서히 변하고
potential work의 증분은 potential Energy의
증분 ΔU 와 같을 것이다.

$$\vec{P}_i \cdot \Delta \vec{S}_i = \Delta U$$

$$\begin{aligned} \vec{P}_i \cdot \Delta \vec{S}_i &= P_i \Delta S_i \cos \theta = (P_i \cos \theta) \Delta S_i \\ &= f_i \Delta S_i \end{aligned}$$

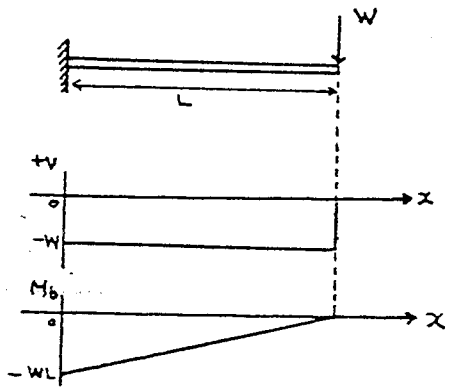
$$\therefore f_i \Delta S_i = \Delta U$$

$$\text{따라서, } \Delta S_i \text{ 값이 매우 작다면 } f_i = \frac{\partial U}{\partial S_i}$$

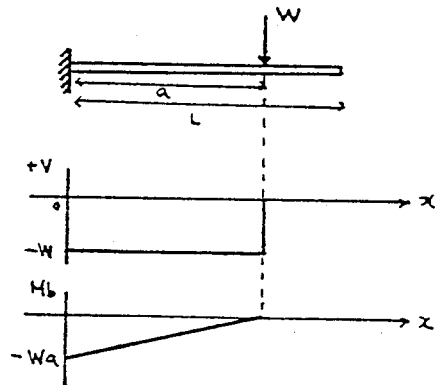
— Q.E.D —

CHAPTER 3

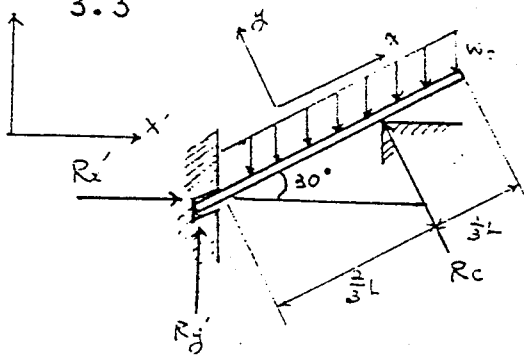
3.1



3.2



3.3



$$\Sigma F_{x'} = 0 \quad R_{x'} = R_c \sin 30^\circ$$

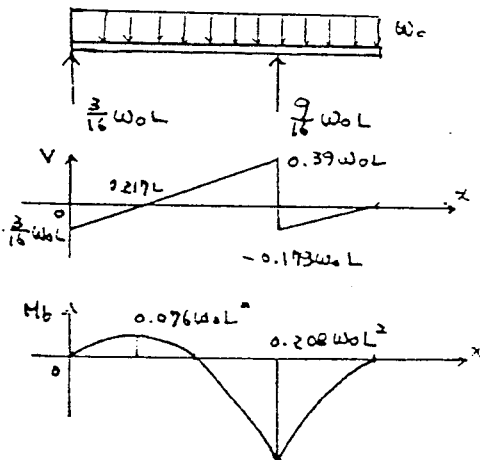
$$\Sigma F_{y'} = 0 \quad R_{y'} + R_c \cos 30^\circ = w_0 L \cos 30^\circ$$

$$\Sigma M_c = 0 \quad \frac{2}{3} R_c L = \frac{1}{2} w_0 (L \cos 30^\circ)^2$$

$$\therefore R_c = \frac{9}{16} w_0 L$$

$$R_{x'} = \frac{9}{32} w_0 L$$

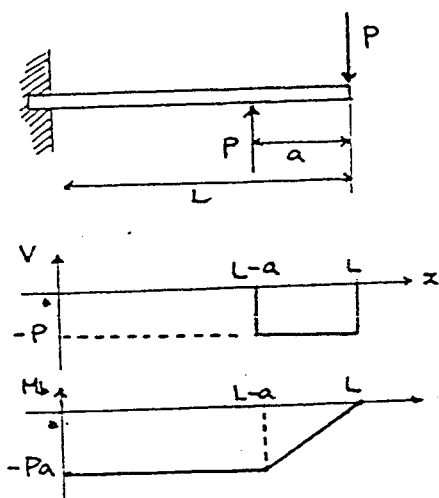
$$R_{y'} = \frac{2\sqrt{3}}{32} w_0 L$$



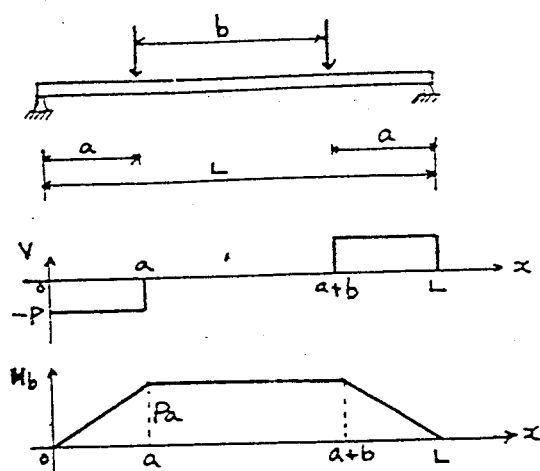
$$R_y = R_{y'} \cos 30^\circ - R_{x'} \sin 30^\circ$$

$$= \frac{3}{16} w_0 L$$

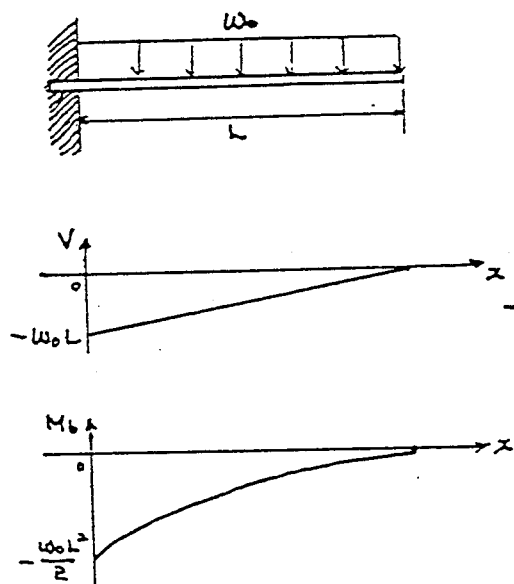
3.4



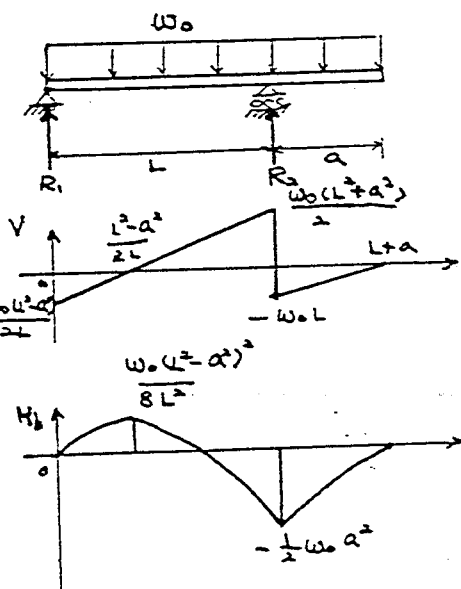
3.5



3.6

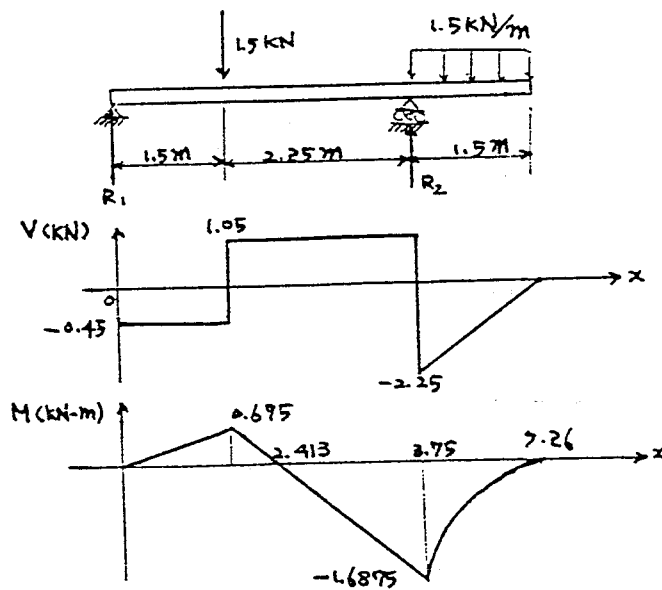


3.7



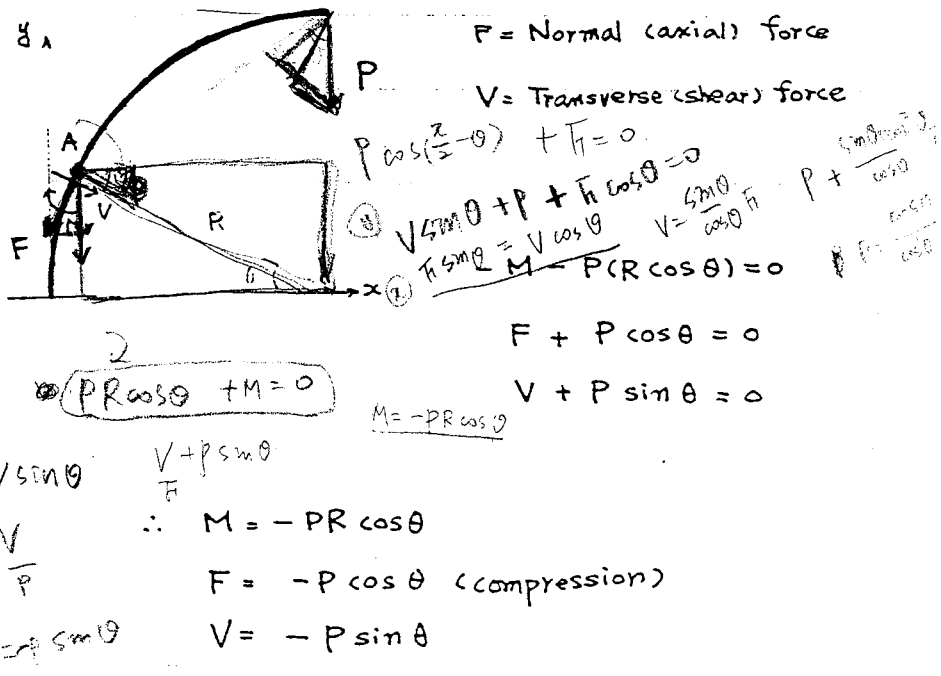
$$R_1 = \frac{w_0(L^2-a^2)}{2L}, \quad R_2 = \frac{w_0(L+a)^2}{2L}$$

3.8

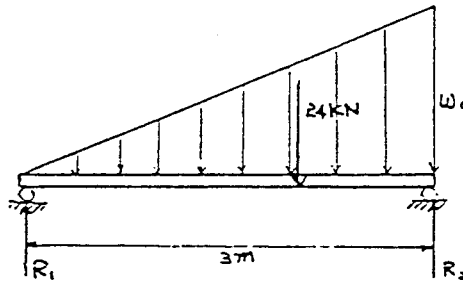


$$R_1 = 0.45 \text{ kN} \quad R_2 = 3.3 \text{ kN}$$

3.9



3.10

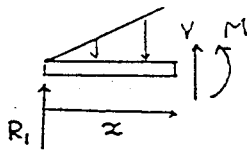


$$R_1 = 8000 \text{ N}$$

$$R_2 = 16000 \text{ N}$$

$$\int_0^3 w_0 \frac{x}{L} dx = 24,000$$

$$= \frac{1}{2} w_0 \frac{x^2}{L} \Big|_0^3$$



$$V = -R_1 + \frac{1}{2} x \left(w_0 \frac{x}{L} \right) \quad \therefore w_0 = 16000 \text{ (N/m)}$$

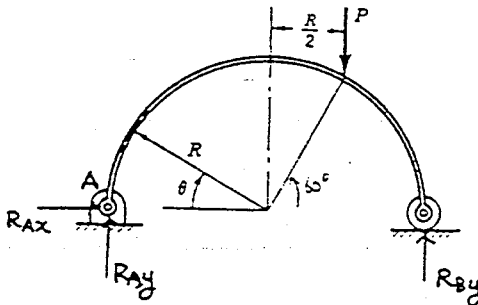
$$= -8000 + \frac{8000}{3} x^2$$

$$M = R_1 x - \frac{1}{2} \left(w_0 \frac{x}{L} \right) \cdot x \cdot \left(\frac{x}{3} \right)$$

$$= R_1 x - \frac{w_0 x^3}{6L}$$

$$= 8000 x - \frac{8000}{9} x^3$$

3.11



$$\sum (M_z)_B = 0 : R_{By} (2R) - P \left(\frac{3}{2} R \right) = 0$$

$$\sum F_x = 0 : R_{Ax} = 0$$

$$\sum F_y = 0 : R_{Ay} + R_{By} - P = 0$$

$$\therefore R_{By} = \frac{3}{4} P \quad R_{Ay} = \frac{P}{4}$$

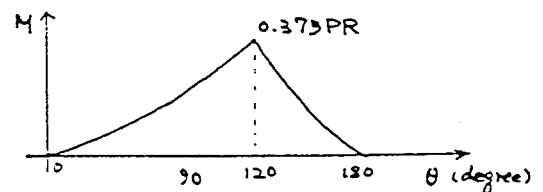
i) $\theta < 120^\circ$

$$M - \frac{P}{4} (R(1 - \cos \theta)) = 0$$

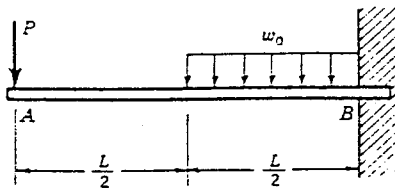
$$M = \frac{PR}{4} (1 - \cos \theta)$$

ii) $\theta > 120^\circ$

$$M = \frac{3PR}{4} (1 + \cos \theta)$$



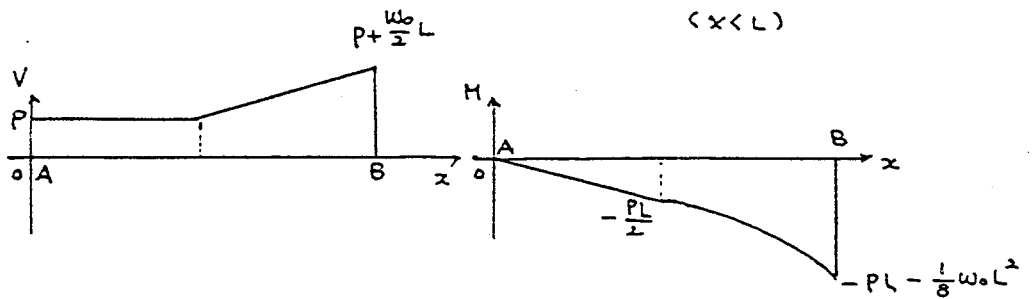
3.12



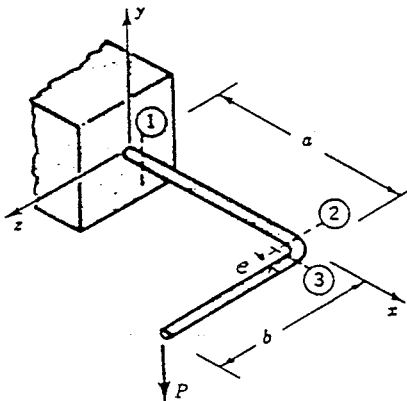
$$q(x) = -P \langle x \rangle^{-1} - w_0 \langle x - \frac{L}{2} \rangle^0$$

$$V = - \int_{-\infty}^x q dx = \{ P \langle x \rangle^0 + w_0 \langle x - \frac{L}{2} \rangle^1 \}$$

$$M = - \int_{-\infty}^x V dx = \{ -P \langle x \rangle^1 - \frac{w_0}{2} \langle x - \frac{L}{2} \rangle^2 \}$$



3.13



Section 3 $V_{y3} = -P$
 $M_{x3} = -Pb$

Section 2.

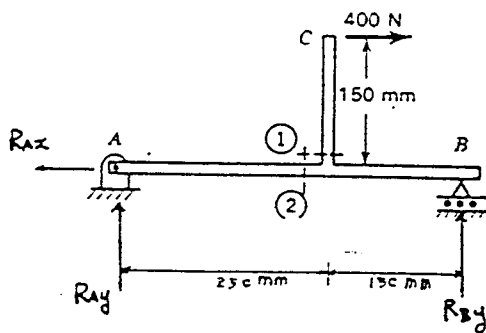
$e \ll b$ 라고 가정 하고 무시 하면

$V_{y2} = -P$
 $M_{xz2} = Pb$

Section 3 $V_{y1} = -P$
 $M_{bz1} = -Pa$
 $M_{tx1} = Pb$

Ans) $V_{y1} = -P, M_{bz} = -Pa, M_{tx1} = Pb$
 $V_{y2} = -P, M_{tx2} = Pb, V_{y3} = -P, M_{bx3} = -Pb$

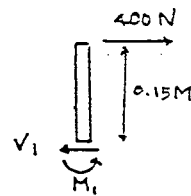
3.14



$$R_{Ay} = -R_{By} = -150 \text{ N}$$

$$R_{Ax} = 400 \text{ N}$$

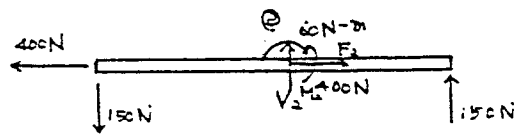
Section 1.



$$V_1 = 400 \text{ N}$$

$$M_1 = -60 \text{ N-m}$$

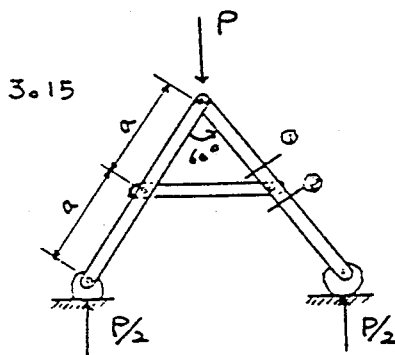
Section 2.



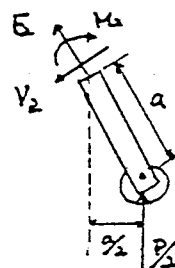
$$F_2 = 400 \text{ N}$$

$$V_2 = 150 \text{ N}$$

$$M_2 = -375 \text{ N-m}$$



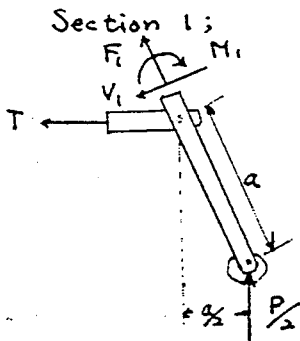
Section 2



$$M_2 = \frac{P_0}{2}$$

$$F_2 = -\frac{P}{2} \sin 30^\circ = -\frac{P}{4}$$

$$V_2 = \frac{P}{2} \cos 30^\circ = \frac{\sqrt{3}}{4} P$$



$$\Sigma M = 0 \quad T a \sin 60^\circ = \frac{P}{2} (2a) \cos 60^\circ$$

$$\therefore T = \frac{Pa}{4}$$

$$M_i = \frac{Pa}{4}$$

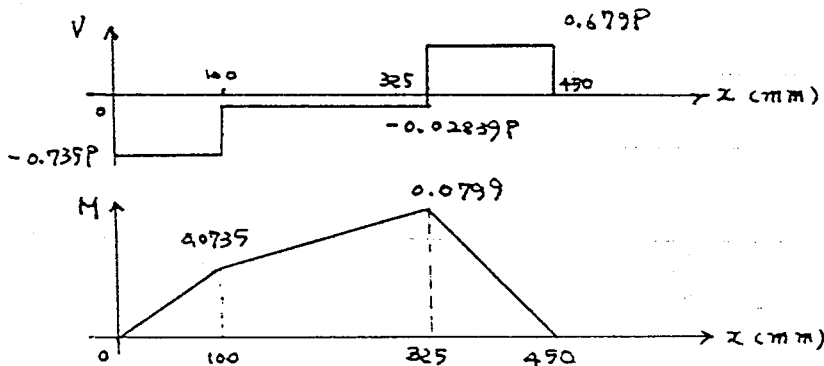
$$V_i = \frac{P}{2} \sin 30^\circ - T \sin 60^\circ = -\frac{P}{4}$$

$$F_i = -\frac{P}{2} \cos 30^\circ - T \cos 60^\circ = -\frac{\sqrt{3}}{2} P$$

8.16

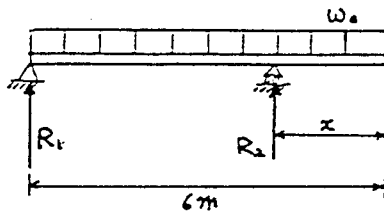
$$\Sigma M_B = 0 ; \quad R_A \cdot 450 = (P \cos 45^\circ \cdot 350) + P(\cos 45^\circ \cdot 13) + (P \sin 45^\circ \cdot 125) - (P \sin 45^\circ \cdot 20)$$

$$\therefore R_A = 0.735 P \quad \therefore R_D = \sqrt{2} P - R_A = 0.679 P$$



B와 C 이 붙어있는 곳쪽은 중간부분의 Bending Moment로
무산시켜 즉 max. Bending Moment의 값을 낮게 해주는
효과.

3.17



$$R_2(6-x) = \frac{1}{2} w_0(6)^2$$

$$\therefore R_2 = \frac{18 w_0}{6-x}$$

$$R_1 = 6 w_0 - \frac{18 w_0}{6-x}$$

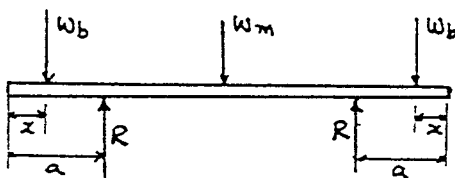
$$= \frac{18 w_0 - 6 w_0 x}{6-x}$$

$$R_1 \times 4.2 = \frac{w_0 \times 4.2^2}{2} \quad \text{여기서 } R_1 \text{ 을代入}$$

$$\frac{18 w_0 - 6 w_0 x}{6-x} \cdot 4.2 = \frac{w_0 \times 4.2^2}{2}$$

$$\therefore x = 1.385 \text{ (m)}$$

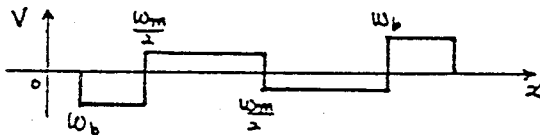
3.18



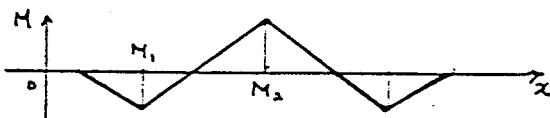
$$R = w_b + \frac{w_m}{2}$$

$$|M_1| = w_b(a-x)$$

$$|M_2| = (w_b + \frac{w_m}{2})(\frac{L}{2} - a) - w_b(\frac{L}{2} - x)$$



$|M_1| = |M_2|$ 인 경우에 max bending moment 가 최소로 될 것이므로

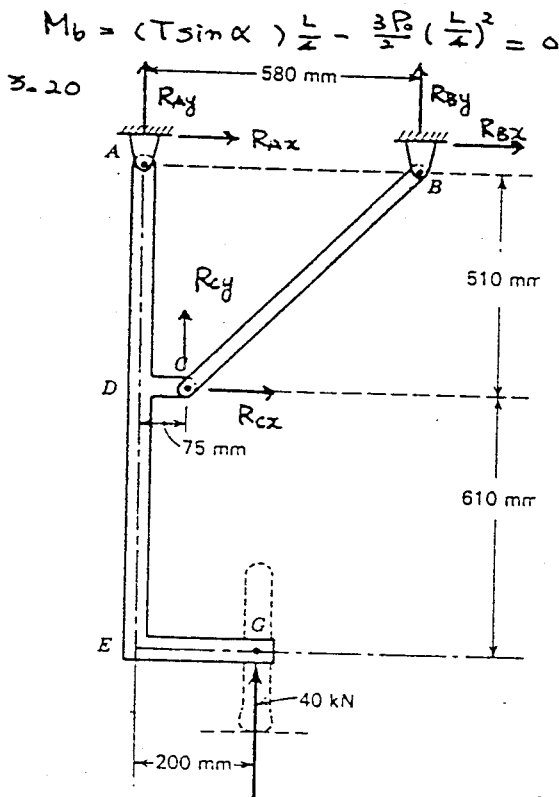
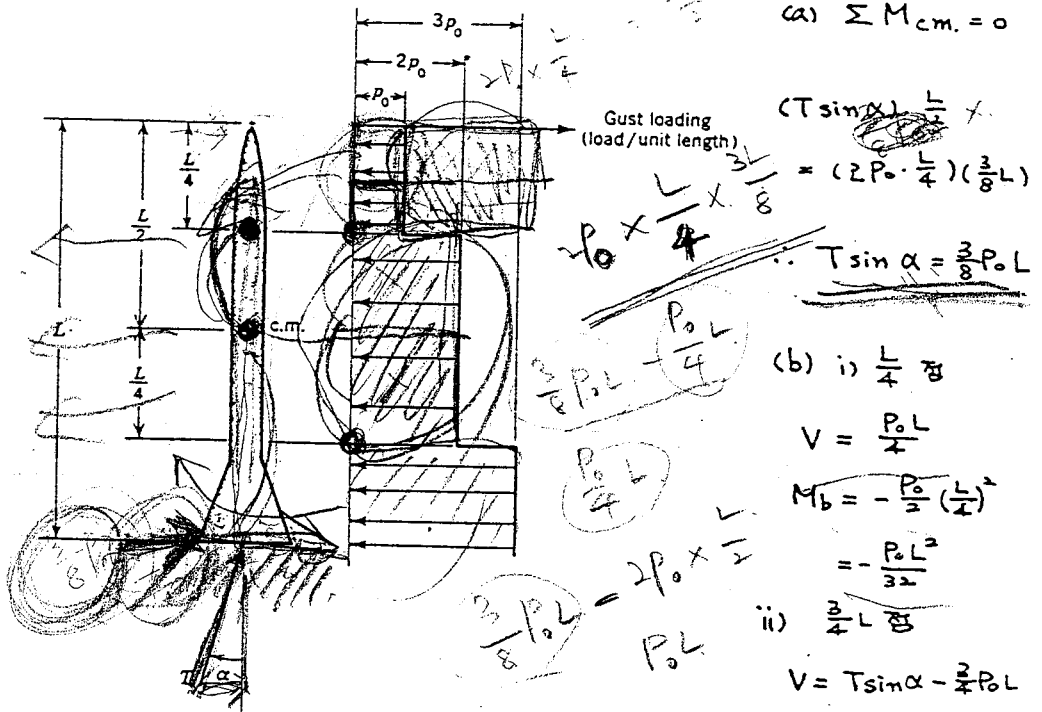


$$|M_1| = |M_2| \text{ 에서 } x \text{ 를 구하면}$$

$$x = a - \frac{w_m}{8 w_b} (L - 2a)$$

즉 w_b 를 $a - \frac{w_m}{8 w_b} (L - 2a)$ 되는 지점에 놓아야 bending moment 의 최대 값이 최소가 되므로 목수의 생각은 틀렸다.

3.19



a)

$$\sum M_B = 0;$$

$$580 R_{Ay} + 40000 \times 200 = 0$$

$$\therefore R_{Ay} = -13793.1 \text{ N}$$

$$R_{By} = -26206.9 \text{ N}$$

CD 가 없는 원형 부재와의 free Body로 생각하면

$$R_{Ay} + R_{Cy} + 40000 = 0$$

$$\therefore R_{Cy} = -26206.9 \text{ N}$$

$$\Sigma M_A = 0; 75 R_{cy} + 510 R_{cx} + 40000 \times 200 = 0$$

$$\therefore R_{cx} = -11832.3 \text{ N}$$

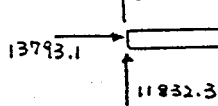
$$R_{Ax} + R_{cx} = 0 \text{ since}$$

$$\therefore R_{Ax} = 11832.3 \text{ N}$$

$$R_{Ax} + R_{Bx} = 0 \text{ since}$$

$$R_{Bx} = -11832.3 \text{ N}$$

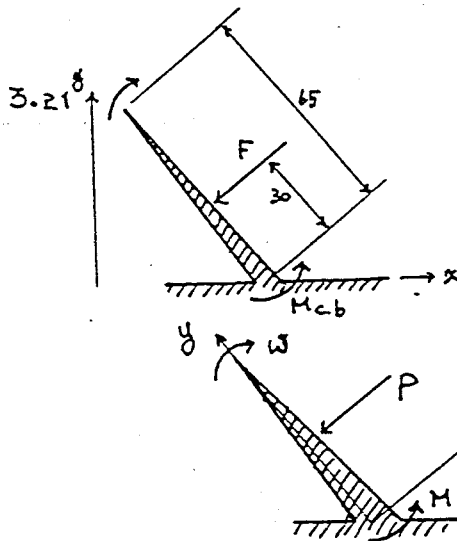
b) AD y'



$$\text{Axial Force} = 13793.1 \text{ (N) (comp)}$$

$$V = -11832.3 \text{ (N)}$$

$$M_b = 11832.3 \times \text{ (N-m)}$$

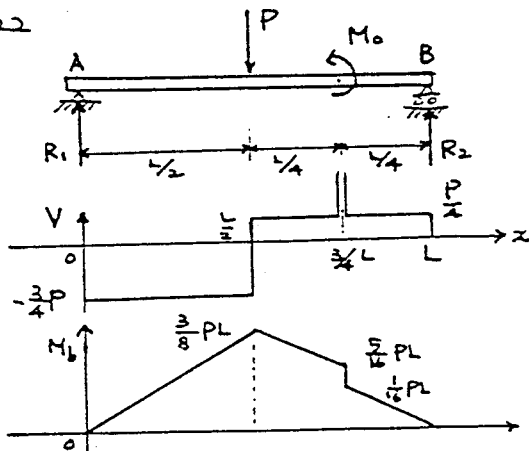


$$\begin{aligned} \text{(a) } M_{c,b} &= -30 \mu\text{m} \times 2.2 \text{ nN} \\ &= -6.6 \times 10^{-14} \text{ (N-m)} \end{aligned}$$

$$\begin{aligned} \text{(b) } M &= -\int_0^L P y \, dy = c \omega \mu f(\theta) \cdot \frac{L^2}{2} \\ &= -(9.154 \times 10^{-4}) c \omega \mu f(\theta) \end{aligned}$$

$$\text{CC; } \frac{\pi}{2}$$

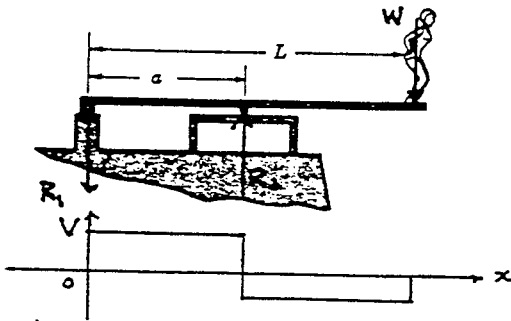
3.22



$$R_1 = \frac{2}{4} P$$

$$R_2 = \frac{1}{4} P$$

3.24



$$R_2 = \frac{L}{a} W \quad R_1 = \frac{L-a}{a} W$$

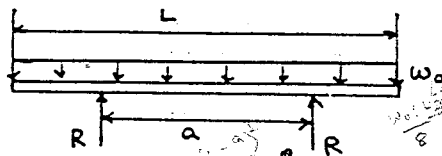
$x=a$ 에서 Max. bending moment

가 발생한다는 것을 알 수 있다

$$(\because \frac{\partial M}{\partial x} = 0 \text{ 이 } V = 0)$$

$$\therefore M_a = -R_1 a = -\frac{(L-a)W}{a} \cdot a = -(L-a)W = \text{const}$$

3.25



$$R = \frac{L w_0}{2}$$

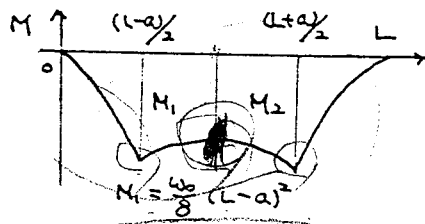
$$f(x) = -w_0 \langle x \rangle^0 + \frac{1}{2} w_0 L \langle x - \frac{L-a}{2} \rangle_{-1} + \frac{w_0 L}{2} \langle x - \frac{L+a}{2} \rangle_{-1}$$

$$M_b(x) = -\frac{w_0}{2} \langle x \rangle^2 + \frac{1}{2} w_0 L \langle x - \frac{L-a}{2} \rangle^1 + \frac{w_0 L}{2} \langle x - \frac{L+a}{2} \rangle^1$$

$$M_b(\frac{L}{2}) = -\frac{w_0}{8} L^2 + \frac{w_0 L}{4} a = 0 \quad \text{에서 } L = 2a$$

$\therefore a \leq \frac{L}{2}$ 과 $a > \frac{L}{2}$ 의 구간을 나누어서 생각한다.

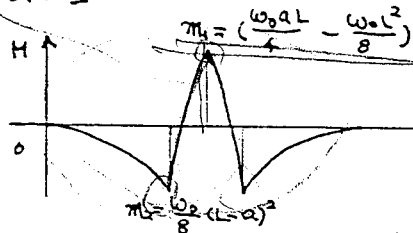
CASE I] $a \leq \frac{L}{2}$



$$(M_1)_{\min} \text{ 은 } M_2 = 0 \text{ 즉 } a = \frac{L}{2} \text{ 일 때}$$

$$(M_{\max})_{\text{CASE I}} = \frac{w_0 L^2}{32}$$

CASE II] $a > \frac{L}{2}$



$M_1 = M_2$ 때 $(M_{\max})_{\min}$ 이 된다

$$\therefore \left(\frac{w_0 a L}{4} - \frac{w_0 L^2}{8} \right) = \frac{w_0}{8} (L-a)^2$$

$$\text{전제하면 } 2aL - L^2 = L^2 - 2aL + a^2$$

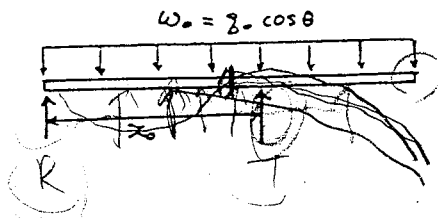
$$\therefore a^2 - 4aL + 2L^2 = 0$$

$$\therefore a = (2 - \sqrt{2})L = 0.5859L$$

$$\therefore (M_{\max})_{\text{CASE I}} = \frac{w_0}{8} (1 - 0.5859^2) L^2$$

$$(M_{\max})_{\text{CASE II}} < (M_{\max})_{\text{CASE I}} \therefore a = 0.5859L \quad = \frac{w_0 L^2}{46.65}$$

8.26

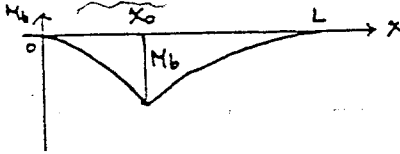


$$R_1 = g_0 L \cos \theta \left(1 - \frac{L}{2x_0}\right)$$

$$T = \frac{g_0 L^2 \cos \theta}{2x_0}$$

$$g = -g_0 \cos \theta \langle x \rangle^0 + R_1 \langle x \rangle^{-1} + T \langle x - x_0 \rangle^{-1}$$

$$M_b = -\frac{g_0 \cos \theta}{2} \langle x \rangle^2 + R_1 \langle x \rangle' + T \langle x - x_0 \rangle'$$

CASE I > $x_0 \leq \frac{L}{2}$ 

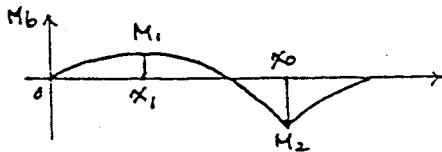
$$M_b = g_0 L \cos \theta \left(x_0 - \frac{L}{2}\right) - \frac{g_0 \cos \theta}{2} x_0^2$$

$$= -\frac{g_0 \cos \theta}{2} (x_0^2 - 2Lx_0 + L^2)$$

$$= -\frac{g_0 \cos \theta}{2} (x_0 - L)^2$$

$$\therefore |M_b \max|_{\min} \hat{=} x_0 = \frac{L}{2} \text{ at}$$

$$(M_b \max)_{\text{CASE I}} = \frac{g_0 L^2 \cos \theta}{8}$$

CASE II > $x_0 > \frac{L}{2}$ 

$$V=0; 0 = -g_0 \cos \theta x_1 + g_0 \cos \theta \left(1 - \frac{L}{2x_0}\right)$$

$$\therefore x_1 = L \left(1 - \frac{L}{2x_0}\right)$$

$$\therefore M_1 = -\frac{g_0 \cos \theta}{2} L^2 \left(1 - \frac{L}{2x_0}\right)^2$$

$$+ g_0 L \cos \theta \left(1 - \frac{L}{2x_0}\right)^2 L$$

$$\therefore M_1 = \frac{1}{2} g_0 L^2 \left(1 - \frac{L}{2x_0}\right)^2 \cos \theta$$

$$M_2 = \frac{1}{2} g_0 (L - x_0)^2 \cos \theta$$

$$|M_1| = |M_2| \text{ at } (M_{\max})_{\min} \quad \frac{1}{2} g_0 L^2 \left(1 - \frac{L}{2x_0}\right)^2 \cos \theta = \frac{1}{2} g_0 (L - x_0)^2 \cos \theta$$

$$L^2 \left(1 - \frac{L}{x_0} + \frac{L^2}{4x_0^2}\right) = L^2 - 2Lx_0 + x_0^2 \rightarrow 4\left(\frac{x_0}{L}\right)^4 - 8\left(\frac{x_0}{L}\right)^3 + 4\left(\frac{x_0}{L}\right) - 1 = 0$$

$$\text{Let } \frac{x_0}{L} = k \quad 4k^4 - 8k^3 + 4k - 1 = 0 \quad \therefore (2k^2 - 1)(2k^2 - 4k + 1) = 0$$

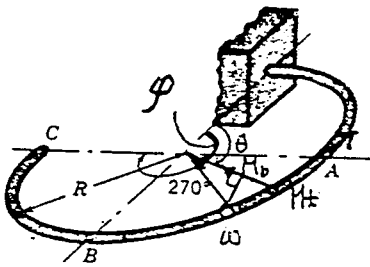
$$2k^2 - 4k + 1 > 0 \text{ or } 2 \quad k = \frac{\sqrt{2}}{2} \quad \therefore x_0 = 0.707L \text{ at}$$

$$(M_b \max)_{\text{CASE II}} = \frac{1}{2} g_0 L^2 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \cos \theta = 0.043 g_0 L^2 \cos \theta$$

$$\therefore (M_b \max)_{\text{CASE II}} < (M_b \max)_{\text{CASE I}} \quad \therefore x_0 = 0.707L = 53 \text{ (ft)}$$

$$\therefore x_0 = 53 \text{ ft}$$

3.27



$$M_b = WR \sin(\varphi - \theta)$$

$$M_t = WR \{1 - \cos(\varphi - \theta)\}$$

Load의 위치가 다음과 같을 때

1) A점 : $\varphi = 90^\circ$

$$M_t = WR(1 - \sin \theta) \rightarrow (M_t)_{\max} = WR \text{ at } \theta = 0$$

$$M_b = WR \cos \theta \rightarrow (M_b)_{\max} = WR \text{ at } \theta = 0$$

2) B점 : $\varphi = 180^\circ$

$$M_t = WR(1 + \cos \theta) \rightarrow (M_t)_{\max} = 2WR \text{ at } \theta = 0$$

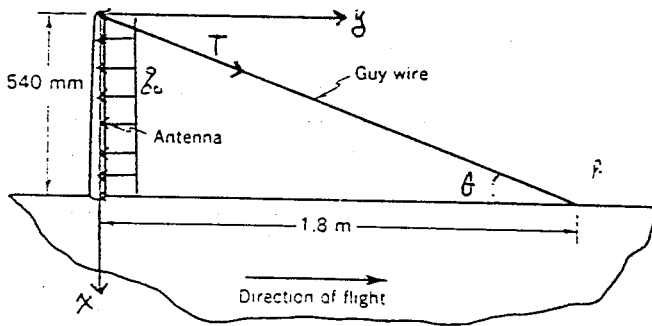
$$M_b = WR \sin \theta \rightarrow (M_b)_{\max} = WR \text{ at } \theta = 90^\circ$$

3) C점 : $\varphi = 270^\circ$

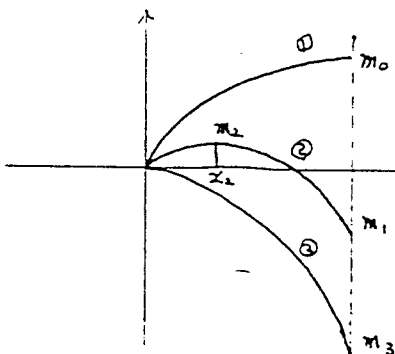
$$M_t = WR(1 + \sin \theta) \rightarrow (M_t)_{\max} = 2WR \text{ at } \theta = 90^\circ$$

$$M_b = -WR \cos \theta \rightarrow (M_b)_{\max} = WR \text{ at } \theta = 180^\circ, 0^\circ$$

3.28



$$0.54 \ell_0 = D$$



$$y = T \cos \theta \langle x \rangle^{-1} - \ell_0 \langle x \rangle^0$$

$$-V = T \cos \theta \langle x \rangle^0 - \ell_0 \langle x \rangle^1$$

$$M_b = T \cos \theta \langle x \rangle^1 - \frac{1}{2} \ell_0 \langle x \rangle^2$$

$$\textcircled{1} \quad T \cos \theta - g_0 L = 0 \quad (\because V=0 \text{ at } x=L)$$

$$M_b = T \cos \theta L - \frac{1}{2} g_0 L^2 = \frac{1}{2} g_0 L^2 = 0.27 D$$

$$\textcircled{2} \quad T \cos \theta - g_0 x_2 = 0 \quad (\because V=0 \text{ at } x=x_2) \therefore x_2 = \frac{T}{g_0} \cos \theta$$

$$M_b)_2 = T \cos \theta \frac{T \cos \theta}{g_0} - \frac{1}{2} g_0 \left(\frac{T \cos \theta}{g_0} \right)^2 = \frac{T^2 \cos^2 \theta}{2 g_0}$$

$$M_b)_1 = |T \cos \theta \cdot L - \frac{1}{2} g_0 L^2|$$

$$M_b)_1 = M_b)_2 \text{ 에서 } (M_b)_{\max} |_{\min}$$

$$\therefore \frac{T^2 \cos^2 \theta}{2 g_0} = \frac{1}{2} g_0 L^2 - T L \cos \theta, \quad \cos^2 \theta T^2 + 2 D \cos \theta T - D^2 = 0$$

$$\therefore T = \frac{-2 D \cos \theta + \sqrt{4 D^2 \cos^2 \theta + 4 D^2 \cos^2 \theta}}{2 \cos^2 \theta} = \frac{(\sqrt{2}-1) D}{\cos \theta} = 0.432 D$$

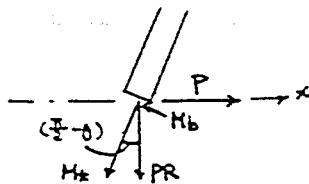
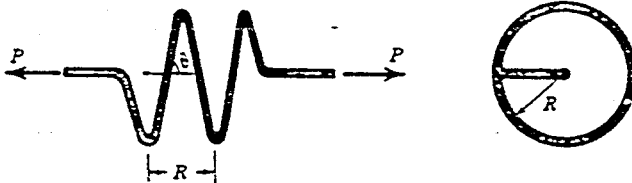
$$\therefore M_b)_{\max} = \frac{T^2 \cos^2 \theta}{2 g_0} = \frac{(T \cos \theta)^2}{2 D} \cdot L = 0.0463 D$$

$$\textcircled{3} \quad M_b)_{\max} = \frac{1}{2} g_0 L^2 = 0.27 D$$

①, ②, ③ 의 경우를 비교하면 ②의 경우가 $M_b)_{\max}$ 가 최소

$$\therefore T = 0.432 D$$

3.29



$$M_1 = PR \cos (90^\circ - \theta) = PR \sin \theta$$

$$M_b = PR \sin (90^\circ - \theta) = PR \cos \theta$$

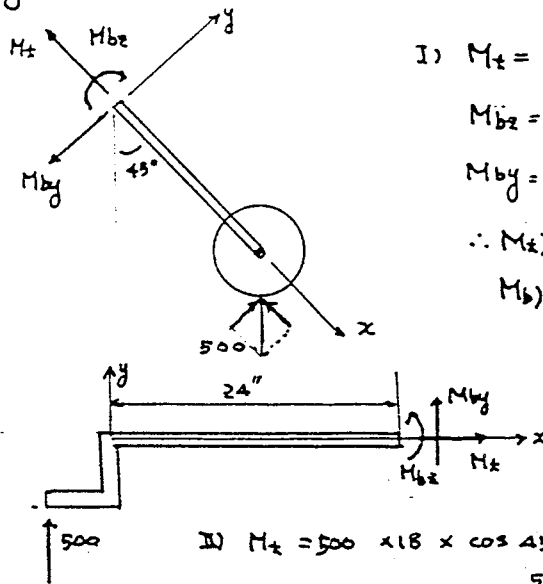
$$2\pi R \cos \theta = R$$

$$\cos \theta = \frac{1}{2\pi} \therefore \theta = 80.9^\circ$$

$$\therefore M_b)_{\max} = PR \cos 80.9^\circ = 0.158 PR$$

$$M_1)_{\max} = PR \sin 80.9^\circ = 0.985 PR$$

3.30



$$I) \quad M_{bz} = -\frac{500}{\sqrt{2}} \times 6 = -2121.3 \text{ (lb-in)}$$

$$M_{by} = \frac{500}{\sqrt{2}} \times 18 = 6364.0 \text{ (lb-in)}$$

$$M_{bz} = -\frac{500}{\sqrt{2}} \times 6 = -2121.3 \text{ (lb-in)}$$

$$\therefore M_{bz} \max = 2121.3 \text{ (lb-in)}$$

$$M_{by} \max = \sqrt{(6364)^2 + (2121.3)^2}$$

$$= 6708.2 \text{ (lb-in)}$$

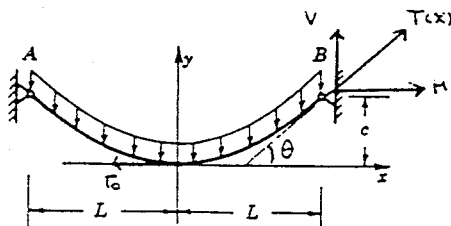
$$II) \quad M_{bz} = 500 \times 18 \times \cos 45^\circ = 6364 \text{ (lb-in)}$$

$$M_{by} = 500 \times 24 + 2 \times \frac{500}{\sqrt{2}} \times 6 \times \cos 45^\circ = 15000 \text{ (lb-in)}$$

$$\therefore M_{bz} \max = 6364 \text{ (lb-in)}$$

$$M_{by} \max = 15000 \text{ (lb-in)}$$

3.31



$$\sum F_x = 0; H = T_0 = \text{const.}$$

$$\sum F_y = 0; V = H \tan \theta = w_0 x$$

$$\therefore \tan \theta = \frac{w_0 x}{H}$$

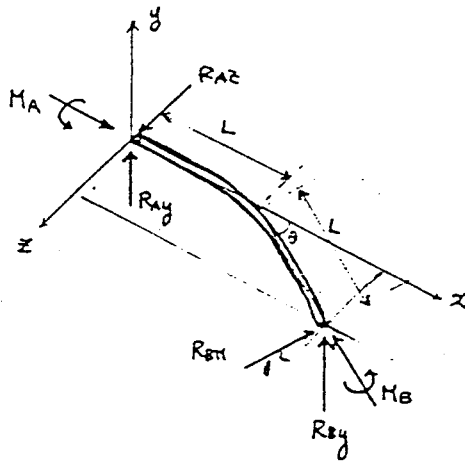
$$\frac{1}{H} \frac{dy}{dx} = \frac{w_0 x}{H} \quad \therefore y = \frac{w_0 x^2}{2H} + c$$

$$x=0; y=0 \quad \therefore c=0$$

$$\therefore y = \frac{w_0 x^2}{2H}$$

$$x=L; y=c \quad \therefore H = \frac{w_0 L^2}{2c}$$

3.32



$$\sum F_x = 0; R_{Bx} \sin \theta = 0 \therefore \underline{R_{Bx} = 0}$$

$$\sum F_y = 0; R_{Ay} + R_{By} = 0 \quad \text{--- (1)}$$

$$\sum F_z = 0; R_{Az} - R_{By} \cos \theta = 0$$

$$\therefore \underline{R_{Az} = 0}$$

$$\sum (M_x)_A = 0;$$

$$M_A - M_B \cos \theta - R_{By} L \sin \theta = 0 \quad \text{--- (2)}$$

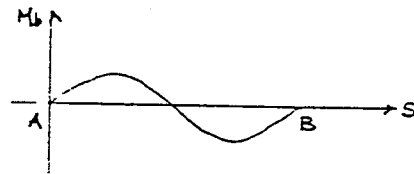
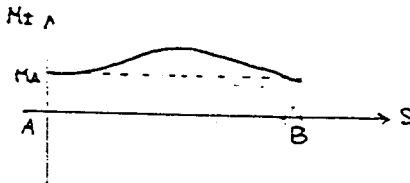
$$\sum (M_y)_A = 0; \text{ identically satisfied}$$

$$\sum (M_z)_A = 0; M_B \sin \theta + R_{By} L (1 + \cos \theta) = 0 \quad \text{--- (3)}$$

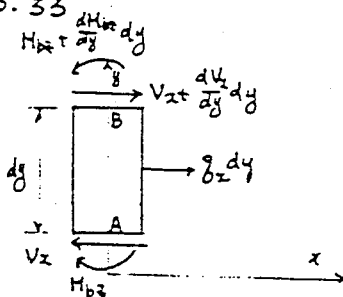
(1), (2), (3) 을 풀면

$$\underline{R_{Ay} = -\frac{M_A}{L} \tan\left(\frac{\theta}{2}\right) = -R_{By}}$$

$$\underline{M_B = M_A}$$



3.33



dy or dx에 Free body diagram을 그려보면

$$\sum F_x = 0; V_x + \frac{dV_x}{dy} dy - V_x + g_x dy = 0$$

$$\therefore \underline{\frac{dV_x}{dy} + g_x = 0}$$

$$\sum (M_z)_B = 0;$$

$$M_{b2} + \frac{dM_{b2}}{dy} dy - V_x dy + g_x dy \frac{dy}{2} - M_{b2} = 0$$

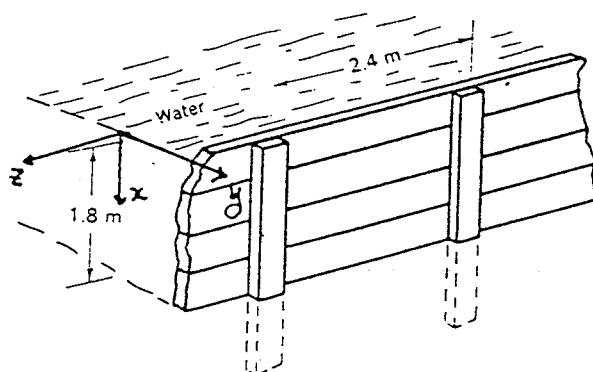
$$\therefore \underline{\frac{dM_{b2}}{dy} - V_x = 0}$$

direction triplet = (direction of member) (direction of loading)
(direction of moment vector) 라고 하면,

이 4의 right hand rule 에 일치하면 (a)의 , 그렇지 않으면 (b)의 이 된다.

direction of member	loading	triplet	equation type
x	z	xzy	(b)
y	z	yzx	(a)
z	x	zxy	(a)
z	y	zyx	(b)

3.34



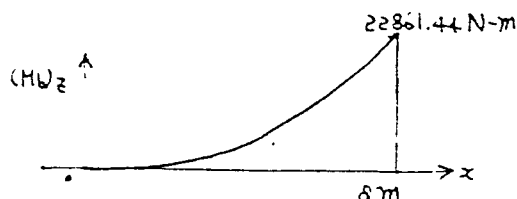
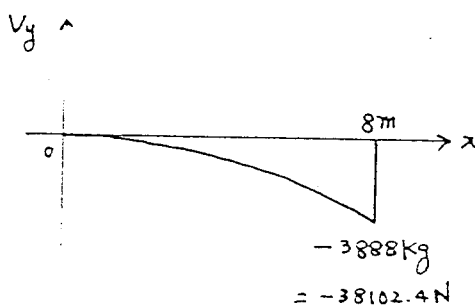
1개의 upright는
2.4m 폭의 수압을
받게 되므로

$$dF = \rho g b y dy$$

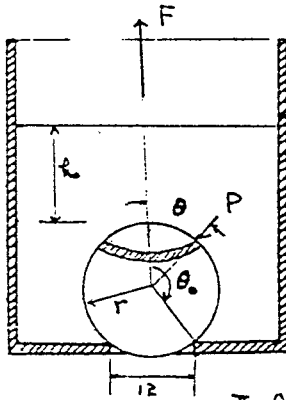
$$\therefore F = \frac{1}{2} \rho g b y^2 = -V_x$$

$$M_b = - \int_0^y V_x dy$$

$$= \frac{1}{2} \rho g b y^3$$



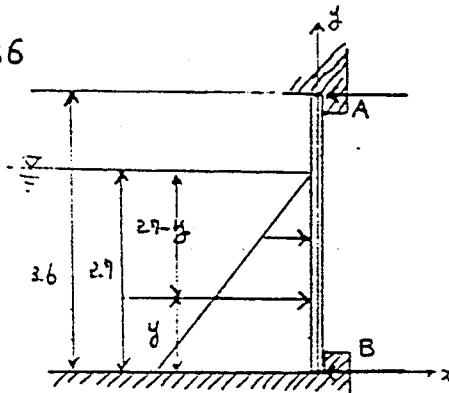
3.35



$$\pi - \theta_0 = \tan^{-1} \frac{3}{4}$$

$$\begin{aligned} F &= W + 2\pi r^2 \rho g \int_0^{\cos \theta_0} x (h_0 + r - r x) (-dx) \\ &= W + 2\pi r^2 \rho g \left[\frac{r}{3} (\cos^3 \theta_0 - 1) + \frac{1}{2} (h_0 + r) \sin^2 \theta_0 \right] \\ &= 160 + 2\pi \left(\frac{10}{12}\right)^2 62.4 \left[\frac{10}{3 \times 12} (-0.8^3 - 1) + \frac{1}{2} \left(\frac{18+10}{12}\right) 0.6^2 \right] \\ &= 160 \text{ (lb)} \end{aligned}$$

3.36



$$\sum M_{Bz} = 0 ;$$

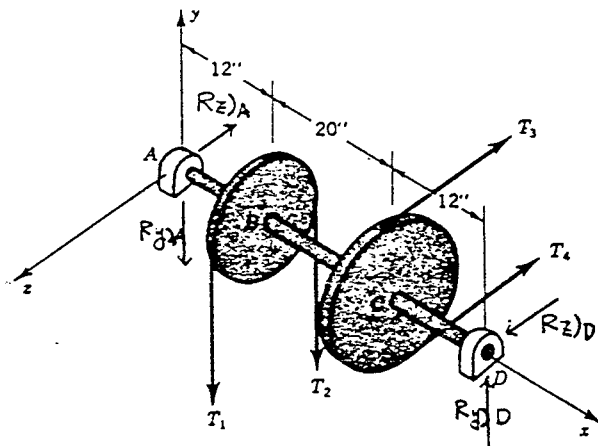
$$3.6 \times R_A = \int_0^{2.7} \rho g (2.7 - y) y \cdot b dy$$

$$= 23520 \left[\frac{2.7}{2} y^2 - \frac{y^3}{3} \right]_0^{2.7}$$

$$= 77157.36$$

$$\therefore R_A = 21432.6 \text{ (N)}$$

3.37



$$\frac{T_1}{T_2} = \frac{T_3}{T_4} = 3$$

$$2\pi \times (T_1 - T_2) \times 1750 \times \frac{1}{3} = 25 \times 33000 ; \text{rotational power}$$

$$\therefore T_1 - T_2 = 225.09 \quad \therefore T_1 = 337.636 \text{ lb}, T_2 = 112.545 \text{ lb}$$

$$2\pi \times (T_3 - T_4) \times \frac{1}{2} \times 1750 = 25 \times 33000$$

$$\therefore T_3 - T_4 = 150.06 \quad \therefore T_3 = 225.091 \text{ lb}, T_4 = 75.031 \text{ lb}$$

$$\Sigma F_y = 0; P_y = T_1 + T_2 + R_y \quad \text{--- ①}$$

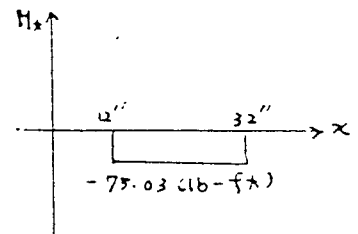
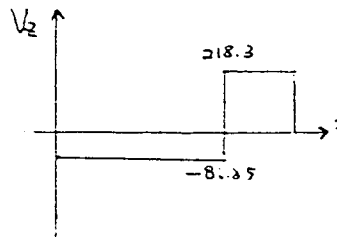
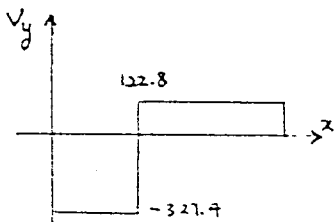
$$\Sigma M_{Az} = 0; (T_1 + T_2) \times 12 = 44 P_y \quad \text{--- ②}$$

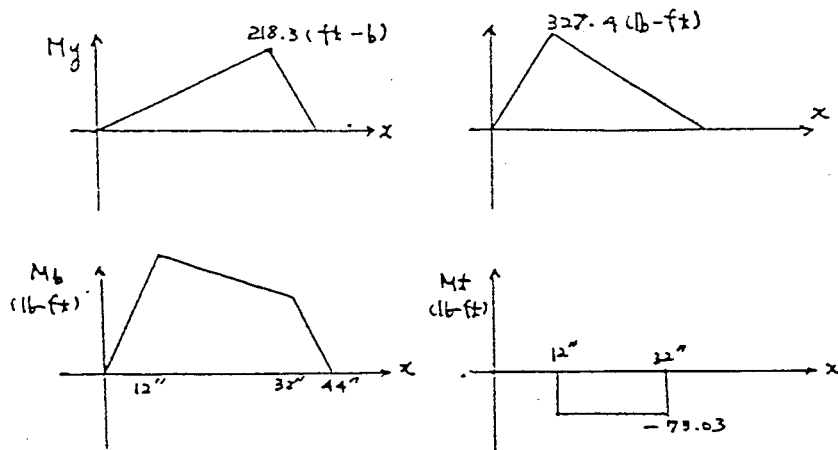
$$\text{①, ② solve, } P_y = 122.777 \text{ lb}, R_y = -327.41 \text{ lb}$$

$$\Sigma F_z = 0; P_z = (T_3 + T_4) + R_z \quad \text{--- ③}$$

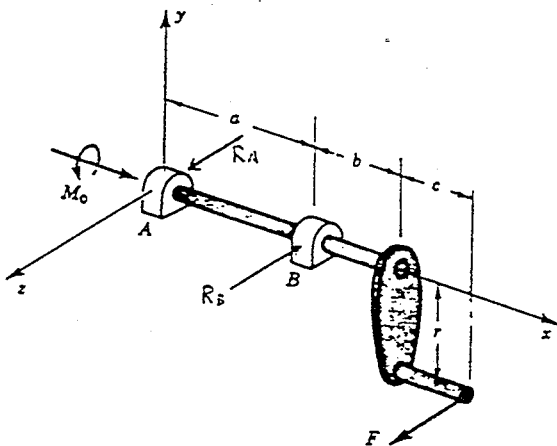
$$\Sigma M_{Ay} = 0; 32(T_3 + T_4) = 44 P_z \quad \text{--- ④}$$

$$\text{③, ④ solve, } P_z = 218.27 \text{ lb}, R_z = -81.85 \text{ lb}$$





3.38

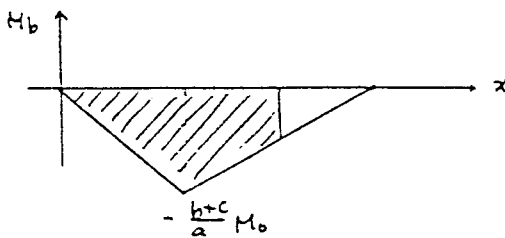
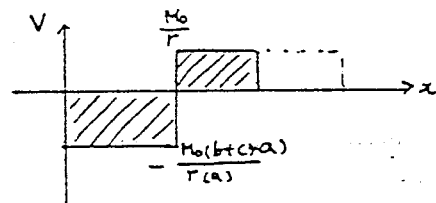
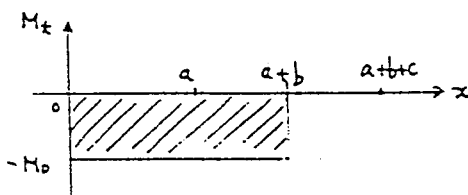


$$\Sigma M_x = 0 ; F = \frac{M_0}{r}$$

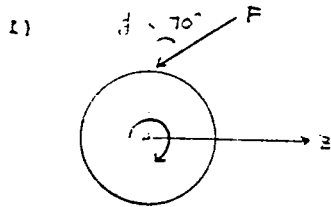
$$\Sigma (M_y)_A = 0 ; R_B = \frac{M_0}{r} \frac{(a+b+c)}{a}$$

$$\Sigma (M_y)_B = 0 ; R_A = \frac{M_0}{r} \frac{(b+c)}{a}$$

$\Sigma F_z = 0$; 위의 식으로
자음 만족



3.39



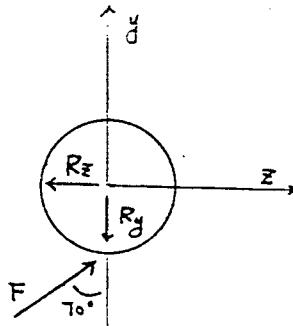
0.25 m 7101

$$\sum M_z = 0$$

$$130 = F \sin 70^\circ \frac{0.25}{2}$$

$$\therefore F = 6640.5 \text{ N}$$

II)



$$\sum F_y = 0: R_y = F \cos 70^\circ = 2271.2 \text{ (N)}$$

$$\sum F_z = 0: R_z = F \sin 70^\circ = 6240 \text{ (N)}$$

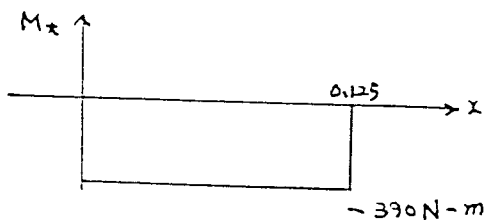
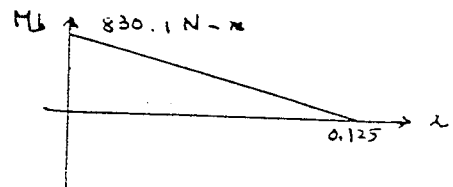
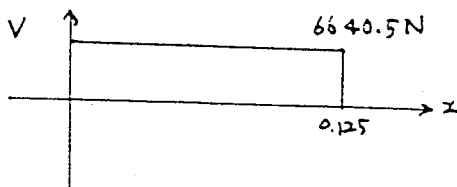
$$M_z = -F \sin 70^\circ \frac{0.125}{2}$$

$$= -390 \text{ (N-m)}$$

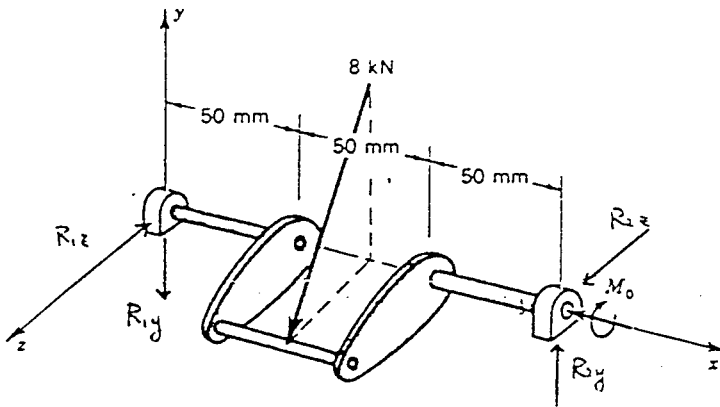
$$M_b)_y = - (0.125 - x) F \sin 70^\circ = 6240 (0.125 - x)$$

$$M_b)_z = (0.125 - x) F \cos 70^\circ = 2271.2 (0.125 - x)$$

$$\therefore M_b = 6640.5 (0.125 - x) \text{ (N-m)}$$



5.40



$$2\ell = 0.075$$

$$\therefore Q = 0.0375 \text{ m}$$

$$\cos \theta = \frac{37.5}{125}$$

$$\therefore \theta = 72.54^\circ$$

$$\Sigma F_z = 0; R_{2z} - R_{1z} = -8000 \cos \theta$$

$$R_{17} = 1200 \text{ N}$$

$$\sum M_y = 0 ; 150 \cdot R_{2x} + 8000 \cos 67.5^\circ = 0$$

$$R_2 = -1200 \text{ N}$$

$$\Sigma F_y = 0; \quad 8000 \sin \theta + R_{1y} - R_{2y}$$

$$R_y = -3817.8 \text{ N}$$

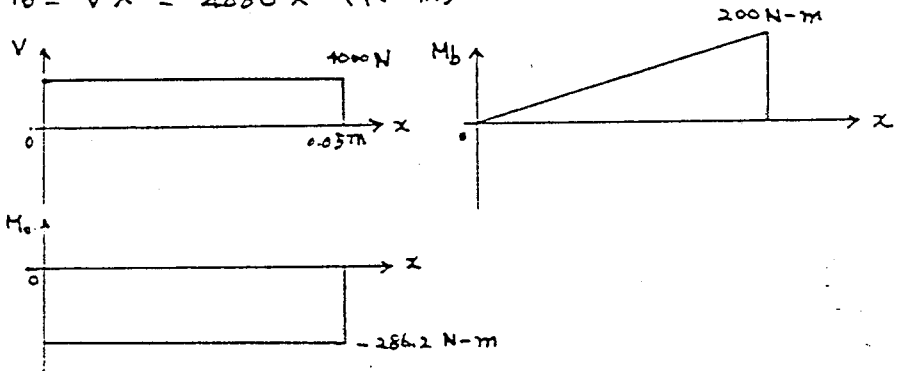
$$\Sigma M_z = 0; 150 R_{zy} = 8000 \sin \theta \cdot 7.5$$

$$R_{2y} = 3815.8 \text{ N}$$

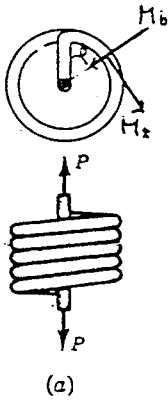
$$M_0 = -8000 \sin \theta \cdot 0.0375 = -280.2 \text{ (N-m)}$$

$$V = \sqrt{R_x^2 + R_y^2} = \sqrt{R_x^2 + R_y^2} = 4000 \text{ (N)}$$

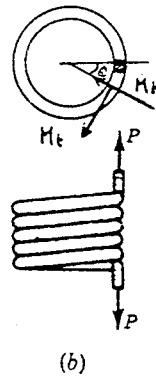
$$M_b = Vx = 4000x \cdot (N - \pi)$$



3.41

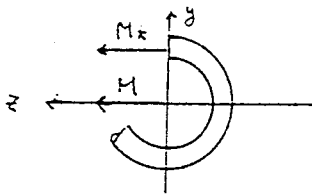


$$\begin{aligned} M_b &= 0 \\ M_x &= PR \\ V &= P \end{aligned}$$



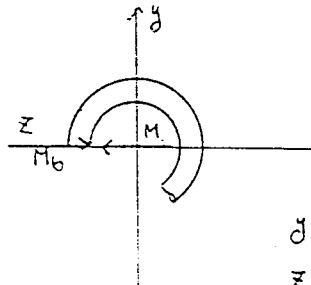
$$\begin{aligned} M_b &= PR \sin \theta \\ M_x &= PR (1 - \cos \theta) \\ V &= P \end{aligned}$$

3.42 I) A점



$$\begin{aligned} M_x &= M \\ M_b &= 0 \\ V &= 0 \end{aligned}$$

II) B점



$$\begin{aligned} M_b &= M \\ M_x &= 0 \\ V &= 0 \end{aligned}$$

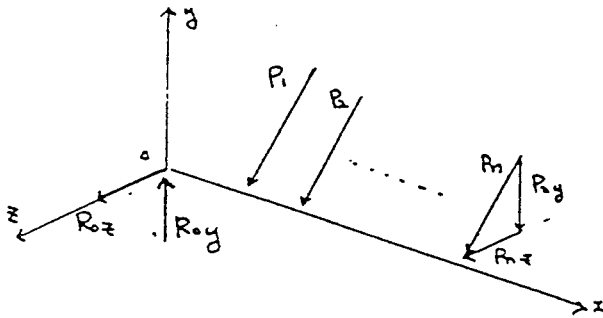
y축상; Pure twisting moment

z축상; Pure bending moment

그의 모든점: Bending + Twisting

Shear 는 아무데도 없다.

3.43



자중을 의한 bending-moment
는 무시한다.

$x_n < x < x_{n+1}$ 의 구간에서

$$M_{by} = \sum_{i=1}^n P_{iz} (x - x_i) + R_{0z} \cdot x = \left(\sum_{i=1}^n P_{iz} + R_{0z} \right) x - \sum_{i=1}^n P_{iz} x_i$$

$$= Ax + B$$

$$M_{bz} = \sum_{i=1}^n P_{iy} (x - x_i) + R_{0y} \cdot x = \left(\sum_{i=1}^n P_{iy} + R_{0y} \right) x - \sum_{i=1}^n P_{iy} x_i$$

$$= Cx + D$$

A, B, C, D 상수 $\therefore M_{by}, M_{bz} \in \text{linear}$ 하고 max는
 x_n or x_{n+1} 에서 일어날 것이다.

$$M_b = \sqrt{(M_{by})^2 + (M_{bz})^2} = \sqrt{(A^2 + C^2)x^2 + 2(AB + CD)x + (B^2 + D^2)}$$

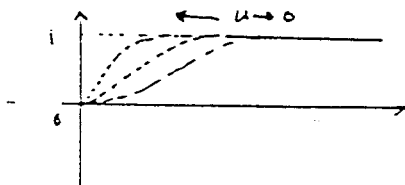
$$= \sqrt{ax^2 + bx + c} \geq 0$$

$$\frac{d^2 M_b}{dx^2} = \frac{4ac - b^2}{4(ax^2 + bx + c)^{3/2}} \quad \text{또는} \quad \begin{cases} ax^2 + bx + c \geq 0 & a \geq 0 \\ b^2 - 4ac \leq 0 \end{cases}$$

$$\therefore \frac{d^2 M_b}{dx^2} \geq 0 \quad - \text{Q.E.D.}$$

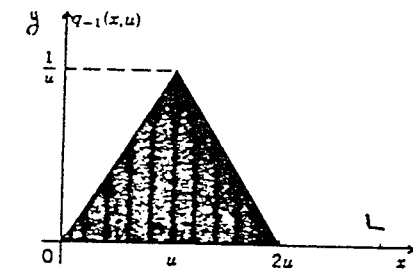
3.44

$$p = \int_0^{\infty} g_1(x, u) dx = \int_0^u \frac{x}{u^2} dx + \int_u^{\infty} \left(\frac{2}{u} - \frac{x}{u^2} \right) dx = 1$$



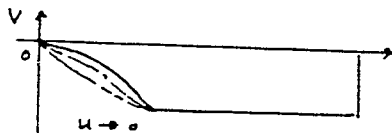
$$\therefore u \rightarrow 0 \text{ 일수록 } g_1(x, u)$$

는 unit concentrated force



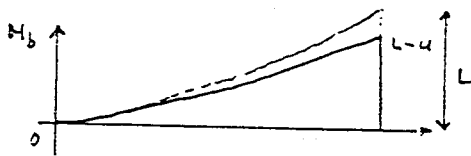
$$u \rightarrow 0 \text{ cm } V(x) = -\langle x \rangle^0$$

$$\delta = -\frac{dV}{dx} = \langle x \rangle_{-1}$$

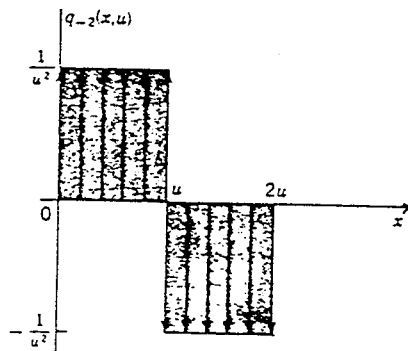


$$u \rightarrow 0 \text{ cm } M(x) = \langle x \rangle^1$$

$$\delta = \frac{d^2 M}{dx^2} = \langle x \rangle_{-1}$$

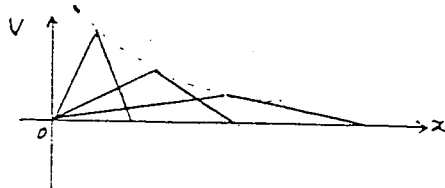


3.45



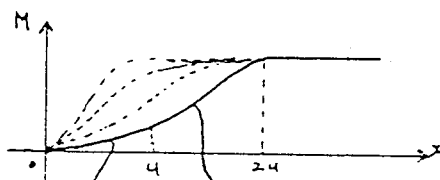
$$u \rightarrow 0 ; V_x = -\langle x \rangle_{-1}$$

$$\therefore \delta = -\frac{dV}{dx} = \langle x \rangle_{-2}$$



$$u \rightarrow 0 ; M_x = \langle x \rangle^0$$

$$\therefore \delta = \frac{d^2 M}{dx^2} = \langle x \rangle_{-2}$$



$$M = \frac{1}{2} \delta x^2$$

$$M = -\frac{1}{2} \delta x^2 + 2\delta u x - \delta u^2$$

3.46

i) $x < u$

$$\int_{-\infty}^x g_{-2}(x, u) dx = \int_0^x \frac{1}{u^2} dx = \frac{x}{u^2} = g_{-1}(x, u)$$

ii) $u < x < u$

$$\int_{-\infty}^x g_{-2}(x, u) dx = \int_u^x \left(-\frac{1}{u^2}\right) dx + \int_0^u \frac{1}{u^2} dx$$

$$= \frac{x}{u} - \frac{x}{u^2} = g_{-1}(x, u)$$

(V.C.)

$$\therefore \int_{-\infty}^x g_{-2}(x, u) dx = g_{-1}(x, u)$$

अतः $\lim_{u \rightarrow 0} g_{-2}(x, u) = \langle x \rangle_{-2}$, $\lim_{x \rightarrow u} g_{-1}(x, u) = \langle x \rangle_{-1}$

$$\begin{aligned} \lim_{x \rightarrow u} \int_{-\infty}^x g_{-1}(x, u) dx &= \int_{-\infty}^x \lim_{x \rightarrow u} g_{-1}(x, u) dx \\ &= \int_{-\infty}^x \langle x \rangle_{-1} dx \end{aligned}$$

$$\lim_{u \rightarrow 0} \int_{-\infty}^x g_{-2}(x, u) dx = \lim_{u \rightarrow 0} g_{-1}(x, u) = \langle x \rangle_{-1}$$

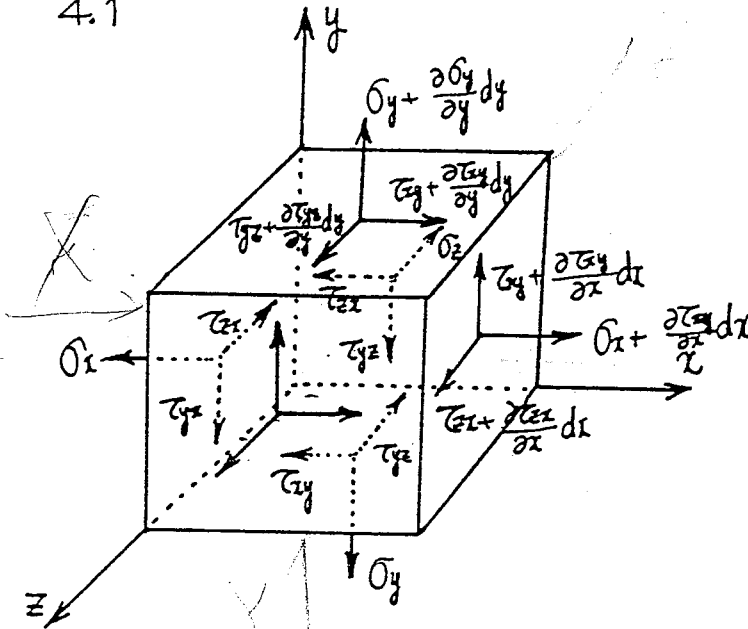
$$\therefore \int_{-\infty}^x \langle x \rangle_{-2} dx = \langle x \rangle_{-1}$$

$$\therefore \int_{-\infty}^x g_{-2}(x-a, u) dx = g_{-1}(x-a, u)$$

- Q.E.D. -

CHAPTER. 4

4.1



$$\Sigma F_x = 0 ; \left(\sigma_x + \frac{\partial \sigma_x}{\partial x} dx \right) dy dz - \sigma_x dy dz + \left(\tau_{xy} + \frac{\partial \tau_{xy}}{\partial y} dy \right) dz dz - \tau_{xy} dz dz + \left(\tau_{xz} + \frac{\partial \tau_{xz}}{\partial z} dz \right) dx dy - \tau_{xz} dx dy = 0$$

$$\therefore \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right) dx dy dz = 0$$

$$\text{따라서, } \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} = 0$$

같은 방법으로,

$$\Sigma F_y = 0 ; \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} = 0$$

$$\Sigma F_z = 0 ; \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} = 0$$

- A.E.D -

4.2 문제 4.1과 관련하여 질의 평형식에 body force 항이 더 첨가 되어진다

$$\therefore \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + X = 0$$

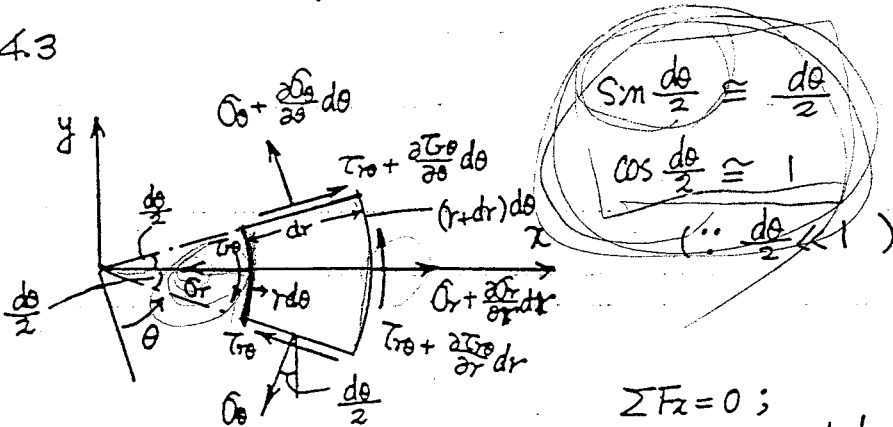
(단, X는 단위체적당 body force를 나타냄)

같은 방법으로

$$\frac{\partial \tau_{zy}}{\partial z} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + Y = 0$$

$$\frac{\partial \tau_{zx}}{\partial z} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + Z = 0$$

4.3



$$\sum F_x = 0;$$

$$(\sigma_r + \frac{\partial \sigma_r}{\partial r} dr)(r + dr) d\theta - \sigma_r r d\theta - \tau_{r\theta} dr - (\sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} d\theta) \frac{dr d\theta}{2} + (\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta) dr - (\sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} d\theta) \frac{dr d\theta}{2} = 0$$

Third term 0을 무시하면

$$\frac{\partial \sigma_r}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\sigma_r - \sigma_\theta}{r} = 0$$

$$\sum F_y = 0; \{ \sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} (d\theta) - \sigma_\theta \} dr \cos \frac{d\theta}{2} + (\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial r} dr)$$

$$(r + dr) d\theta - \tau_{r\theta} r d\theta + (\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta + \tau_{r\theta}) dr \sin \frac{d\theta}{2} = 0$$

∴ third term 0을 무시하여 정리하면

$$\frac{1}{r} \frac{\partial \sigma_\theta}{\partial \theta} + \frac{\partial \tau_{r\theta}}{\partial r} + 2 \frac{\tau_{r\theta}}{r} = 0$$

- Q.E.D -

4.4 1) r과 θ의 관계

$$F_z = [(\sigma_r + \frac{\partial \sigma_r}{\partial r} dr)(r + dr) d\theta - \sigma_r r d\theta - \tau_{r\theta} dr - \sigma_\theta dr \frac{d\theta}{2} + (\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta) dr - (\sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} d\theta) dr \frac{d\theta}{2}] dz$$

$$F_z \cong \left(r \frac{\partial \sigma_r}{\partial r} + \sigma_r - \sigma_\theta + \frac{\partial \tau_{r\theta}}{\partial \theta} \right) (dr d\theta dz)$$

(이와 마찬가지로 나머지 보았음)

$$\begin{aligned} F_y &= \left[(\sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} d\theta) dr + (\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta) dr \frac{d\theta}{2} - \sigma_\theta dr + \tau_{r\theta} dr \frac{d\theta}{2} \right. \\ &\quad \left. + (\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial r} dr)(r+dr) d\theta - \tau_{r\theta}(r d\theta) \right] dz \\ &\cong \left(\frac{\partial \sigma_\theta}{\partial \theta} + \tau_{r\theta} + r \frac{\partial \tau_{r\theta}}{\partial r} + \tau_{r\theta} \right) (dr d\theta dz) \end{aligned}$$

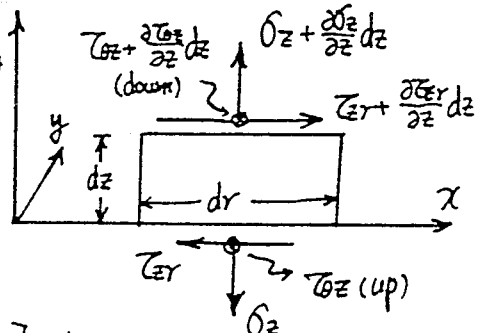
$$\begin{aligned} F_z &= \left[(\tau_{rz} + \frac{\partial \tau_{rz}}{\partial r} dr)(r+dr) d\theta - \tau_{rz} r d\theta + (\sigma_z + \frac{\partial \sigma_z}{\partial z} dz) dr \right. \\ &\quad \left. - \sigma_z dr \right] dz \cong \left(r \frac{\partial \tau_{rz}}{\partial r} + \tau_{rz} + \frac{\partial \sigma_z}{\partial z} \right) (dr d\theta dz) \end{aligned}$$

2) z 방향에 대해서

$$\begin{aligned} F_x &= \left[(\tau_{rz} + \frac{\partial \tau_{rz}}{\partial z} dz) dr - \tau_{rz} dr \right] r d\theta \\ &= \left(r \frac{\partial \tau_{rz}}{\partial z} \right) dr d\theta dz \end{aligned}$$

$$\begin{aligned} F_y &= \left[(\sigma_z + \frac{\partial \sigma_z}{\partial z} dz) dr - \sigma_z dr \right] r d\theta \\ &= \left(r \frac{\partial \sigma_z}{\partial z} \right) dr d\theta dz \end{aligned}$$

$$F_z = \left[(\sigma_z + \frac{\partial \sigma_z}{\partial z} dz) dr - \sigma_z dr \right] r d\theta = \left(r \frac{\partial \sigma_z}{\partial z} \right) dr d\theta dz$$



$$\Sigma F_x = 0 ;$$

$$\left(r \frac{\partial \sigma_r}{\partial r} + \sigma_r - \sigma_\theta + \frac{\partial \tau_{r\theta}}{\partial \theta} + r \frac{\partial \tau_{rz}}{\partial r} \right) dr d\theta dz = 0$$

$$\Sigma F_y = 0 ;$$

$$\left(\frac{\partial \sigma_\theta}{\partial \theta} + \tau_{r\theta} + r \frac{\partial \tau_{r\theta}}{\partial r} + \tau_{r\theta} + r \frac{\partial \sigma_z}{\partial z} \right) dr d\theta dz = 0$$

$$\Sigma F_z = 0 ;$$

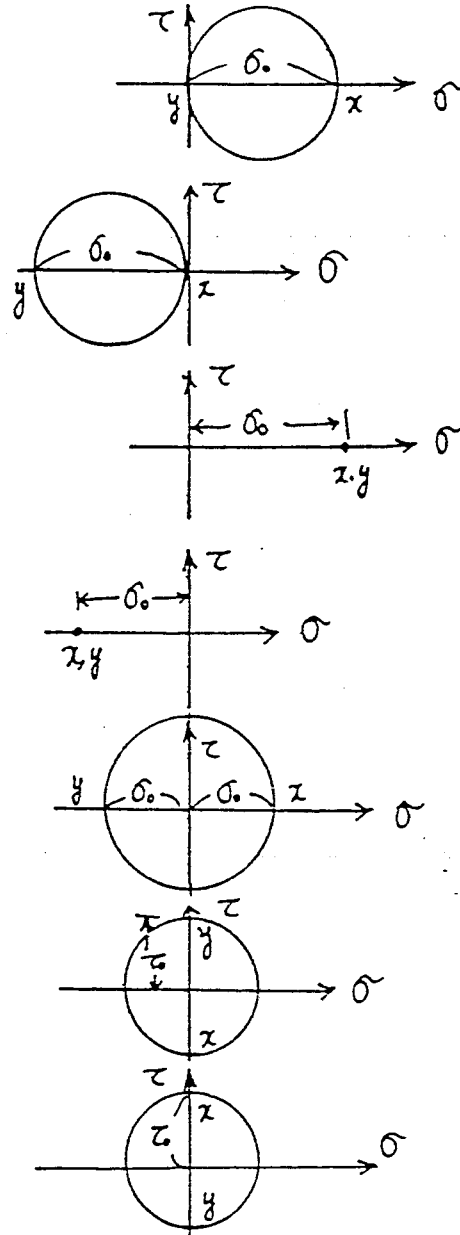
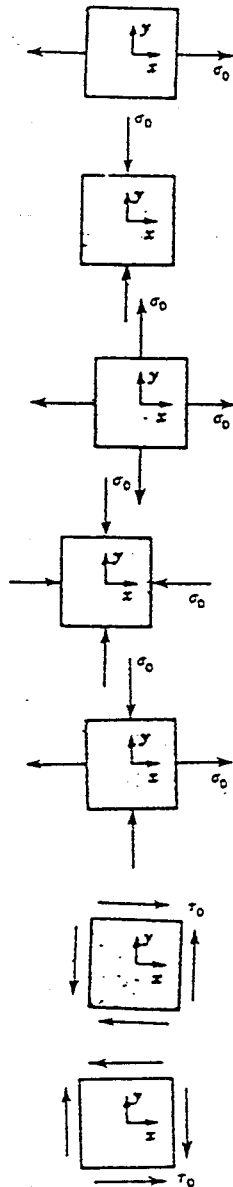
$$\left(r \frac{\partial \tau_{rz}}{\partial r} + \tau_{rz} + \frac{\partial \sigma_z}{\partial z} + r \frac{\partial \sigma_z}{\partial z} \right) dr d\theta dz = 0$$

정리하면 4변수만 남고 끝

$$* \begin{cases} \frac{\partial \sigma_r}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\partial \tau_{rz}}{\partial r} + \frac{(\sigma_r - \sigma_\theta)}{r} = 0 \\ \frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_\theta}{\partial \theta} + \frac{\partial \sigma_z}{\partial z} + \frac{2 \tau_{r\theta}}{r} = 0 \\ \frac{\partial \tau_{rz}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{rz}}{r} = 0 \end{cases}$$

4.5 α 축의 방향 전환은 physical plane에서 α 축이 180° 회전한 것과 같다. 이는 Mohr's Circle plane에서 $180^\circ \times 2 = 360^\circ$ 를 회전한 것과 같다.
따라서, 원래의 위치와 마찬가지로 되며 응력 성분의 방향과 크기도 같을 수밖에 없다.

4.6



$$\begin{aligned}
 4.7 \quad \sigma_x' &= \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \\
 \sigma_y' &= \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta \\
 \tau_{xy}' &= -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta \\
 \text{---} \quad \tau_{xy}' &= 0 \text{ 일때, } \tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}
 \end{aligned}$$

$$(a) \quad \theta_p = \frac{1}{2} \tan^{-1} \frac{2 \times 80}{40 - 0} = 37.98^\circ$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = 20 \pm \sqrt{20^2 + 80^2}$$

$$\therefore \sigma_1 = 102.462 \text{ MN/m}^2, \quad \sigma_2 = -62.462 \text{ MN/m}^2, \quad \theta_p = 37.98^\circ$$

$$(b) \quad \theta_p = \frac{1}{2} \tan^{-1} \frac{2 \times (-60)}{140 - 20} = -22.5^\circ$$

$$\sigma_{1,2} = \frac{140 + 20}{2} \pm \sqrt{\left(\frac{140 - 20}{2}\right)^2 + (-60)^2}$$

$$\therefore \sigma_1 = 164.85 \text{ MN/m}^2, \quad \sigma_2 = -4.85 \text{ MN/m}^2, \quad \theta_p = 22.5^\circ$$

$$(c) \quad \theta_p = \frac{1}{2} \tan^{-1} \left(\frac{2 \times 100}{-120 - 150} \right) = -24.82^\circ$$

$$\sigma_{1,2} = -35 \pm \sqrt{85^2 + 100^2}$$

$$\therefore \sigma_1 = 96.24 \text{ MN/m}^2, \quad \sigma_2 = -166.24 \text{ MN/m}^2, \quad \theta_p = -24.82^\circ$$

$$(d) \quad \theta_p = \frac{1}{2} \tan^{-1} \frac{2 \times 8000}{10000 + 4000} = 24.41^\circ$$

$$\sigma_{1,2} = 3000 \pm \sqrt{7000^2 + 8000^2}$$

$$\therefore \sigma_1 = 13630 \text{ psi}, \quad \sigma_2 = -7630 \text{ psi}, \quad \theta_p = 24.41^\circ$$

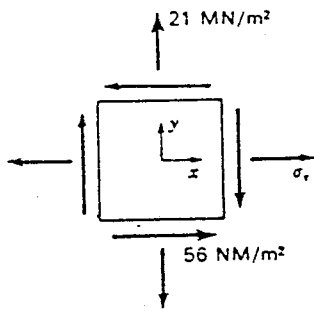
$$(e) \quad \theta_p = \frac{1}{2} \tan^{-1} \frac{2 \times (-6000)}{-10000 - 20000} = -79.10^\circ$$

$$\sigma_{1,2} = 5000 \pm \sqrt{15000^2 + 6000^2}$$

$$\therefore \sigma_1 = 21155.5 \text{ psi}, \quad \sigma_2 = -11155.5 \text{ psi}$$

$$\theta_p = -79.10^\circ$$

4.8



minimum principal stress

$$\sigma_2 = -7 \text{ MN/m}^2$$

$$\sigma_2 = \frac{\sigma_x + 21}{2} - \sqrt{\left(\frac{\sigma_x - 21}{2}\right)^2 + 56^2} = -7$$

$$\left(\frac{\sigma_x - 21}{2}\right)^2 + 56^2 = \left(\frac{\sigma_x + 35}{2}\right)^2$$

$$112\sigma_x = 112^2 + 21^2 - 35^2 \quad \therefore \sigma_x = 105$$

$$\sigma_x = 105 \text{ MN/m}^2, \quad \theta_p = \frac{1}{2} \tan^{-1} \frac{2 \times (-56)}{105 - 21} = -26.565^\circ$$

Ans.) $\sigma_x = 105 \text{ MN/m}^2, \quad \theta_p = -26.565^\circ$

4.9 $\sigma_x = -120 \text{ MN/m}^2, \quad \sigma_y = 50 \text{ MN/m}^2, \quad \tau_{xy} = 100 \text{ MN/m}^2$
 $\theta = 30^\circ$

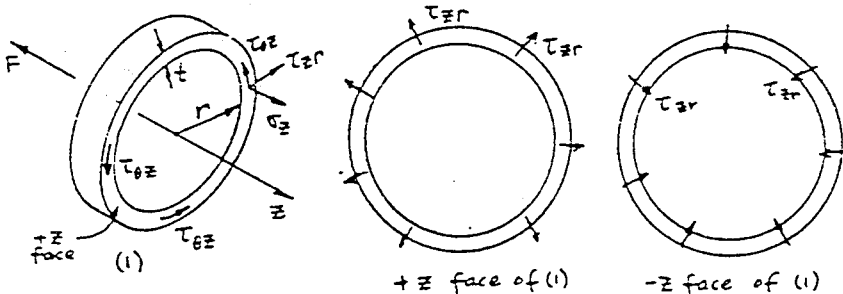
$$\sigma'_x = \frac{-120 + 50}{2} + \frac{-120 - 50}{2} \cos 60^\circ + 100 \sin 60^\circ = 9.1025 \text{ (MN/m}^2)$$

$$\sigma'_y = -35 + 85 \cos 60^\circ - 100 \sin 60^\circ = -79.1025 \text{ (MN/m}^2)$$

$$\tau'_{xy} = 85 \sin 30^\circ + 100 \cos 60^\circ = 123.6122 \text{ (MN/m}^2)$$

Ans.) $\left[\begin{array}{l} \sigma'_x = 9.1025 \text{ MN/m}^2, \quad \sigma'_y = -79.1025 \text{ MN/m}^2 \\ \tau'_{xy} = 123.6122 \text{ MN/m}^2 \end{array} \right.$

4.10



- i) +z 축에 대하여 극 대칭이므로 응력은 θ 값에 무관하다
 +z 면에서 τ_{zr} 이 상수 값을 가지며 방향이 outward라 하자
 F의 작용점에서 먼 거리에 있는 점의 응력은 St. Venant's

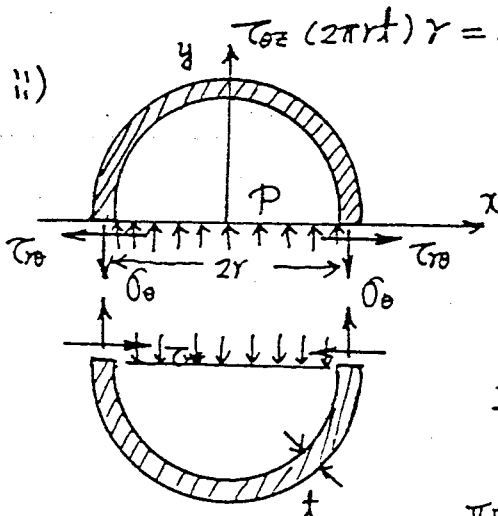
Principle에 의해서 Z 와는 무관하다. 또한 $-Z$ 면에서의 τ_{rz} 은 $+Z$ 면의 응력과 크기는 같고 방향이 반대가 될 것이다 (작용과 반작용의 원리). 그런데 $\oplus \ominus$ 쪽의 선택은 임의적 이므로 같은 요소에 대한 같은 물리적 응력 상태를 가져야 하므로, 결국 $\tau_{rz}=0$ 일수 밖에 없다.

평형식 ; $\Sigma F_z = 0$; $F = \sigma_z 2\pi r t \therefore \sigma_z = \frac{F}{2\pi r t}$

$\Sigma M_z = 0$;

$\tau_{\theta z} (2\pi r t) r = 0 \therefore \tau_{\theta z} = 0$

ii)



i)와 마찬가지로 circular symmetry가 있고 x, y축 선택이 임의이므로 $\tau_{r\theta} = 0$

Equilibrium ;

$\Sigma F_y = 0$; $2 [\sigma_\theta (t)(1)] = p(2r)(1)$

$\therefore \sigma_\theta = \frac{pr}{t}$

표면 안쪽 응력 $\sigma_r = -P$

표면 바깥쪽 응력 $\sigma_r = 0$ 이므로

벽 내부의 응력은 $-P \leq \sigma_r \leq 0$ 이다.

$(r/t) \gg 1$ 이면 $\sigma_\theta \gg \sigma_r$, $\sigma_z \gg \sigma_r$ 이고

σ_r 은 무시될 수 있다 $\therefore \sigma_r \approx 0$

* 요약 ; $\left[\begin{array}{l} \tau_{r\theta} = \tau_{\theta z} = \tau_{rz} = 0 \quad \therefore r, \theta, z = \text{주축} \\ \sigma_r \approx 0 ; \sigma_\theta = \frac{pr}{2t}, \sigma_z = \frac{F}{2\pi r t} \end{array} \right.$

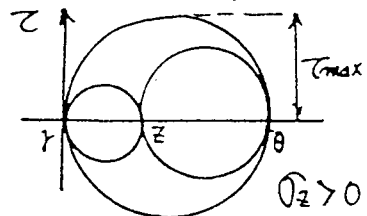
4.11 문제 4.10 으로 부터

$\sigma_r \approx 0$

$\sigma_\theta = \frac{pr}{2t}$

$\sigma_z = \frac{F}{2\pi r t}$

shear stress = 0



$$\tau_{max} = \frac{1}{2} |\sigma_{max} - \sigma_{min}| = \text{radius of largest circle}$$

a) i) 만약 $F > 0$ 이면 $\sigma_z > 0$ 이다

$$\tau_{max} = \frac{1}{2} \sigma_z \text{ or } \frac{1}{2} \sigma_\theta, \quad \sigma_{max} = \sigma_z \text{ or } \sigma_\theta$$

$$\therefore \tau_{max} \neq \sigma_{max} \quad (\text{앞 문제의 그림 참조})$$

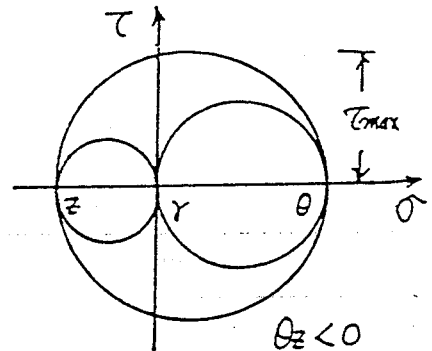
ii) $F < 0$ 이면 $\sigma_z < 0$

$$\begin{aligned} \tau_{max} &= \frac{1}{2} (\sigma_\theta - \sigma_z) \\ &= \frac{1}{2} \left(\frac{pr}{r} + \frac{F}{2\pi r t} \right) \end{aligned}$$

$$\sigma_{max} = \sigma_\theta = \frac{pr}{r}$$

$$\therefore \frac{1}{2} \left(\frac{pr}{r} + \frac{F}{2\pi r t} \right) = \frac{pr}{r}$$

$$\therefore F = 2\pi r^2 p$$



b) $F > 0$ 이면

$$\tau_{max} = \frac{1}{2} \sigma_{max} \quad (\text{항상 성립})$$

$$F \leq 0 \text{ 이면 } \sigma_z \leq 0 \quad \frac{1}{2} \left(\frac{pr}{r} + \frac{F}{2\pi r t} \right) = \frac{1}{2} \left(\frac{pr}{r} \right)$$

$$\therefore F = 0. \quad \therefore F \geq 0 \text{ 일때 성립}$$

$$\text{Ans.) (a) } F = 2\pi r^2 p \quad (b) \quad F \geq 0$$

4.12

원형 자유물체의 평형식;

$$\sum F_z = 0:$$

$$F = p\pi r^2$$

* Circular symmetry와

+ z축의 선장이 일치하므로

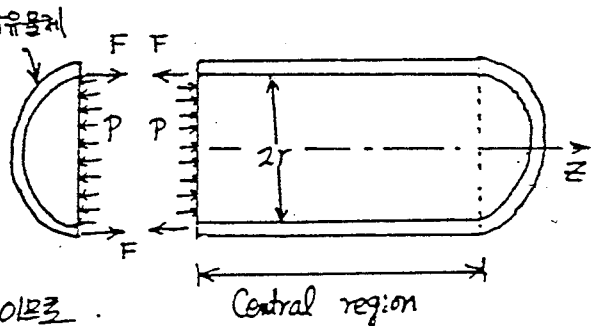
$$\tau_{zr} = 0 \quad (\text{Pro 4.10 참조})$$

$$\tau_{\theta r} = 0$$

$$\sum M_z = 0; \quad \tau_{\theta z} = 0$$

$$\therefore r, \theta, z \text{ 는 주축.}$$

σ_z, σ_θ 를 표현하면



$$\Sigma F_z = 0 ; 2\pi r t \sigma_z = \pi r^2 p \quad \therefore \sigma_z = \frac{rp}{2t}$$

$$\Sigma F_r = 0 ; brp = \sigma_\theta b t \quad \therefore \sigma_\theta = \frac{pr}{t}$$

$$-p \leq \sigma_r \leq 0 \quad r/t \gg 1 \text{ 이면 } \sigma_\theta \gg \sigma_r$$

$$\therefore \sigma_r \approx 0$$

$$\text{Ans.) } \sigma_r \approx 0, \sigma_\theta = \frac{pr}{t}, \sigma_z = \frac{rp}{2t}$$

$$4.13 \quad \sigma_r = 0, \sigma_\theta = \frac{pr}{t}, \sigma_z = \frac{pr}{2t}$$

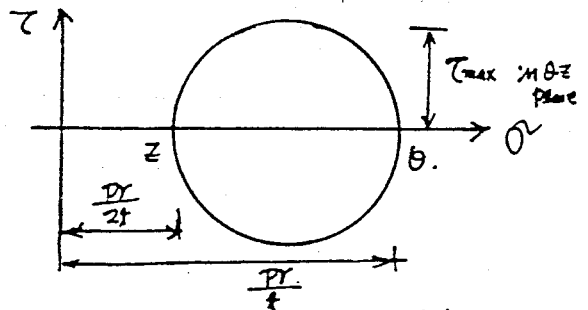
θ - z plane에서 Mohr 원을 그리면. ($\tau_{\theta\theta} = \tau_{zr} = 0$)

$$(\tau_{\max})_{\theta z} = \frac{1}{2}(\sigma_\theta - \sigma_z)$$

$$= \frac{pr}{4t}$$

$$(\sigma_{\max})_{\theta z} = \sigma_\theta = \frac{pr}{t}$$

$$\therefore (\tau_{\max})_{\theta z} = \frac{1}{4}(\sigma_{\max})_{\theta z}$$



< 주의 ; $(\tau_{\max})_{\theta z}$ 는 θ - z plane에서 최대치이지만
좌대 전단 응력은 아님에 주의 할 것 >

4.14

< 문제 4.10 > 을 이용하여
 r, ϕ, θ 가 주어진 것을 안다.

평형식 ; $\Sigma F_x = 0 ;$

$$-p\pi r^2 + \sigma_\phi(2\pi r t) = 0$$

$$\therefore \sigma_\phi = \frac{pr}{2t}$$

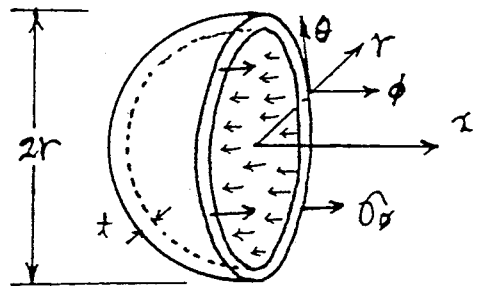
$$\text{마찬가지 방법으로 } \sigma_\theta = \frac{pr}{2t}$$

표면 안쪽 $\sigma_r = -p$, 표면 바깥쪽 $\sigma_r = 0$

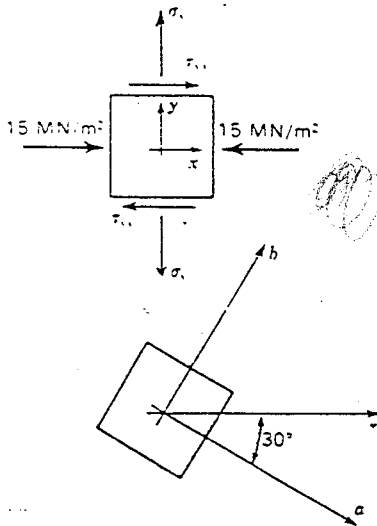
따라서 표면내에서 $-p \leq \sigma_r \leq 0$ 이고 $r/t \gg 1$ 이면

$$\sigma_r \ll \sigma_\theta, \sigma_\phi \quad \therefore \sigma_r \approx 0$$

$$\text{Ans.) } \sigma_r = 0, \sigma_\phi = \sigma_\theta = \frac{pr}{2t}$$



4.15

Shear stress = 0. $\tau_{zy} = 0$

$$\sigma_y = 75 \text{ MN/m}^2, \quad \sigma_x = -15 \text{ MN/m}^2$$

$$\sigma_a = \frac{-15 + 75}{2} + \frac{-15 - 75}{2} \cos(-60^\circ) = 7.5 \text{ (MN/m}^2\text{)}$$

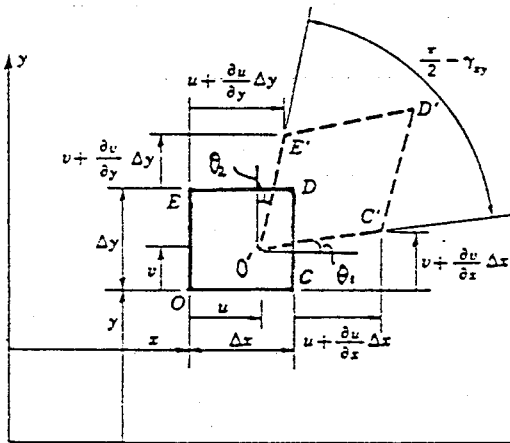
$$\sigma_b = 30 + 45 \cos(-60^\circ) = 52.5 \text{ (MN/m}^2\text{)}$$

$$\tau_{ab} = 45 \sin(-60^\circ)$$

$$= -38.97 \text{ (MN/m}^2\text{)}$$

$$\text{Ans.) } \begin{cases} \sigma_a = 7.5 \text{ MN/m}^2 \\ \sigma_b = 52.5 \text{ MN/m}^2 \\ \tau_{ab} = -38.97 \text{ MN/m}^2 \end{cases}$$

4.16



$$\epsilon_x = \frac{O'C' - OC}{OC} \cong \frac{(\Delta x + u + \frac{\partial u}{\partial x} \Delta x - u) - \Delta x}{\Delta x} = \frac{\partial u}{\partial x}$$

$$\epsilon_y = \frac{O'E' - OE}{OE} \cong \frac{(\Delta y + v + \frac{\partial v}{\partial y} \Delta y - v) - \Delta y}{\Delta y} = \frac{\partial v}{\partial y}$$

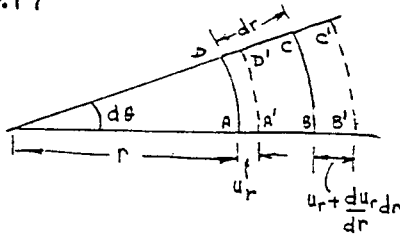
$$\gamma_{xy} = \theta_1 + \theta_2 \cong \frac{(v + \frac{\partial v}{\partial x} \Delta x) - v}{\Delta x} + \frac{(u + \frac{\partial u}{\partial y} \Delta y) - u}{\Delta y} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

< 참고; z 방향 처짐 w를 고려하면 A'B'는 x-y 평면과 작은 각을 이루나, 이에 대한 고려는 무시하여도 무방하다)
같은 방법으로 다른 항도 구하면 다음과 같다.

$$\begin{cases} \epsilon_x = \frac{\partial u}{\partial x} \\ \epsilon_y = \frac{\partial v}{\partial y} \\ \epsilon_z = \frac{\partial w}{\partial z} \end{cases}$$

$$\begin{cases} \gamma_{xy} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \\ \gamma_{yz} = \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \\ \gamma_{zx} = \frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} \end{cases} \quad \text{--- Q.E.D.}$$

4.17



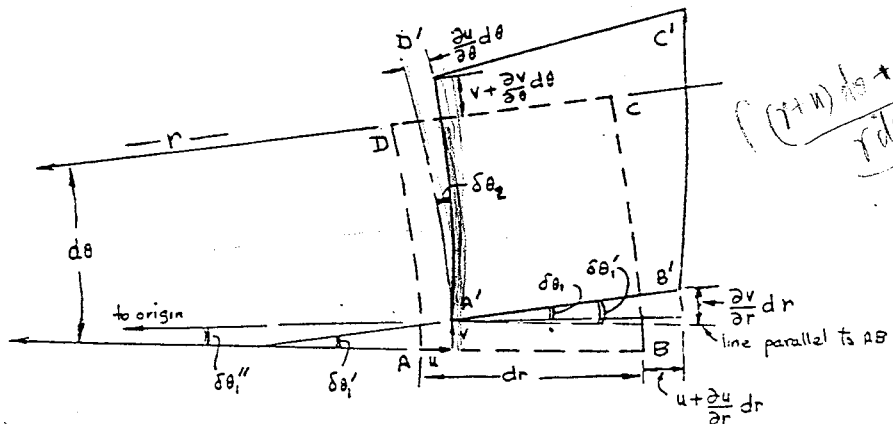
$$\epsilon_r = \frac{A'B' - AB}{AB} = \frac{(dr + u_r + \frac{du_r}{dr}dr - u_r) - dr}{dr} = \frac{du_r}{dr}$$

$$\epsilon_\theta = \frac{A'B' - AD}{AD}$$

$$\gamma_{re} = \angle DAB \text{의 각변화} = 0 \quad \left| \quad = \frac{(r + u_r)d\theta - r d\theta}{r d\theta} = \frac{u_r}{r} \right.$$

- Q.E.D -

4.18



$$\epsilon_r = \frac{A'B' - AB}{AB} = \frac{[(dr + u + \frac{\partial u}{\partial r}dr) - u] - [dr]}{dr} = \frac{\partial u}{\partial r}$$

$$\epsilon_\theta = \frac{A'B' - AD}{AD} = \frac{[(r + u)d\theta + v + \frac{\partial v}{\partial \theta}d\theta - v] - [rd\theta]}{rd\theta} = \frac{u}{r} + \frac{1}{r} \frac{\partial v}{\partial \theta}$$

$$\gamma_{re} = d\theta_1 + d\theta_2 = (d\theta_1' - d\theta_1'') + d\theta_2 = \left(\frac{\partial v}{\partial r}dr - \frac{v}{r} \right) + \frac{\partial u}{\partial \theta}d\theta$$

$$= \frac{\partial v}{\partial r} - \frac{v}{r} + \frac{1}{r} \frac{\partial u}{\partial \theta}$$

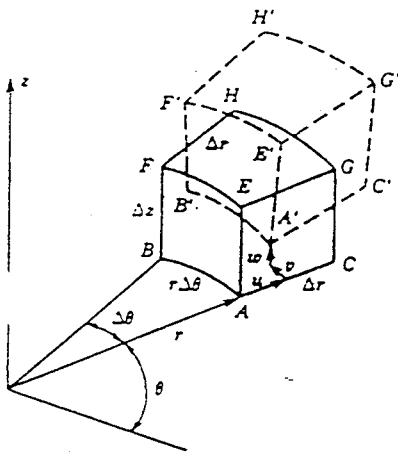
- Q.E.D -

4.19 그림의 B'F'E'A'를 r-θ 평면에 투영하고

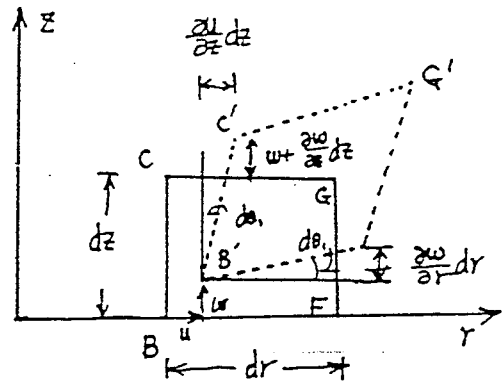
< 문제 4.18 >의 결과를 이용한다

$$\epsilon_r = \frac{\partial u}{\partial r}, \quad \epsilon_\theta = \frac{1}{r} \frac{\partial v}{\partial \theta} - \frac{u}{r}$$

$$\gamma_{re} = \frac{\partial v}{\partial r} + \frac{1}{r} \frac{\partial u}{\partial \theta} - \frac{v}{r}$$



$B'F'G'C'$ 를 $BFGC$ 에 평행한 평면에 투영시키고 아들이 이루는 각이 매우 작으므로 이각을 무시한다.



$$\epsilon_z = \frac{B'C' - BC}{BC} = \frac{[dz + w + \frac{\partial w}{\partial z} dz - w] - dz}{dz} = \frac{\partial w}{\partial z}$$

$$\gamma_{zr} = \delta\theta_1 + \delta\theta_2 = \frac{\partial w}{\partial r} dr + \frac{\partial u}{\partial z} dz = \frac{\partial w}{\partial r} + \frac{\partial u}{\partial z}$$

마찬가지로, $A'B'C'D'$ 를 $ABCD$ 를 포함하는 평면에 투영하여 문제를 풀면 $\gamma_{\theta z} = \frac{\partial u}{\partial z} + \frac{1}{r} \frac{\partial w}{\partial \theta}$ 를 얻을 수 있다.

4.20 $\epsilon_x = 800 \times 10^{-6}$, $\epsilon_y = 100 \times 10^{-6}$, $\gamma_{xy} = -800 \times 10^{-6}$

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

$$\epsilon_1 = 981.5 \times 10^{-6}, \quad \epsilon_2 = -81.5 \times 10^{-6}$$

$$\theta_p = \frac{1}{2} \tan^{-1} \frac{\gamma_{xy}}{\epsilon_x - \epsilon_y} = -24.4^\circ$$

Ans.) $\epsilon_1 = 981.5 \times 10^{-6}$, $\epsilon_2 = -81.5 \times 10^{-6}$, $\theta_p = -24.4^\circ$

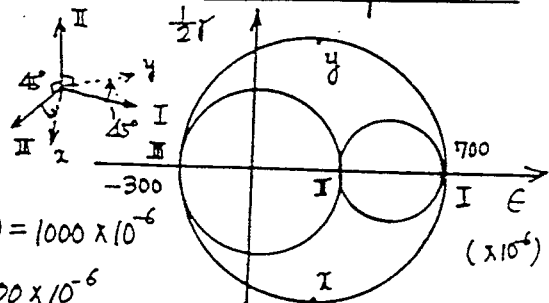
4.21 $\epsilon_I = 700 \times 10^{-6}$

$$\epsilon_{II} = 300 \times 10^{-6}$$

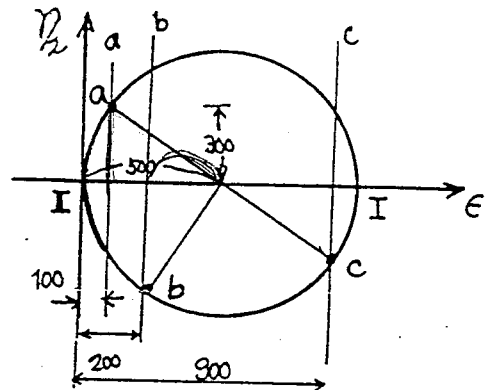
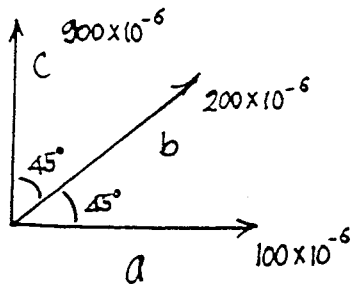
$$\epsilon_{III} = -300 \times 10^{-6}$$

$$\gamma_{max} = 2 \left(\frac{1}{2} |\epsilon_I - \epsilon_{III}| \right) = 1000 \times 10^{-6}$$

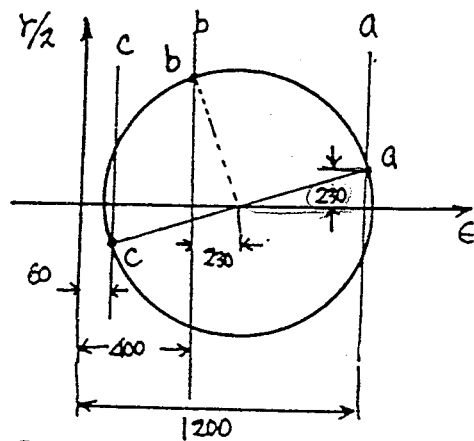
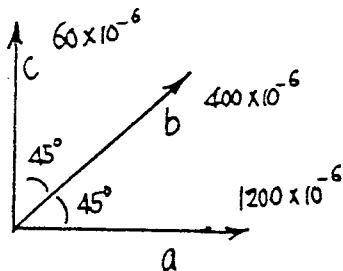
Ans.) $\gamma_{max} = 1000 \times 10^{-6}$



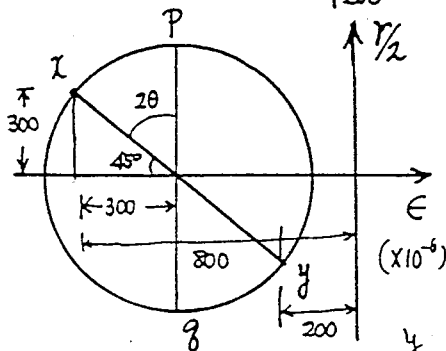
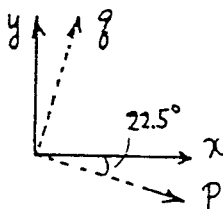
4.22 (a) $\epsilon_a = 100 \times 10^{-6}$, $\epsilon_b = 200 \times 10^{-6}$, $\epsilon_c = 900 \times 10^{-6}$



(b) $\epsilon_a = 1200 \times 10^{-6}$
 $\epsilon_b = 400 \times 10^{-6}$
 $\epsilon_c = 60 \times 10^{-6}$

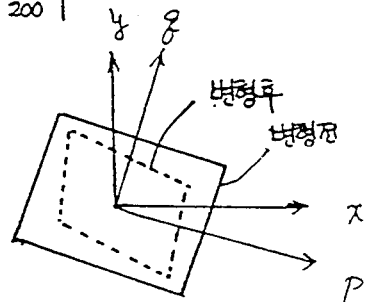


4.23 $\epsilon_x = -800 \times 10^{-6}$
 $\epsilon_y = -200 \times 10^{-6}$
 $\gamma_{xy} = -600 \times 10^{-6}$



$2\theta = 45^\circ$
 $\therefore \theta = 22.5^\circ$

P, q ; max shear axes
 [γ_{pq} ; negative]



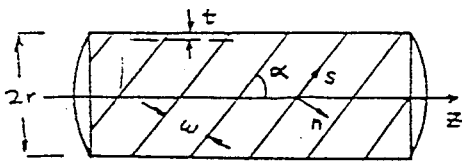
4.24 $u=w=0$. $v = Cr$

<문제 4.19>에서의 변형도 성분을 이용하여 풀면

$$\left\{ \begin{array}{l} \epsilon_r = \frac{\partial u}{\partial r} = 0 \\ \epsilon_\theta = \frac{1}{r} \frac{\partial v}{\partial \theta} + \frac{u}{r} = 0 \\ \epsilon_z = \frac{\partial w}{\partial z} = 0 \end{array} \right. \quad \left\{ \begin{array}{l} \gamma_{r\theta} = \frac{\partial v}{\partial r} + \frac{1}{r} \frac{\partial u}{\partial \theta} - \frac{v}{r} = C - C = 0 \\ \gamma_{rz} = \frac{1}{r} \frac{\partial w}{\partial \theta} + \frac{\partial v}{\partial z} = 0 \\ \gamma_{zr} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} = 0 \end{array} \right.$$

변위 $v = Cr$ 로 부터 유발되는 Strain은 없고
이 운동은 각축에 관한 회전 운동이다.

4.25



가하중 조건; $2\pi r \cos \alpha = W$

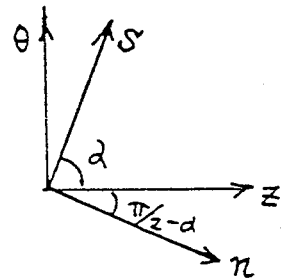
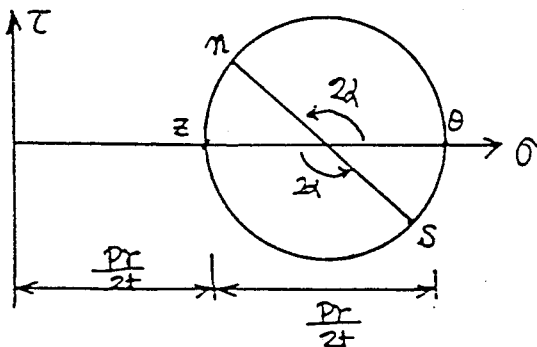
<문제 4.12>로 부터

$\sigma_r \approx 0$, Shear stress = 0

$\sigma_\theta = \frac{Pr}{t}$, $\sigma_z = \frac{Pr}{2t}$

θ - z 평면을 n - s 평면으로

바꾸기 위하여 Mohr 원을 이용하면



$$\sigma_n = \frac{\sigma_\theta + \sigma_z}{2} - \frac{\sigma_\theta - \sigma_z}{2} \cos(\pi - 2\alpha)$$

$$= \frac{3Pr}{4t} + \frac{Pr}{4t} \cos 2\alpha$$

$\sigma_{max} = \sigma_\theta = \frac{Pr}{t}$, 만약 $\sigma_n = 0.8 \sigma_{max}$ 이라면

$$\sigma_n = 0.8 \frac{Pr}{t} = \frac{3Pr}{4t} + \frac{Pr}{4t} \cos 2\alpha$$

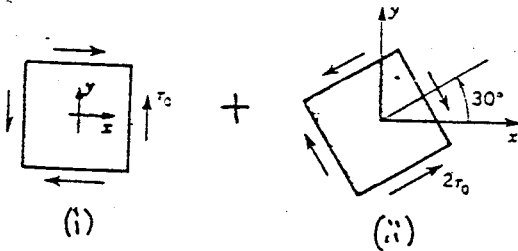
$$\therefore \cos 2\alpha = 0.2, \quad \alpha = 39.23^\circ, \quad \cos \alpha = 0.775$$

$$W = 2\pi r \cot \alpha \cdot \sin \alpha = 2\pi r \cos \alpha$$

$$\therefore \frac{W}{r} = 2\pi \cos 39.23^\circ \approx 4.86$$

$$\text{Ans.) } \frac{W}{r} \approx 4.86$$

4.26 (a)



Mohr's circle를 이용하여

(ii)를 x-y 축의

응력으로 변환시키면

$$\sigma_x = -2\tau_0 \sin(-60^\circ) = \sqrt{3}\tau_0$$

$$\sigma_y = 2\tau_0 \sin(-60^\circ) = -\sqrt{3}\tau_0$$

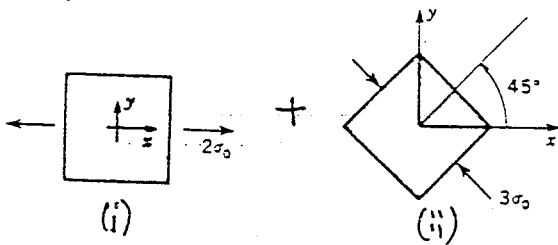
$$\tau_{xy} = -2\tau_0 \cos(-60^\circ) = -\tau_0$$

이를 (i)과 합하면

$$\sigma_x = \sqrt{3}\tau_0, \quad \sigma_y = -\sqrt{3}\tau_0, \quad \tau_{xy} = \tau_0 - \tau_0 = 0$$

따라서, 합한 결과는 x-y 축이 주축이다

(b)



마찬가지로 (ii)를

x-y 축의 응력으로 나타내면

$$\sigma_x = \frac{-3\sigma_0}{2} + \frac{3\sigma_0}{2} \cos(-90^\circ)$$

$$= -1.5\sigma_0$$

$$\sigma_y = -\frac{3\sigma_0}{2} - \frac{3\sigma_0}{2} \cos(-90^\circ)$$

$$= -1.5\sigma_0$$

$$\tau_{xy} = (-\frac{3\sigma_0}{2}) = 1.5\sigma_0$$

(i), (ii)를 합하면

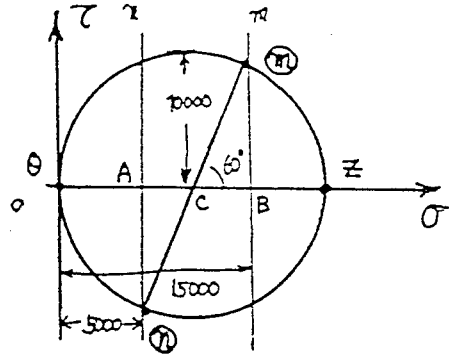
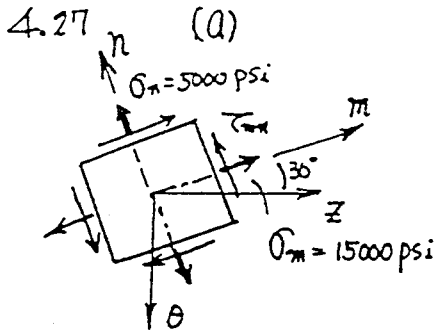
$$\sigma_x = 2\sigma_0 - 1.5\sigma_0 = 0.5\sigma_0, \quad \sigma_y = -1.5\sigma_0$$

$$\tau_{xy} = 1.5\sigma_0$$

따라서 주축 방향은 다음과 같다.

$$\theta_p = \frac{1}{2} \tan^{-1} \frac{2 \times 1.5}{0.5 - (-1.5)} = 28.15^\circ$$

$$\text{Ans.) (a) } \theta_p = 0, \quad (b) \theta_p = 28.15^\circ$$



<문제 4.10>의 결과를 이용하면 $\sigma_z = \frac{F}{2\pi r t}$, $\sigma_\theta = \frac{Pr}{t}$,
 주축 θ, z 를 30° 회전시키면 $\sigma_m, \sigma_n, \tau_{mn}$ 을 구할 수 있다.

$$\sigma_m = \frac{\sigma_z + \sigma_\theta}{2} + \frac{-\sigma_z + \sigma_\theta}{2} \cos 60^\circ = \frac{3F}{8\pi} + 25P = 15000$$

$$\sigma_n = \frac{\sigma_z + \sigma_\theta}{2} - \frac{-\sigma_z + \sigma_\theta}{2} \cos 60^\circ = \frac{F}{8\pi} + 75P = 5000$$

위 두식을 연결하여 풀면 $P=0$, $F=40000\pi$ (lb)

$$\tau_{mn} = \frac{\sigma_z - \sigma_\theta}{2} \sin 60^\circ = -8670 \text{ psi}$$

(b) $\sigma_m = 15,000$, $\sigma_n = 15,000$

$$\left. \begin{aligned} \frac{3F}{8\pi} + 25P &= 15000 \\ \frac{F}{8\pi} + 75P &= 15000 \end{aligned} \right\} \begin{aligned} P &= 150 \text{ (psi)} \\ F &= 30000\pi \text{ (lb)} \end{aligned}$$

$$\tau_{mn} = -\frac{\sigma_z - \sigma_\theta}{2} \sin 60^\circ = 0$$

Ans.) [(a) $P=0$, $F=40000\pi$ lb, $\tau_{mn} = -8670$ psi
 (b) $P=150$ psi, $F=30000\pi$ lb, $\tau_{mn} = 0$]

4.28 $\sigma_1 = 20 \text{ MN/m}^2$, $\sigma_2 = -45 \text{ MN/m}^2$

$$\sigma_1 = \frac{1}{2}(\sigma_x + \sigma_y) + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + 20^2} = 20 \quad \dots \textcircled{1}$$

$$\sigma_2 = \frac{1}{2}(\sigma_x + \sigma_y) - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + 20^2} = -45 \quad \dots \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} ; \quad \sigma_x + \sigma_y = -25 \quad \dots \textcircled{3}$$

$$\textcircled{1} - \textcircled{2} ; \quad 65 = 2\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + 20^2} \quad \therefore (\sigma_x - \sigma_y)^2 = 2625$$

$$\therefore \sigma_x - \sigma_y = \pm 51.235 \dots \textcircled{4}$$

③ ④로부터 2개의 값을 구한다.

$$\begin{aligned} \text{i)} \quad & \left(\begin{aligned} \sigma_x &= 13.118 \text{ MN/m}^2 \\ \sigma_y &= -38.118 \text{ MN/m}^2 \end{aligned} \right. & \text{ii)} \quad & \left(\begin{aligned} \sigma_x &= -38.118 \text{ MN/m}^2 \\ \sigma_y &= 13.118 \text{ MN/m}^2 \end{aligned} \right. \end{aligned}$$

$$\text{Ans.)} \quad \left[\begin{aligned} \text{i)} \quad & \sigma_x = 13.118 \text{ MN/m}^2, \sigma_y = -38.118 \text{ MN/m}^2 \\ \text{ii)} \quad & \sigma_x = -38.118 \text{ MN/m}^2, \sigma_y = 13.118 \text{ MN/m}^2 \end{aligned} \right.$$

$$4.29 \quad u(x,y) = (\cos\beta - 1)x - \sin\beta y$$

$$v(x,y) = \sin\beta x + (\cos\beta - 1)y$$

$$\epsilon_x = \frac{\partial u}{\partial x} = \cos\beta - 1, \quad \epsilon_y = \frac{\partial v}{\partial y} = \cos\beta - 1$$

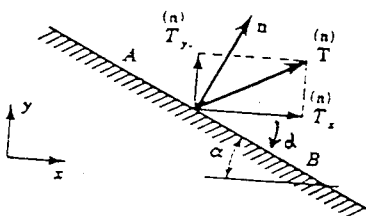
$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = -\sin\beta + \sin\beta = 0$$

$$w_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (\sin\beta + \sin\beta) = \sin\beta$$

$$\left\{ \begin{aligned} \epsilon_x &= \cos\beta - 1 \\ \epsilon_y &= \cos\beta - 1 \\ \gamma_{xy} &= 0 \\ w_z &= \sin\beta \end{aligned} \right. \Rightarrow \beta \rightarrow 0 \quad \left\{ \begin{aligned} \epsilon_x &= 0 \\ \epsilon_y &= 0 \\ \gamma_{xy} &= 0 \\ w_z &= \beta \end{aligned} \right. \quad \begin{aligned} &\beta \rightarrow 0 \text{ 일때} \\ &(\because \sin\beta \approx \beta) \end{aligned}$$

$$\text{Ans.)} \quad \epsilon_x = \epsilon_y = \gamma_{xy} = 0, \quad w_z = \beta$$

4.30



$$\sigma'_x = \frac{T_x + T_y}{2} + \frac{T_x - T_y}{2} \cos(-2\alpha)$$

$$\sigma'_y = \frac{T_x + T_y}{2} - \frac{T_x - T_y}{2} \cos(-2\alpha)$$

$$\tau'_{xy} = \frac{T_x - T_y}{2} \sin(-2\alpha)$$

위식의 σ'_x, σ'_y 가 둘이 Zero가

되려면 $T_x + T_y = 0$ 이고 $\cos 2\alpha = 0$ 인 $\alpha = 45^\circ$ 일때
이때, $\tau'_{xy} = T_x$ 이다

$$\text{Ans.)} \quad T_x + T_y = 0, \quad \alpha = 45^\circ$$

$$4.31 \quad \sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

극값을 찾기 위해서 $\frac{\partial \sigma_{x'}}{\partial \theta} = 0$ 인 θ 를 구하면, 다음과 같다.

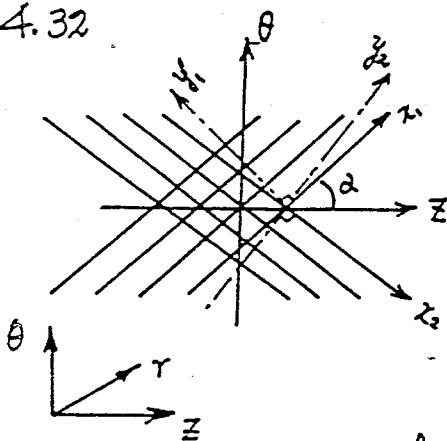
$$\frac{\partial \sigma_{x'}}{\partial \theta} = -(\sigma_x - \sigma_y) \sin 2\theta + 2\tau_{xy} \cos 2\theta = 0$$

$$\therefore \tan 2\theta = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

위 값은 $\tau_{xy}' = 0$ 로 만드는 θ 값에 해당한다.

즉, $\sigma_{x'}$ 의 극값은 $\tau_{xy}' = 0$ 로 되는 각에서 값으며
그 각은 주축의 방향각과 일치한다 - Q.E.D -

4.32



Filament는 축방향 장력만 견뎌

축 방향 응력을 σ 라 하고

filament 응력을 θ - z 축으로 변환한다.

i) x_1 방향 filament

$$\sigma_{x_1}' = \frac{\sigma}{2} + \frac{\sigma}{2} \cos(-2\alpha)$$

$$\sigma_{y_1}' = \frac{\sigma}{2} - \frac{\sigma}{2} \cos(-2\alpha)$$

$$\tau_{xy_1}' = -\frac{\sigma}{2} \sin(-2\alpha)$$

ii) x_2 방향 filament.

$$\sigma_{x_2}' = \frac{\sigma}{2} + \frac{\sigma}{2} \cos 2\alpha$$

$$\sigma_{y_2}' = \frac{\sigma}{2} - \frac{\sigma}{2} \cos 2\alpha$$

$$\tau_{xy_2}' = -\frac{\sigma}{2} \sin 2\alpha$$

위 두 응력 성분을 합하면,

$$\sigma_z = \sigma_{x_1}' + \sigma_{x_2}' = \sigma(1 + \cos 2\alpha)$$

$$\sigma_\theta = \sigma_{y_1}' + \sigma_{y_2}' = \sigma(1 - \cos 2\alpha), \quad \tau_{\theta z} = 0 \text{ 다.}$$

연립하면 $\frac{1 + \cos 2\alpha}{1 - \cos 2\alpha} = \frac{1}{2}$

$$\therefore 3 \cos 2\alpha = -1.$$

$$\cos 2\alpha = -\frac{1}{3}$$

$$\therefore \alpha = 54.7^\circ$$

Ans.) $\alpha = 54.7$

$$4.33 \quad V = \int_{-h/2}^{h/2} \tau_{xy} dy, \quad M = - \int_{-h/2}^{h/2} y \sigma_x dy$$

$$Q = \sigma_y(y = \frac{h}{2}) - \sigma_y(y = -\frac{h}{2})$$

(a) $\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} = 0$ 를 이용하자.

$$\int_{-h/2}^{h/2} \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} \right) dy = \frac{\partial}{\partial x} \int_{-h/2}^{h/2} \tau_{xy} dy + \int_{-h/2}^{h/2} \frac{\partial \sigma_y}{\partial y} dy$$

$$d\sigma_y = \frac{\partial \sigma_y}{\partial x} dx + \frac{\partial \sigma_y}{\partial y} dy$$

그런데, 평면을 보의 두께 방향으로 수행하므로 $dx = 0$

$$\therefore d\sigma_y = \frac{\partial \sigma_y}{\partial y} dy$$

$$\int_{-h/2}^{h/2} \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} \right) dy = \frac{\partial V}{\partial x} + \int_{-h/2}^{h/2} d\sigma_y$$

$$= \frac{\partial V}{\partial x} + \left\{ \sigma_y(y = \frac{h}{2}) - \sigma_y(y = -\frac{h}{2}) \right\} = \frac{\partial V}{\partial x} + Q = 0$$

$$\frac{\partial V}{\partial x} + Q = 0$$

(b) $\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0$ 를 이용하자.

$$\int_{-h/2}^{h/2} \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right) y dy = \frac{\partial}{\partial x} \int_{-h/2}^{h/2} \sigma_x y dy + \int_{-h/2}^{h/2} \frac{\partial \tau_{xy}}{\partial y} y dy$$

$$d\tau_{xy} = \frac{\partial \tau_{xy}}{\partial x} dx + \frac{\partial \tau_{xy}}{\partial y} dy$$

위와 마찬가지로 $dx = 0 \quad \therefore d\tau_{xy} = \frac{\partial \tau_{xy}}{\partial y} dy$

$$\int_{-h/2}^{h/2} \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right) y dy = -\frac{\partial M}{\partial x} + \int_{-h/2}^{h/2} y d(\tau_{xy})$$

$$= -\frac{\partial M}{\partial x} + \int_{-h/2}^{h/2} d(y \tau_{xy}) - \int_{-h/2}^{h/2} \tau_{xy} dy$$

($\because$ 둘째 항을 부분적분 수행)

$$= -\frac{\partial M}{\partial x} + \frac{h}{2} (\tau_{xy})_{y=\frac{h}{2}} - \frac{h}{2} (\tau_{xy})_{y=-\frac{h}{2}} - V$$

$$(\tau_{xy})_{y=\frac{h}{2}} = (\tau_{xy})_{y=-\frac{h}{2}} = 0 \quad \text{O.B.B.}$$

$$\int_{-\frac{h}{2}}^{\frac{h}{2}} \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right) y dy = - \frac{\partial M}{\partial x} - V = 0$$

$$\therefore \frac{\partial M}{\partial x} + V = 0$$

- Q.E.D -

$$4.34 \quad \sigma_\theta = \sigma_\phi = \frac{pr}{2t}$$

$$V = \frac{4}{3} \pi r^3 = 35 \times 10^{-5} \times 10^{-9} \text{ (m}^3\text{)}$$

$$r^3 = \frac{3 \times 35}{4\pi} \times 10^{-12} \quad \therefore r = 0.4372 \times 10^{-3} \text{ (m)}$$

$$\sigma_\theta = \sigma_\phi = \frac{150 \times \frac{101325}{760}}{2 \times (1 \times 10^{-6})} \times 0.4372 \times 10^{-3}$$

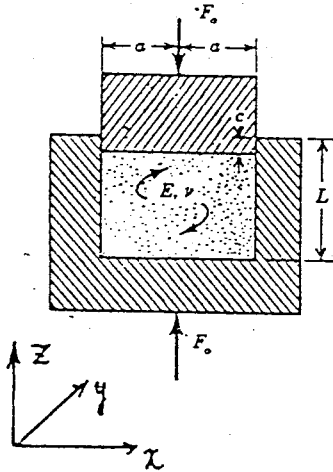
$$= 4371640 \text{ N/m}^2 = 4.37 \text{ MN/m}^2$$

$$\underline{\text{Ans.) } \sigma_\theta = \sigma_\phi = 4.37 \text{ MN/m}^2}$$

CHAPTER 5.

$C = \epsilon E$
 $\epsilon = \frac{C}{E}$

5.1



응력-변형도 관계식으로부터

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_z + \sigma_x)]$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$$

그러나 x, y 방향 변형이 의제되었으므로

$$\epsilon_x = \epsilon_y = 0$$

$$\text{따라서 } \sigma_x = \nu(\sigma_y + \sigma_z)$$

$$\sigma_y = \nu(\sigma_x + \sigma_z)$$

$$\sigma_x = \sigma_y \text{ 이고}$$

$$\text{위 식을 연결하면 } \sigma_x = \sigma_y = \frac{\nu}{1-\nu} \sigma_z$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)] = \frac{1}{E} [\sigma_z - \frac{2\nu^2}{1-\nu} \sigma_z] = \frac{C}{L}$$

$$\therefore \sigma_z = \frac{(1-\nu)}{(1-\nu-2\nu^2)} \frac{EC}{L}$$

$$F_0 = 4a^2 \times \sigma_z = \frac{4a^2 EC}{L} \frac{(1-\nu)}{(1-\nu-2\nu^2)}$$

$$\text{Ans.) } F_0 = \frac{4a^2 EC}{L} \frac{(1-\nu)}{(1-\nu-2\nu^2)}$$

$$5.2 \quad \epsilon = \frac{\Delta V}{V} = \frac{V' - V}{V} = \frac{V'}{V} - 1 = (1 + \epsilon_x)(1 + \epsilon_y)(1 + \epsilon_z) - 1$$

$$= \epsilon_x + \epsilon_y + \epsilon_z + \epsilon_x \epsilon_y + \epsilon_y \epsilon_z + \epsilon_z \epsilon_x + \epsilon_x \epsilon_y \epsilon_z$$

ϵ 값은 상당히 작으므로 ϵ 의 2차항 이상을 무시하여도 무방

$$\therefore \epsilon_v \cong \epsilon_x + \epsilon_y + \epsilon_z$$

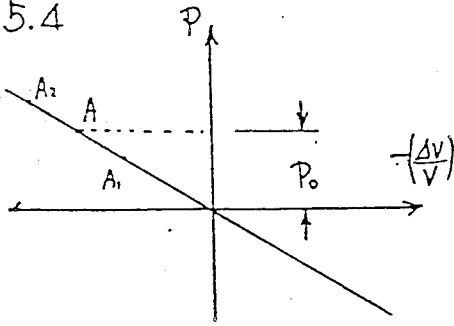
5.3 (a) 5.2의 식을 이용하면

$$\begin{aligned} \frac{\Delta V}{V} &= \epsilon_x + \epsilon_y + \epsilon_z = \frac{1}{E} [\sigma_x + \sigma_y + \sigma_z - 2\nu(\sigma_x + \sigma_y + \sigma_z)] \\ &= \frac{1}{E} (\sigma_x + \sigma_y + \sigma_z) (1 - 2\nu) \end{aligned}$$

$$\begin{aligned} (b) \quad B &= \frac{P}{(-\Delta V/V)} = \frac{-\frac{1}{3}(\sigma_x + \sigma_y + \sigma_z)}{-\frac{1}{E}(\sigma_x + \sigma_y + \sigma_z)(1 - 2\nu)} \quad (\because p = -\frac{1}{3}(\sigma_x + \sigma_y + \sigma_z)) \\ &= \frac{E}{3(1 - 2\nu)} \quad \text{또한 5.3(a)로부터} \end{aligned}$$

** isotropic ; 재료의 거동이 어느 방향에 대해서도 일정한 성질 (동방성)

5.4



Poisson's ratio $\nu > \frac{1}{2}$ 이면 $B < 0$ 이고

압력 P 와 체적 감소율 $-(\frac{\Delta V}{V})$ 와의
관계는 원뿔 그림과 같다.

물체가 P_0 의 압력에서 평형상태를
아리고 있다고 하면 물체의 상태는
 A_2 로 표시된다.

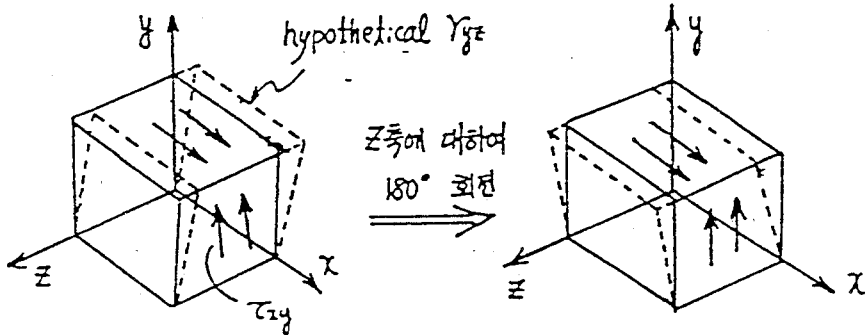
만약, 열등으로 인한 약간의 상하변화로 물체의 체적이 증가하였다면
물체 상태는 A_2 로 변하고 따라서 압력이 증가하며, 체적이 증가한
방향으로 변화를 가속시키는 압력차가 존재하게 된다.

또한, 체적이 약간 감소하게 되게 되면 물체의 압력이 감소하며
체적 감소가 가속될 것이며 결국 zero volume 상태가 될 것이다.

따라서, 이러한 물질은 불안정하다.

실제로, 현존 물질은 $B > 0$ 이며 따라서 $\nu < \frac{1}{2}$ 인 범위에 있다.

5.5

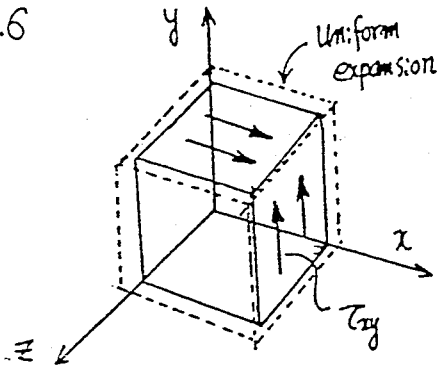


Z축에 대하여 180° 회전한 후 τ_{yz} 는 변하지 않은 반면
 γ_{yz} 는 방향이 바뀌었다. 이 물질은 등방성이므로 좌표계에
무관하게 shear-strain behavior가 동일하여야 하므로

$\gamma_{yz} = 0$ 이다.

즉, 전단응력 성분 τ_{yz} 로 인한 전단 변형도 성분 γ_{yz} 를
유발시키지는 못한다.

5.6



y축에 광하여 180° 회전을 시키면
 Uniform expansion은 불변이지만
 τ_{xy} 의 방향이 반대가 된다.
 expansion과 전단 응력이 선형
 관계라고 가정한다면,
 $(\text{expansion}) = C(\tau_{xy})$ 이다.
 유사해서 같은 팽창 현상에 대해서

C가 양, 음 부호를 전부 생략시킬 수 없으므로 $C=0$ 이어야
 또한 expansion 역시 zero 이어야 한다.

따라서, 등방성인 물질에서는 전단 응력에 의하여 발생하는
 균일 팽창 및 contraction은 존재치 않는다.

$$5.7 \quad \epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] + \alpha(T - T_0) \quad \dots ①$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)] + \alpha(T - T_0) \quad \dots ②$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)] + \alpha(T - T_0) \quad \dots ③$$

②를 σ_y 에 대하여 표현하면

$$\sigma_y = \frac{[\epsilon_y - \alpha(T - T_0)]E}{1 - \nu} + \nu(\sigma_x + \sigma_z) \quad \dots ④$$

$$\begin{aligned} ④ \rightarrow ① \quad \epsilon_x &= \frac{1}{E} [\sigma_x - \nu\{\sigma_z + \nu(\sigma_x + \sigma_z) + \frac{[\epsilon_y - \alpha(T - T_0)]E}{1 - \nu}\}] + \alpha(T - T_0) \\ &= \frac{1}{E} (1 - \nu^2)\sigma_x - \frac{1}{E}\nu(1 + \nu)\sigma_z - \nu\epsilon_y + \alpha(1 + \nu)(T - T_0) \quad \dots ⑤ \end{aligned}$$

$$④ \rightarrow ③ \quad \epsilon_z = \frac{1}{E} (1 - \nu^2)\sigma_z - \frac{1}{E}\nu(1 + \nu)\sigma_x - \nu\epsilon_y + \alpha(1 + \nu)(T - T_0) \quad \dots ⑥$$

⑤, ⑥ 으로부터

$$\epsilon_x(1 - \nu) + \nu\epsilon_z = \frac{1}{E}(1 + \nu)(1 - 2\nu)\sigma_x - \nu\epsilon_y + \alpha(1 + \nu)(T - T_0)$$

$$\therefore \sigma_x = \frac{E}{(1 + \nu)(1 - 2\nu)} [(1 - \nu)\epsilon_x + \nu(\epsilon_y + \epsilon_z) - \alpha(1 + \nu)(T - T_0)]$$

$$= \frac{E}{1 + \nu} \epsilon_x + \frac{\nu E}{(1 + \nu)(1 - 2\nu)} (\epsilon_x + \epsilon_y + \epsilon_z) - \frac{\alpha E (T - T_0)}{1 - 2\nu}$$

따라서 σ_y, σ_z 를 변형도 성분과 T 로 표현할 수 있다

$$\sigma_y = \frac{E}{1+\nu} \epsilon_y + \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_x + \epsilon_y + \epsilon_z) - \frac{\alpha E (T - T_0)}{1-2\nu}$$

$$\sigma_z = \frac{E}{1+\nu} \epsilon_z + \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_x + \epsilon_y + \epsilon_z) - \frac{\alpha E (T - T_0)}{1-2\nu}$$

$$\text{Ans.) } \sigma_i = \frac{E}{1+\nu} \epsilon_i + \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_x + \epsilon_y + \epsilon_z) - \frac{\alpha E (T - T_0)}{1-2\nu} \quad (i=x, y, z)$$

5.8 $\sigma_x = 130 \text{ MN/m}^2$, $\sigma_y = -70 \text{ MN/m}^2$, $\tau_{xy} = 80 \text{ MN/m}^2$

($\nu = 0.3$, $E = 205 \text{ GN/m}^2$, $G = 78.8 \text{ GN/m}^2$ 이라 하자)

$$\epsilon_x = \frac{1}{E} (130 + 70\nu) = 0.737 \times 10^{-3}$$

$$\epsilon_y = \frac{1}{E} (-70 - 130\nu) = -0.532 \times 10^{-3}$$

$$\epsilon_z = \frac{1}{E} (-130\nu - 70\nu) = -0.0878 \times 10^{-3}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{80}{78.8 \times 10^3} = 1.015 \times 10^{-3}$$

$$\theta = \frac{1}{2} \tan^{-1} \frac{\gamma_{xy}}{\epsilon_x - \epsilon_y} = \frac{1}{2} \tan^{-1} \frac{1.015}{0.737 + 0.001} = 19.3^\circ$$

$$\epsilon_{1,2} = \frac{\epsilon_x + \epsilon_y}{2} \pm \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \frac{\gamma_{xy}^2}{4}} \quad \begin{cases} = 1.447 \times 10^{-3} \\ = -0.178 \times 10^{-3} \end{cases}$$

Ans.) $\epsilon_1 = 1.447 \times 10^{-3}$, $\epsilon_2 = -0.178 \times 10^{-3}$, $\epsilon_3 = \epsilon_z = -0.0878 \times 10^{-3}$

$\theta = 19.3^\circ$

5.9 $\sigma_x = 145 \text{ MN/m}^2$, $\tau_{xy} = 42 \text{ MN/m}^2$, $\epsilon_z = -36 \times 10^{-4}$

$$\epsilon_z = \frac{1}{E} [-\nu(\sigma_x + \sigma_y)] \quad \text{따라서}$$

$$\sigma_y = -\left(\frac{E}{\nu} \epsilon_z + \sigma_x\right) = -\left[\frac{200 \times 10^9}{0.3} \times (-3.6 \times 10^{-4}) + 145 \times 10^6\right] = 95 \text{ MN/m}^2$$

Ans.) $\sigma_y = 95 \text{ MN/m}^2$

5.10 $\epsilon_1 = 3.2 \times 10^{-4}$, $\epsilon_2 = -5.4 \times 10^{-4}$

$$\epsilon_x = \frac{1}{E} (\sigma_x - \nu \sigma_y)$$

$$\epsilon_y = \frac{1}{E} (\sigma_y - \nu \sigma_x)$$

$$\tau_{xy} = \frac{\tau_{xy}}{G}$$

$$\frac{1}{2} \tan^{-1} \frac{\tau_{xy}}{\epsilon_x - \epsilon_y} = 60^\circ, \quad (\epsilon_x - \epsilon_y) \tan 120^\circ = \tau_{xy} \dots \textcircled{1}$$

$$\frac{\epsilon_x + \epsilon_y}{2} + \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\tau_{xy}}{2}\right)^2} = 3.2 \times 10^{-4} \dots \textcircled{2}$$

$$\frac{\epsilon_x + \epsilon_y}{2} - \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\tau_{xy}}{2}\right)^2} = -5.4 \times 10^{-4} \dots \textcircled{3}$$

$$\textcircled{2} - \textcircled{3} \quad \sqrt{(\epsilon_x - \epsilon_y)^2 + \tau_{xy}^2} = 8.6 \times 10^{-4} = \tau_{xy} \sqrt{\left(\frac{1}{\tan 120^\circ}\right)^2 + 1}$$

$$\therefore \tau_{xy} = 7.448 \times 10^{-4}$$

$$\begin{aligned} \textcircled{2} + \textcircled{3} ; \quad \epsilon_x + \epsilon_y &= -2.2 \times 10^{-4} \\ \textcircled{1} ; \quad \epsilon_x - \epsilon_y &= -4.3 \times 10^{-4} \end{aligned} \Rightarrow \epsilon_x = -3.25 \times 10^{-4}$$

$$\epsilon_y = 1.05 \times 10^{-4}$$

$$\text{알루미늄} \quad \begin{cases} E = 69 \text{ GN/m}^2 \\ \nu = 0.33 \end{cases}, \quad G = 26 \text{ GN/m}^2$$

$$\nu \epsilon_x + \epsilon_y = \frac{1-\nu^2}{E} \sigma_y \text{에 대해}$$

$$\sigma_y = \frac{E}{1-\nu^2} (\epsilon_y + \nu \epsilon_x) = \frac{69 \times 10^9}{1-0.33^2} (1.05 \times 10^{-4} - 0.33 \times 3.25 \times 10^{-4})$$

$$= -0.1742 \text{ (MN/m}^2\text{)}$$

$$\sigma_x = \frac{69 \times 10^9}{1-0.33^2} (1.05 \times 10^{-4} \times 0.33 - 3.25 \times 10^{-4})$$

$$= -22.48 \text{ (MN/m}^2\text{)}$$

$$\tau_{xy} = G \gamma_{xy} = 26 \times 10^9 \times 7.448 \times 10^{-4} = 19.36 \text{ (MN/m}^2\text{)}$$

$$\text{Ans.) } \sigma_x = -22.48 \text{ MN/m}^2, \quad \sigma_y = -0.1742 \text{ MN/m}^2$$

$$5.11 \quad \epsilon_x = \ln \frac{L_f}{L_0} \quad \tau_{xy} = 19.36 \text{ MN/m}^2$$

$$A L_f = A_0 L_0 \rightarrow \epsilon_x = \ln \frac{A_0}{A} \quad \therefore \frac{A_0}{A} = e^{\epsilon_x}$$

$$\Delta A = \frac{A_0 - A}{A_0} = 1 - \frac{A}{A_0} = 1 - e^{-\epsilon_x} \quad - \text{a.e.d. -}$$

$$5.12 \quad \sigma_\theta = \frac{Pr}{t}, \quad \sigma_z = \frac{Pr}{2t}, \quad \sigma_r = 0$$

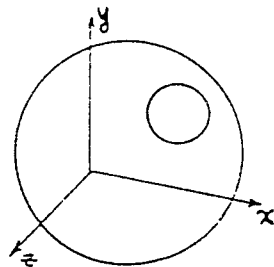
$$\epsilon_0 = \frac{1}{E} (\sigma_z - \nu \sigma_\theta) = \frac{Pr}{2Et} (1-2\nu) \quad \therefore p = \frac{2Et\epsilon_0}{r(1-2\nu)}$$

$$\text{Ans.) } p = \frac{2Et\epsilon_0}{r(1-2\nu)}$$

5.13 열이 가해진 후에 plate는
일정한 strain을 갖는 stress free의
상태로 될 것이다.

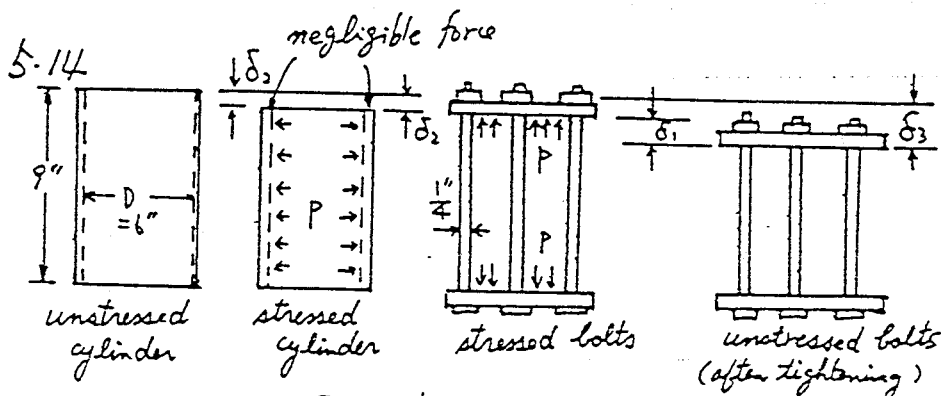
$$\epsilon_x = \epsilon_y = \epsilon_z = \alpha (T - T_0)$$

$$\gamma_{xy} = \gamma_{yz} = \gamma_{zx} = 0$$



(이러한 해는 equilibrium, compatibility
stress-strain-temperature relation, boundary
condition을 모두 만족한다.)

따라서 변위는 크기만 일정하게 변한다. 그러므로 원은
원의 형태를 그대로 유지하면서 크기만 변한다.



$$\delta_1 + \delta_2 = \delta_3 = \frac{1}{\alpha} \left(\frac{1}{20} \right) = 0.025'' \quad (1)$$

평형 조건; Cylinder; $\sigma_\theta = p \cdot \frac{D}{2t} = p \cdot \frac{6}{2 \cdot \frac{1}{16}} = 48p \text{ psi}$

$$\sigma_r = \sigma_z = 0$$

Bolts; Force = $p \cdot \frac{\pi}{4} (D^2) = p \cdot \frac{\pi}{4} (6)^2 = 28.3p \text{ lb}$

stress $\sigma_z = \frac{28.3P}{3} \cdot \frac{1}{\frac{\pi}{4} \left(\frac{1}{4} \right)^2} = 192P \text{ psi}$

결과 변형률의 관계;

Cylinder; $\epsilon_z = \frac{1}{E_B} [\sigma_z - \nu_B (\sigma_\theta + \sigma_r)] = -\frac{\nu_B \sigma_\theta}{E_B}$

$$\text{Bolts ; } \epsilon_z = \frac{\delta_z}{E_s}$$

$$\text{따라서 } \delta_2 = (-\epsilon_z)_{\text{cylinder}} \cdot (L)_{\text{cyl.}} = -\left(-\frac{\nu_B}{E_B} 48P\right) \cdot 9$$

$$\delta_1 = (\epsilon_z)_{\text{bolt}} \cdot (L)_{\text{bolt}} = \left(\frac{1920P}{E_s}\right) \cdot 10$$

$$(1) \text{식에서 } \frac{(9)(48)\nu_B P}{E_B} + \frac{1920P}{E_s} = 0.025$$

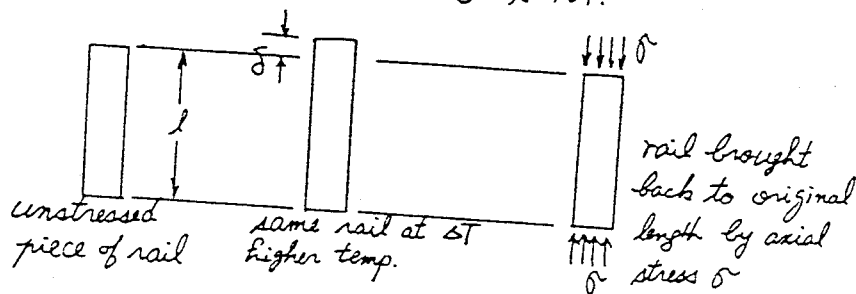
$$\text{한편, } \nu_B = \nu_{\text{BRASS}} = 0.34 ; E_B = 15.2 \times 10^6 \text{ psi}$$

$$E_s = E_{\text{steel}} = 30 \times 10^6 \text{ psi}$$

$$\text{그러므로 } \frac{(9)(48)(0.34)(P)}{15.2 \times 10^6} + \frac{1920P}{30 \times 10^6} = 0.025$$

$$\therefore P = 338 \text{ psi} \quad \underline{\text{Ans } P = 338 \text{ psi}}$$

5.15 rail의 양단이 움직일 수 없다면, 열이 가해질 때 thermal stress가 발생되게 될 것이다.



$$\delta = \alpha \cdot l \cdot \Delta T \quad \therefore \sigma = E \cdot \frac{\delta}{l} = E \alpha (\Delta T)$$

따라서, $E \alpha \Delta T$ 만큼의 stress가 rail의 양단에서 발생할 것이다. rail이 점점 더 굽어지면, 양단에서의 힘이 점점 커지므로 rail은 buckling을 일으키려 할 것이다. buckling을 방지하기 위해서는 lateral constraints가

필요하며, lateral support의 간격은 rail의 단면과 탄성계수에 의해 정해진다. rail이 curve를 이루지 않는 한, rail 양단에서의 힘이 인장 (ΔT 가 negative)인 경우에는 buckling 문제는 일어나지 않는다. rail이 curve인 경우에는 안히려 펴지려 할 것이다.

5.16 서로 다른 열팽창에 의해서 응력이 발생한다. cladding의 두께가 plate의 두께에 비해서 상당히 작다면, plate에서의 응력은 작으므로 무시한다.

대칭 조건; shear stress = 0 (양단에서)

Geometry; $(\epsilon_x)_{clad} = (\epsilon_x)_{plate}$
 $(\epsilon_y)_{clad} = (\epsilon_y)_{plate}$ (1)

Stress-strain-temp. relation;

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu (\sigma_y + \sigma_z)] + \alpha \Delta T \dots$$

평형 조건; $\sigma_z = 0$

(notation; subscript p for plate; c for cladding)

plate; 모든 응력 = 0. $\therefore (\epsilon_x)_p = \alpha_p \Delta T = (\epsilon_y)_p$

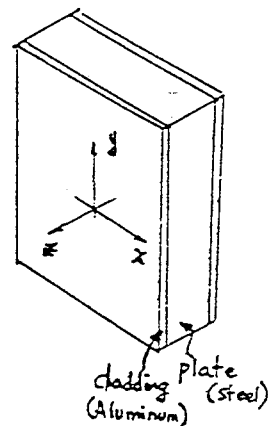
cladding; (1) 식으로 부터

$$(\epsilon_x)_c = \frac{1}{E_c} [(\sigma_x)_c - \nu_c (\sigma_y)_c] + \alpha_c \Delta T = \alpha_p \Delta T$$

$$(\epsilon_y)_c = \frac{1}{E_c} [(\sigma_y)_c - \nu_c (\sigma_x)_c] + \alpha_c \Delta T = \alpha_p \Delta T$$

이 식을 풀면 $(\sigma_x)_c = (\sigma_y)_c = \frac{E_c (\alpha_p - \alpha_c) \Delta T}{1 - \nu_c}$

이것이 두개의 주응력이며, 세 번째 주응력은 $\sigma_z = 0$.



Tresca yield criterion을 이용하면,

$$\tau_{max} = \frac{1}{2} \tau$$

$$\text{cladding ; } \tau_{max} = \frac{1}{2} |(\sigma_{max} - \sigma_{min})| = \frac{1}{2} \left[\frac{E_c(\alpha_p - \alpha_c)\Delta T}{1 - \nu_c} - 0 \right]$$

material property ; $\alpha_c = 12 \times 10^{-6} \text{ } ^\circ\text{F}^{-1}$, $\alpha_p = 6 \times 10^{-6} \text{ } ^\circ\text{F}^{-1}$

$$\nu_c = 0.33 ; E_c = 11 \times 10^6 \text{ psi } \quad Y_c = 5000 \text{ psi}$$

$$\therefore \frac{5000}{2} = \frac{11 \times 10^6 (12 - 6) (10^{-6}) (\Delta T)}{2 (1 - 0.33)} \quad \therefore \Delta T = 51^\circ\text{F}$$

Ans $\Delta T = 51^\circ\text{F}$ (Von Mises yield criterion gives the same answer)

5.17 Pulley를 가열하는 것이 더 쉽다해도, pulley를 가열하거나 shaft를 냉각시키거나 같은 효과를 가져올 것이다. 그러나, 실제로는 pulley가 shaft end에서 어느 정도 떨어져 고정되어 있다면 pulley를 가열하는 것이 더 편하다.

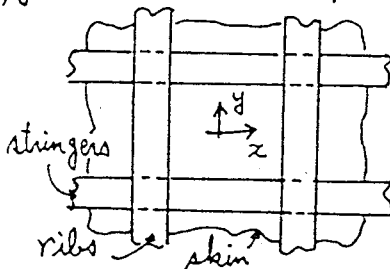
pulley hole의 직경이 shaft의 직경보다 0.05 mm 작고, assembling하는 동안에 clearance 0.025 mm를 필요로 하므로, 열팽창은 다음과 같아야 한다.

$$\text{열팽창} = 0.05 + 0.025 = 0.075 \text{ (mm)}, \quad \alpha = 11 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$$

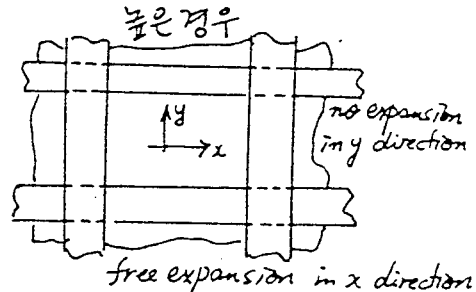
$$\therefore \epsilon = \frac{0.075}{25} = \alpha \Delta T \quad \therefore \Delta T = \frac{0.075}{25 \times 11 \times 10^{-6}} = 272.7^\circ\text{C}$$

Ans $\Delta T = 272.7^\circ\text{C}$

5.18 uniform temp.



skin 이 ribs 보다 50°F



i) Rib에 의해 lateral strain은 없다.

Geometry ; $\epsilon_y = 0$

평형 조건 ; $\sigma_x = \sigma_z = \text{shear stress} = 0$.

Stress-strain-temp. relation ;

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)] + \alpha \Delta T = 0$$

$$\therefore \sigma_y = -E \alpha \Delta T = -(11 \times 10^6)(12 \times 10^{-6})(50) = -6600 \text{ psi}$$

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] + \alpha \Delta T = -\frac{(0.33)(-6600)}{11 \times 10^6} + (12 \times 10^{-6})(50) = 800 \times 10^{-6}$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)] + \alpha \Delta T = 800 \times 10^{-6}$$

ii) Rib에 의한 lateral restraint가 없다면

all stresses = 0. $\epsilon_x = \epsilon_y = \epsilon_z = \alpha \Delta T = 600 \times 10^{-6}$

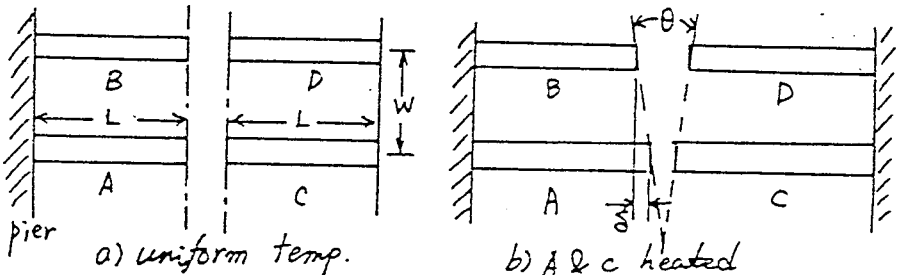
Ams. i) $\sigma_x = \sigma_z = 0$, $\sigma_y = -6600 \text{ psi}$

$$\epsilon_x = \epsilon_z = 800 \times 10^{-6}, \quad \epsilon_y = 0$$

$$\text{ii) } \sigma_x = \sigma_y = \sigma_z = 0, \quad \epsilon_x = \epsilon_y = \epsilon_z = 600 \times 10^{-6}$$

5.19 Z개의 member로써 truss를 표시할 수 있다.

그리고 태양은 한쪽 member에만 열을 주게 될 것이다.



$$A \text{ 와 } C \text{ 의 팽창} = (\alpha \Delta T) \cdot L = \delta$$

Z span의 양 끝은 θ 만큼의 각 차이를 갖게 될 것이다.

$$\theta \approx \frac{\alpha \delta}{w} = \alpha (\Delta T) \left(\frac{L}{w} \right)$$

Estimate ; $L = 200 \text{ ft}$; $w = 30 \text{ ft}$; $\Delta T = 50^\circ \text{F}$

$\alpha = 6 \times 10^{-6} \text{ } ^\circ \text{F}^{-1}$ (for steel)

$$\therefore \theta = 6 \times 10^{-6} (50) \left(\frac{200}{30} \right) = 0.004 \text{ (rad)} = 0.2 \text{ (degree)}$$

Ans $\theta = 0.2 \text{ degree}$

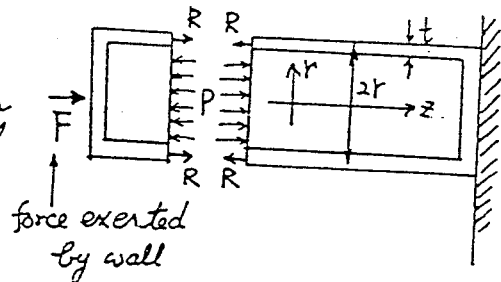
5.20

Geometry ; $\epsilon_z = 0$ in cylinder wall

평형 조건 ; (외압이 없다)

$$F + R = p \pi r^2$$

$$\therefore R = p \pi r^2 - F$$



$$\therefore \text{Axial stress in cylinder} = \frac{R}{2 \pi r t} = \frac{p \pi r^2 - F}{2 \pi r t} = \tilde{\sigma}_z$$

$$\text{Circumferential stress} = \tilde{\sigma}_\theta = \frac{p r}{t}, \quad \tilde{\sigma}_r = 0$$

Stress-strain relation에 대해

$$\epsilon_z = \frac{1}{E} [\tilde{\sigma}_z - \nu (\tilde{\sigma}_r + \tilde{\sigma}_\theta)] = 0$$

$$\therefore \tilde{\sigma}_z = \nu (\tilde{\sigma}_r + \tilde{\sigma}_\theta) \quad \text{즉,} \quad \frac{p \pi r^2 - F}{2 \pi r t} = \nu \cdot \frac{p r}{t}$$

$$\text{따라서} \quad F = p \pi r^2 (1 - 2\nu) \quad \text{Ans. } F = p \pi r^2 (1 - 2\nu)$$

5.21 tank와 cavity의 벽 사이에는 마찰력이 없다고 가정하고, p_c = contact pressure between tank and cavity로 놓자.

Geometry ; $\epsilon_\theta = 0$ (tank의 벽에 대해)

평형 조건 ;

$$\sum F_z = 0 ; p\pi r^2 = R_1 = \sigma_z (2\pi r t)$$

$$\therefore \sigma_z = \frac{p \cdot r}{2t}$$

$$\sum F_{vertical} = 0 ;$$

$$2\sigma_\theta t = (p - p_c)(2r)$$

$$\therefore \sigma_\theta = (p - p_c) \frac{r}{t}$$

그리고 σ_r 은 $-p$ 과 $-p_c$ 사이에 있다.

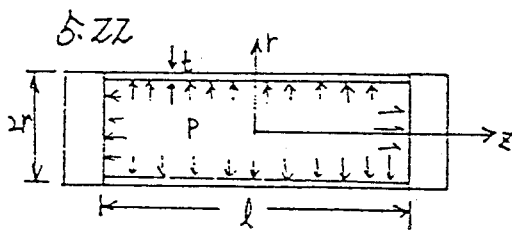
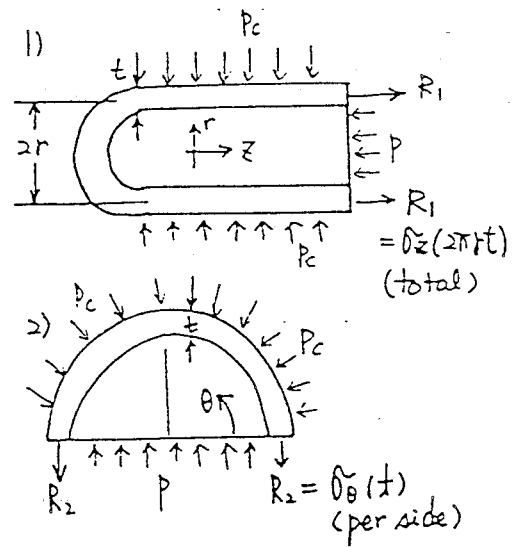
p_c 가 p 과 같은 order라고 생각하면

즉, $(\frac{p_c}{p}) \ll 1$ 이라면 σ_r 은 무시할 수 있다.

$$\text{Stress-strain relation ; } \epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)] = 0$$

$$\therefore \sigma_\theta = \nu(\sigma_r + \sigma_z) = \nu(\sigma_z) = \frac{\nu p r}{2t}$$

$$\text{Ans } \sigma_\theta = \nu p r / 2t$$



Geometry ; Rigid end 는 tangential strain (ϵ_θ) 를 방지한다. 그러나 (l/r) 이 상당히 크다면 이러한 end effect 는 무시할 수 있다.

$$\text{평형 조건 ; } \sigma_z = \frac{p \cdot r}{2t} ; \sigma_\theta = \frac{p \cdot r}{t} ; \sigma_r = 0$$

$$\text{Stress-strain relation ; } \epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_\theta + \sigma_r)] = \frac{p r}{2 E t} (1 - 2\nu)$$

$$\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)] = \frac{p r}{2 E t} (2 - \nu)$$

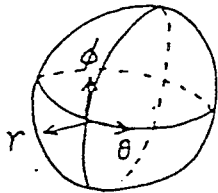
$$\delta_D = D \cdot \epsilon_\theta = 2r \cdot \epsilon_\theta , \quad \delta_L = l \epsilon_z$$

$$\therefore \frac{\delta_l}{\delta_D} = \frac{l}{2r} \frac{(1-2\nu)}{(2-\nu)}$$

$$\text{Ans. } (\delta_l/\delta_D) = \frac{l}{2r} \frac{(1-2\nu)}{(2-\nu)}$$

5.23

평형조건; $\sigma_\theta = \sigma_\phi = \frac{Pr}{2t}$, $\sigma_r \approx 0$



internal pressure = p
radius = r

Stress-strain relation;

$$\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_\phi + \sigma_r)] = \frac{Pr}{2Et} (1-\nu)$$

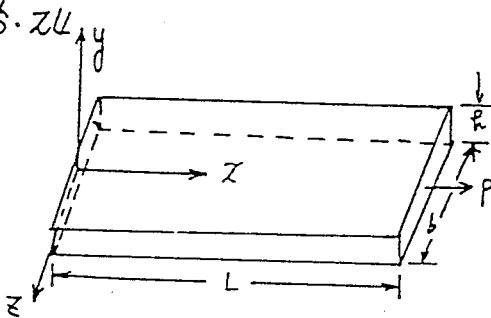
$$\delta_D = (D) \cdot \epsilon_\theta = 2r \epsilon_\theta = \frac{Pr^2}{Et} (1-\nu)$$

문제 (5.22)에 $\ll$ $(\delta_D)_{\text{cylinder}} = \frac{Pr^2}{Et} (2-\nu)$

$$\therefore \frac{(\delta_D)_{\text{cylinder}}}{(\delta_D)_{\text{sphere}}} = \left(\frac{Pr^2}{Et} (2-\nu) \right) / \left(\frac{Pr^2}{Et} (1-\nu) \right) = \frac{2-\nu}{1-\nu}$$

Ans $(\delta_D)_{\text{cylinder}} / (\delta_D)_{\text{sphere}} = (2-\nu)/(1-\nu)$

5.24



Bounding surface 상의

다음 응력들은 전체적인 평형조건을 만족한다.

$$\sigma_x = \frac{P}{bh}; \sigma_y = \sigma_z = 0$$

$$\tau_{xy} = \tau_{yz} = \tau_{zx} = 0$$

Stress-strain relation;

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] = \frac{P}{Eb h}, \quad \epsilon_y = \epsilon_z = \frac{-\nu P}{Eb h} \quad (1)$$

All shear strains = 0

따라서 $\frac{\partial u}{\partial x} = \frac{P}{Eb h}; \frac{\partial v}{\partial y} = -\frac{\nu P}{Eb h} = \frac{\partial w}{\partial z}; \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \text{etc} = 0$

$u=v=w=0$ 인 원점에 대하여 적분해 주면,

$$u = \frac{Px}{EbL} ; v = -\frac{\nu Py}{EbL} ; w = -\frac{\nu Pz}{EbL}$$

i) Equilibrium ; stress의 derivatives가 모두 0이므로 만족.

ii) Geometric compatibility ; 변위가 모두 single value이며 연속이므로 만족.

iii) Stress-strain relation ; (1)식에서 만족

따라서 위의 해는 모든 principle of elasticity를 만족함.

Ans. $u = \frac{Px}{EbL} , v = -\frac{\nu Py}{EbL} , w = -\frac{\nu Pz}{EbL}$

5.25 처음 yielding이 일어나는 온도 이상으로 100°F 까지 가열

했을 때, cladding은 $\sigma_x = \sigma_y = Y$ 로

소성이 될 것이다. 온도가 다시 낮아짐에

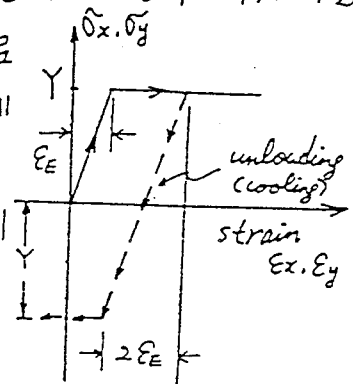
따라서 cladding은 탄성적으로 하중이

줄어들게 된다. 냉각하는 도중에 strain이

$2\epsilon_E$ (그림참조)를 넘게되면 다시 소성이

될 것이다. 약 50°F 의 온도는 ϵ_E 만큼의

strain을 발생시킨다. (문제(5.18)).



plate가 주변보다 150°F 높은 온도에서부터 냉각될 때, strain은 $2\epsilon_E$ 보다 큰 $3\epsilon_E$ 이 될 것이다. 그러므로, cladding은 $\sigma_x = \sigma_y = -Y$ 로 주변 온도에서 소성이 될 것이다.

5.26 서로 다른 열팽창은 다음과 같은 일을 야기시킬 것이다.

i) 두 개의 hoop의 지름의 차이, 접촉면에서의 contact pressure p

ii) 서로 다른 lateral expansion.

만일 hoop 사이에 마찰력이 크거나, 2개의 hoop가 서로 접촉되어 있다면, lateral expansion은 axial stress나 shear stress로 나타날 것이다. 따라서 문제의 편의상 hoop는 마찰력이 없고 자유롭게 움직인다고 가정한다. 그러면 lateral expansion은 무시되지 않을 것이다.

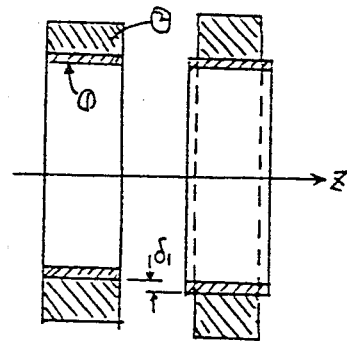
각각의 hoop에서 (t/r) 은 작기 때문에, 2개의 hoop에 대하여 같은 반경을 사용한다. ($= 300 \text{ mm}$). 가열한 후에 반경은 δ_1 만큼 증가한다.

Geometry; tangential strain ϵ_θ
in both hoops = $\frac{\delta_1}{300 \times 10^{-3}}$

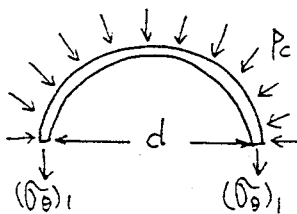
Symmetry; 접촉면에서 $\tau_{r\theta} = 0$

평형 조건; $\sigma_z = 0$,

$\sigma_r = 0$ 으로 가정.



i) inner hoop; $\sum F_{\text{vertical}} = 0$;



$$-P_c (0.6)(0.06) - (\sigma_\theta)_1 (0.06)(0.003)(z) = 0$$

$$\therefore (\sigma_\theta)_1 = -100 P_c$$

ii) outer hoop;

$$\sum F_{\text{vertical}} = 0;$$

$$P_c (0.6)(0.06) - (\sigma_\theta)_2 (0.06)(0.006)(z) = 0$$

$$\therefore (\sigma_\theta)_2 = 50 P_c$$

Stress-strain relation;

$$\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_z + \sigma_r)] + \alpha \Delta T = \frac{\sigma_\theta}{E} + \alpha \Delta T$$

따라서 inner hoop; $(\epsilon_\theta)_1 = -\frac{100 P_c}{E_1} + \alpha_1 \Delta T = \delta_1 / 300 \times 10^{-3}$

$$\text{outer loop; } (\epsilon_{\theta})_2 = \frac{50P_c}{E_2} \times \alpha_2 \Delta T = \delta_1 / 300 \times 10^{-3}$$

$$\therefore -\frac{100P_c}{E_1} + \alpha_1 \Delta T = \frac{50P_c}{E_2} + \alpha_2 \Delta T \quad \text{--- ①}$$

material property (① (brass); $E_1 = 100 \times 10^9$, $\alpha_1 = 20.4 \times 10^{-6}$
 ② (steel); $E_2 = 200 \times 10^9$; $\alpha_2 = 11.4 \times 10^{-6}$)

따라서 ①식이 material property를 대입하면

$$P_c \left[\frac{100}{100 \times 10^9} + \frac{50}{200 \times 10^9} \right] = (20.4 - 11.4) \times 10^{-6} \times 85$$

$$\therefore P_c = 6.12 \times 10^5 \text{ (N/m}^2\text{)}$$

Brass hoop (inner); $\tilde{\sigma}_{\theta} = -100P_c = -61.2 \text{ (MN/m}^2\text{)}$

Steel hoop (outer); $\tilde{\sigma}_{\theta} = 50P_c = 30.6 \text{ (MN/m}^2\text{)}$

Ans. $(\tilde{\sigma}_{\theta})_{\text{brass}} = -61.2 \text{ MN/m}^2$ $(\tilde{\sigma}_{\theta})_{\text{steel}} = 30.6 \text{ MN/m}^2$

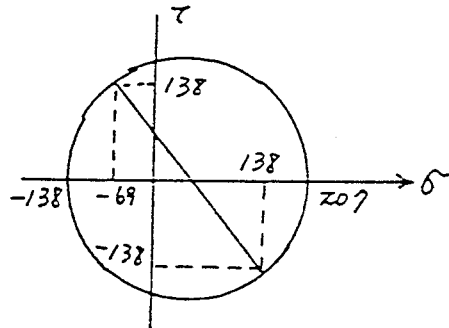
5.27 우선 x-y 평면에서 Mohr's circle을 이용하여 principal stress를 구한다.

principal stress;

$$\tilde{\sigma}_1 = 207 \text{ MN/m}^2$$

$$\tilde{\sigma}_2 = -138 \text{ MN/m}^2$$

$$\tilde{\sigma}_3 = \tilde{\sigma}_z = 0$$



a) Mises yield criterion

$$\begin{aligned} Y = \bar{\sigma} &= \sqrt{\frac{1}{2} \{ (\tilde{\sigma}_1 - \tilde{\sigma}_2)^2 + (\tilde{\sigma}_2 - \tilde{\sigma}_3)^2 + (\tilde{\sigma}_3 - \tilde{\sigma}_1)^2 \}} \\ &= 300.764 \text{ (MN/m}^2\text{)} < 330 \text{ MN/m}^2 \end{aligned}$$

따라서 yielding이 발생하지 않는다.

b) Max. shear stress criterion; $|\tilde{\sigma}_{\max} - \tilde{\sigma}_{\min}| = Y$

$$\tilde{\sigma}_{\max} = \tilde{\sigma}_1 = 207 \text{ MN/m}^2, \quad \tilde{\sigma}_{\min} = \tilde{\sigma}_2 = -138 \text{ MN/m}^2$$

$$\therefore Y = |207 - (-138)| = 345 \text{ (MN/m}^2\text{)} > 330 \text{ MN/m}^2$$

따라서 yielding 이 발생한다.

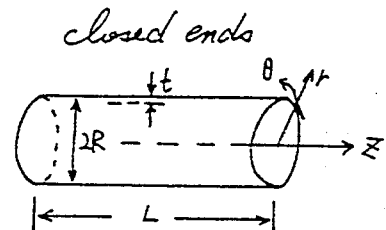
Ans. a) Mises yield criterion 으로는 yielding 이 발생하지 않음.

b) Max. shear stress criterion 으로는 yielding 이 생김.

5.28 $p = \text{internal pressure}$

평형조건; $\sigma_\theta = \frac{PR}{t}$; $\sigma_z = \frac{PR}{2t} = \frac{1}{2}\sigma_\theta$

$$\sigma_r \approx 0$$



Yield criterion (Max. shear stress criterion)

$$|\sigma_{\max} - \sigma_{\min}| = Y ; \sigma_{\max} = \sigma_\theta ; \sigma_{\min} = 0$$

1020 HR steel 에 대한 항복 응력 $Y = 250 \text{ MN/m}^2$

$$\therefore \sigma_\theta = Y = 250 \times 10^6 \text{ N/m}^2, \quad \sigma_z = \frac{1}{2}\sigma_\theta = 125 \times 10^6 \text{ N/m}^2$$

elastic 이나 stress-strain relation;

$$\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)] = \frac{1}{200 \times 10^9} (250 \times 10^6 - 0.3 \times 125 \times 10^6)$$

$$= 1.0625 \times 10^{-3}$$

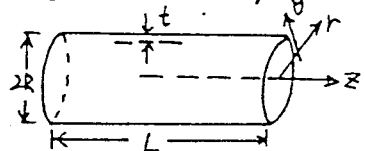
$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_r + \sigma_\theta)] = \frac{(125 - 0.3 \times 250) \times 10^6}{200 \times 10^9} = 0.25 \times 10^{-3}$$

Ans. length (ϵ_z); 0.025%, circumferential (ϵ_θ); 0.106%.

5.29 평형조건; $\sigma_z = \frac{PR}{2t}$; $\sigma_\theta = \frac{PR}{t}$; $\sigma_r \approx 0$ closed ends, internal pressure: p

plastic 이나 stress-strain relation;

$$d\epsilon_z^p = \frac{d\bar{\epsilon}}{\bar{\sigma}} [\sigma_z - \frac{1}{2}\sigma_\theta] = 0$$



$$d\epsilon_\theta^p = \frac{d\bar{\epsilon}}{\bar{\sigma}} \left[\sigma_\theta - \frac{1}{2}\sigma_z \right] = \left(\frac{3}{4}\right)\left(\frac{PR}{t}\right) \frac{d\bar{\epsilon}}{\bar{\sigma}}$$

$$\text{따라서 } \epsilon_z^p = \int d\epsilon_z^p = 0; \quad \epsilon_\theta^p = \int d\epsilon_\theta^p = \left(\frac{3}{4}\right)\left(\frac{PR}{t}\right) \frac{d\bar{\epsilon}}{\bar{\sigma}} \neq 0 = \epsilon_\theta.$$

elastic strain 을 무시하면 $\epsilon_z = \epsilon_z^p$, $\epsilon_\theta = \epsilon_\theta^p$

$$\therefore \frac{\text{axial strain}}{\text{tangential strain}} = \frac{\epsilon_z}{\epsilon_\theta} = \frac{0}{\epsilon_\theta} = 0$$

Ans (axial strain / tangential strain) = 0

5.30 실제 test에 있어서 시편의 폭 w 는, 두께가 작기 때문에 두께보다 상당히 클 것이다.

이러한 경우 해석은 가능하나, stress σ_z 가 무시될 수 없기 때문에 상당히 복잡해 진다.

이러한 경우는 일반화된 plane strain의 경우이다. 문제를 단순하게 하기 위해서

w 가 t_1 에 비하여 상당히 작다는 비현실적인 가정을 한다.

이러한 가정하에서는 stress σ_z , τ_{xy} , τ_{xz} 는 edge를 따라서

만일 "0"이다. 그러므로 그러한 응력들은 전체적으로 0으로 취할 수 있다.

Symmetry ; $\tau_{yz} = 0$, 평형 조건 ; $\sigma_z = 0$

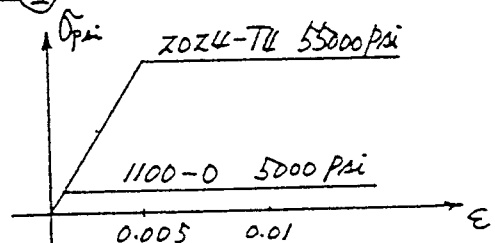
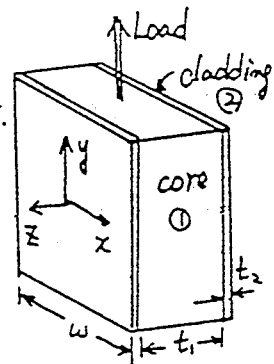
따라서 stress는 σ_y 만이 남는다.

하중을 제거한 후에 y방향의 순수한 힘은 0이다.

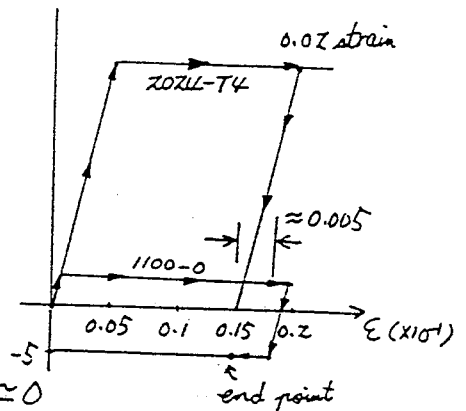
$$\sum F_y = 0; \quad (\sigma_y)_1 t_1 + 2(\sigma_y)_2 t_2 = 0 \quad \therefore (\sigma_y)_1 = -2\left(\frac{t_2}{t_1}\right)(\sigma_y)_2 \quad \text{--- ①}$$

$$\text{Geometry ; } (\epsilon_y)_1 = (\epsilon_y)_2 \quad \text{--- ②}$$

stress-strain data;
elastic perfectly plastic
model은 오른쪽 그림과 같은
형태를 이룬다.



a) ① 식에서 second term을 무시하면, $(\sigma_y)_1 = 0$. 이것은 core에 대한 stress-strain history의 end point를 결정해 준다. 또, ② 식을 이용하여 cladding에 대한 end point를 결정할 수 있다. 이것은 오른쪽처럼 graph로 나타낼 수 있다.

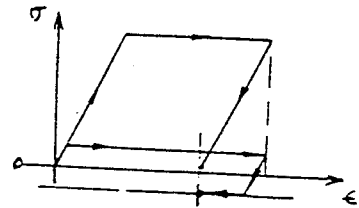


따라서 residual stress (core ≈ 0 , cladding $= -5000$ psi)

b) $t_1 = 0.041$ " , $t_2 = 0.005$ " 인 경우 ① 식은 다음과 같이 된다. $(\sigma_y)_1 = -0.244 (\sigma_y)_2$, 이 경우에 양 end point는 strain axis 상의 오른쪽으로 이동해 갈 것이다. 이 이동이 약 0.005보다 작다고 가정하면 $(\sigma_y)_2 = -5000$ psi

$(\sigma_y)_1 = -0.244(-5000) = 1220$ psi
[end point의 이동 $\approx \frac{1220}{E} < 0.005$]

따라서 stress는 (core 1220 psi, residual cladding -5000 psi)

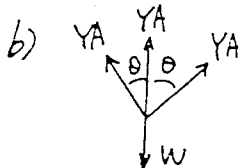


Ans. a) $(\sigma_y)_1 = 0$, $(\sigma_y)_2 = -5000$ psi b) $(\sigma_y)_1 = 1220$ psi $(\sigma_y)_2 = -5000$ psi

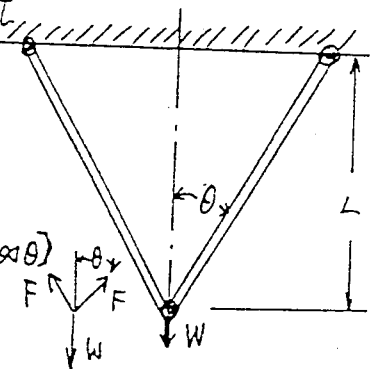
5.31 a) 미소 변위로 가정하면 θ 는 constant

평형조건; $2F \cos \theta = W \therefore F = \frac{W}{2 \cos \theta}$

yielding; $F = Y A \therefore W = 2 Y A \cos \theta$



평형조건; $W = Y A [1 + 2 \cos \theta]$
 $\therefore \text{increase} = Y A$

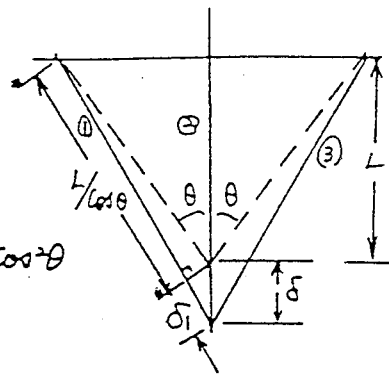


c) Geometry ; $\delta_1 = \delta_3 = \delta \cos \theta$
 $\delta_2 = \delta$

rod에 발생하는 strain은;

$$\epsilon_1 = \frac{\delta_1}{L/\cos \theta} = \frac{\delta \cos \theta}{L/\cos \theta} = \frac{\delta}{L} \cos^2 \theta$$

$$\epsilon_2 = \frac{\delta}{L}$$



stress-strain relation (elastic) $F = A(E\epsilon)$

$$F_1 = F_3 = EA \frac{\delta}{L} \cos^2 \theta ; F_2 = EA \frac{\delta}{L}$$

평형조건 ; $W = 2F_1 \cos \theta + F_2 = EA \frac{\delta}{L} [1 + 2\cos^3 \theta]$

$$\therefore \delta = \frac{WL/EA}{1 + 2\cos^3 \theta}$$

d) c)에서 하중 W에 의해 생기는 stress는 다음과 같다.

$$\sigma_1 = \sigma_3 = \frac{F_1}{A} = E \frac{\delta}{L} \cos^2 \theta = \frac{(W/A) \cos^2 \theta}{1 + 2\cos^3 \theta} ; \sigma_2 = \frac{(W/A)}{1 + 2\cos^3 \theta}$$

b)에서 소성 하중 $= W_p = YA [1 + 2\cos \theta]$,

elastic unloading을 가정하고, $-W_p$ 에 의한 stress를 구하여 yield stress에 더하라. 그러면 $W = -W_p$ 인 경우에

$$\sigma_1 = \sigma_3 = -\frac{Y(1 + 2\cos \theta) \cos^2 \theta}{1 + 2\cos^3 \theta} ; \sigma_2 = -\frac{Y(1 + 2\cos \theta)}{1 + 2\cos^3 \theta}$$

Residual stress ;

rod ①, ③ ; $Y - \frac{Y(1 + 2\cos \theta) \cos^2 \theta}{1 + 2\cos^3 \theta} = Y \left(\frac{1 - \cos^2 \theta}{1 + 2\cos^3 \theta} \right)$

rod ② ; $Y - \frac{Y(1 + 2\cos \theta)}{1 + 2\cos^3 \theta} = -\frac{2Y \sin^2 \theta \cos \theta}{1 + 2\cos^3 \theta}$

$\therefore$ rod ①②③ 의 residual stress는 모두 yield stress

Y 보다 작다. 따라서 elastic unloading 이라는 가정은 옳다.
 * $\theta = 0$ 인 경우; 3개의 rod에 uniaxial tension 이 걸린다.
 따라서 residual stress 는 0. 윗 식은 이것을 만족한다.
 $\theta = \pi/2$ 인 경우; ①③은 무관한 rod ②에 단축한 tension 만이 작용한다. 따라서 residual stress 는 0. rod ②에 대한 구한 식이 이것을 만족한다.

Ams. a) $W = 2YA \cos \theta$ b) 하중의 공분 = YA
 c) $\delta = \frac{WL/EA}{1+2\cos^3 \theta}$ d) rod ①③; $Y \left(\frac{1-\cos^2 \theta}{1+2\cos^3 \theta} \right)$
 rod ②; $-\frac{2Y \sin^2 \theta \cos \theta}{1+2\cos^3 \theta}$

5.32 모든 joint 는 pin joint 이고
 θ 를 미소각으로 이상화 하자.

Geometry; $\delta_1 = c\theta$; $\delta_2 = 3c\theta$
 $\therefore \epsilon_1 = \frac{c\theta}{L}$, $\epsilon_2 = \frac{3c\theta}{L}$

평형조건; $\sum M_{\theta} = 0$
 $F_1(c) + F_2(3c) = P(2c)$
 $\therefore F_1 + 3F_2 = 2P$

stress-strain relation (elastic)

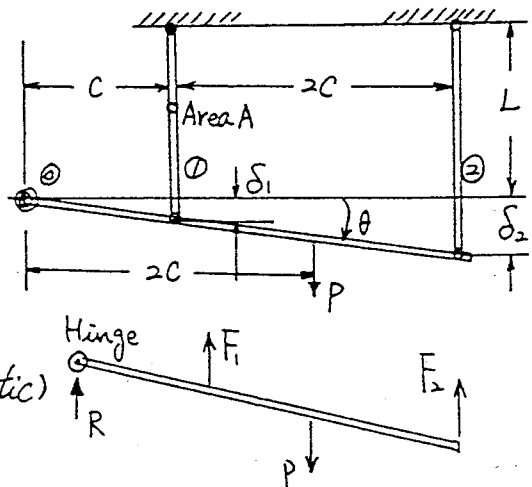
$F = \sigma A = E \epsilon A$

$F_1 = \frac{EAc\theta}{L}$; $F_2 = \frac{3EAc\theta}{L}$

$\therefore \left(\frac{EAc\theta}{L} \right) + 3 \left(\frac{3EAc\theta}{L} \right) = 2P \therefore F_1 = \frac{P}{5}$; $F_2 = \frac{3P}{5}$
 $\sigma_1 = \frac{P}{5A}$; $\sigma_2 = \frac{3P}{5A}$

rod ①이 수직이 되면, 다른 rod 들도 수직이 된다.

$F_1 = F_2 = YA$, $P = (F_1 + 3F_2)/2 = 2YA$



reverse plastic flow가 생기지 않는 elastic unloading을 가정하면, $-2YA$ 에 의한 elastic stress는;

$$\sigma_1 = -\frac{2}{5}Y \quad ; \quad \sigma_2 = -\frac{6}{5}Y$$

Residual stress = plastic stress + change in stress on unloading

∴ Residual stress;

$$\text{rod ①} \quad ; \quad Y - \frac{2}{5}Y = \frac{3}{5}Y$$

$$\text{rod ②} \quad ; \quad Y - \frac{6}{5}Y = -\frac{1}{5}Y$$

모두 Y 보다 작은 값이다.
따라서 elastic unloading이라는 가정은 옳다.

rod ①이 막 심장이 되었을 때;

$$\sigma_1 = E\left(\frac{\theta L}{L}\right) = Y \quad \therefore \theta = \frac{YL}{EC}$$

elastic unloading을 할 때;

$$\frac{EAC\theta}{L} = \frac{P}{5}, \quad P = -2YA \quad \therefore \theta = -\frac{2}{5} \frac{YL}{EC}$$

$$\therefore \text{residual angle} = \frac{YL}{EC} - \frac{2}{5} \frac{YL}{EC} = \frac{3}{5} \left(\frac{YL}{EC} \right)$$

$$\text{Ans. rod ①} = \frac{3}{5}Y, \quad \text{rod ②} = -\frac{1}{5}Y \quad \text{residual angle} = \frac{3}{5} \left(\frac{YL}{EC} \right)$$

5.33 1020 CR max. uniform strain ≈ 0.01

1020 HR max. uniform strain ≈ 0.20

$$\therefore \frac{(\Delta l)_{CR}}{(\Delta l)_{HR}} \approx \frac{L(0.01)}{L(0.20)} = 0.05$$

fracture 때의 true strain; 1020 CR $\epsilon_f = 0.70$

1020 HR $\epsilon_f = 0.90$

$$\therefore \frac{(\epsilon_f)_{CR}}{(\epsilon_f)_{HR}} = \frac{0.7}{0.9} = 0.78$$

fracture에 대한 average strain의 비로써 두개의 material의 relative ductility DR 를 결정하자.

대단히 긴 시편에서는 average strain $\approx$ uniform strain

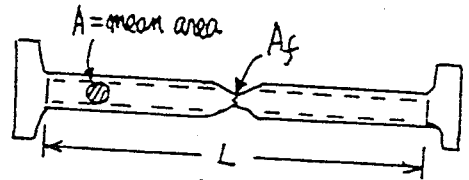
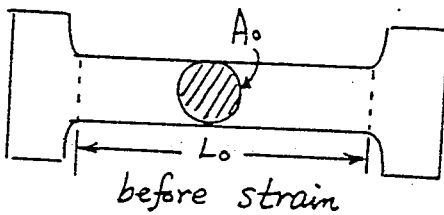
$$D_R = \frac{1020 \text{ CR ductility}}{1020 \text{ HR ductility}} = 0.05$$

상당히 짧은 시험에서는 average strain $\approx$ fracture strain

$$D_R = 0.78$$

$$\text{Ans. } (\epsilon_f)_{CR} / (\epsilon_f)_{HR} = 0.78$$

5.34



elastic strain은 무시한다.

$$\% \text{ elongation} = PE = 100 \left(\frac{L - L_0}{L_0} \right); \% \text{ reduction in Area} = RA$$

$$= 100 \left(\frac{A_0 - A_f}{A_0} \right)$$

Volume constancy (plastic flow) $A_0 L_0 = A L$

그런데 necking이 없을 때; $A_f = A$, necking의 경우; $A_f < A$

$$\therefore A_f \leq A$$

$$\text{따라서 } A_0 L_0 \geq A_f L; L = L_0 \left[1 + \frac{PE}{100} \right]; A_f = A_0 \left[1 - \frac{RA}{100} \right]$$

$$\therefore A_0 L_0 \geq A_0 \left[1 - \frac{RA}{100} \right] L_0 \left[1 + \frac{PE}{100} \right] \quad \text{즉 } \left[1 - \frac{RA}{100} \right] \left[1 + \frac{PE}{100} \right] \leq 1$$

만일, $\left[1 - \frac{RA}{100} \right] \left[1 + \frac{PE}{100} \right]$ 가 1 보다 작다면, fracture 전 necking이 생기므로, 1 이라면 necking 없이 fracture가 발생한다.

5.35 σ_x 를 제외한 다른 stress는 0.

$$\bar{\sigma} = \sigma_x, \quad d\epsilon_x^p = \frac{d\bar{\epsilon}^p}{\bar{\sigma}} [\sigma_x - 0] = \frac{\partial \bar{\epsilon}^p}{\partial \bar{\sigma}} \sigma_x$$

$$\therefore \bar{\epsilon}^p = \int d\bar{\epsilon}^p = \int d\epsilon_x^p = \epsilon_x^p \quad - Q.E.D -$$

$$5.36 \quad u = Ar + \frac{B}{r}$$

$$\text{Equilibrium ; } \frac{d\sigma_r}{dr} + \frac{\sigma_r - \sigma_\theta}{r} = 0 \quad \text{--- ①}$$

Strain - displacement ;

$$\epsilon_r = \frac{du}{dr} = A - \frac{B}{r^2} ; \epsilon_\theta = \frac{u}{r} = A + \frac{B}{r^2} ; \epsilon_z = \frac{dw}{dz} \quad \text{--- ②}$$

Stress - strain ;

$$\epsilon_r = \frac{1}{E} [\sigma_r - \nu(\sigma_\theta + \sigma_z)] , \quad \epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_z + \sigma_r)] \quad \text{--- ③}$$

$$\epsilon_r - \epsilon_\theta = \frac{1}{E} [(\sigma_r - \sigma_\theta)(1 + \nu)] = (A - \frac{B}{r^2}) - (A + \frac{B}{r^2}) = -\frac{2B}{r^2}$$

$$\therefore \sigma_r - \sigma_\theta = -\frac{E}{1+\nu} \cdot \frac{2B}{r^2} \quad \text{③식인 대입}$$

$$\frac{d\sigma_r}{dr} = \frac{E}{1+\nu} \cdot \frac{2B}{r^3} \quad \therefore \sigma_r = -\frac{EB}{1+\nu} \cdot \frac{1}{r^2} + C$$

$$\begin{aligned} \text{at } r=r_i ; \sigma_r &= -P_i \\ \text{at } r=r_o ; \sigma_r &= -P_o \end{aligned} \quad \therefore \begin{cases} -P_i = -\frac{EB}{1+\nu} \cdot \frac{1}{r_i^2} + C \\ -P_o = -\frac{EB}{1+\nu} \cdot \frac{1}{r_o^2} + C \end{cases}$$

$$\therefore P_o - P_i = \frac{EB}{1+\nu} \cdot \frac{r_i^2 - r_o^2}{r_i^2 r_o^2}$$

$$\text{따라서 } B = \frac{(1+\nu)(P_o - P_i) r_i^2 r_o^2}{E(r_i^2 - r_o^2)}$$

$$\text{②③에서 } A - \frac{B}{r_i^2} = \frac{1}{E} [\sigma_r - \nu(\sigma_\theta + \sigma_z)] , \quad A + \frac{B}{r_i^2} = \frac{1}{E} [\sigma_\theta - \nu(\sigma_z + \sigma_r)]$$

$$\therefore A - \frac{B}{r_i^2} = \frac{1}{E} [-P_i - \nu(\sigma_\theta + \sigma_z)] , \quad A + \frac{B}{r_i^2} = \frac{1}{E} [\sigma_\theta - \nu(-P_i + \sigma_z)]$$

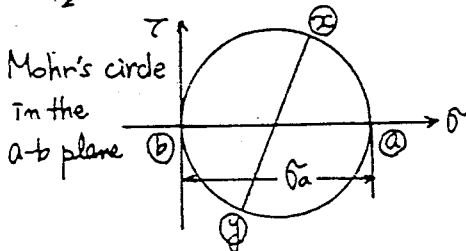
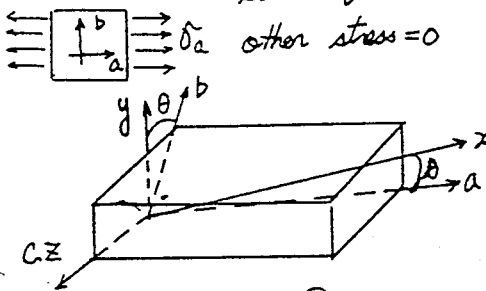
$$\text{위 두식에서 } \sigma_\theta \text{를 소거하면 } (1+\nu)(A - \frac{B}{r_i^2}) = \frac{1}{E} [(\nu^2 - 1)P_i - \nu^2 \sigma_z]$$

$$\therefore A = \frac{1}{E(1+\nu)} [(\nu^2 - 1)P_i - \nu^2 \sigma_z] + \frac{B}{r_i^2} \quad \text{여기서 B를 대입하면}$$

A를 얻는다.

5.37 문제 (5.36)의 결과에 $P_o = 0$, $\sigma_z = 0$ 을 대입하여 얻는다.

5.38



$$\sigma_x = \frac{\sigma_a}{2} (1 + \cos 2\theta)$$

$$\sigma_y = \frac{\sigma_a}{2} (1 - \cos 2\theta)$$

$$\tau_{xy} = -\frac{\sigma_a}{2} \sin 2\theta$$

other stress = 0

stress-strain ;

$$E_x = S_{11} \sigma_x + S_{12} (\sigma_y + \sigma_z)$$

$$= \frac{a}{2} [(S_{11} + S_{12}) + (S_{11} - S_{12}) \cos 2\theta]$$

$$E_y = S_{11} \sigma_y + S_{12} (\sigma_x + \sigma_z)$$

$$= \frac{a}{2} [(S_{11} + S_{12}) - (S_{11} - S_{12}) \cos 2\theta]$$

$$\gamma_{xy} = S_{44} \tau_{xy} = -\frac{\sigma_a}{2} [S_{44} \sin 2\theta]$$

이러한 strain을 (a, b, c) coordinate로 다시 바꿔주기 위해 Mohr's circle을 이용하면

$$\gamma_{ab} = (E_x - E_y) \sin 2\theta + \gamma_{xy} \cos 2\theta$$

$$= \frac{\sigma_a}{2} [2(S_{11} - S_{12}) \cos 2\theta] \sin 2\theta - \frac{\sigma_a}{2} [S_{44} \sin 2\theta] \cos 2\theta$$

$$\therefore \frac{\gamma_{ab}}{\sigma_a} = [S_{11} - S_{12} - \frac{1}{2} S_{44}] \sin 2\theta \cdot \cos 2\theta \quad \text{--- Q.E.D. ---}$$

한 값을 더 나아가기 ;

$$\frac{\gamma_{ab}}{E_a} = \frac{[S_{11} - S_{12} - \frac{1}{2} S_{44}] \sin 2\theta \cos 2\theta}{(S_{11} + S_{12}) + (S_{11} - S_{12}) \cos^2 2\theta + \frac{1}{2} S_{44} \sin^2 2\theta}$$

$$(Table 5.7) \text{에 } S_{11} - S_{12} - \frac{1}{2} S_{44} = 0.419 \times 10^{-7} \text{ in}^2/\text{lb}$$

$$\frac{\gamma_{ab}}{\sigma_a} = 0.419 \times 10^{-7} \cdot \frac{1}{2} \sin 4\theta$$

$$\therefore \left(\frac{\gamma_{ab}}{\sigma_a} \right)_{\max} = 0.2095 \times 10^{-7} \text{ in}^2/\text{lb}$$

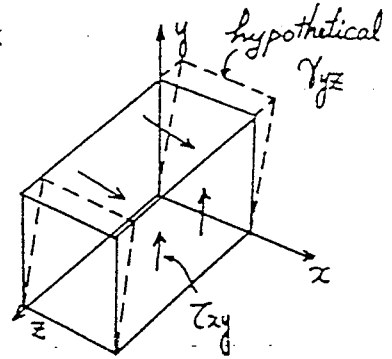
$$\text{Ans. } \left(\frac{\gamma_{ab}}{\sigma_a} \right)_{\max} = 0.2095 \times 10^{-7} \text{ in}^2/\text{lb}$$

5.39 τ_{xy} 가 γ_{yz} 를 발생시킨다고 하자.

z 축에 대해 180° 회전시키면 γ_{yz} 는 반대가 되지만, τ_{xy} 는 그렇지 않다.

x, y, z 축은 회전축이므로 structural axis 이므로, 회전에 의해서 stress-strain 의 관계는 변함이 없다.

τ 와 γ 의 관계는 선형적이어야 한다. 현대 $\gamma_{yz} = C \tau_{yz}$ 에서 C 는 위의 단위에 의해 $C=0$ 이다. 따라서 τ_{xy} 에 의한 shear strain γ_{yz} 는 없다. γ_{zx} 도 같은道理.



5.40 pipe가 충분히 길다고 생각하면 중간부분에서는 lateral restraint가 없으므로 $\sigma_r = \sigma_\theta = 0$, z 방향은 구속되었으므로 $\epsilon_z = 0$ 으로 놓을 수 있다.

stress-strain relation;

$$\epsilon_z = \alpha \Delta T + \frac{1}{E} [\sigma_z - \nu(\sigma_\theta + \sigma_r)] = 0 \quad \therefore \sigma_z = -E \alpha \Delta T$$

$$\sigma_z = -(200 \times 10^9)(12 \times 10^{-6})(-35^\circ) = 84 \times 10^6 \text{ (N/m}^2\text{)}$$

$$\epsilon_\theta = \alpha \Delta T + \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)]$$

$$\epsilon_r = \alpha \Delta T + \frac{1}{E} [\sigma_r - \nu(\sigma_\theta + \sigma_z)]$$

$$\therefore \epsilon_\theta = \epsilon_r = \alpha \Delta T - \frac{\nu}{E} \sigma_z = \alpha \Delta T + \nu \alpha \Delta T$$

$$= \alpha \Delta T (1 + \nu) = (12 \times 10^{-6})(-35)(1 + 0.3)$$

$$= -5.46 \times 10^{-4}$$

Ans. $\sigma_r = \sigma_\theta = 0$, $\sigma_z = 84 \times 10^6 \text{ N/m}^2$; $\epsilon_\theta = \epsilon_r = -5.46 \times 10^{-4}$, $\epsilon_z = 0$

5.41 Bolt의 displacement = turn(δ_1) + 온도 증가(δ_2) + A의 팽창에 의한 displacement(δ_3) = δ_b

$$\delta_1 = \frac{1}{4} \times \frac{1}{16} = \frac{1}{64} \text{ (압축)}, \quad \delta_2 = \alpha_s \cdot L \cdot \Delta T \text{ (인장)}$$

$$\delta_3 = \frac{F \cdot L}{E_s A_s} \text{ (인장)} \quad (F: A_s \text{의 팽창에 의한 인장력})$$

Sleeve 의 displacement $\delta_s =$ 온도증가 (δ_2') + A_l 의 팽창에 의한 (즉. A_l 과 F_e 만의 팽창 계수의 차이에 의해 생기는 힘)

displacement (δ_3')

$$\delta_2' = \alpha_a \cdot L \cdot \Delta T \text{ (인장)}, \quad \delta_3' = \frac{F \cdot L}{E_a A_a} \text{ (압축)}$$

$$\text{Geometry; } \delta_b = \delta_s$$

$$\therefore -\frac{1}{64} + \alpha_s \cdot L \cdot \Delta T + \frac{F \cdot L}{E_s A_s} = \alpha_a \cdot L \cdot \Delta T - \frac{F \cdot L}{E_a A_a}$$

$$F = \left\{ (\alpha_a - \alpha_s) L \cdot \Delta T + \frac{1}{64} \right\} / \left(\frac{L}{E_s A_s} + \frac{L}{E_a A_a} \right)$$

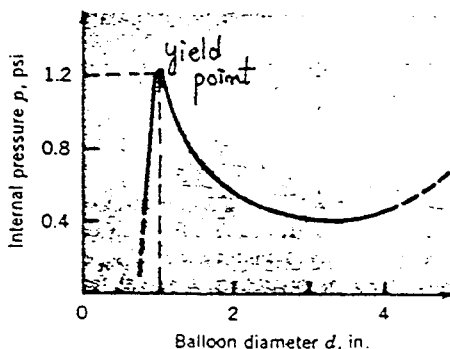
$$= \left\{ (12-6) \times 10^{-6} \times 4 \times 40 + \frac{1}{64} \right\} / \left(\frac{4}{(30 \times 10^6) \left(\frac{\pi}{4} \right) \left(\frac{1}{2} \right)^2} + \frac{4}{(10 \times 10^6) \left(\frac{\pi}{4} \right) \left(\frac{5}{16} \right)^2} \right)$$

$$= 7183.4 \text{ (lb)}$$

$$\therefore \sigma_s = F/A_s = 36580 \text{ psi (tension)}$$

$$\sigma_a = F/A_a = 29270 \text{ psi (compression)} \quad) \text{ Ans.}$$

5.42



rubber 는 비압축성

$$\therefore 4\pi r^2 t = \text{const} = C'$$

$$r^2 t = C''$$

$$\sigma_\theta = \sigma_\phi = \frac{p \cdot r}{2t} = \frac{p \cdot r}{2 \cdot \frac{C}{r^2}}$$

$$= C p d^3$$

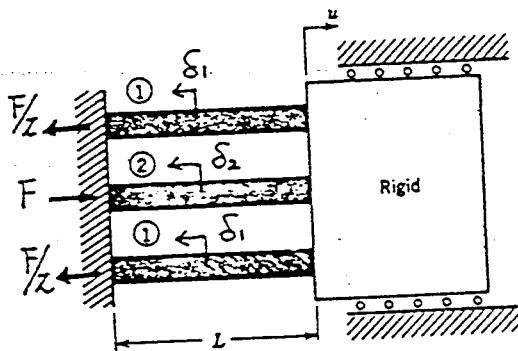
$$\text{그래프에 대략 } p = 4d - 1$$

따라서 $\sigma_\theta = \sigma_\phi = c(4d^4 - d^3)$
 그림에서 대략 직경이 1 in 압력이 1.2 psi 에서 항복이
 일어남을 알 수 있다.

Ans. $\sigma_\theta = \sigma_\phi = c(4d^4 - d^3)$, $c = \text{const}$

5.43 (a) 변형된 온도: T'

Geometry 의 의해서 그림과 같이 힘이 작용하게 된다.



$$\therefore \delta_1 = \alpha_1 LT' + \frac{(F/2) \cdot L}{E_1 A}$$

$$= \alpha LT' + \frac{F \cdot L}{2EA}$$

$$\delta_2 = \alpha_2 LT' - \frac{F \cdot L}{E_2 A}$$

$$= 2\alpha LT' - \frac{FL}{2EA}$$

$$\delta_1 = \delta_2 \text{ 이므로 하므로 } \alpha LT' + \frac{F \cdot L}{2EA} = 2\alpha LT' - \frac{FL}{2EA}$$

$$\therefore F = EA \alpha T'$$

$$\text{material ①; } \sigma_1 = (F/2)/A = \frac{1}{2} E \alpha T' \Rightarrow \sigma_y = \frac{1}{2} E \alpha T_y$$

$$\therefore T_y = \frac{2\sigma_y}{E\alpha} \quad (\sigma_y; \text{yield stress})$$

$$(b) \text{ 전체 변형 } \delta = \alpha LT' + \frac{F \cdot L}{2EA} = \frac{3}{2} \alpha LT'$$

$$\therefore u = \alpha LT_y + \frac{\alpha LT_y}{2} = \frac{3}{2} \alpha LT_y = \frac{3L\sigma_y}{E}$$

$$(c) F = 2\sigma_y AE, \quad T' = 2T_y$$

$$\therefore u = \alpha L \cdot 2T_y + \frac{2EA\sigma_y \cdot L}{2EA} = \frac{7L\sigma_y}{E}$$

$$(d) U_{plastic} = U_{total} - U_{elastic} \\ = \frac{7}{2} \alpha L T_y - \frac{3}{2} \alpha L T_y = 2 \alpha L T_y = \frac{4 L \sigma_y}{E}$$

Ams. a) $T_y = \frac{2 \sigma_y}{E \alpha}$ b) $U = \frac{3 L \sigma_y}{E}$ c) $U = \frac{7 L \sigma_y}{E}$ d) $U_p = \frac{4 L \sigma_y}{E}$

5.44 문제 (5.41) 과 같은 요령. (압축(-), 인장(+))

i) Bolt ; $\delta_b = \text{axial strain} + \text{온도변화} + \text{인장력}$
 $= -(0.0005)(150+1.5) + (12 \times 10^{-6})(151.5)(95)$
 $+ \frac{P(151.5)}{(200 \times 10^9)(310 \times 10^{-6})} = 2.44 \times 10^{-6} P + 0.09696$

ii) washer ; $\delta_w = \text{온도변화} + \text{압축력}$
 $= (12 \times 10^{-6})(1.5)(95) + \frac{(-P)(1.5)}{(200 \times 10^9)(625 \times 10^{-6})} = 1.2(-P) \times 10^{-8} + 1.71 \times 10^{-3}$

iii) sleeve ; $\delta_s = \text{온도변화} + \text{압축력}$
 $= (24 \times 10^{-6})(150)(95) + \frac{-P(150)}{(70 \times 10^9)(625 \times 10^{-6})} = -3.43 \times 10^{-6} P + 0.342$

Geometry ; $\delta_b = \delta_w + \delta_s$ i) ii) iii) 을 대입 $\therefore P = 41353.7 \text{ N}$

$\sigma_b = P/A_b = 1.33 \times 10^8 \text{ (N/m}^2\text{)}, \sigma_w = P/A_w = 6.62 \times 10^7 \text{ (N/m}^2\text{)}$

Ams $\sigma_b = 1.33 \times 10^8 \text{ N/m}^2$ (인장); $\sigma_w = 6.62 \times 10^7 \text{ N/m}^2$ (압축)

5.45 대기압 P_0 라 놓으면

Equilibrium ; $k A \sigma = (A - k A) P - P_0 A$

$\therefore \sigma = \frac{1-k}{k} P - \frac{1}{k} P_0$

yielding ; $280 \times 10^6 = \left(\frac{1-k}{k}\right) P_y - \frac{P_0}{k} \therefore P_y = \frac{P_0 + 280 \times 10^6 k}{1-k}$

lateral 방향의 stress는 없다고 가정하면, uniaxial loading이 된다. $\sigma_1 = \sigma$, $\sigma_2 = \sigma_3 = 0$ — ①

i) Mises yielding criterion ;

$$\bar{\sigma} = \sqrt{\frac{1}{2} \{ (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \}}$$

①을 대입하면 $\bar{\sigma} = \sigma = Y$

ii) Max. shear stress criterion ;

$$\bar{\tau} = \frac{1}{2} | \sigma_{\max} - \sigma_{\min} | = \frac{1}{2} | \sigma - 0 | = \frac{1}{2} Y$$

$$\therefore Y = \sigma$$

Ans. $\sigma = \frac{1-k}{k} p - \frac{P_0}{k}$, $P_y = \frac{P_0 + 280 \times 10^6 k}{1-k}$

i) ii) 같아 서로 같다.

5.46(a) δ_s 가 δ_c 보다 residual stress에 의한 displacement 만큼 더 커야 한다. $\rightarrow$ copper

따라서 $\alpha_s L \Delta T + \frac{L}{E_c} (\sigma_{\text{resi.}})_c = \alpha_c L \cdot \Delta T$

$$\therefore \Delta T = \frac{(\sigma_{\text{resi.}})_c}{(\alpha_c - \alpha_s) E_c} = \frac{21 \times 10^6}{(16.2 - 10.8) (10^{-6}) (118 \times 10^9)} = 33^\circ \text{C}$$

(b) copper 의 residual stress에 의해서 신장되는 만큼을 tensile stress를 가해서 늘려놓고, electroplating을 하면 copper에는 residual stress가 남지 않는다.

$$\delta = \frac{L}{E_c} (\sigma_{\text{resi.}})_c = \frac{L}{E_s A_s} = \left(\frac{P}{A_s} \right) \cdot \frac{L}{E_s} = \sigma_s \cdot \frac{L}{E_s}$$

$$\therefore \sigma_s = \frac{A_s}{E_s} (\sigma_{\text{resi.}})_c = \frac{200 \times 10^9}{118 \times 10^9} \times (21 \times 10^6)$$

$$= 35.6 \times 10^6 \text{ (N/m}^2\text{)}$$

(a)의 열을 가해주는 방법 보다는 (b)의 방법이 수월하다.

Ans. a) $\Delta T = 33^\circ\text{C}$ b) $35.6 \times 10^6 \text{ N/m}^2$

5.47 $P_o = 0, P_i = p$

Thin walled tube ; $t/r_i \ll 1$ 이면 σ_r 은 σ_z, σ_θ 이
비하여 미소하다. 따라서 $\sigma_r \approx 0$. — ①

Eg. (5.9) 에서 $P_o = 0$ 인 경우이므로.

$$\sigma_r = -\frac{P[(r_o/r)^2 - 1]}{(r_o/r_i)^2 - 1} = -\frac{P[(r_o/r)^2 - 1]}{[(r_i + t)/r_i]^2 - 1}$$

$$= -p \cdot \frac{r_i}{2t} [(r_o/r)^2 - 1] \quad (\because \frac{t}{r_i} \ll 1 \text{ 이면 } (\frac{t}{r_i})^2 \approx 0)$$

꺾쇠식에서 $r_i < r < r_o$ 이면 $-p \leq \sigma_r \leq 0$ 따라서 적합조건 만족.

$\lim_{(t/r_i) \rightarrow 0} \sigma_r \approx 0$ 따라서 ①과 일치한다.

Thin walled tube 에서 $\sigma_\theta \approx \frac{Pr}{t}$ — ②

Eg. (5.10) 에서 $P_o = 0, P_i = p$ 이므로

$$\sigma_\theta = p \frac{(r_o/r)^2 - 1}{(r_o/r_i)^2 - 1} \approx \frac{p \cdot r_i}{2t} [(r_o/r)^2 + 1]$$

꺾쇠식에서 t 가 아주 작으면 $r_i \approx r_o \approx r \therefore \sigma_\theta = \frac{Pr}{t}$

따라서 ②식과 일치한다.

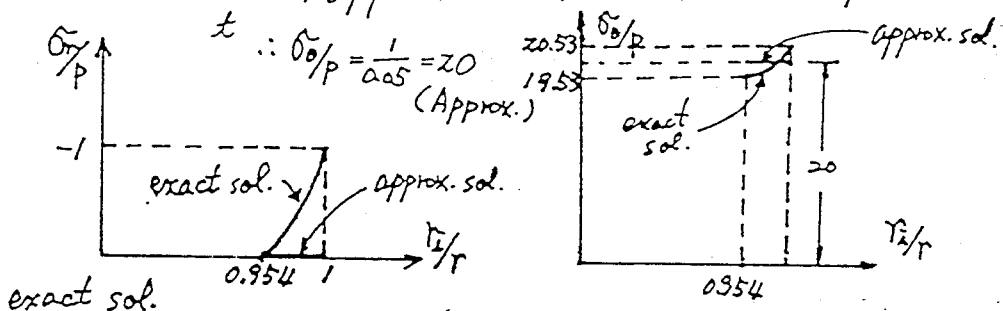
$\frac{t}{r_i} = 0.05$ 인 경우. 꺾쇠식에서

i) $\sigma_r = -\frac{P[(r_o/r)^2 - 1]}{(0.05)^2 - 1} = -9.756 p [(\frac{1.05 r_i}{r})^2 - 1]$; exact sol.

$\sigma_r \approx 0$; approximate sol.

ii) $\sigma_\theta = p \frac{(r_o/r)^2 - 1}{(r_o/r_i)^2 - 1} = 9.756 p [(\frac{1.05 r_i}{r})^2 + 1]$; exact sol.

$\sigma_\theta = \frac{Pr_i}{t}$; approximate sol. $r \gg r_i \therefore \frac{r_i}{r} \leq 1$



exact sol. $(\sigma_r/p) = -9.756[1.05^2(\frac{r_i}{r})^2 - 1]$, $(\sigma_\theta/p) = 9.756[1.05^2(\frac{r_i}{r})^2 + 1]$

5. 48

(a) $r_i = c$. $r_o = a$. $P_o = P$. $P_i = 0$

axial force는 없으므로

$\sigma_z = 0$, shear stresses = 0

$$\sigma_r = - \frac{P[(a/c)^2 - (a/r)^2]}{(a/c)^2 - 1}$$

$$= -p \left[\frac{1 - (c/r)^2}{1 - (c/a)^2} \right] \quad \text{--- ①}$$

$$\text{②, } \sigma_\theta = -p \left[\frac{1 + (c/r)^2}{1 - (c/a)^2} \right] \quad \text{--- ③}$$

$$\sigma_{1,2} = \frac{\sigma_r + \sigma_\theta}{2} \pm \sqrt{\left(\frac{\sigma_r - \sigma_\theta}{2}\right)^2}$$

$c/a \rightarrow 0$ 이면 $\sigma_r = -p[1 - (c/r)^2]$, $\sigma_\theta = -p[1 + (c/r)^2]$

$\therefore \sigma_{1,2} = -p \pm p(\frac{c}{r})^2$ 따라서 $\sigma_{max} = \left| -p \pm p(\frac{c}{r})^2 \right|_{max}$

$$= p + p(\frac{c}{r})^2 = p(1 + (\frac{c}{r})^2)$$

최대값은 $r = c$ 에서 생긴다. $\sigma_{max} = 2p$

(b) $c = 0$ 이면 ①③에서 $\sigma_r = \sigma_\theta = -p \therefore \sigma_{max} = p$

따라서 hole에 의한

응력 집중계수 ; $\frac{2p}{p} = 2$

Ans. (a) $\sigma_{max} = z\rho$. (b) 응력집중 계수 = 2

5.49 (Tensor Analysis)

$$\sigma_{ij,i} = \frac{\partial \sigma_{ij}}{\partial i}, \quad u_{i,j} = \frac{\partial u_i}{\partial j}, \quad u_{j,i} = \frac{\partial u_j}{\partial i}$$

$$\text{Kronecker delta } \delta_{ij} = \begin{cases} 0 & (i \neq j) \\ 1 & (i = j) \end{cases}$$

위의 관계를 이용하여 i, j 이 x, y, z 를 순차적으로 대입하면 된다.

$$\sigma_{ij,i} + X_i = 0 \quad ; \text{ Equilibrium condition}$$

$$\epsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i}) \quad ; \text{ strain-displacement relation}$$

$$\epsilon_{ij} = \frac{1+\nu}{E} \sigma_{ij} - \frac{\nu}{E} \delta_{ij} \theta + \alpha \delta_{ij} \Delta T \quad ; \text{ stress-strain relation}$$

< tensor 에 관한 수학책 참조 >

5.50 Strain energy ; $U = \frac{1}{2} \int (\sigma_{ij} \epsilon_{ij}) dV$

i 에 x, y, z , j 이 x, y, z 를 순차적으로 대입.

$$U = \frac{1}{2} \int (\sigma_{xx} \epsilon_{xx} + \sigma_{xy} \epsilon_{xy} + \sigma_{xz} \epsilon_{xz} + \sigma_{yx} \epsilon_{yx} + \sigma_{yy} \epsilon_{yy} + \sigma_{yz} \epsilon_{yz} + \sigma_{zx} \epsilon_{zx} + \sigma_{zy} \epsilon_{zy} + \sigma_{zz} \epsilon_{zz}) dV$$

여기에서 $\sigma_{xx} = \sigma_x$, $\sigma_{yy} = \sigma_y$, $\sigma_{zz} = \sigma_z$, $\sigma_{xy} = \sigma_{yx}$, $\sigma_{yz} = \sigma_{zy}$

$\sigma_{zx} = \sigma_{xz}$ 로 바꿔준다. (E도 마찬가지로)

$$\text{또 } \tau_{xy} = 2 \epsilon_{xy}, \quad \tau_{yz} = 2 \epsilon_{yz}, \quad \tau_{zx} = 2 \epsilon_{zx}$$

또 전단응력 σ_{xy} , σ_{yz} , σ_{zx} 는 τ_{xy} , τ_{yz} , τ_{zx}

이다 같이 우리가 상용하던 notation 으로 바꿔주면.

$$U = \frac{1}{2} \int (\sigma_x \epsilon_x + \sigma_y \epsilon_y + \sigma_z \epsilon_z + \tau_{xy} \gamma_{xy} + \tau_{yz} \gamma_{yz} + \tau_{zx} \gamma_{zx}) dV$$

- Q. E. D. -

$$5.51 \quad U = \frac{1}{2} \int (\sigma_x \epsilon_x + \sigma_y \epsilon_y + \sigma_z \epsilon_z + \tau_{xy} \gamma_{xy} + \tau_{yz} \gamma_{yz} + \tau_{zx} \gamma_{zx}) dV$$

이 식을 모두 stress의 term으로 바꿔주기 위해 (5.2)식의 stress-strain의 관계식을 이용하면

$$U = \frac{1}{2} \int \left[\frac{1}{E} (\sigma_x^2 + \sigma_y^2 + \sigma_z^2) - \frac{2\nu}{E} (\sigma_x \sigma_y + \sigma_y \sigma_z + \sigma_z \sigma_x) + \frac{1}{G} (\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2) \right] dV \quad - Q.E.D. -$$

윗 식을 모두 strain으로 바꿔주기 위해 문제 (5.7)의

결과를 이용하면, $\gamma = \frac{\tau}{G}$ 와

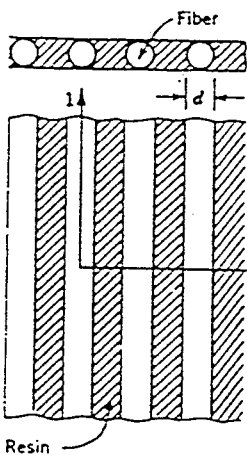
$$\sigma_i = \frac{E}{1+\nu} \epsilon_i + \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_x + \epsilon_y + \epsilon_z) - \frac{\nu E (T - T_0)}{(1-2\nu)}, \quad (i=x, y, z)$$

$$U = \int \left[\frac{E(1-\nu)}{2(1+\nu)(1-2\nu)} (\epsilon_x + \epsilon_y + \epsilon_z)^2 - \frac{E}{(1+\nu)} \{ \epsilon_x \epsilon_y + \epsilon_y \epsilon_z + \epsilon_z \epsilon_x - \frac{1}{2} (\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{zx}^2) \} \right] dV \quad (\because G = \frac{E}{2(1+\nu)})$$

- Q.E.D. -

5.52 (a) ① 방향의 길이의 변화

$$\frac{PL}{EA} = \frac{P_f \cdot L}{E_f \cdot A_f} = \frac{P_r \cdot L}{E_r \cdot A_r} \quad \text{①} \quad P = P_f + P_r$$



$$\text{따라서 } P = \left(1 + \frac{E_r A_r}{E_f A_f} \right) P_f$$

①식에 대입.

$$\frac{L}{E_i A} \left(1 + \frac{E_r A_r}{E_f A_f} \right) P_f = \frac{P_f \cdot L}{E_f A_f}$$

$$\therefore \frac{1}{E_i A} = \frac{1}{E_f A_f + E_r A_r}$$

$$\therefore E_i = \frac{1}{A} (E_f A_f + E_r A_r)$$

$$A_f/A = 40\% = 0.4 \quad \therefore A_r/A = 0.6$$

$$\text{따라서 } E_i = 0.4 E_f + 0.6 E_r = (0.4)(5 \times 10^7) + (0.6)(5 \times 10^5)$$

$$= 2.03 \times 10^7 \text{ (psi)}$$

② 방향의 길이의 변화;

$$\frac{PL}{E_2 A} = \frac{PL_f}{E_f \cdot A} + \frac{PL_r}{E_r \cdot A} \quad \text{따라서} \quad \frac{1}{E_2} = \frac{0.4}{E_f} + \frac{0.6}{E_r}$$

$$\therefore E_2 = 1 / \left(\frac{0.4}{E_f} + \frac{0.6}{E_r} \right) = \frac{E_r \cdot E_f}{0.4 E_r + 0.6 E_f}$$

$$= (5 \times 10^5) \cdot (5 \times 10^7) / (0.4)(5 \times 10^5) + (0.6)(5 \times 10^7)$$

$$= 8.278 \times 10^5 \text{ (psi)}$$

(b) 기하학적으로 대칭이므로 $E_1' = E_2' = E$, $p = p_1 + p_2$

first layer 에 걸리는 힘; p_1 ; second layer; p_2

늘어난 길이 (① ② 방향 모두 같다)

$$\frac{PL}{E(2A)} = \frac{p_1 L}{E_1 A} = \frac{p_2 L}{E_2 A} \quad \text{--- ① } p_1 + p_2 = p \text{ 이므로}$$

$$p_1 \left(1 + \frac{E_2}{E_1} \right) = p$$

$$\therefore \frac{L}{L(2A)} \cdot \frac{E_1 + E_2}{E} p_1 = \frac{p_1 L}{E_1 A} \quad \text{따라서 } E = (E_1 + E_2) / 2$$

$$\therefore E = (2.03 \times 10^7 + 8.278 \times 10^5) / 2 = 1.0564 \times 10^7 \text{ (psi)}$$

(c) ① ② 방향이 서로 같은 material property 를 갖고,

기하학적으로 대칭이므로 isotropy 이다.

Ans. (a) $E_1 = 2.03 \times 10^7 \text{ psi}$, $E_2 = 8.278 \times 10^5 \text{ psi}$ (b) $E = 1.0564 \times 10^7$

$$5.53 \quad \epsilon_{ij} = S_{ji} \tau_i \quad S_{ij} = S_{ji} = 1, \dots, 6$$

$$\text{where } \epsilon_j = \epsilon_x \dots \tau_{zx}, \quad \tau_i = \sigma_x \dots \tau_{zx}$$

$$ij = x, \quad S_{ij} = S_{ji} = 1, \dots, 6, \quad \tau_i = \sigma_x \dots \tau_{zx} \text{로 변화시키면}$$

$$\epsilon_x = S_{11} \sigma_x + S_{12} \sigma_y + S_{13} \sigma_z + S_{14} \tau_{xy} + S_{15} \tau_{yz} + S_{16} \tau_{zx}$$

$$\text{나머지도 같은 과정} \quad - Q.E.D -$$

5.54 $\bar{\sigma}_{mean} = 30.000 \text{ psi}$, $\bar{\sigma}_{ultimate} = 130.000 \text{ psi}$
 endurance limit을 60.000 psi 로 가정하고, reduction factor K_f
 를 3으로 놓으면,

$$K_f = \frac{60.000}{\bar{\sigma}_0} = 3 \quad \therefore \bar{\sigma}_0 = 20.000$$

비례관계; $130.000 : 20.000 = 10.000 : \bar{\sigma}_a$

$$\therefore \bar{\sigma}_a = \frac{10}{13} \times 20.000 = 15384.6 \text{ psi}$$

Ams 15384.6 psi

5.55 endurance limit; $\bar{\sigma}_e = 40.000 \text{ psi}$

$$\bar{\sigma}_a = 0.15 \bar{\sigma}_m \text{ --- ① 안정계수 } 3, K_f = 3$$

load line curve; $\bar{\sigma}_a = 0.15 \bar{\sigma}_m$ 과 안정계수를 3으로
 하고 fatigue-strength reduction factor K_f 를 더해준

$\bar{\sigma}_a$ curve와의 만나는 점이 $\bar{\sigma}_m$ 이 된다.

$$\text{위의 설명에 따라 } \bar{\sigma}_a = 4.444 - \frac{4.444}{33.333} \bar{\sigma}_m \text{ --- ②}$$

$$\text{① \& ②에 따라 } \bar{\sigma}_m = 15685 \text{ psi}$$

Ams 15685 psi

5.56 HMZ1A-T8 (mean stress ; 12000 psi
 alternating stress ; 17000 psi

Fig 5-42b에 따라 endurance limit ; 약 10000 psi

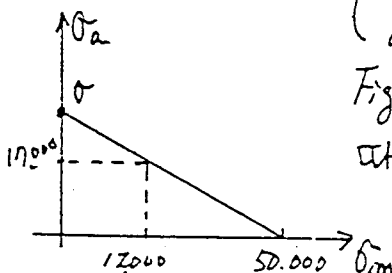
ultimate strength ; 약 50.000 psi

Fig 5-42b는 $\bar{\sigma}_{mean} = 0$ 인 경우이다

따라서 $\bar{\sigma} \approx 22368 \text{ psi}$

Fig 5.42b에 따라 대략 Cycle $\approx 14000 \text{ (cycles)}$

Ams 약 14000 cycle



CHAPTER 6

$$6.1 \quad \tau_{max} = \frac{M_t \cdot r_o}{I_z} = \frac{1100 \times \frac{5 \times 10^{-2}}{2}}{\frac{\pi}{32} (5 \times 10^{-2})^4} = 4.482 \times 10^7 \text{ (N/m}^2\text{)}$$

$$\phi = \frac{M_t \cdot L}{G \cdot I_z} = \frac{1100 \times 1.5}{(80 \times 10^9) \cdot \frac{\pi}{32} (5 \times 10^{-2})^4} = 0.0336 \text{ (rad)} = 1.926^\circ$$

Ans. $\tau_{max} = 4.482 \times 10^7 \text{ N/m}^2$ $\phi = 1.926^\circ$

$$6.2 \quad \tau_{max} = \frac{M_t \cdot r_o}{I_z} = 25000 \times \frac{d_o/2}{\frac{\pi}{32} (d_o^4 - d_i^4)} = 82 \times 10^6$$

따라서 $d_o = 644 (d_o^4 - d_i^4)$

$$\phi = \frac{M_t \cdot L}{G \cdot J} = \frac{25000 \times 2.5}{(80 \times 10^9) \cdot \frac{\pi}{32} (d_o^4 - d_i^4)} = \frac{\pi}{90} \text{ (rad)}$$

따라서 $d_o^4 - d_i^4 = 2.2797 \times 10^{-4}$

②를 ①에 대입하면 $d_o = 0.1468 \text{ m}$ $d_i = 0.124 \text{ m}$

Ans. $d_i = 124 \text{ mm}$ $d_o = 146.8 \text{ mm}$

$$6.3 \quad \sigma_r = \sigma_\theta = \sigma_z = \tau_{r\theta} = \tau_{rz} = 0$$

$$\tau_{\theta z} = G_1 r \frac{d\phi}{dz} \quad (0 < r < r_i) \quad , \quad \tau_{\theta z} = G_2 r \frac{d\phi}{dz} \quad (r_i < r < r_o)$$

한편, twisting moment $M_t = \int_A r \tau_{\theta z} dA$

$$= \frac{d\phi}{dz} \left\{ 2\pi G_1 \int_0^{r_i} r^3 dr + 2\pi G_2 \int_{r_i}^{r_o} r^3 dr \right\}$$

$$= \frac{\pi}{2} \frac{d\phi}{dz} \left\{ (G_1 - G_2) r_i^4 + G_2 r_o^4 \right\}$$

이 식은 다음과 같이 쓸 수 있다.

$$M_t = \frac{d\phi}{dz} \{G_1 I_1 + G_2 I_2\} \quad \text{where} \quad \begin{cases} I_1 = \frac{\pi}{2} r_i^4 \\ I_2 = \frac{\pi}{2} (r_o^4 - r_i^4) \end{cases}$$

$$\therefore \text{회전각의 총합} : \phi = \frac{M_t \cdot L}{G_1 I_1 + G_2 I_2}$$

$$\tau_{\theta z} = G_1 \cdot r \frac{d\phi}{dz} = \frac{G_1 M_t \cdot r}{G_1 I_1 + G_2 I_2}, \quad 0 < r < r_i$$

$$\tau_{\theta z} = G_2 \cdot r \frac{d\phi}{dz} = \frac{G_2 M_t \cdot r}{G_1 I_1 + G_2 I_2}, \quad r_i < r < r_o$$

6.4 4130 HT steel ; $G = 11.9 \times 10^6 \text{ psi}$, 항복응력 $Y = 200,000 \text{ psi}$

$$\tau_{\theta z} = \frac{1}{2} Y = 100,000 \text{ psi}$$

$$\phi_r = \frac{M_t \cdot L}{G \cdot J} = \left(\frac{M_t \cdot r_o}{J} \right) \cdot \left(\frac{L}{G r_o} \right) = \tau_{\theta z} \cdot \frac{L}{G r_o}$$

$$= 100,000 \times \left(\frac{2}{11.9 \times 10^6 \times \frac{5}{4} \times \frac{1}{12} \times \frac{1}{2}} \right) = 0.3227 \text{ (rad)} = 18.49^\circ$$

Ans. $\phi_r = 18.49^\circ$

6.5 1020 CR steel ; $G = 11.5 \times 10^6 \text{ psi}$ $Y = 86,000 \text{ psi}$

$$\tau_{\theta z} = \frac{1}{2} Y = 43,000 \text{ psi} \quad d_o = 2"$$

(a) $M_t = 2500 \text{ ft-lb}$

$$\tau_{\max} = \frac{M_t \cdot r_o}{J} = 43,000 = \frac{2500 \times 12 \times 1}{\frac{\pi}{32} (d_o^4 - d_i^4)} \quad \therefore d_i = 1.727"$$

(b) $M_t = 5000 \text{ ft-lb}$

$$\text{(a)와 마찬가지로 } 43,000 = \frac{5000 \times 12 \times 1}{\frac{\pi}{32} (d_o^4 - d_i^4)} \quad \therefore d_i = 1.1562"$$

Ans. (a) $d_i = 1.727 \text{ in}$ (b) $d_i = 1.1562 \text{ in}$

6.6

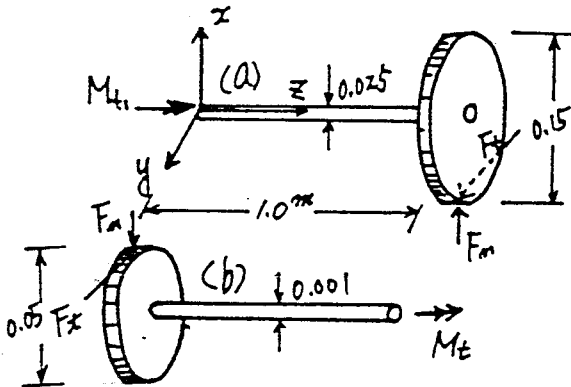
$$M_t = 10 (\text{N}\cdot\text{m}), \Sigma M_z = 0, M_{t1} = 0.075 F_t$$

$$M_t = 0.025 F_t$$

$$\text{따라서 } M_{t1} = 3M_t = 30 (\text{N}\cdot\text{m})$$

부재 ①의 비틀림각

$$\begin{aligned} \phi_1 &= \frac{M_{t1} \cdot L_1}{G \cdot J_1} = \frac{30 \times 1}{(80 \times 10^9) \frac{\pi}{32} (0.025)^4} \\ &= 9.7785 \times 10^{-3} (\text{rad}) \\ &= 0.56027^\circ \end{aligned}$$



$$\text{따라서 부재 ②의 비틀림각 } \phi'_1 = \frac{0.015}{0.05} \phi_1 \therefore \phi'_1 = 1.68081^\circ$$

외력에 의한 부재 ②의 비틀림각

$$\begin{aligned} \phi_2 &= \frac{M_t \cdot L_2}{G \cdot J_2} = \frac{10 \times 0.67}{(80 \times 10^9) \frac{\pi}{32} (0.01)^4} = 0.085307 (\text{rad}) \\ &= 4.88773^\circ \end{aligned}$$

$$\therefore \text{total torsion angle } \phi = \phi'_1 + \phi_2 = 6.57^\circ$$

$$\text{Ans. } \phi = 6.57^\circ$$

$$6.7 \quad M_{t1} = 3M_t$$

$$\tau_1 = \frac{M_{t1} \cdot r_1}{J_1} = \frac{3 \times 0.0125}{\frac{\pi}{32} (0.025)^4} M_t = 977848 M_t$$

$$\tau_2 = \frac{M_t \cdot r_2}{J_2} = \frac{0.005}{\frac{\pi}{32} (0.01)^4} M_t = 5092958 M_t$$

$$\therefore \tau_2 > \tau_1$$

$$\therefore 275 \times 10^6 = 5092958 M_t$$

$$\text{Ans. } M_t = 53.996 (\text{N}\cdot\text{m})$$

6.8 아래쪽 gear의 비틀림각을 ϕ 라 하면, 기하학적 적합조건에 의해 위쪽 gear의 비틀림각은 $\frac{3}{5}\phi$ 이다.

따라서 아랫쪽 gear의 최종 비틀림각은 $(\frac{3\pi}{180} - \phi)$ rad.

F_A 를 gear가 접하는 점의 tangential force라 놓으면

$$\text{lower shaft; } M_{tL} = F_A \times 125 = \frac{(\frac{3\pi}{180} - \phi) G \cdot I_L}{L} = \frac{(\frac{3\pi}{180} - \phi) G (\frac{125}{2})^4}{L} \quad \text{--- ①}$$

$$\text{upper shaft; } M_{tU} = F_A \times 15 = \frac{\phi \cdot G \cdot I_U}{L} = \frac{\phi \cdot G \cdot \frac{\pi}{2} (\frac{75}{2})^4}{L} \quad \text{--- ②}$$

$$G = 80 \times 10^9$$

$$\text{① ② 식에서 } \theta = 0.0252 \text{ (rads)}$$

$$(\tau_{\theta z})_{\max, \text{upper}} = \frac{2 M_{tU}}{\pi (r_{ou})^3} \quad (\tau_{\theta z})_{\max, \text{lower}} = \frac{2 M_{tL}}{\pi (r_{ol})^3}$$

① ② 식을 이용하면

$$\text{Ans. } (\tau_{\theta z})_{\max, \text{upper}} = 45.9 \text{ MN/m}^2$$

$$(\tau_{\theta z})_{\max, \text{lower}} = 33.6 \text{ MN/m}^2$$

6.9 gear의 접촉점 A, B에서의 tangential force를 F_{AT} , F_{BT} 라 놓으면

$$\text{upper shaft; } F_{AT} \times 50 + F_{BT} \times (75/2) = 0 \quad \text{--- ①}$$

$$\text{lower shaft; } F_{AT} \times 50 + F_{BT} \times (125/2) = 0 \quad \text{--- ②}$$

$$\text{① ② 식에서 } F_{AT} = F_{BT} = 0$$

$$\text{Ans. } \text{shear stress} = 0$$

$$6.10 \text{ 평형 조건 } \sum M_x = 0 ; -M_A - M_C + M_o = 0 \quad \text{--- ①}$$

$$\text{가하화된 적합조건 ; } \phi_{AB} = \phi_{BC} \quad \text{--- ②}$$

$$M_A = \frac{\phi_{AB}}{L_1} G_1 I_{AB} \quad I_{AB} = \frac{\pi}{2} \left(\frac{d_1}{2} \right)^4$$

$$M_B = \frac{\phi_{BC}}{L_2} G_2 I_{BC} \quad I_{BC} = \frac{\pi}{2} \left(\frac{d_2}{2} \right)^4$$

$$G_1 = G_2$$

① ② 에서

$$\text{Ans. } M_A = \frac{M_0}{1 + \frac{L_1}{L_2} \left(\frac{d_2}{d_1} \right)^4} \quad M_C = \frac{M_0}{1 + \frac{L_2}{L_1} \left(\frac{d_1}{d_2} \right)^4}$$

6.11 문제 6.10과 마찬가지로

$$\phi_{AB} = \frac{2M_A}{\pi G_1} \left(\frac{L_1}{r_1^4} \right) \quad \phi_{BC} = \frac{2M_C}{\pi G_2} \left(\frac{L_2}{r_2^4} \right)$$

$$\phi_{AB} = \phi_{BC}, \quad M_A = M_C$$

$$\text{Ans. } \left(\frac{G_2}{G_1} \right) \left(\frac{L_1}{L_2} \right) \left(\frac{d_1}{d_2} \right)^4 = 1$$

$$6.12 \quad \tau = \frac{M_t \cdot r}{J} \quad \text{--- ①}$$

$$M_t = 12 \times 5000 + 10P = 60,000 + 10P$$

6개의 Bolt를 1개의 Bolt로 치환하기 위해

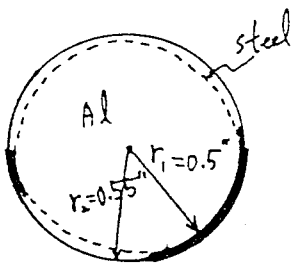
$$J = \sum (r^2 dA + A l^2) = 6 \times \frac{\pi}{32} \times 0.75^4 + \frac{\pi}{4} \times \left(\frac{3}{4} \right)^2 (4 \times 25 + 2 \times 9)$$

$$= 87.66 \text{ (in}^4) \quad \text{--- ②}$$

$$\therefore \tau_{\max} = \frac{60000 + 10P}{87.66} \times \sqrt{45} = 10,000 \quad \therefore P = 7068 \text{ (lb)}$$

$$\text{Ans. } P = 7068 \text{ lb}$$

6.13



$$\text{Steel; } G_{st} = 7.8 \times 10^{10} \text{ N/m}^2$$

$$Y_{st} = 1360 \text{ MN/m}^2$$

$$\text{Al; } G_{Al} = 26 \times 10^9 \text{ N/m}^2$$

$$Y_{Al} = 100 \text{ MN/m}^2$$

$$(a) \quad \frac{(\tau_{\theta z})_{r_2}}{(\tau_{\theta z})_{r_1}} = \frac{G_{st} \cdot r_2}{G_{Al} \cdot r_1} \quad \frac{(\tau_{\theta z})_{r_2} / Y_{st}}{(\tau_{\theta z})_{r_1} / Y_{Al}} = \frac{G_{st} \cdot Y_{Al} \cdot r_2}{G_{Al} \cdot Y_{st} \cdot r_1}$$

$$\frac{(\tau_{\theta z})_E / Y_{st}}{(\tau_{\theta z})_E / Y_{Al}} = \frac{78}{26} \times \frac{100}{1360} \times \frac{27.5}{25} =$$

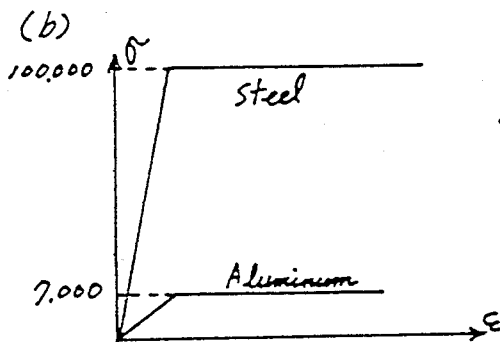
$r = r_1$ 에서 Al 이 먼저 항복을 일으킨다.

composite shaft에 대하여

$$(\tau_{\theta z})_{r_1} = \frac{2 \cdot M_t \cdot G_{Al} \cdot r_1}{\pi \{ G_{Al} \cdot r_1^4 + G_{St} (R^4 - r_1^4) \}} = \frac{1}{2} Y_{Al} = 50 \times 10^6$$

$$\therefore M_t = \frac{\pi \times 0.5 \times 100 \times \{ 26 \times 10^3 \times (0.025/z)^4 + 78 \times 10^3 \times (\left(\frac{0.0275}{z} \right)^4 - (0.025/z)^4) \}}{26 \times 10^3 \times (0.025/z)}$$

$$\cong 367 \text{ (N.m)}$$



shaft가 완전히 항복되었다고 하면

$$M_t = 50 \int_0^{r_1} 2\pi r^2 dr + 680 \int_{r_1}^{r_2} 2\pi r^2 dr = 1155.2 \text{ (N.m)}$$

$$\frac{d\phi}{dz} = \frac{\tau_{\theta z}}{G \cdot r} = \frac{50 \times 10^6}{26 \times 10^9 \times (0.025/z)} = 0.23 \text{ (rad)} = 13.22^\circ$$

Ans. (a) 366.9 N.m (b) 1155.2 N.m, 13.22°

6.14 정방단면의 거리를 x 라 잡으면 $M_t = \int_0^L \frac{q_0}{L} x dz$

$$\therefore M_{tx} = M_x - \int_0^x \frac{q_0}{L} x dx = M_x - \frac{q_0}{2L} x^2$$

$$d\phi = \frac{M_{tx}}{GJ} dx \therefore \phi = \int_0^L \frac{1}{GJ} \left(M_t - \frac{q_0}{2L} x^2 \right) dx$$

$$= \frac{1}{GJ} \left(M_t \cdot L - \frac{q_0}{6L} L^3 \right) \quad M_t = \frac{q_0 \cdot L}{2}$$

$$\therefore \phi = \frac{2M_t L}{3GJ} ; \text{Ans.}$$

6.15 구속되어 있으므로 회전각은 "0"이다. 구속되지 않을 때의 회전각; ϕ
 $\phi = \frac{2M_t \cdot L}{3GJ}$ 구속에 의한 회전각은 방향이 반대 이므로 $\phi' = -\frac{M_t \cdot L}{GJ}$

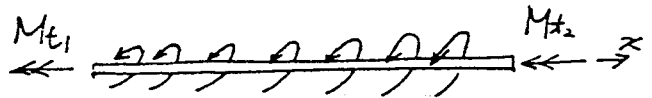
$$\phi + \phi' = \frac{2M_t \cdot L}{3GJ} + \left(-\frac{M_t \cdot L}{GJ}\right) = 0 \quad \text{--- ①}$$

평형 조건; $\sum M_x = 0$; $M_{t1} + M_{t2} = M_t$ --- ②

① ② 에서

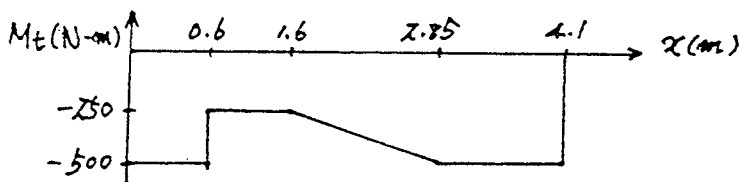
$$M_{t1} = \frac{1}{3}M_t$$

$$M_{t2} = \frac{2}{3}M_t$$



Ans. $M_{t1} = \frac{1}{3}M_t$, $M_{t2} = \frac{2}{3}M_t$

6.16 beam A의 B.M.D를 그려라.



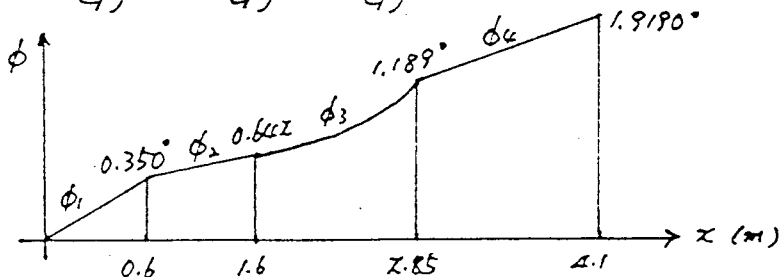
$$\phi = \int \frac{M_t}{GJ} dx$$

$$\phi_1 = \int \frac{500}{GJ} dx = \frac{500}{GJ} x \quad (0 \leq x \leq 0.6)$$

$$\phi_2 = \int \frac{250}{GJ} dx = \frac{250}{GJ} (x - 0.6) + \frac{500 \times 0.6}{GJ} \quad (0.6 \leq x \leq 1.6)$$

$$\phi_3 = \frac{100x^2}{GJ} + \frac{570x}{GJ} + \frac{500 \times 0.6}{GJ} + \frac{250 \times 1}{GJ} - \frac{570 \times 1.6}{GJ} \quad (1.6 \leq x \leq 2.85)$$

$$\phi_4 = \frac{500}{GJ} x + \frac{1018.17}{GJ} + \frac{500 \times (-2.85)}{GJ} \quad (2.85 \leq x \leq 4.1)$$



$$6.17 \quad \tau = \frac{M_t \cdot r}{J} \quad J = \frac{\pi}{32} \{ d_o^4 - d_i^4 \} = \frac{\pi}{32} \left\{ d_o^4 - \left(\frac{d_o}{2} \right)^4 \right\} = \frac{15\pi}{512} d_o^4$$

$$M_t = \frac{\tau_{max} \cdot J}{r_o} = \frac{15\pi}{256} d_o^3 \cdot \tau_{max}$$

$$HP = \frac{M_t \omega}{76} = \frac{1}{76} \cdot \frac{15\pi}{256} d_o^3 \cdot \tau_{max} \cdot \frac{2\pi \cdot rpm}{60} = (2.54 \times 10^{-4} rpm)(\tau_{max})(d_o^3)$$

Ans. $HP = 2.54 \times 10^{-4} (rpm)(\tau_{max})(d_o^3)$

6.18 두께를 미소하다고 가정하면 shear flow는 항상 일정하다.

$q = \text{shear flow}$, minimum thickness = $t_{min} = 0.15''$

Tresca의 yield criteria를 적용하면

$$\text{max. shear stress} = \frac{q}{t_{min}} = \frac{\tau}{2} = \frac{35000}{2}$$

$$\therefore q = 0.15 \times \frac{35000}{2}$$

$$(M_t)_{max} \div 2qA = 2 \times 0.15 \times \frac{35000}{2} \times \frac{\pi}{4} (5^2 - 4.5^2) \\ = 19586 \text{ lb-in}$$

Ans. $(M_t)_{max} = 19586 \text{ lb-in}$

6.19 일반적이지 thin-walled tube 이라 $I_p \approx 2\pi r^3 \cdot t$

$$\tau = \frac{M_t \cdot r}{I_p}$$

$$(a) \text{ outer tube; } M_{t_o} = \frac{\tau_y \cdot I_p}{2r} = \frac{\tau_y \cdot 2\pi (2r)^3 \cdot t}{2r} = 8\pi r^2 t \tau_y$$

$$\text{inner tube; } \tau = \frac{1}{2} \tau_y \quad M_{t_i} = \frac{\frac{1}{2} \tau_y \cdot 2\pi r^3 \cdot t}{r} = \pi r^2 t \tau_y$$

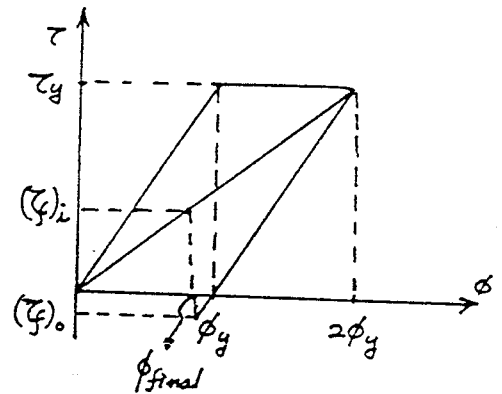
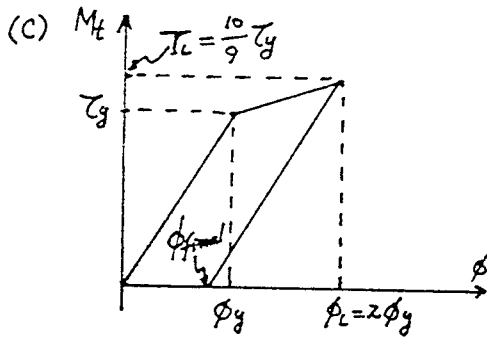
$$\therefore T_y = M_{t_o} + M_{t_i} = 9\pi r^2 t \tau_y$$

$$\phi_y = \frac{M_t \cdot L}{GI_p} = \frac{\tau_y \cdot L}{2Gr}$$

(b) outer & inner tube 이끼의 stress 는 τ_y

$$M_{t_o} = 8\pi r^2 t \tau_y \quad M_{t_i} = 2\pi r^2 t \tau_y \quad \therefore T_L = 10\pi r^2 t \tau_y$$

$$\text{inner tube 이끼 stress} \frac{1}{2} \text{ 이라하면 } \phi_L = 2\phi_y = \frac{\tau_y \cdot L}{Gr}$$



$$(M_t)_{outer} = T_{outer} \times 4\pi r^2 t$$

$$(M_t)_{inner} = T_{inner} \times \pi r^2 t$$

$$(M_t)_{outer} + (M_t)_{inner} = 0$$

$\therefore$ shear stress에 대하여 $T_{outer} = -\frac{1}{4} T_{inner}$

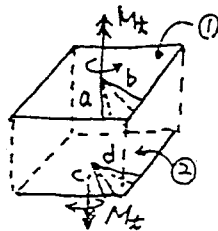
$$\text{그림으로 부터 } (T_{outer})_{final} = -\frac{1}{9} T_y \quad (T_{inner})_{final} = \frac{4}{9} T_y$$

Ans. (a) $T_y = 9\pi r^2 t T_y$ $\phi_y = T_y \cdot L / 2Gr$ (b) $T_L = 10\pi r^2 t T_y$ $\phi_L = T_y L / Gr$

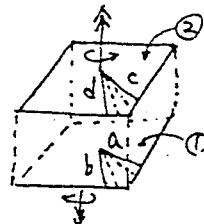
(c) $\phi_f = 4T_y L / 9Gr$ $T_{of} = -T_y / 9$ $T_{if} = 4T_y / 9$

6.20

(a)

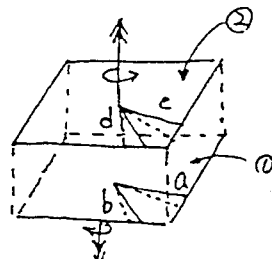
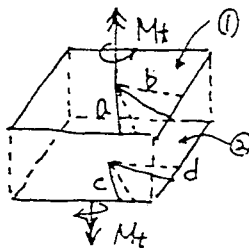


inverting



위의 두 그림 같은 면을 비교하면, 변형이 대각선에 관하여 대칭이며 양쪽 대각선에 대하여 비슷한 위치에 있는 점은 같은 변형을 한다. 따라서 기하학적으로 가능한 변형 형태이다.

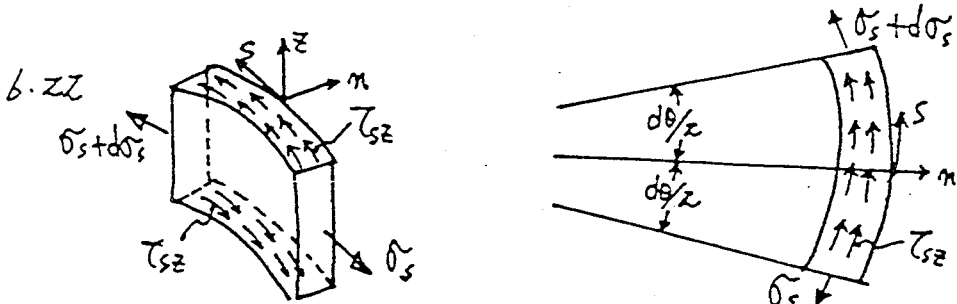
(b)



대칭축에 대하여 대칭을 이루지 않으므로 이러한 변형 형태는 기하학적으로 불가능하다.

6.21 (a) 대각선에 관해서 대칭이 아니므로, 이것은 가하확적으로 불가능한 변형 형태이다.

(b) 대각선에 관해 대칭이므로, 가능한 변형 형태이다.



표면에 작용하는 하중은 없다. 이러한 shaft에 대해서는, 위 아랫 면에서의 τ_{sz} 는 s-coordinate의 주어진 값과 같아야만 한다.

$$\text{평형 조건; } \sum F_n = 0; \left[\sigma_s \frac{d\theta}{2} + (\sigma_s + d\sigma_s) \frac{d\theta}{2} \right] dn dz = 0$$

$$\therefore \sigma_s = 0$$

6.23 승객이 타기 전; shaft의 torque $M_{t1} = 3000 \text{ lb-ft}$

승객이 탄 후; shaft의 torque $= (3000 - 3W) \text{ lb-ft}$

$\therefore$ Torque의 감소량 $= 3W \text{ lb-ft}$

$$\text{따라서 } W \text{에 의한 각 변위 } \phi = \frac{M_{t1} L}{GJ} = \frac{3W \times 12 \times 60}{12 \times 10^6 \times \frac{\pi}{32} \times 2.5^4} = 7.15 W \times 10^{-6} \text{ (rad)}$$

$$\therefore \delta_1 = 7.15 W \times 10^{-6} \times 36 \text{ (in)}$$

W 에 의한 tension에 의한 cable의 길이의 증가량

$$\delta_2 = \frac{F \cdot L}{AE} = \frac{W}{(1/2)} \times \frac{200 \times 12}{30 \times 10^6} = 13.3 W \times 10^{-6} \times 12 \text{ (in)}$$

$$\text{따라서 총 변위 } \delta = \delta_1 + \delta_2 = 420 W \times 10^{-6} \text{ in} = 0.2 \text{ (in)}$$

$$\therefore W = 475 \text{ lb} \quad \text{Ans. } 475 \text{ lb}$$

6.24 switch에서 l만큼 떨어진 점의 torsion moment (torque)

$$(M_t)_l = M_0 + \int_0^l dM_t = M_0 + \frac{dM_t}{dl} \int_0^l dl = M_0 + l \frac{dM_t}{dl} \quad \left(M_0: \text{switch의 torque} \right)$$

$$\therefore (M_t)_l = 0.33 + 0.45l$$

$$(\tau_{\theta z})_{max} = \frac{T \cdot r}{J} = \frac{(0.33 + 0.45l) \times 0.0015}{\frac{\pi}{32} \times (0.003)^4} = 280 \times 10^6$$

$$\therefore l = 2.565 \text{ (m)}$$

$$\frac{d\phi}{dl} = \frac{(M_t)_l}{GI_p} ; \quad \phi_l = \frac{(M_t)_l \cdot l}{GI_p} = \int_0^{2.565} \frac{(0.33 + 0.45l) dl}{(80 \times 10^9) \cdot \frac{\pi}{32} (0.003)^4} = 3.657 \text{ (rad)}$$

$$\therefore \phi_{total} = 2\phi_l = 7.315 \text{ (rad)} = 1.164 \text{ (revolution)}$$

Ans. $l = 2.565 \text{ (m)}$ total play = 1.164 (revolutions)

6.25 $w = \text{foil 위의 물의 무게} = 2 \times (2 \times 3) \times 10^{-8} \times 1 \text{ (grams)}$
 $= 12 \times 10^{-8} \text{ grams}$

즉 AB에 생기는 moment $g_w = 2w = 24 \times 10^{-8} \text{ gram-cm}$

$M_B = DE$ 를 수평으로 되돌리는데 필요한 moment

$M_A = A$ 를 지지함으로써 A에 생기는 moment = 0 ($\because \phi_{AC} = 0$)

평형을 이루기 위해 $M_B = g_w = 24 \times 10^{-8} \text{ gram-cm}$

$$\therefore \phi_B = \frac{M_t \cdot L}{GI_p} = \frac{24 \times 10^{-8} \times 4}{2.7 \times 10^8 \cdot \frac{\pi}{32} (0.001)^4} = 0.036 \text{ (rad)} = 2.06^\circ$$

Ans. $\phi_B = 2.06^\circ$

6.26 shear flow에 일정하므로. 2024-T4 Al ; $\gamma = 400 \text{ MN/m}^2$

$$\tau_{AB} = \tau_{DC} = \frac{1}{2} \tau_{AD} = \frac{1}{2} \tau_{BC}$$

$$M_t = 2Aq = 2A \times (\tau)_{BC} \quad \text{--- ①}$$

$$A = (52 - 2.6) \times (39 - 13) \times 10^{-6} = 1.86238 \times 10^{-3} \text{ (m}^2\text{)}$$

$$q = (\tau)_{BC} = \frac{\gamma}{2} \cdot t_{BC} = (200 \times 10^{-6}) \cdot (1.3 \times 10^{-3}) = 260000 \text{ N/m}^2$$

값식을 ①에 대입

$$\therefore M_t = 2 \times 1.862 \times 10^{-3} \times 260000 = 968.4 \text{ (N-m)}$$

Ans. $M_t = 968.4 \text{ (N-m)}$

6.27 (a) shear flow는 일정하기 때문에

$$1.3 T_y = (2.6 - x) T_y - x T_y \quad \therefore x = 0.6 \text{ mm}$$

(b) $(M_t)_L = 2Aq$

$$= 2 \cdot (52 - 2.6) (39 - 1.3) \times 10^{-6} \cdot (1.3 \times 10^{-3} T_y)$$

$$T_y = \frac{1}{2} \tau = 200 \times 10^6 \text{ N/m}^2$$

$$\therefore (M_t)_L = 968.4 \text{ (N-m)}$$

Thin wall의 모든 부분에서 shear flow가 양의 문제라 같기 때문에 $(M_t)_L$ 도 같아진다. 실제로는 $(M_t)_L$ 이 $(M_t)_Y$ 보다 약간 큰 데 그 이유는, 항복이 생기는 단면에서의 shear stress가 일정하다는 가정이 옳지 않기 때문이다.

6.28

$$\text{contact area} = \pi a^2$$

$$\text{pressure} = 30 \text{ psi}$$

$$\text{Normal load} = 1000 \text{ (lb)} = 30 \pi a^2$$

$$\therefore a = 3.25 \text{''}$$

contact area의 면적 원소

$$\text{생기는 friction force } df = 0.6 p dA = 0.6 p \times r d\theta \cdot dr$$

$$\text{Torque} = df \times r = 0.6 p r^2 d\theta dr$$

$$\text{Total torque} = \iint_A 0.6 p r^2 d\theta dr = \int_{r=0}^a \int_{\theta=0}^{2\pi} 0.6 p r^2 d\theta dr = 0.6 p \cdot 2\pi \cdot \frac{a^3}{3}$$

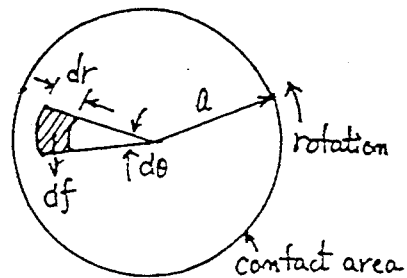
$$= \frac{0.6 \times 30 \times 2\pi \times (3.25)^3}{3} = 1300 \text{ (lb-in)}$$

$$\therefore \text{두 개의 tire에 의한 torque} = 2600 \text{ lb-in}$$

Torque in steering column (due to 20:1 reduction gear)

$$= \frac{2600}{20} = 130 \text{ (lb-in)}, \quad T_{\max} = \frac{T \cdot r}{I_t} = \frac{2T}{\pi r^3} = \frac{2 \times 130}{\pi \times (\frac{3}{8})^3} = 1570 \text{ (psi)}$$

$$\underline{\text{Ans. } T_{\max} = 1570 \text{ psi}}$$



6.29 중첩 이론을 적용하자.

하중이 의해서 생기는 stress 의 분포는
구속의 그림과 같다. 여기에 linear
reserved stress 를 중첩시킨다.

이때의 torque 는 $-M_t$ 이다.

maximum 중첩 응력 (superposed stress) T_m 이

$2T_y$ 보다 작다면, linearity 를 가정하는 것이 가능하다. 왜냐하면

최대 잔류 응력 (max residual stress) $T_r = T_m - T_y \leq T_y$

중첩된 응력은 우편과 같다.

최초의 torque ; $\phi_i = 2\phi_y$

$$M_t = \frac{4}{3} T_y \left[1 - \frac{1}{4} \left(\frac{1}{3} \right)^2 \right] = 1.29 T_y$$

elastic moment T_y 는 비틀림 각 ϕ_y 를
발생시킨다.

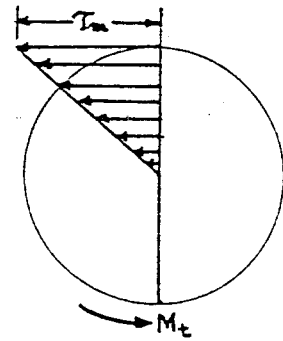
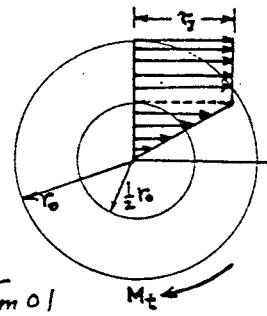
중첩된 elastic moment 는 $1.29 T_y$

따라서 중첩된 springback 의 변형각은 $1.29 \phi_y$

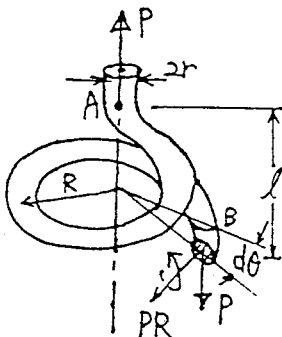
$$\therefore \phi_2 = \phi_i - 1.29 \phi_y = 0.71 \phi_y = 0.355 \phi_i$$

[note ; $T_m < 2T_y$, 왜냐하면 Torque = $1.29 T_y < 2T_y$]

$$\underline{\text{Ans. } \phi_{\text{residual}} = 0.355 \phi_i}$$



6.30 Spring 의 한 점을 wire 에 수직하게 자르는 pitch 가
작다고 가정하면 equilibrium condition.



$$\sum M = 0 \quad \therefore M_t = PR, \quad \sum F = 0 \quad \therefore P = V$$

$$\phi = \int \frac{M_t \cdot L}{GJ} = \int_0^{2\pi\pi} \frac{PR \cdot R d\theta}{GJ} = \frac{2\pi\pi \cdot PR^2}{GJ}$$

$$\delta = R\phi = \frac{2\pi\pi PR^3}{G \left(\frac{\pi}{2} \right) r^4} = \frac{4\pi\pi PR^3}{Gr^4}$$

6.31 Estimate; Twisting Torque = 5 N-m Compressive Force = 15 N

$$\text{Compressive stress} = \frac{F}{A} = \frac{15}{\frac{\pi}{4} \cdot (0.006)^2} = 530516 \text{ (N/m}^2\text{)}$$

$$\text{Shear stress} = \frac{M_t \cdot \frac{d}{2}}{J} = \frac{16 M_t}{\pi d^3} = \frac{16 \times 5}{\pi \times (0.006)^3} = 117892550 \text{ (N-m)}$$

$$\frac{\text{Compressive stress}}{\text{Shear stress}} = 4.5 \times 10^{-3}$$

Ans. $4.5 \times 10^{-3} (\sigma_c / \tau)$

6.32 (a) loading 을 완전한 elastic 이라고 가정하면, 1100-O 와 2024-T4 는 같은 elastic modulus 를 갖는다.

$$T = \frac{M_t \cdot r}{I_z} = \frac{120 \times 0.0067}{\frac{\pi}{32} (11.3^4 - 6.7^4) \times (10^{-3})^4} = 5.731 \times 10^8 \text{ N/m}^2$$

$$\tau_{\max} = \frac{\sigma_y}{2} = 17.5 \times 10^6 \text{ N/m}^2$$

$\tau > \tau_{\max}$ 따라서 elastic loading 은 아니다.

(b) cladding 은 plastic, core 는 elastic 이라 가정하면

$$\text{moment due to cladding} = M_c = \sum \frac{T}{r} \tau_y$$

$$= \sum 2\pi r^2 \tau_y = 2\pi \left\{ (6.7 + 0.4)(10^{-3}) \right\}^2 (0.8 \times 10^{-3}) (17.5 \times 10^6) + 2\pi \left\{ (11.3 - 0.4)(10^{-3}) \right\}^2 (0.8 \times 10^{-3}) (17.5 \times 10^6) = 14.885 \text{ (N-m)}$$

applied moment = 120

$$\therefore \text{moment due to core} = M_A = 120 - 14.885 = 105.115 \text{ (N-m)}$$

$$J_{\text{core}} = \frac{\pi}{2} [(10.5)^4 - (7.5)^4] (10^{-3})^4 = 1.4123 \times 10^{-8} \text{ (m}^4\text{)}$$

$$\therefore \tau = \frac{M_A \cdot r}{J_{\text{co}}} = \frac{105.115 \times r}{1.4123 \times 10^{-8}} = 7.443 \times 10^9 r$$

$$\therefore (\tau_a)_{\text{core}} = 55.821 \text{ MN/m}^2 \quad (\tau_o)_{\text{core}} = 78.149 \text{ MN/m}^2$$

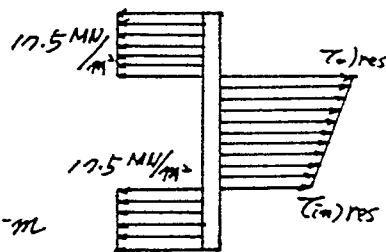
(c) 잔류응력을 구하기 위해, 역 방향으로 120 N-m 의 moment 에 기인한 stress 와 위의 stress 와 중첩해야 한다.

앞의 계산으로 부터 역 방향의 moment 를 가할 때, cladding 에서는 역방향의 yielding stress 이

도달한다. cladding 이 의한

moment 는 $-14.885 \text{ N}\cdot\text{m}$

따라서 core 의 moment $= 14.885 \text{ N}\cdot\text{m}$



$$\therefore \tau = \frac{M_A \cdot r}{J_{co}} = \frac{14.885 \times r}{1.4123 \times 10^{-8}} = 1.054 \times 10^9 \cdot r$$

$$\therefore (\tau_i)_{\text{residual}} = 7.905 \text{ MN/m}^2, (\tau_o)_{\text{residual}} = 11.066 \text{ MN/m}^2$$

Ans. (pure aluminum ; $\tau_{\text{residual}} = -14.885 \text{ N}\cdot\text{m}$

aluminum alloy ; $(\tau_i)_{\text{res}} = 7.905 \text{ MN/m}^2$

$(\tau_o)_{\text{res}} = 11.066 \text{ MN/m}^2$

6.33 hole 의 길이 = L (ft)

물의 부력과 무게는 무시한다.

$$\begin{aligned} \text{weight of metal supported} &\approx 12L \times \frac{\pi}{4} (3^2 - 2^2) \cdot \rho \quad (\rho = \frac{16}{\text{in}^3}) \\ &= 12L \times \frac{\pi}{4} \times 5 \times 0.28 = 13.2L \text{ lb} \end{aligned}$$

단면 전체에 걸쳐 uniform tensile stress 를 가정하면

$$\text{tensile stress } \sigma_z = \frac{13.2L}{\frac{\pi}{4} (3^2 - 2^2)} = 3.36L \text{ psi}$$

$$\text{shear stress } \tau_{\theta z} = \frac{M_z \cdot r_o}{I_z} = \frac{2000 \times 12 \times \frac{3}{2}}{\frac{\pi}{32} (3^4 - 2^4)} = 5650 \text{ psi}$$

Mohr's circle 이라 하자

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_z}{2}\right)^2 + \tau_{\theta z}^2} = \sqrt{5650^2 + \left(\frac{3.36L}{2}\right)^2} = \frac{50000}{L}$$

$$\therefore L = 14500 \text{ ft} \quad (\because \tau_{\max} = \frac{Y}{2})$$

Ans. $L = 14500 \text{ ft}$

6.34 spring constant $= 125 \frac{\text{lb}}{\text{in}}$

$$\text{Torque} = P \cdot x \quad \phi = \frac{(Px)L}{GI} = \frac{Px \cdot L}{G \cdot \frac{\pi}{32} d^4} \quad (d; \text{wire diameter})$$

load에 의한 deflection $= \phi x$

$$\text{spring constant } k = \frac{P}{\phi x} = \frac{G \pi d^4}{32 x^2 L} = \frac{12 \times 10^6 \pi}{32} \frac{d^4}{x^2 L}$$

$$\frac{d^4}{x^2 L} = \frac{32 \times 125}{12 \times 10^6 \pi} = \frac{1}{3000 \pi} \quad \text{--- ①}$$

static load 1000 lb

$$\text{dynamic load } \pm 6'' \times 125 \frac{\text{lb}}{\text{in}} = \pm 750 \text{ lb}$$

$$\therefore \text{maximum load} = 1000 + 750 = 1750 \text{ lb}$$

$$(M_x)_{\max} = 1750 \text{ lb-in}$$

$$\tau = \frac{(1750 x) \frac{d}{2}}{\frac{\pi}{32} d^4} = 8900 \frac{x}{d^3}$$

$$\text{따라서 } \frac{x}{d^3} \cdot 8900 \leq 50,000 \text{ or } \frac{x}{d^3} \leq 5.62 \quad \text{--- ②}$$

$$x_{\max} = 30'' \quad L_{\max} = 120''$$

$$\text{① 식으로 부터 } d^4 = \frac{900 \times 120}{3000 \pi} = 11.4 \quad \therefore d = 1.84''$$

$$\text{② 식에 대입하면 } \frac{x}{d^3} = \frac{30}{61\pi} = 4.85 < 5.62 \quad \text{② 식을 만족한다.}$$

$$\tau = \frac{4.85}{5.62} \times 50,000 = 43,000 \text{ psi}$$

$$\text{① 식은 다음과 같이 쓸 수 있다. } L d^2 = \frac{3000 \pi}{x^2 / d^6}$$

$$\text{② 식에 대입하면 } L d^2 \geq \frac{3000 \pi}{(5.62)^2}$$

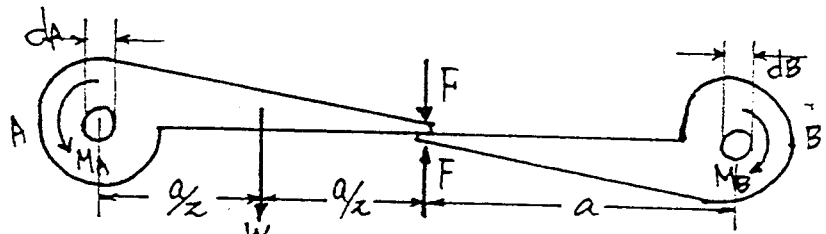
$L d^2$ 은 부피에 비례하고, cost를 최소화하기 design 해야 한다.

$$L d^2 = \frac{3000 \pi}{(5.62)^2} \quad \left(\begin{array}{l} L = 120'' \text{ 이면 } d = 1.57'' \quad x = 21.4'' \\ x = 30'' \text{ 이면 } d = 1.75'' \quad L = 98.0'' \end{array} \right)$$

두 경우 모두 $\tau = 50,000 \text{ psi}$ 이다.

$$\text{Ans. } L = 120'' ; d = 1.57'' , \quad x = 30'' ; d = 1.75''$$

$$6.35 \quad \phi_A = \frac{M_A \cdot L}{G J_A} \quad \phi_B = \frac{M_B \cdot L}{G J_B} \quad \text{--- ①}$$



그림에서 $M_B = Fa$ $M_A = W \cdot \frac{a}{2} - Fa$

기하학적 적합조건에서 $\phi_A = \phi_B$, $\frac{M_A}{M_B} = \left(\frac{d_A}{d_B}\right)^4 = \frac{W}{2F} - 1$

$$\therefore F = \frac{W}{2 \left[1 + \left(\frac{d_A}{d_B} \right)^4 \right]}$$

0.4에 대입하면

$$\phi_A = \phi_B = \frac{1}{G \cdot \frac{\pi}{32} \cdot d_B^4} \cdot \frac{W \cdot a \cdot L}{\left[1 + \left(\frac{d_A}{d_B} \right)^4 \right]} = \frac{16 W \cdot a \cdot L}{\pi G (d_A^4 + d_B^4)}$$

Ans. $F = W/2 \left[1 + \left(\frac{d_A}{d_B} \right)^4 \right]$, $\phi_A = \phi_B = 16 W \cdot a \cdot L / \pi G (d_A^4 + d_B^4)$

6.36 $\epsilon_r = \frac{\partial u}{\partial r} = 0$, $\epsilon_\theta = \frac{\partial v}{r \partial \theta} + \frac{u}{r} = 0$, $\epsilon_z = \frac{\partial w}{\partial z} = 0$.

$$\gamma_{r\theta} = \frac{\partial v}{\partial r} + \frac{1}{r} \frac{\partial r}{\partial \theta} - \frac{v}{r} = \phi - \phi = 0, \quad \gamma_{zr} = \frac{\partial w}{\partial r} + \frac{\partial u}{\partial z} = 0$$

$$\gamma_{\theta z} = \frac{\partial v}{\partial z} + \frac{1}{r} \frac{\partial w}{\partial \theta} = r \frac{d\phi}{dz}$$

$\therefore u=0$ $v=r\phi$ $w=0$ 일 때 $\epsilon_r = \epsilon_\theta = \epsilon_z = \gamma_{r\theta} = \gamma_{zr} = 0$

$\gamma_{\theta z} = r \frac{d\phi}{dz}$ 을 만족한다.

i) Hooke's law 에 대입.

$$\epsilon_r = \frac{1}{E} [\sigma_r - \nu(\sigma_\theta + \sigma_z)] = 0, \quad \gamma_{r\theta} = \frac{1}{G} \tau_{r\theta} = 0$$

$$\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)] = 0, \quad \gamma_{rz} = \frac{1}{G} \tau_{rz} = 0$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_r + \sigma_\theta)] = 0, \quad \gamma_{\theta z} = \frac{1}{G} \tau_{\theta z} = r \frac{d\phi}{dz}$$

$$\therefore \tau_{\theta z} = G r \frac{d\phi}{dz}$$

ii) 평형 조건식의 대입.

$$\frac{\partial \sigma_r}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_r - \sigma_\theta}{r} = 0$$

$$\begin{aligned} \frac{\partial \tau_\theta}{\partial r} + \frac{1}{r} \frac{\partial \sigma_\theta}{\partial \theta} + \frac{\partial \tau_{\theta z}}{\partial z} + \frac{\partial \tau_{r\theta}}{\partial r} &= \frac{\partial \tau_{\theta z}}{\partial z} = Gr \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} \right) \\ &= Gr \frac{\partial}{\partial z} \left(\frac{M_x}{GL_z} \right) = 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial \tau_z}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta z}}{\partial \theta} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{zr}}{r} &= \frac{1}{r} \frac{\partial \tau_{\theta z}}{\partial \theta} = G \frac{\partial}{\partial \theta} \left(\frac{d\phi}{dz} \right) \\ &= G \frac{\partial}{\partial \theta} \left(\frac{M_x}{GL_z} \right) = 0 \end{aligned}$$

따라서 $\sigma_r = \sigma_\theta = \sigma_z = \tau_{r\theta} = \tau_{rz} = 0$, $\tau_{\theta z} = \frac{T \cdot r}{J}$ 의 해는 elastic theory 의 범위 내에서 exact solution 이 된다.

6.37) 대칭 조건; $\tau_{r\theta} = \tau_{rz} = 0$. $\frac{\partial}{\partial \theta} = 0$. $\tau_{\theta z} = r \frac{d\phi}{dz}$

$\epsilon_r = \epsilon_\theta = 0$ 으로 가정하자.

i) Hooke's law 에서 다음 식을 얻을 수 있다.

$$\sigma_r = \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_r + \epsilon_\theta + \epsilon_z) + \frac{E}{1+\nu} \epsilon_r$$

$$\sigma_\theta = \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_r + \epsilon_\theta + \epsilon_z) + \frac{E}{1+\nu} \epsilon_\theta$$

$$\sigma_z = \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_r + \epsilon_\theta + \epsilon_z) + \frac{E}{1+\nu} \epsilon_z$$

$$\therefore \sigma_r = \sigma_\theta = \frac{\nu E \epsilon_z}{(1+\nu)(1-2\nu)}$$

$$\sigma_z = \left[\frac{\nu E}{(1+\nu)(1-2\nu)} + \frac{E}{1+\nu} \right] \epsilon_z = \frac{(1-\nu)E}{(1+\nu)(1-2\nu)} \epsilon_z$$

$$\tau_{r\theta} = \tau_{rz} = 0, \quad \tau_{\theta z} = Gr \frac{d\phi}{dz}$$

ii) 평형 조건

$$\frac{\partial \sigma_r}{\partial r} + \frac{\partial \tau_{r\theta}}{r \partial \theta} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_r - \sigma_\theta}{r} = \frac{\nu E}{(1+\nu)(1-2\nu)} \cdot \frac{\partial \epsilon_z}{\partial r} = 0$$

$$\frac{\partial \tau_{r\theta}}{\partial r} + \frac{\partial \sigma_\theta}{r \partial \theta} + \frac{\partial \tau_{\theta z}}{\partial z} + \frac{\tau_{r\theta}}{r} = \frac{\nu E}{(1+\nu)(1-2\nu)} \cdot \frac{1}{r} \frac{\partial \epsilon_z}{\partial \theta} = 0$$

$$\frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta z}}{\partial \theta} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{zr}}{r} = \frac{(1-\nu)E}{(1+\nu)(1-2\nu)} \frac{\partial \epsilon_z}{\partial z} = 0$$

$$\frac{\partial \epsilon_z}{\partial r} = \frac{1}{r} \frac{\partial \epsilon_z}{\partial \theta} = \frac{\partial \epsilon_z}{\partial z} = 0 \quad \therefore \epsilon_z = \text{constant}$$

$$\therefore \sigma_z = \frac{(1-\nu)\nu E}{(1+\nu)(1-2\nu)} \epsilon_z = \text{constant}$$

Surface loading of $\square \times \square \times \square$, $\sigma_z = 0$. $\therefore \epsilon_z = 0$.

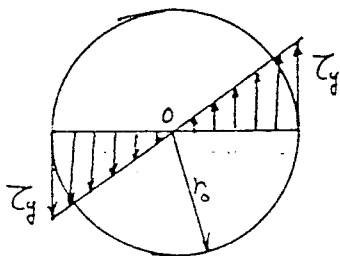
$$6.38 \quad \tau_i = \frac{T_i \cdot r}{J} = \frac{T_i \left(\frac{1.9}{2} \times 10^{-3} \right)}{\frac{\pi}{32} (1.9 \times 10^{-3})^4}$$

$$\bar{\tau} = \frac{1}{4} \sum_i \tau_i = \frac{1}{4} \left[\frac{(0.95 \times 10^{-3})}{\frac{\pi}{32} (1.9 \times 10^{-3})^4} \right] \times (0.1 + 0.125 + 0.1 + 0.11)$$

$$= 80749264 \text{ N/m}^2 = 80.75 \text{ MN/m}^2$$

Ans. $\bar{\tau} = 80.75 \text{ MN/m}^2$

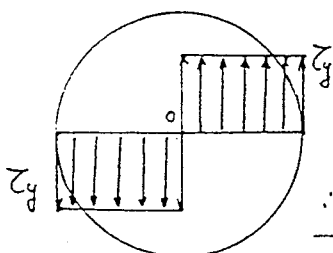
6.39 Efficiency of spring = U_y / U_L



$$M_x = \int_0^r \left(\frac{\tau_y}{r_0} \cdot r \right) r (2\pi r dr)$$

$$= \frac{1}{2} \pi r_0^3 \tau_y$$

$$\therefore U_y = \frac{M_x^2 L}{2GJ} = \left(\frac{\pi}{2} r_0^3 \tau_y \right)^2 \frac{L}{2GJ}$$



$$T_L = \int_0^{r_0} \tau_y \cdot r (2\pi r dr) = \frac{\pi}{3} \pi r_0^3 \tau_y$$

$$\therefore U_L = \frac{T_L^2 L}{2GJ} = \left(\frac{2}{3} \pi r_0^3 \tau_y \right)^2 \frac{L}{2GJ}$$

$$\therefore \text{Efficiency} = U_y / U_L = 9/16 = 56.25\%$$

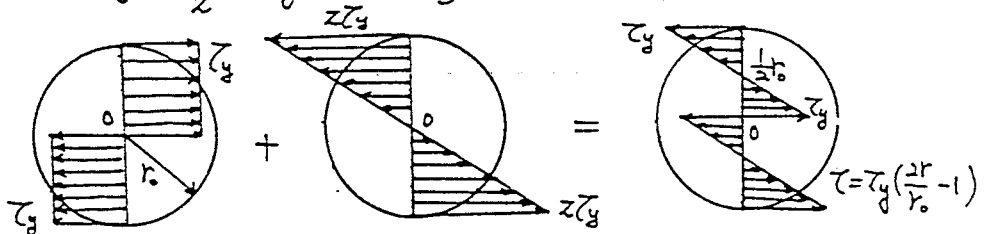
$$6.40 \quad \gamma = k\tau^2 \quad \gamma_{\theta z} = \gamma \frac{d\phi}{dz} \quad \therefore k\tau^2 = r \frac{d\phi}{dz} \quad (\text{from Eq. (6.3)})$$

$$\begin{aligned} \therefore M &= \int_A r\tau dA = \int r \cdot 2\pi r \left(\frac{r}{k}\right)^{\frac{1}{2}} \left(\frac{d\phi}{dz}\right)^{\frac{1}{2}} dr \\ &= \int_0^{r_0} \left(\frac{1}{k} \cdot \frac{d\phi}{dz}\right)^{\frac{1}{2}} (2\pi r^{\frac{5}{2}}) dr = 2\pi \cdot \left(\frac{1}{k} \frac{d\phi}{dz}\right)^{\frac{1}{2}} \cdot \left(\frac{2}{7} r_0^{\frac{7}{2}}\right) \end{aligned}$$

$$\therefore \frac{d\phi}{dz} = \frac{49 k M_x^2}{16 \pi^2 r_0^7}, \quad \tau_{\max} = \left(\frac{r_0}{k} \cdot \frac{d\phi}{dz}\right)^{\frac{1}{2}} = \frac{7 M_x}{4 \pi r_0^3}$$

$$\text{Ans.} \quad \frac{d\phi}{dz} = \frac{49 k M_x^2}{16 \pi^2 r_0^7}, \quad \tau_{\max} = \frac{7 M_x}{4 \pi r_0^3}$$

$$6.41 \quad T_y = \frac{\pi}{2} r_0^3 \tau_y, \quad T_L = \frac{2}{3} \pi r_0^3 \tau_y = \frac{4}{3} T_y$$



Elastic theory에서 어느 한쪽 방향으로 완전 선형으로 twisting된 상태에서 구속을 풀면, 그리고 반대방향으로 moment를 증가시켜 가면 반대 방향의 항복 응력에 도달할 때까지는 선형적으로 변한다. 그러나 일단 반대방향의 항복 응력이 도달하게 되면, plastic하게 되는 선형적인 변화를 하지 않는다. 선형을 벗어날 때의 moment를 구해보면, 그 때의 shear stress는 $\tau = \left(\frac{2r}{r_0} - 1\right) \tau_y$ 이다.

$$\therefore (M_x)_y' = \int_0^r \left(\frac{2r}{r_0} - 1\right) \tau_y \cdot r (2\pi r) dr = \left[2\pi \tau_y \left(\frac{r_0^3}{3} - \frac{r_0^3}{3}\right) \right]$$

$$= \frac{1}{3} \pi r_0^3 \tau_y = \frac{2}{3} T_y$$

따라서 M_x 가 $\frac{2}{3} T_y$ 의 값보다 크게될 때부터 선형이 아니다.
- Q.E.D. -

즉, 반대 방향으로 moment를 증가시켜 갈때는, $M_x = T_y \frac{\phi}{\phi_y}$ 의 관계를 유지하며 선형적으로 변한다. Ans. $M_x = T_y (\phi / \phi_y)$

6.42 Fig (6.28) (6.29)에

$$U = \frac{1}{2G} \iiint \tau_{xz}^2 \, d\mathbf{n} \, ds \, dz \quad M_x = z \tau_{xz} \cdot t \cdot A \text{ 를 대입.}$$

$$U = \frac{1}{2G} \iiint \frac{M_x^2}{4t^2 A^2} \, d\mathbf{n} \, ds \, dz \quad \text{또는 Castigliano's theorem 적용}$$

$$\therefore \phi = \frac{\partial U}{\partial M_x} = \frac{1}{2G} \iiint \frac{M_x}{2t^2 A^2} \, d\mathbf{n} \, ds \, dz$$

$$= \frac{M_x}{4A^2 G} \iiint \frac{1}{t^2} \, d\mathbf{n} \, ds \, dz = \frac{M_x}{4A^2 G} \iint \left[\frac{\pi}{t^2} \right]_{-\frac{t}{2}}^{\frac{t}{2}} \, ds \, dz$$

$$= \frac{M_x}{4A^2 G} \iint \frac{1}{t} \, dz \, ds = \frac{M_x \cdot L}{4A^2 G} \int \frac{ds}{t} \quad - Q.E.D. -$$

$$6.43 (a) \phi_1 = \frac{M_x \cdot L_i}{G J_1} = \frac{M_x \cdot L_i}{G \cdot \frac{\pi}{32} (2R_i)^4} = \frac{2M_x L_i}{\pi R_i^4 G}$$

$$\phi_2 = \frac{M_x L_o}{G J_2} = \frac{M_x L_o}{G \cdot \frac{\pi}{32} (R_o^4 - R_i^4)} = \frac{2M_x L_o}{\pi (R_o^4 - R_i^4) G}$$

$$\text{중첩의 원리; } \phi = \phi_1 + \phi_2 = \frac{2M_x}{\pi G} \left(\frac{L_i}{R_i^4} + \frac{L_o}{R_o^4 - R_i^4} \right)$$

$$\therefore k_t = \frac{M_x}{\phi} = \frac{\pi G}{2} \left[\frac{1}{L_i/R_i^4 + L_o/(R_o^4 - R_i^4)} \right]$$

$$(b) \tau_i = \frac{M_x \cdot r_i}{J_i} = \frac{M_x \cdot R_i}{\frac{\pi}{32} (2R_i)^4} = \frac{2M_x}{\pi R_i^3}$$

$$\tau_o = \frac{M_x \cdot R_o}{\frac{\pi}{32} (R_o^4 - R_i^4)} = \frac{2M_x R_o}{\pi (R_o^4 - R_i^4)}$$

$$\tau_i = \tau_o; \quad 1 = (1/R_i^3) / (R_o / (R_o^4 - R_i^4))$$

$$\therefore \left(\frac{R_o}{R_i}\right)^4 - 1 = \left(\frac{R_o}{R_i}\right)$$

trial error method에 의해서 $\frac{R_o}{R_i}$ 의 값을 구하면

$$\frac{R_o}{R_i} \approx 1.22$$

Ans. (a) $k_t = \frac{\pi G}{2[L_i/R_i^4 + L_o/(R_o^4 - R_i^4)]}$

(b) $\left(\frac{R_o}{R_i}\right)^4 - 1 = \left(\frac{R_o}{R_i}\right), \frac{R_o}{R_i} = 1.22$

6.44 $M_A = W \cdot b + M_B - M_D$

$$\phi_{AB} = \frac{M_A \cdot a}{GJ} = \frac{a}{GJ} (W \cdot b + M_B - M_D), \phi_{BC} = \frac{a}{GJ} (W \cdot b - M_D)$$

중첩의 원리; $\phi_{AC} = \phi_{AB} + \phi_{BC} = \frac{a}{GJ} (2W \cdot b + M_B - 2M_D) = 0$

따라서 $2M_D - M_B = 2W \cdot b \quad \text{--- ①}$

$$\phi_{BD} = \frac{za}{GJ} (W \cdot b - M_D) \therefore \phi_{DE} = \frac{za}{GJ} W \cdot b$$

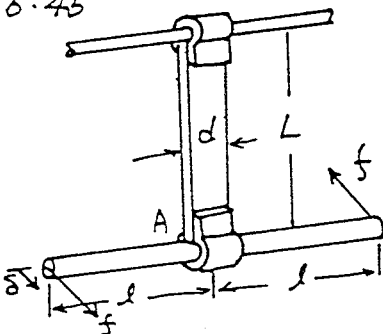
중첩의 원리; $\phi_{AE} = \phi_{AB} + \phi_{BD} + \phi_{DE} = \frac{a}{GJ} (5W \cdot b + M_B - 3M_D) = 0$

따라서 $3M_D - M_B = 5W \cdot b \quad \text{--- ②}$

① & ②에 의해 $M_D = 3W \cdot b, M_B = 4W \cdot b$

Ans. $M_B = 4W \cdot b, M_D = 3W \cdot b$

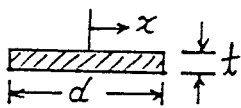
6.45



Al 봉은 bending이 일어나지 않는다
가정한다.

$$f = G_0 \frac{Mm}{r^2}, M_t = G_0 \frac{Mm}{r^2} \cdot 2l \quad \text{--- ①}$$

where $\left\{ \begin{array}{l} G_0 : \text{두 질점간의 만유인력} \\ M : \text{Bucket의 무게} \\ m : \text{Bucket의 갯수} \end{array} \right.$



Mylar tape의 A부분을 보면, tape의 양쪽면에서 전단 변형이 max. 이 된다.

($\because \tau = \frac{VQ}{bI}$ 이니 Q_{max} 이 되기 때문)

$$\tau = G \frac{x\phi}{L} \quad \therefore M_x = \tau \int_{-\frac{d}{2}}^{\frac{d}{2}} x dx \cdot t = \tau \int_{-\frac{d}{2}}^{\frac{d}{2}} G \cdot \frac{x\phi}{L} \cdot x \cdot t dx$$

$$= \frac{2G\phi t}{L} \int_0^{\frac{d}{2}} x^2 dx = \frac{G\phi d^3 t}{12L}$$

따라서 $\phi = \frac{12 \cdot M_x \cdot L}{G d^3 t}$

$$\delta = \phi \cdot l = \frac{12 M_x \cdot L \cdot l}{G d^3 t} = \frac{24 G_0 M_m L l^2}{G d^3 t \cdot r^2}$$

($\because$ ①에서 $M_x = G_0 \frac{M \cdot m}{r^2}$)

$$= \frac{24 \times (6.67 \times 10^{-11}) (100) (5) (10) (1)^2}{\frac{1.4 \times 10^9}{2(1+0.4)} \times (25.4 \times 10^{-3})^3 (0.254 \times 10^{-3}) (0.4)^2} = 2.4 \times 10^{-5} (m)$$

($\because G = \frac{E}{2(1+\nu)}$)

Ans. $\delta = 2.4 \times 10^{-5} m$

CHAPTER 7

$$7.1 \quad \bar{y} = \frac{\int y dA}{\int dA} = \frac{\int y dA_1 + \int y dA_2 + \int y dA_3}{\int dA_1 + \int dA_2 + \int dA_3} \quad \int y dA = \bar{y}_i A_i, \dots$$

$$\therefore \bar{y} = \frac{\bar{y}_1 A_1 + \bar{y}_2 A_2 + \bar{y}_3 A_3}{A_1 + A_2 + A_3} = \frac{\sum \bar{y}_i A_i}{\sum A_i} \quad \text{and similarly} \quad \bar{z} = \frac{\sum \bar{z}_i A_i}{\sum A_i}$$

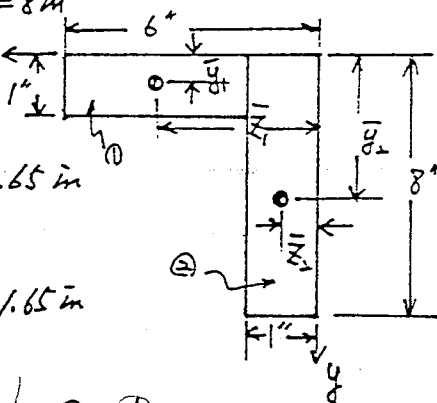
$$7.2 \quad A_1 = (6-1)(1) = 5 \text{ in}^2, \quad A_2 = (8)(1) = 8 \text{ in}^2$$

$$\bar{y}_1 = 0.5 \text{ in}, \quad \bar{z}_1 = 3.5 \text{ in}$$

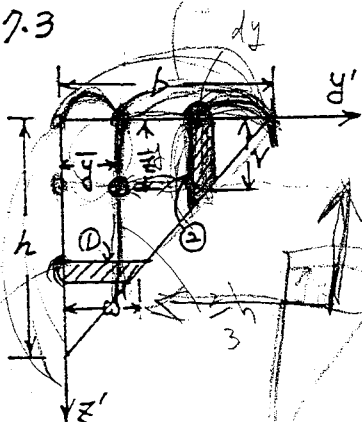
$$\bar{y}_2 = 4 \text{ in}, \quad \bar{z}_2 = 0.5 \text{ in}$$

$$\bar{y} = \frac{\sum \bar{y}_i A_i}{\sum A_i} = \frac{(0.5)(5) + (4)(8)}{13} = 2.65 \text{ in}$$

$$\bar{z} = \frac{\sum \bar{z}_i A_i}{\sum A_i} = \frac{(3.5)(5) + (0.5)(8)}{13} = 1.65 \text{ in}$$



7.3



For the area ①

$$I_{yy} = \int y^2 dA = \int_{-2b/3}^{b/3} y^2 \left(\frac{2b}{3} + \frac{b}{h} y \right) dy = \frac{b^3}{36}$$

For the area ②

$$I_{zz} = \int z^2 dA = \int_{-2b/3}^{b/3} z^2 \left(\frac{2h}{3} + \frac{h}{b} z \right) dz = \frac{b^3 h}{36}$$

For the area ③ $dA = dy \cdot dz$

$$I_{yz} = \int yz dA = \int dy \int dz (yz)$$

$$= \int_{-2b/3}^{b/3} dy \int_{-(b/3 + \frac{b}{h}y)}^{b/3} dz (yz) = \int_{-2b/3}^{b/3} y \cdot \frac{1}{2} \left[\left(\frac{b}{3} \right)^2 - \left(\frac{b}{3} + \frac{b}{h} y \right)^2 \right] dy$$

$$= \int_{-2b/3}^{b/3} \left(-\frac{1}{2} \right) \left(\frac{b}{h} y^2 \right) \left(\frac{2b}{3} + \frac{b}{h} y \right) dy = -\frac{1}{2} \left(\frac{b}{h} \right) \left[\frac{2b}{9} b y^3 + \frac{b}{h} \cdot \frac{y^4}{4} \right]_{-2b/3}^{b/3}$$

$$= -\frac{b^3 h^2}{72} \quad - \text{Q.E.D.} -$$

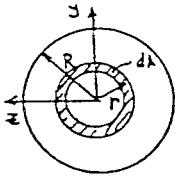
$$7.4 \quad I_x = \int r^2 dA = \int (y^2 + z^2) dA = \int y^2 dA + \int z^2 dA = I_{yy} + I_{zz}$$

symmetry ; $I_{yy} = I_{zz}$

$$\therefore I_{yy} = I_{zz} = \frac{1}{2} I_x$$

$$I_x = \int r^2 dA = \int_0^R r^2 (2\pi r dr) = \frac{\pi R^4}{2}$$

$$\therefore I_{yy} = I_{zz} = \frac{\pi}{4} R^4$$



- Q.E.D -

$$7.5 \quad I_{y'y'} = \int y'^2 dA = \int (y + \bar{y})^2 dA$$

$$= \int y^2 dA + \int 2y\bar{y} dA + \int \bar{y}^2 dA$$

$$= I_{yy} + 2\bar{y} \int y dA + \bar{y}^2 A$$

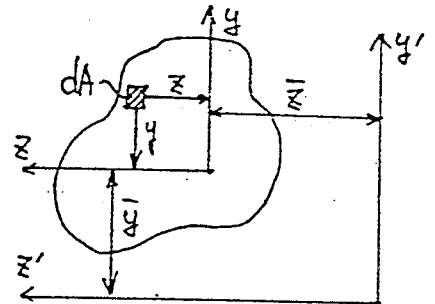
$\int y dA = 0$ ($\because y$ - z 축이 centroid를 통과하기 때문)

$$\therefore I_{y'y'} = I_{yy} + \bar{y}^2 A \quad \text{같은 방법으로 } I_{z'z'} = I_{zz} + \bar{z}^2 A$$

$$I_{y'z'} = \int y'z' dA = \int (y + \bar{y})(z + \bar{z}) dA = \int yz dA + \int \bar{y}z dA + \int y\bar{z} dA + \int \bar{y}\bar{z} dA$$

$$\int y dA = \int z dA = 0$$

$$\therefore I_{y'z'} = I_{yz} + \bar{y}\bar{z}A$$



$$7.6 \quad \text{Area ①} \quad I_{yy_1} = \frac{1}{12} (5)(1)^3 = 0.417 \text{ in}^4$$

$$I_{zz_1} = \frac{1}{12} (1)(5)^3 = 10.417 \text{ in}^4$$

$$I_{yz_1} = 0, \quad A = 5 \text{ in}^2$$

$$\therefore (I_{yy})_1 = 0.417 + (2.15)^2 (5) = 23.5 \text{ in}^4$$

$$(I_{zz})_1 = 10.417 + (1.85)^2 (5) = 27.5 \text{ in}^4$$

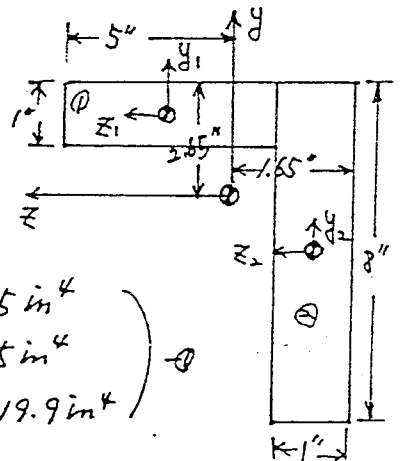
$$(I_{yz})_1 = 0 + (2.15)(1.85)(5) = 19.9 \text{ in}^4$$

Area ② 대해서,

$$(I_{yy})_2 = 57.3 \text{ in}^4, \quad (I_{zz})_2 = 11.3 \text{ in}^4, \quad (I_{yz})_2 = 12.4 \text{ in}^4 \quad \text{--- ②}$$

$$\text{①} + \text{②} \quad I_{yy} = 80.8 \text{ in}^4 \quad I_{zz} = 38.8 \text{ in}^4 \quad I_{yz} = 32.3 \text{ in}^4$$

$$\text{Ans. } I_{yy} = 80.8 \text{ in}^4 \quad I_{zz} = 38.8 \text{ in}^4 \quad I_{yz} = 32.3 \text{ in}^4$$



$$7.7 \quad \bar{y} = \frac{\bar{y}_1 A_1 + \bar{y}_2 A_2}{A} = \frac{-(0)(at) - (a)(2at)}{3at} = -\frac{2}{3}a$$

$$\bar{z} = \frac{-(\frac{a}{2})(at) - (0)(2at)}{3at} = -\frac{a}{6}$$

평행축 정리 이용.

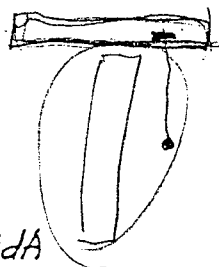
$$I_{yy} = [\frac{1}{12}at^3 + (\frac{4a^2}{P})at] + [\frac{1}{12}8a^3t + (\frac{a^2}{P})2at] = \frac{4}{3}a^3t$$

$$I_{zz} = [\frac{1}{12}ta^3 + (\frac{a^2}{P})at] + [\frac{1}{12}t^3 \cdot 2a + (\frac{a^2}{P})2at] = \frac{1}{4}a^3t$$

$$I_{yz} = [0 + (\frac{2a}{3})(-\frac{a}{3})at] + [0 + (-\frac{a}{3})(\frac{a}{6})2at] = -\frac{1}{3}a^3t$$

$$7.8 \quad I_{zz} = \frac{1}{12}(5.75)(200 - z \times 7.75)^3 + z[(130)(7.75)(100 - \frac{7.75}{z})^2 + \frac{1}{12}(130)(7.75)^3] = 21638087.83 = 2.164 \times 10^7 (\text{mm}^4)$$

Ans. $I_{zz} = 2.164 \times 10^7 \text{ mm}^4$



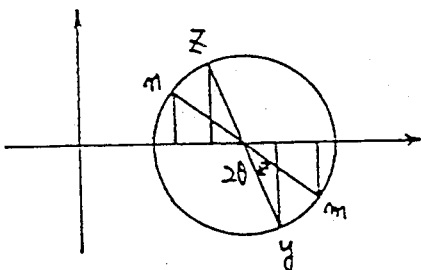
$$7.9 \quad m = z \cos \theta - y \sin \theta$$

$$m = z \sin \theta + y \cos \theta$$

$$I_{mm} = \int m^2 dA = \int (z \sin \theta + y \cos \theta)^2 dA \\ = \int z^2 \sin^2 \theta dA + \int y^2 \cos^2 \theta dA + \int 2yz \sin \theta \cos \theta dA = I_{zz} \sin^2 \theta + I_{yy} \cos^2 \theta + 2I_{yz} \sin \theta \cos \theta$$

I_{mx}, I_{my} 도 같은 방법으로 얻어진다.

$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$, $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$, $2 \sin \theta \cos \theta = \sin 2\theta$ 를 이용하면 (4.23) (4.24) 식이 얻어진다.



$$I_{mm} = 0 \text{ 이라 } \tan 2\theta = \frac{-2I_{yz}}{I_{yy} - I_{zz}}$$

$$I_{1,2} = \frac{I_{yy} + I_{zz}}{2} \pm \sqrt{\left(\frac{I_{yy} - I_{zz}}{2}\right)^2 + I_{yz}^2}$$

- Q.E.D -

$$7.10 \quad I_{yy} = 80.8 \text{ in}^4, I_{zz} = 38.8 \text{ in}^4, I_{yz} = 32.3 \text{ in}^4$$

$$\tan 2\theta = \frac{32.3}{\frac{1}{2}(80.8 - 38.8)} = 1.54 \quad \therefore \theta = 28.5^\circ$$

$$I_{11} = \frac{1}{2}(38.8 + 80.8) + \sqrt{\left(\frac{38.8 - 80.8}{2}\right)^2 + (32.3)^2} = 98.33 \text{ (in}^4\text{)}$$

$$I_{22} = \frac{1}{2}(38.8 + 80.8) - \sqrt{\left(\frac{38.8 - 80.8}{2}\right)^2 + (32.3)^2} = 21.27 \text{ (in}^4\text{)}$$

7.11 $\sum M_i = 0$ $I_{yy} = b^3 h^3 / 36$, $I_{zz} = b^3 h^3 / 36$, $I_{yz} = -b^3 h^3 / 72$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{b^3 h^3 / 72}{\frac{1}{2} \left[\frac{b^3 h^3}{36} - \frac{b^3 h^3}{36} \right]} \right) = \frac{1}{2} \tan^{-1} \left(\frac{b h}{h^2 - b^2} \right)$$

$$I_{11} = \frac{1}{2} \left(\frac{b^3 h^3}{36} + \frac{b^3 h^3}{36} \right) + \sqrt{\frac{1}{4} \left(\frac{b^3 h^3}{36} - \frac{b^3 h^3}{36} \right)^2 + \left(\frac{b^3 h^3}{72} \right)^2}$$

$$= \frac{b^3 h^3}{72} \left[(h^2 + b^2) + \sqrt{(h^2 - b^2)^2 + h^2 b^2} \right]$$

$$I_{22} = \frac{b^3 h^3}{72} \left[(h^2 + b^2) - \sqrt{(h^2 - b^2)^2 + h^2 b^2} \right] \quad - \text{Q.E.D.}$$

7.12 $I_{11} = \frac{1}{12}(c)(3c)^3 = \frac{9}{4}C^4$

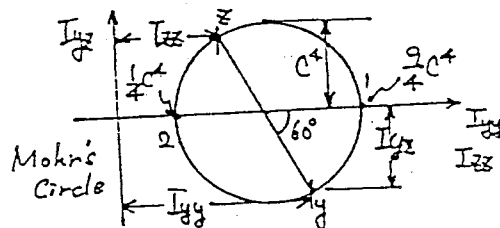
$$I_{22} = \frac{1}{12}(3c)(c)^3 = \frac{1}{4}C^4$$

$$I_{12} = 0$$

$$I_{yy} = \frac{1}{2} \left(\frac{C^4}{4} + \frac{9C^4}{4} \right) + \frac{1}{2} \left(\frac{9C^4}{4} - \frac{C^4}{4} \right) \cos 60^\circ = 1.75C^4$$

$$I_{zz} = \frac{1}{2} \left(\frac{C^4}{4} + \frac{9C^4}{4} \right) - \frac{1}{2} \left(\frac{9C^4}{4} - \frac{C^4}{4} \right) \cos 60^\circ = 0.75C^4$$

$$I_{yz} = (C^4) \sin 60^\circ = 0.866C^4$$



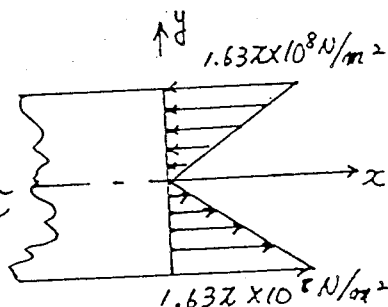
7.13 $I_{zz} = \frac{1}{12} b h^3 = \frac{1}{12} \times 0.025 \times (0.05)^3 = 2.60417 \times 10^{-7} \text{ (m}^4\text{)}$

$$\sigma_{max} = \frac{M y_{max}}{I_{zz}}$$

$$= \frac{(1.7 \times 10^3)(0.025)}{2.60417 \times 10^{-7}}$$

$$= 1.632 \times 10^8 \text{ (N/m}^2\text{)} < \sigma_Y$$

$\therefore \sigma_Y$ is not reached linear stress.



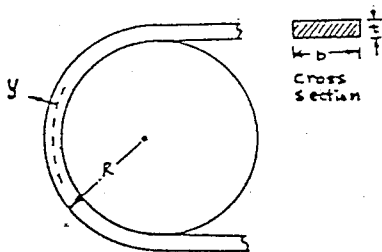
$$7.14 \quad I_{zz} = 2.164 \times 10^7 \text{ mm}^4 = 2.164 \times 10^{-5} \text{ m}^4$$

$$M_{\max} = 5000 \times 6 = 30000 \text{ (N}\cdot\text{m)}$$

$$\therefore \sigma_{\max} = \frac{M_{\max} \cdot y_{\max}}{I_{zz}} = \frac{30000 \times 0.1}{2.164 \times 10^{-5}} = 138632 \times 10^5 \text{ N/m}^2$$

$$\text{Ans. } \sigma_{\max} = 138.632 \text{ MN/m}^2$$

7.15



$$\text{max. strain} = \frac{y_{\max}}{R} = \frac{t/2}{R}$$

$$\text{max. stress} = E \left(\frac{y_{\max}}{R} \right) = \frac{E \cdot t}{2R}$$

$$\frac{E \cdot t}{2R} = 280 \times 10^6$$

$$\therefore t = \frac{2 \cdot \frac{1}{2} (0.3 + t) \times 280 \times 10^6}{200 \times 10^9}$$

$$= 4.206 \times 10^{-4} \text{ m}$$

$$t \text{ 가 } \frac{t}{2} \text{ 로 되면 } \sigma'_{\max} = 280 \times \frac{1}{2} = 140 \text{ MN/m}^2$$

$$\text{Ans. } t = 4.206 \times 10^{-4} \text{ m}, \quad \sigma'_{\max} = 140 \text{ MN/m}^2$$

$$7.16 \quad I_{zz} = \frac{1}{12} (12)(300)^3 + 2 \left[\frac{1}{12} (20)(25)^3 + (20)(25)(1625)^2 \right]$$

$$= 3.048 \times 10^8 (\text{mm}^4) = 3.048 \times 10^{-4} (\text{m}^4)$$

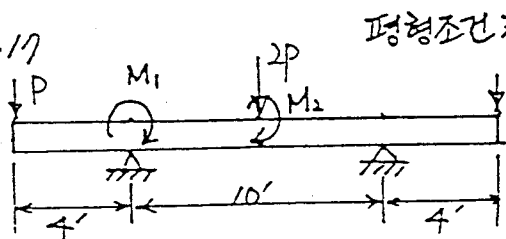
$$M_b = \frac{1}{2} q L x - \frac{1}{2} q x^2$$

$$(M_b)_{\max} = \frac{q}{2} \left(\frac{L^2}{4} \right) = \frac{1}{8} q L^2 = 203906.25 \text{ (N}\cdot\text{m)}$$

$$\therefore \sigma_{\max} = \frac{(M_b)_{\max} \cdot y_{\max}}{I_{zz}} = \frac{203906.25 \times (0.175)}{3.048 \times 10^{-4}} = 117.1 \times 10^6 (\text{N/m}^2)$$

$$\text{Ans. } \sigma_{\max} = 117.1 \text{ MN/m}^2$$

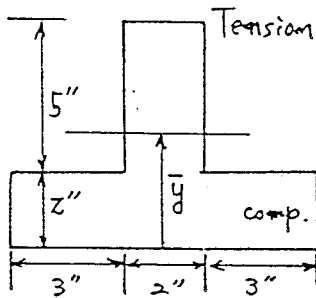
7.17



$$\text{평형조건; } \sum M_1 = 0 \quad M_1 = 4P$$

$$M_2 = 9P - 10P = -P$$

$$\therefore M_{\max} = 4P$$



$$\bar{y} = \frac{(5 \times 8) \times 4.5 + (8 \times 2) \times 1}{26} = 2.346 \text{ (in)}$$

$$I_{zz} = \frac{1}{12} (8)(5)^3 + (8 \times 5)(4.5 - 2.346)^2 + \frac{1}{12} (2)(8)^3 + (2 \times 8)(2.346 - 1)^2 = 101.6 \text{ (in}^4\text{)}$$

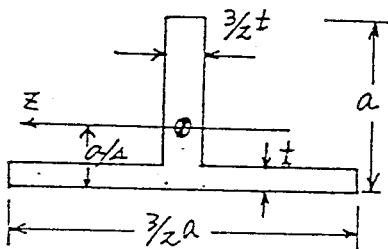
$$\sigma_t = \frac{4P_t}{I_{zz}} (7 - 2.346) = 5000 \quad \therefore P_t = 27280 \text{ (lb)}$$

$$\sigma_c = \frac{4P_c}{I_{zz}} (2.346) = 20,000 \quad \therefore P_c = 216400 \text{ (lb)}$$

$$\therefore P_{\max} = P_t = 27280 \text{ lb (Tension 으로 파괴된다.)}$$

$$\text{Ans. } P_{\max} = 27280 \text{ lb}$$

7.18



$$\bar{y} = \frac{(a)(\frac{3}{2}t)(\frac{a}{2}) + (\frac{3}{2}at)(\frac{a}{2})}{(a)(\frac{3}{2}t) + (\frac{3}{2}at)(t)} = \frac{3a}{4(\alpha + \frac{3}{2})} \quad \text{--- ①}$$

$$(\because t \ll a \text{ 이므로 } t/a \text{ 는 무시})$$

$$(\sigma_t)_{\max} = \frac{(M_b)_{\max} \cdot y_t}{I_{zz}}, (\sigma_c)_{\max} = \frac{(M_b)_{\max} \cdot y_c}{I_{zz}}$$

$$3(\sigma_t)_{\text{failure}} = (\sigma_c)_{\text{failure}}$$

$$\therefore \frac{(\sigma_t)_{\max}}{(\sigma_c)_{\max}} = \frac{1}{3} = \frac{y_t}{y_c} \quad y_t = \bar{y} = \frac{3a}{4(\alpha + \frac{3}{2})}$$

$$y_c = a - \bar{y} = \frac{a(4\alpha + 3)}{4(\alpha + \frac{3}{2})}$$

$$\text{따라서 } \frac{1}{3} = \left(\frac{3a}{4(\alpha + \frac{3}{2})} \right) / \left(\frac{a(4\alpha + 3)}{4(\alpha + \frac{3}{2})} \right) \quad \therefore \alpha = \frac{3}{2}$$

$$I_{zz} = \frac{1}{12} \left(\frac{3}{2}t \right) (a)^3 + \left(\frac{3}{2}at \right) \left(\frac{a}{4} \right)^2 + \frac{1}{12} \left(\frac{3a}{2} \right) t^3 + \left(\frac{3at}{2} \right) \left(\frac{a}{4} \right)^2 = \frac{5}{16} a^3 t$$

$$\alpha = \frac{3}{2} \text{ 을 ①에 대입하면 } y_t = \bar{y} = \frac{a}{4}$$

$$\therefore (\sigma_x)_{\max} = \frac{(M_b)_{\max} \cdot \frac{a}{4}}{\frac{5}{16} a^3 t} = \frac{4}{5} \cdot \frac{(M_b)_{\max}}{a^2 t}$$

$$\text{한편 } (M_b)_{\max} = \frac{1}{8} \omega_0 \cdot l^2 = \frac{1}{8} \omega_0 (10a)^2 = \frac{25}{2} \omega_0 a^2$$

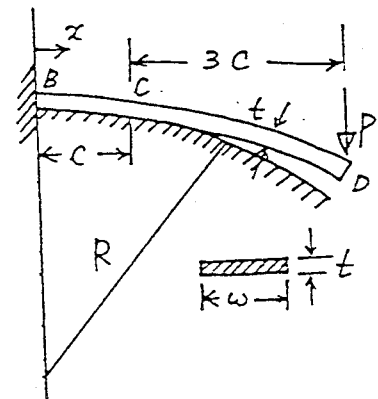
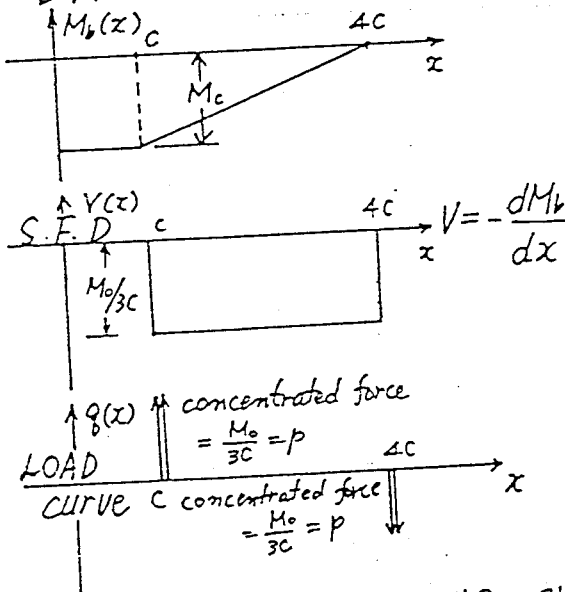
$$\therefore (\sigma_x)_{\max} = \frac{4}{5a^2 t} \left(\frac{25}{2} \omega_0 a^2 \right) = 10 \cdot \frac{\omega_0}{t} \quad \therefore \omega_0 = \frac{t}{10} \sigma_x$$

$$\text{Ans. } \alpha = \frac{3}{2}, \quad \omega_0 = \frac{t}{10} \sigma_x$$

7.19 구간 BC에서 strip의 곡률은 일정하다. 따라서 BC구간의 Bending moment는 일정하다.

$$M_0 = -(EI/R) = -\frac{1}{12} (Ewt^3/R) = -P(3C)$$

$\therefore$ B.M.D



$$q = -\frac{dV}{dx}$$

Strip과 block 사이의 반작용은 집중 하중이다.

$$P = \frac{M_0}{3C} = \left(\frac{1}{3C} \right) \frac{1}{12} \frac{Ewt^3}{R} = \frac{Ewt^3}{36CR}$$

$$\text{Ans. } P = Ewt^3/36CR$$

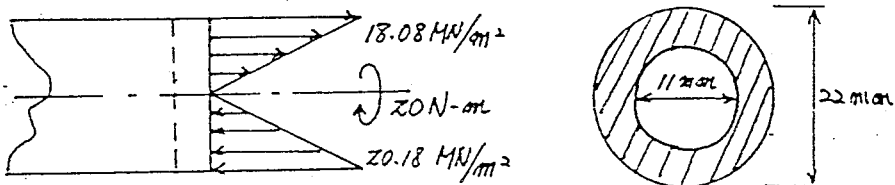
$$7.20 \quad M_b = 400 \times 0.05 = 20 \text{ (N-m)}$$

$$(a) I_{zz} = \frac{\pi d^4}{64} = \frac{\pi}{64} (22 \times 10^{-3})^4 = 1.150 \times 10^{-8} \text{ (m}^4\text{)}$$

$$\text{Bending stress at B } (\sigma_{\max})_B = \frac{20 \times 0.011}{1.15 \times 10^{-8}} = 19.13 \text{ (MN/m}^2\text{)}$$

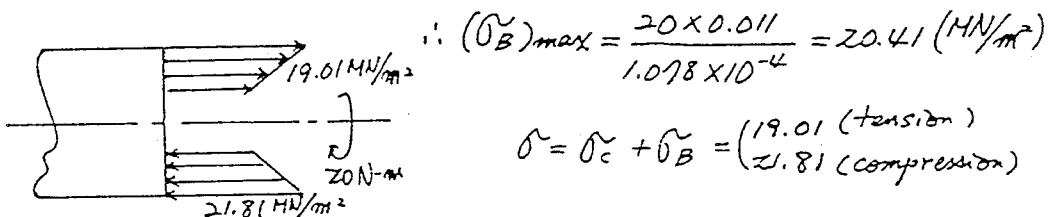
$$\text{Compressive stress } \sigma_c = \frac{-P}{\frac{\pi}{4} d^2} = \frac{-400}{\frac{\pi}{4} (22 \times 10^{-3})^2} = -1.052 \text{ (MN/m}^2\text{)}$$

$$\sigma = \sigma_B + \sigma_c = 18.08 \text{ MN/m}^2 \text{ (tension), } 20.18 \text{ MN/m}^2 \text{ (compression)}$$



$$(b) \sigma_c = \frac{-400}{\frac{\pi}{4} \{ (22 \times 10^{-3})^2 - (11 \times 10^{-3})^2 \}} = -1.403 \text{ (MN/m}^2\text{)}$$

$$I_{zz} = \frac{\pi}{64} \{ (22 \times 10^{-3})^4 - (11 \times 10^{-3})^4 \} = 1.078 \times 10^{-8} \text{ (m}^4\text{)}$$



$$\therefore (\sigma_B)_{\max} = \frac{20 \times 0.011}{1.078 \times 10^{-8}} = 20.41 \text{ (MN/m}^2\text{)}$$

$$\sigma = \sigma_c + \sigma_B = (19.01 \text{ (tension)})$$

$$21.81 \text{ (compression)}$$

$$(c) \text{ tension ; } \left(\frac{19.01 - 18.08}{18.08} \right) \times 100 = 5.144 \text{ (\%)}$$

$$\text{compression ; } \left(\frac{21.81 - 20.18}{20.18} \right) \times 100 = 8.077 \text{ (\%)}$$

Ans. (c) tension 5.144%, compression 8.077%

$$7.21 \quad (M_B)_{\max} = 1.8 \times 26000 = 46800 \text{ (N-m)}, \quad I_{zz} = \frac{b^4}{12}$$

$$\tau_u = \frac{P}{A} = \frac{2600}{b_s^2} = 10 \times 10^6 \quad \therefore b_s = 0.016 \text{ m}$$

$$\sigma_u = \frac{M \cdot y}{I} = \frac{12}{b_t^4} (46800) \cdot \frac{b_t}{2} = 52 \times 10^6 \quad \therefore b_t = 0.01754 \text{ m}$$

b_s or b_t 중이서 큰것을 b 로 택한다. $\therefore b = 0.1754 \text{ m}$

Ans. $b = 0.1754 \text{ m}$

7.22 Bending stress $\sigma_b = -\frac{M \cdot y}{I_{zz}}$ $I_{zz} = \pi r^3 t$, $y = r \sin \theta$

$$\therefore \sigma_b = \frac{M(r \sin \theta)}{\pi r^3 t} = -\frac{M}{\pi r^2 t} \sin \theta$$

Ans. $\sigma_b = -\frac{M}{\pi r^2 t} \sin \theta$

7.23 유체의 비중: γ $\delta = \pi r^2 \cdot \gamma$; 단위 길이당 하중

$$(M_b)_{\max} = \frac{1}{8} \delta L^2 = \frac{1}{8} \pi r^2 \cdot \gamma \cdot L^2$$

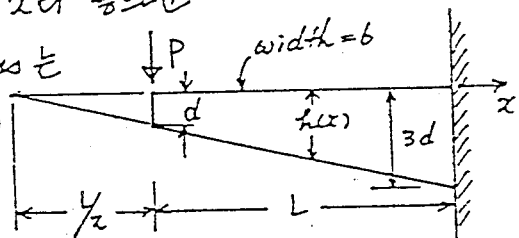
$$\therefore \sigma_{\max} = \frac{\frac{1}{8} \pi r^2 L^2 \cdot \gamma}{\pi r^3 t} \cdot r = \frac{\gamma L^2}{8t}$$

-tank의 반경이
무관하다.
- Q.E.D -

7.24 자유단에서 복합의 거리를 x 라 놓으면

x 점에서의 maximum stress는

다음과 같다. $\frac{M_b(x) \cdot \frac{h(x)}{2}}{I_{yy}(x)}$



$$M_b(x) = P \cdot x$$

$$I_{yy}(x) = \frac{1}{12} b h^3 = \frac{1}{12} b d^3 \left[1 + \frac{2x}{L}\right]^3$$

$$(\sigma_x) = \frac{P \cdot x \cdot \frac{1}{2} d \left(1 + \frac{2x}{L}\right)}{\frac{1}{12} b d^3 \left(1 + \frac{2x}{L}\right)^3} = \frac{6Px}{bd^2 \left(1 + \frac{2x}{L}\right)^2}$$

최대가 되기 위해서 $\frac{d}{dx} \left(\frac{x}{\left(1 + \frac{2x}{L}\right)^2} \right) = 0 \therefore x = \frac{L}{2}$

$$\therefore (\sigma_x)_{\max} = \frac{6P \cdot \frac{L}{2}}{bd^2 (2)^2} = \frac{3PL}{4bd^2}$$

Ans. $x = \frac{L}{2}$ $\sigma_{\max} = \frac{3PL}{4bd^2}$

7.25 plane section은 변형후에도 plane section을 유지한다고 가정하자.

$$\text{그러면, } \sigma_{\max} = \frac{M_b \cdot y_{\max}}{I_{yy}}$$

I_{yy} 의 계산; shell 자체의 2차 moment와

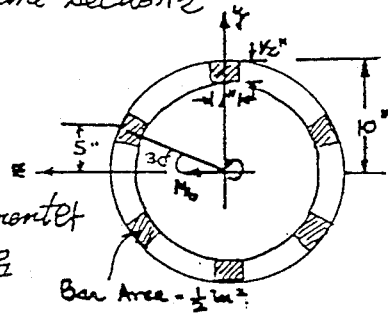
bar 자체의 moment는 미소하므로

무시한다. $\therefore I_{yy} = A \bar{y}^2$

$$I_{yy} = \left(\frac{1}{2}\right) [10^2 + 10^2 + 5^2 + 5^2 + 5^2 + 5^2] = 150 \text{ in}^4$$

$$\therefore \sigma_{\max} = \frac{100,000 \times 10}{150} = 6667 \text{ psi}$$

$$\text{Ans. } \sigma_{\max} = 6667 \text{ psi}$$



7.26 $\rho = 6 \mu\text{m}$, $M_b = 5 \times 10^{-7} \text{ N}\cdot\text{m}$, $(I_{yy})_{\text{total}} = 4 \times 10^{-8} \text{ mm}^4$

$$E = \frac{M\rho}{I} = \frac{(5 \times 10^{-7})(6 \times 10^{-6})}{(4 \times 10^{-8} \times 10^{-12})} = 75 \text{ (MN/m}^2\text{)} \quad (\rho: \text{curvature})$$

$$\text{Ans. } E = 75 \text{ MN/m}^2$$

7.27 $\tau_{yz}(y_1) = \frac{4P}{3\pi r^4} (r^2 - y_1^2)$. 등재가 1 이므로

$$U_x = \frac{1}{2G} \int_V \left(\frac{\tau_y}{J}\right)^2 dV = \frac{1}{2G} \int_A \left(\frac{\tau_y}{J}\right)^2 2\pi y dy \quad (T=PR, J=\frac{\pi r^4}{2})$$

$$\therefore U_x = \frac{1}{2G} \cdot \frac{1}{\pi^2 r^8} P^2 R^2 \int_0^r y^2 2\pi y dy = \frac{P^2 R^2}{\pi G r^4}$$

$$U_s = \int_V \frac{\tau^2}{2G} dV \text{ 이고 등재가 1 이므로, } U_s = \int_A \frac{\tau^2}{2G} dA$$

$$U_s = \frac{1}{2G} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \tau_{yz}^2 (2r \cos \theta) (r \cos \theta) d\theta$$

$$\tau_{yz} = \frac{4P}{3\pi r^4} (r^2 - y^2) = \frac{4P}{3\pi r^4} (r \cos \theta)^2 = \frac{4P}{3\pi r^2} \cos^2 \theta$$

$$\therefore U_s = \frac{r^2}{G} \left(\frac{4P}{3\pi r^2} \right)^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6 \theta d\theta = \frac{5P^2}{9\pi G r^2}$$

$$\text{따라서 } \frac{U_s}{U_t} = \frac{5}{9} \left(\frac{r}{R} \right)^2$$

$$\text{Ans. } U_t = \frac{P^2 R^2}{\pi G r^4}, \quad U_s = \frac{5P^2}{9\pi G r^2}$$

7.28 Trial I $P \approx 1 \text{ (lb)}$; with perfectly sharp point

$$\sigma = \frac{M \cdot y}{I_{yy}}; \quad M = Px$$

$$r = \frac{3}{64} \left(\frac{x}{1/4} \right) = \frac{3}{16} x$$

$$y = r$$

$$I_{yy} = \frac{\pi r^4}{4} = \frac{\pi}{4} \left(\frac{3}{16} \right)^4 x^4$$

$$\therefore \sigma(x) = \frac{(Px) \left(\frac{3}{16} x \right)}{\frac{\pi}{4} \left(\frac{3}{16} \right)^4 x^4} = 192 \frac{P}{x^2}$$

연필심이 날카로우수록 $x \rightarrow 0$ 가 되어 $\sigma \rightarrow \infty$ 로 된다.
따라서 파괴하려는 경향이 농후해 진다.

Trial II

연필 끝은 $1/64$ " 가 편평하고, 하중 1(lb) 라 가정하자.

$$M = P \left(x - \frac{1}{24} \right)$$

r, y 는 I 에 따라 같다.

$$\sigma = \frac{P \left(x - \frac{1}{24} \right) \left(\frac{3}{16} x \right)}{\frac{\pi}{4} \left(\frac{3}{16} \right)^4 x^4}$$

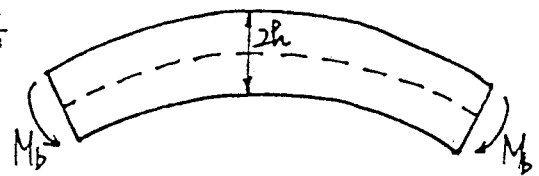
$$= \frac{192P \left(x - \frac{1}{24} \right)}{x^3} \quad \frac{1}{24} \text{ " } \leq x \leq \frac{1}{4} \text{ "}$$

$$x = 3/48 \text{ " 이 } \sigma_{\max} = 16,300 \text{ psi}$$

7.29 2개의 bar 를 soldering 하였으므로 높이가 2인 하나의 bar 가 된다.

따라서 $\frac{d\phi}{ds} = \frac{M_b}{EI_{yy}} = \frac{12}{8} \frac{M_b}{Eb h^3}$

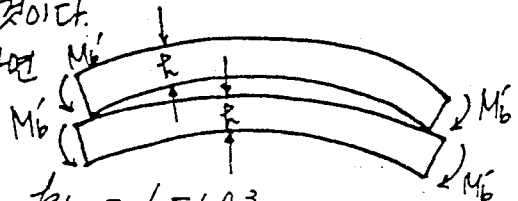
$\frac{M_b}{d\phi/ds} = k_{b1} = \frac{2}{3} Eb h^3$



seperate 되어 있는 경우에는 각각의 curvature $d\phi/ds$ 로 구부러지고 양단에서만 집중할 것이다.

각각의 moment를 M_b' 이라 하면

$M_b' = E \left(\frac{1}{12} b h^3 \right) \frac{d\phi}{ds}$



$M_b = 2M_b' = \frac{1}{6} Eb h^3 \frac{d\phi}{ds} \therefore k_{b2} = \frac{1}{6} Eb h^3$

Ratio of stiffness ($\frac{\text{soldered}}{\text{separate}}\bigg) = \frac{k_{b1}}{k_{b2}} = 4.0$

max. bending stress의 비는 max. bending strain의 비와 같다. 각각의 경우 max. strain은

max. strain = $\frac{d\phi}{ds} \left(\frac{1}{2} \text{ beam height} \right)$

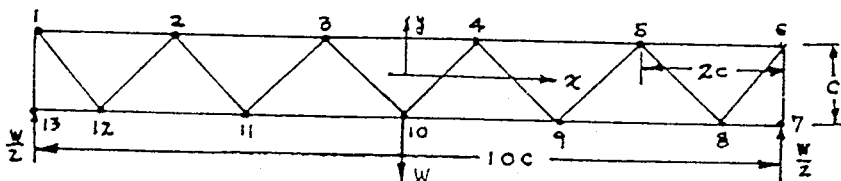
Soldered beam; $\frac{d\phi}{ds} = \frac{M_b}{k_{b1}}; \epsilon_{max} = \frac{2h}{2} \left(\frac{M_b}{k_{b1}} \right) = \frac{h M_b}{k_{b1}}$

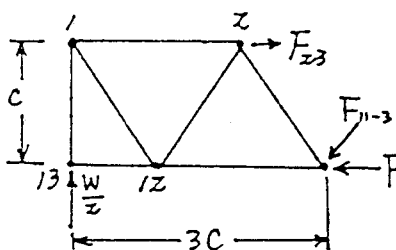
separate beam; $\frac{d\phi}{ds} = \frac{M_b}{k_{b2}}; \epsilon_{max} = \frac{h}{2} \left(\frac{M_b}{k_{b2}} \right) = \frac{h M_b}{2 k_{b2}}$

$\therefore$ Ratio of stresses; $\left(\frac{\text{soldered}}{\text{separate}} \right) = \left(\frac{h M_b / k_{b1}}{h M_b / 2 k_{b2}} \right) = \frac{1}{2}$

Ans. $\left(\frac{\text{Soldered}}{\text{Seperate}} \right)_{\text{stiffness}} = 4. \quad \left(\frac{\text{Soldered}}{\text{Seperate}} \right)_{\text{stress}} = \frac{1}{2}$

7.30





Equilibrium ;

$\sum M_{11} = 0$ (pin-joint transfers no moment)

$$\therefore \frac{W}{2}(3c) + F_{23}(c) = 0$$

$$F_{23} = -\frac{3}{2}W$$

이제 방법 2. $F_{12} = -\frac{W}{2}$; $F_{23} = -\frac{3}{2}W$; $F_{34} = -\frac{5}{2}W$, $F_{45} = -\frac{3}{2}W$

$$F_{56} = -\frac{W}{2} ; F_{78} = 0 ; F_{89} = \frac{2}{2}W ; F_{910} = \frac{4}{2}W ; F_{1011} = \frac{4}{2}W ; F_{1112} = \frac{2}{2}W ;$$

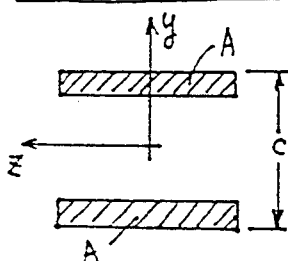
$$F_{1213} = 0$$

따라서 stress ($\sigma = F/A$) 는 다음과 같다.

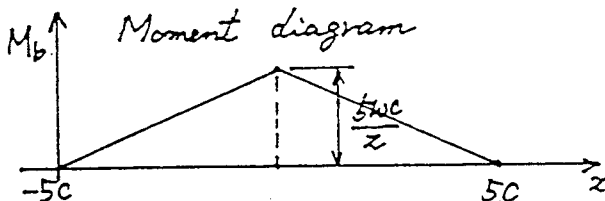
$$\sigma_{12} = -\frac{W}{2A} ; \sigma_{23} = -\frac{3W}{2A} ; \sigma_{34} = -\frac{5W}{2A} ; \sigma_{45} = -\frac{3W}{2A} ; \sigma_{56} = -\frac{W}{2A}$$

$$\sigma_{78} = 0 ; \sigma_{89} = \frac{W}{A} ; \sigma_{910} = \frac{2W}{A} ; \sigma_{1011} = \frac{2W}{A} ; \sigma_{1112} = \frac{W}{A} ; \sigma_{1213} = 0$$

Bending Theory ; 단을 bending으로 가정하고, 대각선 부재는 무시.

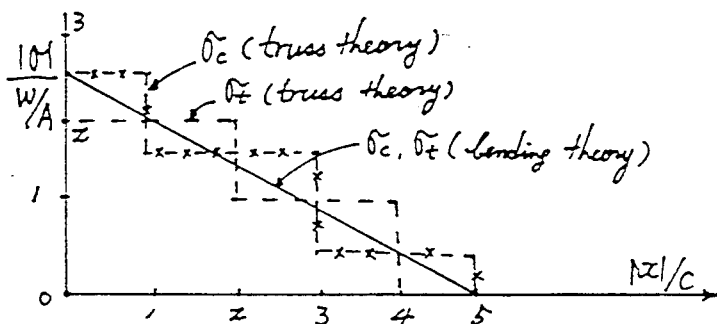


$$I_{yy} \approx 2[A(\frac{c}{2})^2] = \frac{Ac^2}{2}$$



$$M_b = \frac{5Wc}{2} \left[1 - \frac{|z|}{5c} \right]$$

$$\sigma_T = |\sigma_c| = \frac{M_b(\frac{c}{2})}{I_{yy}} = \frac{5W}{2A} \left[1 - \frac{|z|}{5c} \right]$$



대각선 부재는 shear force 를 전달하고, beam 의 곡률에 의해 생기는 y 방향의 작은 힘을 전달한다.

7.31 bending 이므로 strain $\epsilon = \frac{y}{\rho} \therefore \sigma = C|\epsilon|^n = \frac{C}{\rho^n} |y|^n$

$$M_b = 2 \int_0^{\frac{h}{2}} \sigma \cdot b \cdot y \, dy = \frac{2bc}{\rho^n} \int_0^{\frac{h}{2}} y^{n+1} \, dy = \frac{2bc}{(n+2)\rho^n} \left(\frac{h}{2}\right)^{n+2}$$

$$\therefore \frac{C}{\rho^n} = \frac{(n+2)M_b}{2b \left(\frac{h}{2}\right)^{n+2}}$$

$$\sigma_{\max} = \frac{C}{\rho^n} \left|\frac{h}{2}\right|^n = \frac{(n+2)M_b}{2b \left(\frac{h}{2}\right)^{n+2}} \left(\frac{h}{2}\right)^n = \frac{2(n+2)}{bh^2} M_b$$

Ans. $\sigma_{\max} = \frac{2(n+2)}{bh^2} M_b$

7.32 $u = -\frac{xy}{\rho} ; v = \frac{x^2 + \nu(y^2 - z^2)}{2\rho} ; w = \frac{\nu yz}{\rho}$

$$\epsilon_x = \frac{\partial u}{\partial x} = -\frac{y}{\rho} ; \epsilon_y = \frac{\partial v}{\partial y} = \nu \frac{y}{\rho} ; \epsilon_z = \frac{\partial w}{\partial z} = \frac{\nu y}{\rho}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = -\frac{x}{\rho} + \frac{x}{\rho} = 0$$

$$\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = -\frac{\nu z}{\rho} + \frac{\nu z}{\rho} = 0$$

$$\gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} = 0 + 0 = 0.$$

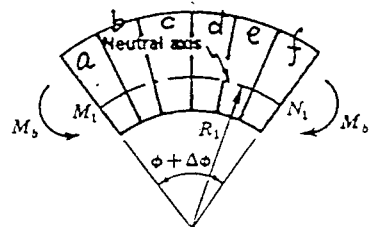
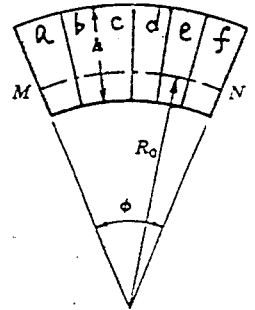
7.33 변형하기 전에 circular segments

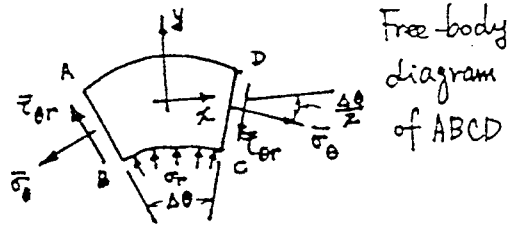
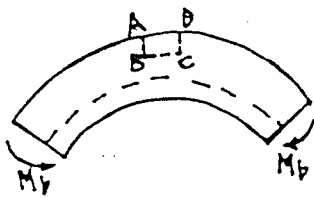
a.b.c.d...는 같은 하중을 받으므로 같은 형태로 변형할 것입니다. y축에 대하여 회전을 해도, 회전하기 전과 변형이 없을 것이다.

$\therefore$ 이것은 curve 면에 수직한 flat end를 갖는 segment 만이 이것을 만족한다.

radial plane section은 변형 후에도 plane을 유지한다. 또 longitudinal strain은 없으므로 증감폭은 변형이 없다.

$$R_0 \phi = R_1 (\phi + \Delta \phi), \quad \frac{1}{R_1} = \frac{\phi + \Delta \phi}{R_0 \phi} = \frac{1}{R_0} + \frac{\Delta \phi}{R_0 \phi} \quad \text{--- ①}$$





Free-body
diagram
of ABCD

$\Sigma F_y = 0$, AB, CD에서의 shear stress $\tau_{\theta r}$ 의 y성분은 서로 상쇄된다.
Normal stress σ_{θ} 는 y방향의 성분이 "0"이 아니다. 이것과 평형을 이루기 위해서 σ_r 이 존재한다.

Tangential strain distribution;

중립축으로부터 y만큼 떨어진 circular arc를 생각해 보면.

$$\text{Length before deformation} = (R_0 + y)\phi$$

$$\text{Length after deformation} = (R_1 + y)(\phi + \Delta\phi)$$

$$\text{Tangential strain } \epsilon_{\theta} = \frac{(R_1 + y)(\phi + \Delta\phi) - (R_0 + y)\phi}{(R_0 + y)\phi}$$

①식을 사용하면

$$\epsilon_{\theta} = \frac{R_0 - R_1}{R_1} \left(\frac{y}{R_0 + y} \right)$$

$R_0 \gg y$ 가 아니면 non-linear이다.

7.34 geometrical results;

$$\epsilon_x = -\frac{1}{\rho}(y - y_N) \quad \text{--- ①}$$

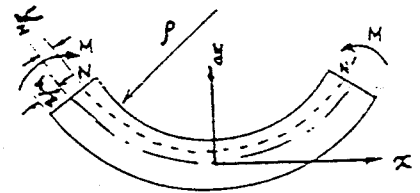
$$\epsilon_z = \text{constant} = \epsilon_0 \quad \text{--- ②}$$

equilibrium; $\sigma_y = 0$ ($k/\rho \ll \sigma_{\theta}$)

$$\Sigma F_z = 0; \int_{-k/2}^{k/2} \sigma_x dy = 0 \quad \text{--- ③}$$

$$\Sigma F_z = 0; \int_{-k/2}^{k/2} \tilde{\sigma}_z dy = 0 \quad \text{--- ④}$$

$$\Sigma M_z = 0; -\int_{-k/2}^{k/2} y \cdot \sigma_x dy \quad \text{--- ⑤}$$



<Neutral Axis는
centerline으로 부터
 y_N 만큼의 거리에 NN'이다>

stress-strain

$$\epsilon_x = \frac{1}{E} (\sigma_x - \nu \sigma_y - \nu \sigma_z) \quad \text{--- ②}$$

$$\epsilon_z = \frac{1}{E} (\sigma_z - \nu \sigma_x - \nu \sigma_y) \quad \text{--- ③}$$

$$\text{①, ②, ③, ④} ; \quad \sigma_x = \frac{E}{1-\nu^2} \left[-\frac{(y-y_N)}{\rho} + \nu \epsilon_0 \right], \quad \sigma_z = \frac{E}{1-\nu^2} \left[\epsilon_0 - \frac{\nu(y-y_N)}{\rho} \right]$$

$$\Sigma F_x = 0, \quad \Sigma F_z = 0, \quad \left(\frac{y_N}{\rho} + \nu \epsilon_0 \right) h = 0, \quad \left(\epsilon_0 + \frac{\nu y_N}{\rho} \right) h = 0$$

$$\therefore y_N = \epsilon_0 = 0$$

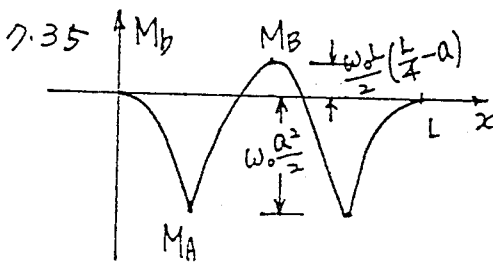
$$\therefore \sigma_x = -\frac{E}{1-\nu^2} \cdot \frac{y}{\rho}$$

$$\Sigma M_z = 0: \quad M = -\int_{-h/2}^{h/2} y \left[-\frac{E}{1-\nu^2} \cdot \frac{y}{\rho} \right] dy = \frac{E h^3}{12(1-\nu^2) \cdot \rho}$$

$$\therefore \frac{d\phi}{ds} = \frac{1}{\rho} = \frac{12(1-\nu^2) \cdot M}{E h^3}$$

$$\sigma_x = -\frac{E y}{1-\nu^2} \left(\frac{1}{\rho} \right) = \frac{12 M y}{h^3}$$

$$\text{Ans. } \sigma_x = \frac{12 M y}{h^3}$$



$$M_A = M_B;$$

$$\frac{\omega_0 \cdot L}{2} \left(\frac{L}{4} - a \right) = \omega_0 \frac{a^2}{2}$$

$$\therefore 4a^2 + 4La - L^2 = 0$$

$$\therefore a = \frac{-2 \pm \sqrt{4+4}}{4} L \quad \text{or } 1/4$$

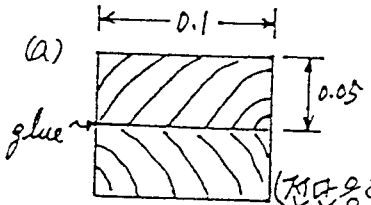
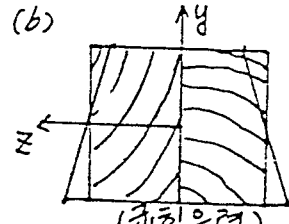
$$a = 0.2071 L = 178.1 \text{ mm} \quad (a > 0)$$

$$\therefore M = \frac{1}{2} \omega_0 a^2 = 0.01586 \omega_0$$

$$\sigma_{\max} = \frac{M y}{I} = \frac{(0.01586 \omega_0)(3 \times 10^{-3})}{\frac{1}{12} (190 \times 10^{-3})(6 \times 10^{-3})^3} = 7 \times 10^6$$

$$\therefore \omega_0 = 503.1 \text{ N/m}$$

$$\text{Ans. } \omega_0 = 503.1 \text{ N/m}$$

7.36 (a)  $\tau_{max} = \frac{VQ}{bI} = \frac{P \times \frac{1}{2} \times (0.1) \times (0.05)^2}{(0.1) \times \frac{1}{12} \times (0.1)^4} = 150P$
 $\therefore \tau_{max} = 150P \text{ (N/m}^2\text{)}$
 (전단응력)
 (b)  $\epsilon_x = \frac{M \cdot y}{E \cdot I_{zz}} = \frac{3P \times (0.05)}{E \times \frac{1}{12} \times (0.1)^4} = 18000 \frac{P}{E}$
 $|\epsilon_z| = |\nu \epsilon_x| = 18000 \frac{\nu P}{E}$
 $\sigma_z = E |\epsilon_z| = 18000 \nu P$
 $\therefore \nu = 0.4$ 로 놓으면 $\sigma_z = 7200 P \text{ (N/m}^2\text{)}$
 Ans. (a) $\tau = 150P \text{ (N/m}^2\text{)}$ (b) $\sigma_z = 7200 P \text{ (N/m}^2\text{)}$

7.37 (a) just above the solder joint

$$Q = 2 \times \left(\frac{1}{4}\right) (4+1) = 2.5 \text{ (in}^3\text{)} \quad \therefore q = \frac{VQ}{I_{zz}} = \frac{6000 \times 2.5}{75} = 200 \text{ (lb/in)}$$

(b) just below solder joint

$$Q = 2.5 + 4.7 = 7.2 \text{ (in}^3\text{)} \quad \therefore q = \frac{6000 \times 7.2}{75} = 575 \text{ (lb/in)}$$

(c) at the neutral surface

$$Q = 6 \times \left(\frac{1}{4}\right) \times 3 + 4.7 = 9.2 \text{ (in}^3\text{)} \quad \therefore q = \frac{6000 \times 9.2}{75} = 740 \text{ (lb/in)}$$

Ans a) 200 lb/in b) 575 lb/in c) 740 lb/in

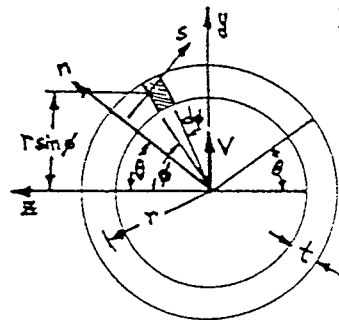
7.38 $I_{yy} = \pi r^3 t \quad (t/r \ll 1)$

$Q =$ 바깥쪽 부분의 1차 moment

$$= 2 \int_0^{\pi/2} (r \sin \phi) (t r d\phi)$$

$$= 2 r^2 t \cos \theta$$

$$q_{sx} = q_{xs} = \frac{1}{2} \left(\frac{VQ}{I_{yy}} \right) = \frac{V}{\pi r} \cos \theta$$



(바깥쪽 부분의 양쪽 면을 곱하므로)

Ans. $q_{sx} = \frac{V}{\pi r} \cos \theta$

7.39 못이 접착면 A에서 shear flow를 전달한다.

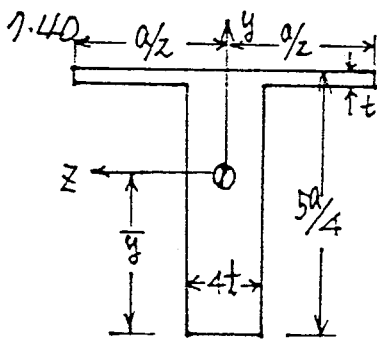
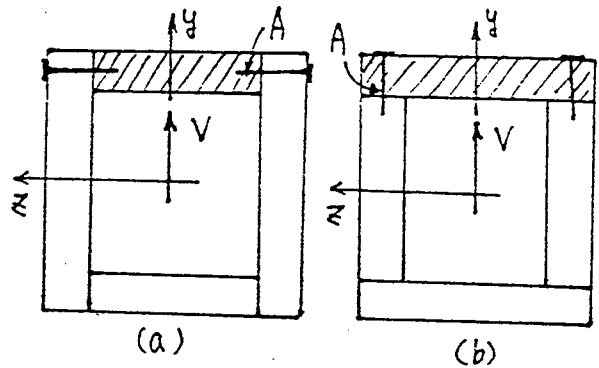
(a), (b)에서 다른 모든 조건은 같기 때문에 전달하는 shear flow가 작을수록 좋다.

$$\text{shear flow} = \frac{VQ}{I_{yy}}$$

Q ; 빔의 면의 1차 moment (a)가 (b)보다 작다.

I_{yy} ; (a), (b) 서로 같다.

Ans. (a)쪽의 설계가 (b)보다 좋다.



$$\bar{y} = \frac{(at)(\frac{5}{8}a) + (4t)(\frac{5}{8}a)(\frac{5}{8}a)}{(at) + (4t)(\frac{5}{8}a)} = 0.73a$$

$$(\sigma_{\max})_{\max} = \frac{M}{I_{yy}} (y_{\max})_{\text{stem}} = \frac{M}{I_{yy}} (0.73a)$$

$$(\sigma_{\max})_{\text{flange}} = \frac{M}{I_{yy}} (y_{\max})_{\text{flange}} = \frac{M}{I_{yy}} (0.52a)$$

$$\therefore \frac{(\sigma_{\max})_{\text{stem}}}{(\sigma_{\max})_{\text{flange}}} = \frac{0.73a}{0.52a} = 1.40$$

$(\tau_{xy})_{\max}$ 은 $y=0$ 이다

$$Q = \frac{1}{2}(4t)(0.73a)(\frac{0.73a}{2}) = 1.06 a^2 t$$

$(\tau_{xy})_{\max}$ 은 flange와 stem의 연결점에서

$$Q(\text{half flange}) = (\frac{a}{2})(t)(0.52a) = 0.26 a^2 t$$

$$\therefore \frac{(\tau_{xy})_{\text{stem}}}{(\tau_{xy})_{\text{flange}}} = \left(\frac{V}{I_{yy}} \frac{1.06 a^2 t}{4t} \right) / \left(\frac{V}{I_{yy}} \frac{0.26 a^2 t}{t} \right) = 1.02$$

Ans. $(\sigma_{xs}/\sigma_{xf})_{\max} = 1.4$

$(\tau_{xys}/\tau_{xyf})_{\max} = 1.02$

7.41 Bolt는 빗금친 부분의 양단에서 shear flow를 전달한다.

$$I_{yy} = \frac{1}{12} [(6)(10)^3 - (4)(8)^3] = 330 \text{ in}^4$$

빗금친 부분의 1차 moment : Q

$$Q = (1)(2)(4.5) = 9 \text{ in}^3$$

q_{zx} 를 각 끝에서의 shear flow라

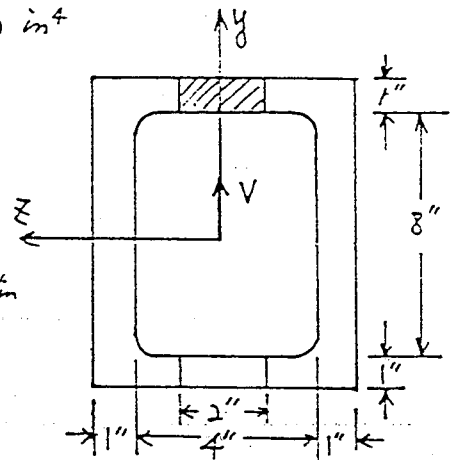
$$\text{하면, } 2q_{zx} = \frac{VQ}{I_{yy}} = \frac{(10,000)(9)}{330} = 272 \text{ lb/in}$$

$$\therefore q_{zx} = 136 \text{ lb/in}$$

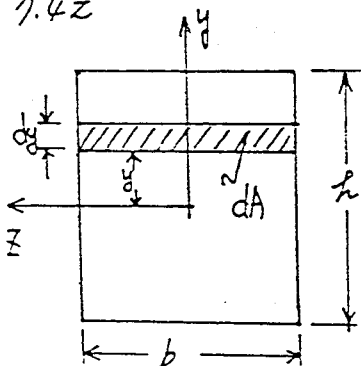
Total shear flow (1'인 S에 대한)은

$$17개의 bolt에 의해서 전달된다 $\therefore 136 \times S = 460 \quad S = 2.95''$$$

$$\text{Ans. } S = 2.95''$$



7.42



$$\tau_{xy} = \frac{V}{2 I_{yy}} \left[\left(\frac{h}{2} \right)^2 - y^2 \right] \quad \left(\tau = \frac{VQ}{bI} \right)$$

$$\text{따라서 Shear force} = \int_A \tau_{xy} dA = \int_{y=-h/2}^{h/2} \tau_{xy} \cdot b dy$$

$$= \frac{Vb}{2 I_{yy}} \int_{-h/2}^{h/2} \left[\left(\frac{h}{2} \right)^2 - y^2 \right] dy = \frac{Vb}{2 I_{yy}} \left[h \left(\frac{h}{2} \right)^2 - \frac{1}{3} \left(\frac{h}{2} \right)^3 \right]$$

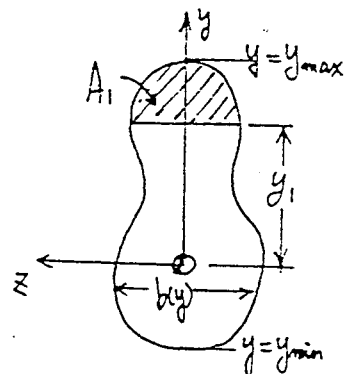
$$= \frac{V}{I_{yy}} \left(\frac{1}{12} b h^3 \right) = \frac{V}{I_{yy}} \cdot I_{yy} = V \quad - Q.E.D.$$

$$7.43 \quad Q = \int_{A_1} y dA = \int_{y_1}^{y_{max}} b y dy = - \int_{y_{max}}^{y_1} b y dy$$

$$\therefore \frac{dQ}{dy_1} = -b y_1$$

$$\int_A \tau_{xy} dA = \int_A \frac{VQ}{b I_{yy}} dA = \frac{V}{I_{yy}} \int \frac{Q}{b} (b dy)$$

($b = b(y)$; a function of y)



$$= \frac{V}{I_{yy}} \int_{y_{\min}}^{y_{\max}} Q dy = \frac{V}{I_{yy}} \left[Qy - \int y \frac{dQ}{dy} dy \right]_{y_{\min}}^{y_{\max}}$$

$$= \frac{V}{I_{yy}} \left[Qy + \int by^2 dy \right]_{y_{\min}}^{y_{\max}} \quad \text{팔레트의 첫 항은 } Q=0 \text{ 이므로}$$

(at $y=y_{\max}$, $y=y_{\min}$)

$$\therefore \int_A \tau_{xy} dA = \frac{V}{I_{yy}} \int_{y_{\min}}^{y_{\max}} y^2 (b dy) = \frac{V}{I_{yy}} \int_A y^2 dA = \frac{V}{I_{yy}} (I_{yy}) = V$$

$$7.44 \quad \sigma_{\max} = \frac{M_b}{I_{zz}} y_{\max} = \frac{M_b}{I_{zz}} \left(\frac{h}{2} \right) = \left(\frac{1}{8} \omega_0 L^2 \cdot h \right) / \left(\frac{bh^3}{12} \cdot \frac{1}{2} \right) = \frac{3}{4} \cdot \frac{\omega_0 L}{bh^2}$$

$$(\tau_{yz})_{\max} = \frac{V \cdot Q_{\max}}{I_{zz} \cdot b_{\min}} = \left(\frac{1}{2} \omega_0 L \cdot \frac{h^2 b}{8} \cdot \frac{1}{b} \right) / \left(\frac{bh^3}{12} \right) = \frac{3}{4} \cdot \frac{\omega_0 L}{bh}$$

$$\therefore (\sigma/\tau)_{\max} = L/h$$

$$\text{Ans. } (\sigma/\tau)_{\max} = L/h$$

$$7.45 \quad Q_{\max} = (0.75h)(2t)\left(\frac{h}{2}\right) + (0.5h)(t)(0.25h)$$

$$= 0.875th^2$$

$$I_{zz} = \frac{1}{12} h^3 + 2 \left(\frac{3}{4} h \right) (2t) \left(\frac{h}{2} \right)^2 = \frac{5}{6} th^3$$

($\because$ flange 의 자중이 22 moment 는 미소하므로 무시)

$$(M_b)_{\max} = \frac{1}{4} Ph \quad V = \frac{1}{2} P$$

$$\sigma_{\max} = \frac{(M_b)_{\max} \cdot y_{\max}}{I_{zz}} = \frac{\frac{1}{4} Ph \cdot \frac{h}{2}}{\frac{5}{6} th^3} = \frac{3Ph}{20h^2t}$$

$$\tau_{\max} = \frac{V \cdot Q_{\max}}{b_{\min} I_{zz}} = \frac{\frac{1}{2} P \cdot 0.875th^2}{t \cdot \frac{5}{6} th^3} = \frac{21}{40} \cdot \frac{P}{h \cdot t}$$

$$\therefore (\sigma/\tau)_{\max} = \frac{2}{7} \cdot \frac{L}{h}$$

$$\text{Ans. } (\sigma/\tau)_{\max} = \frac{2}{7} \cdot \frac{L}{h}$$

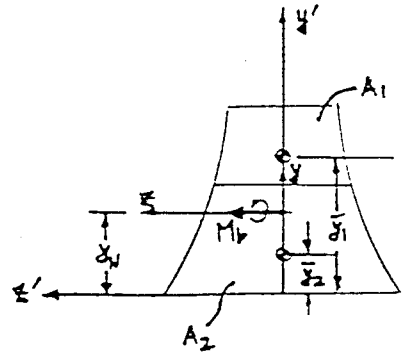
7.46 (a) $\epsilon_x = -\frac{d\phi}{ds} y$

기하학적 적합조건 ; $y = y' - y_N$

평형조건 ; $\sigma_y = 0, \sigma_z = 0$

$$\sum F_x = 0 ; - \int_A \sigma_x dA = 0$$

$$\sum M_z = 0 ; -M_b + \int_A \sigma_x y dA = 0$$



stress-strain relation ;

$$(\sigma_x)_i = E_i (\epsilon_x)_i = -E_i \left(\frac{d\phi}{ds} \right) y \quad (i=1, 2)$$

$$\int_A \sigma_x dA = -E_1 \left(\frac{d\phi}{ds} \right) \int_{A_1} (y' - y_N) dA - E_2 \left(\frac{d\phi}{ds} \right) \int_{A_2} (y' - y_N) dA = 0$$

$$\text{한편 } \int_{A_1} y' dA = \bar{y}_1 A_1, \quad \int_{A_2} y' dA = \bar{y}_2 A_2$$

$$\therefore -E_1 \left(\frac{d\phi}{ds} \right) (\bar{y}_1 - y_N) A_1 - E_2 \left(\frac{d\phi}{ds} \right) (\bar{y}_2 - y_N) A_2 = 0$$

$$\therefore y_N = \frac{E_1 \bar{y}_1 A_1 + E_2 \bar{y}_2 A_2}{E_1 A_1 + E_2 A_2} \quad - Q.E.D. -$$

$$(b) M_b = - \int_A \sigma_x \cdot y dA = E_1 \left(\frac{d\phi}{ds} \right) \int_{A_1} (y' - y_N)^2 dA + E_2 \left(\frac{d\phi}{ds} \right) \int_{A_2} (y' - y_N)^2 dA$$

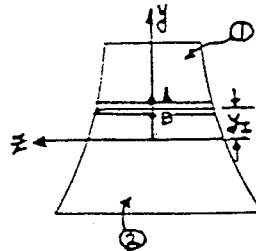
$$= E_1 \left(\frac{d\phi}{ds} \right) (I_{yy})_1 + E_2 \left(\frac{d\phi}{ds} \right) (I_{yy})_2$$

$$\therefore \frac{d\phi}{ds} = \frac{1}{\rho} = \frac{M_b}{E_1 (I_{yy})_1 + E_2 (I_{yy})_2} \quad - Q.E.D. -$$

$$(c) (\sigma_x)_i = -E_i \left(\frac{d\phi}{ds} \right) y = -E_i \frac{M_b \cdot y}{E_1 (I_{yy})_1 + E_2 (I_{yy})_2} \quad - Q.E.D. -$$

7.47 두 재료의 접촉면 근처에

점 A, B를 그림처럼 잡고, lateral strain ϵ_z 를 생각하자. σ_z 를 제각각 모든 응력은 "0"으로 가정.



$$(\tilde{\sigma}_x)_A = -\lambda E_1 y_I ; (\tilde{\sigma}_x)_B = -\lambda E_2 y_I \quad \left(\lambda = \frac{M_b}{E_1 (I_{yy})_1 + E_2 (I_{yy})_2} \right)$$

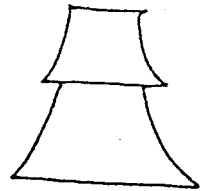
$$\therefore (\tilde{\epsilon}_z)_A = \frac{1}{E} (\tilde{\sigma}_z - \nu \tilde{\sigma}_x - \nu \tilde{\sigma}_y) = \lambda \nu_1 E_1 y_I$$

$$(\tilde{\epsilon}_z)_B = \lambda \nu_2 E_2 y_I$$

따라서 일반적으로 lateral strain은 서로 다르다. 따라서 접촉면에 저항이 없다면 우측 그림과 같이 변형할 것이다.

그러나, 실제로는 기하학적 적합조건을 만족시켜야 한다.

따라서 $\tilde{\sigma}_z$ 가 존재할 것이고 그러므로 $\tilde{\epsilon}_z$, τ_{yz} 도 존재한다. 만약 $E_1 \nu_1 = E_2 \nu_2$ 이면 $\tilde{\sigma}_z$ 만 존재한다.



$$7.48 \quad I_a = 23.70 \times 10^6 \text{ mm}^4 = 23.70 \times 10^{-6} \text{ m}^4$$

$$I_s = 2 \left[(0.00625)(100 \times 10^{-3})(0.1 + 0.00315)^2 + \frac{1}{12}(0.1)(0.00625)^3 \right]$$

$$= 13.7 \times 10^{-6} \text{ m}^4$$

$$\text{Material Property ; } E_a = 75 \text{ GN/m}^2 \quad E_s = 200 \text{ GN/m}^2$$

$$\left(\frac{d\phi}{ds} \right)_1 = \frac{M_b}{E_a I_a} \quad \left(\frac{d\phi}{ds} \right)_2 = \frac{M_b}{E_a I_a + E_s I_s}$$

$$\text{Stiffness ; } (k_b)_1 = E_a I_a \quad (k_b)_2 = E_a I_a + E_s I_s$$

$$\Delta k_b = \frac{(k_b)_2 - (k_b)_1}{(k_b)_1} = \frac{E_s I_s}{E_a I_a} = 1.54$$

$$\tilde{\sigma}_z = -E \cdot y \cdot \left(\frac{d\phi}{ds} \right)$$

$$\text{따라서 } \tilde{\sigma}_a = -E_a y_a \left(\frac{d\phi}{ds} \right)_2, \quad \tilde{\sigma}_s = -E_s y_s \left(\frac{d\phi}{ds} \right)_2$$

$$\therefore (\tilde{\sigma}_s / \tilde{\sigma}_a)_{\max} = \frac{E_s (y_s)_{\max}}{E_a (y_a)_{\max}} = 2.833$$

$$\text{Ans. Ratio of stiffness ; 1.54} \quad (\tilde{\sigma}_s / \tilde{\sigma}_a)_{\max} = 2.833$$

$$7.49 \quad \text{material property ; } E_a = 75 \text{ GN/m}^2 \quad E_s = 200 \text{ GN/m}^2$$

$$(\tilde{\sigma}_y)_a = 109.1 \text{ MN/m}^2 \quad (\tilde{\sigma}_y)_s = 254.5 \text{ MN/m}^2$$

$$I_s = \frac{\pi}{64} (d_o^4 - d_i^4) = \frac{\pi}{64} (0.048^4 - 0.042^4) = 1078 \times 10^{-9} \text{ (m}^4\text{)}$$

$$I_a = \frac{\pi}{64} (0.042^4 - 0.03^4) = 1.13 \times 10^{-9} \text{ (m}^4\text{)}$$

$$k = \frac{M_b}{E_a I_a + E_s I_s} = \frac{M_b}{30035}$$

$$\therefore (\sigma_y)_s = k E_s (y_s)_{\max} = \frac{(M_b)_s}{30035} (200 \times 10^9) (0.024) = 254.5 \times 10$$

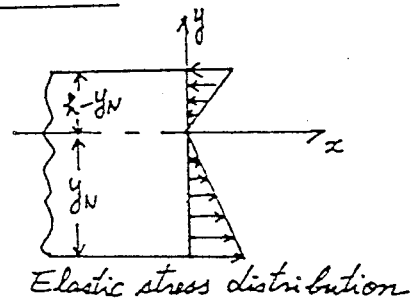
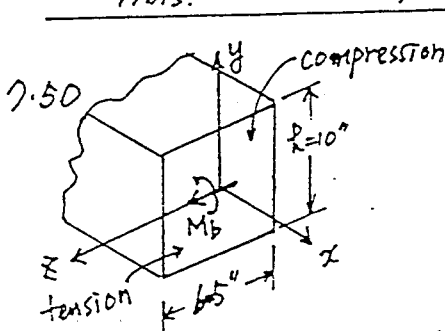
$$\therefore (M_b)_s = 1592 \text{ N-m}$$

$$(\sigma_y)_a = k E_a (y_a)_{\max} = \frac{(M_b)_a}{30035} (75 \times 10^9) (0.021) = 109.1 \times 10^6$$

$$\therefore (M_b)_a = 2081 \text{ N-m}$$

따라서 M_b 는 $(M_b)_s$ 와 $(M_b)_a$ 중의 작은 값이 된다.

$$\text{Ans. } M_b = 1592 \text{ N-m}$$



$$E_T = E_{\text{tension}} = 7.5 \times 10^6 \text{ psi} ; Y_T = 10.000 \text{ psi}$$

$$E_c = E_{\text{compression}} = 30 \times 10^6 \text{ psi} ; Y_c = 20.000 \text{ psi}$$

y_N = height of neutral axis ; ρ = radius of curvature

$$\text{Geometry ; } \epsilon_x = -y/\rho$$

$$\text{Equilibrium ; } \sigma_y = \sigma_z = 0 ; \sum F_x = \int_{-y_N}^{h-y_N} b \sigma_x dy = 0$$

$$\sum M_z = M_b + \int_{-y_N}^{h-y_N} b \sigma_x y dy = 0$$

$$\text{Stress-Strain ; Compression ; } \sigma_x = E_c \epsilon_x$$

$$\text{Tension ; } \sigma_x = E_T \epsilon_x$$

$$\text{Yield Criterion ; Compression ; } |\sigma_x| = Y_c$$

$$\text{Tension ; } |\sigma_x| = Y_T$$

$$\text{તટાકા } \Sigma F_z = -\frac{E_c}{\rho} \int_0^{k-y_N} b y dy - \frac{E_T}{\rho} \int_{-y_N}^0 b y dy = 0$$

$$\therefore, -\frac{E_c \cdot b}{\rho} \frac{1}{2} (k-y_N)^2 + \frac{E_T \cdot b}{\rho} \frac{1}{2} y_N^2 = 0$$

$$\therefore y_N = \frac{2}{3} k$$

$$\text{તટાકા } \Sigma M_z = M_b - \frac{E_c}{\rho} \int_0^{k/3} b y^2 dy + \frac{E_T}{\rho} \int_{-2k/3}^0 b y^2 dy = 0$$

$$\therefore M_b = \frac{b k^3}{81 \rho} [E_c + 8 E_T]$$

$$\text{Max. Compressive stress} = \sigma_c = \frac{E_c (k/3)}{\rho} = E_c \left(\frac{k}{3}\right) \cdot \frac{81 M_b}{b k^3 [E_c + 8 E_T]}$$

$$\text{Max. tensile stress} = \sigma_T = \frac{E_T (2k/3)}{\rho} = E_T \left(\frac{2k}{3}\right) \cdot \frac{81 M_b}{b k^3 [E_c + 8 E_T]}$$

$$\text{તટાકા } M_b = \frac{b k^2}{27} \sigma_c \left[1 + 8 \frac{E_T}{E_c}\right] = \frac{b k^2}{27} \frac{\sigma_T}{2} \left[\frac{E_c}{E_T} + 8\right]$$

$$\frac{b k^2}{27} = \frac{(5)(100)}{27} = 18.5 ; \left[1 + 8 \frac{E_T}{E_c}\right] = 1 + 8 \cdot \frac{2.5}{30} = 3$$

$$\left[\frac{E_c}{E_T} + 8\right] = \frac{30}{2.5} + 8 = 12$$

$$\text{તટાકા, } |\sigma_c| = Y_c ; M_b = (18.5)(20000)(3) = 1.11 \times 10^6 \text{ lb-in}$$

$$\sigma_T = Y_T ; M_b = (18.5)\left(\frac{10000}{2}\right)(12) = 1.11 \times 10^6 \text{ lb-in}$$

$$\text{Ans. } (M_b)_{\max} = 1.11 \times 10^6 \text{ lb-in}$$

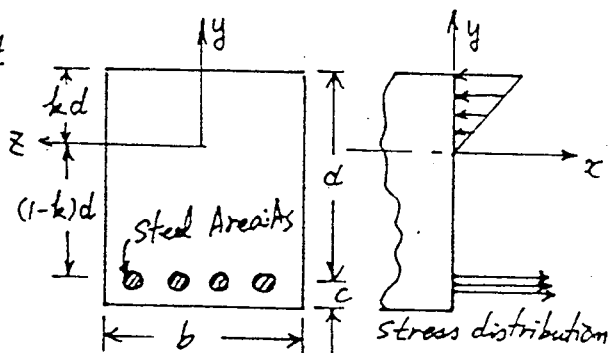
$$7.51 \text{ Geometry ; } \epsilon_x = -\frac{y}{\rho}$$

Equilibrium ;

$$\sigma_y = \sigma_z = 0$$

$$\Sigma F_x = \int \sigma_x dA = 0$$

$$\Sigma M_z = M_b + \int \sigma_x y dA = 0$$



Stress-Strain ; Concrete ; Compression $\sigma_x = E_c \epsilon_x$, Steel ;

Tension $\sigma_x = 0$

$\sigma_x = E_s \epsilon_x$

steel 이 대해서 strain 이 대한 평 균치 를 구 하라.

$$\sum F_x = - \int_0^{kd} \frac{E_c y}{\rho} b dy + \frac{E_s (1-k)d}{\rho} A_s = 0$$

$$E_s A_s (d - kd) - E_c b \frac{(kd)^2}{2} = 0 \quad - (1)$$

$$\sum M_z = M_b - \int_0^{kd} \frac{E_c y}{\rho} b y dy - E_s \frac{(1-k)d}{\rho} A_s (1-k)d = 0$$

$$\text{giving } \frac{1}{\rho} = \frac{M_b}{\frac{1}{3} E_c b k^3 d^3 + E_s A_s (1-k)^2 d^2}$$

$$\sigma_{\text{steel}} = \sigma_s = \frac{E_s (1-k)d}{\rho} = \frac{M_b E_s (1-k)d}{\frac{1}{3} E_c b k^3 d^3 + E_s A_s (1-k)^2 d^2} = \frac{M_b}{A_s d \left[\frac{1}{3} \frac{E_c b k^3 d}{A_s E_s (1-k)} + (1-k) \right]}$$

① 4:3 부터

$$\frac{E_c b d}{E_s A_s} = \frac{2(1-k)}{k^2} ; \therefore \frac{1}{3} \frac{E_c b k^3 d}{A_s E_s (1-k)} = \frac{2}{3} \frac{k}{1-k}$$

$$\sigma_s = \frac{M_b}{A_s d \left[\frac{2}{3} \frac{k}{1-k} + 1 - k \right]} = \frac{M_b}{A_s d \left[1 - \frac{k}{3} \right]} \quad - \text{Q.E.D.} -$$

$$(\sigma_{\text{concrete}})_m = (\sigma_c)_m = E_c \frac{kd}{\rho} = \sigma_s \frac{kd}{(1-k)d} \frac{E_c}{E_s} = \sigma_s \frac{E_c}{E_s} \left(\frac{k}{1-k} \right)$$

① 4:3 부터

$$\frac{E_c}{E_s} = \frac{2A_s(1-k)}{bdk^2} ; \therefore (\sigma_c)_m = \frac{M_b}{A_s d \left[1 - \frac{k}{3} \right]} \frac{2A_s(1-k)}{bdk^2} \frac{k}{1-k} = \frac{2M_b}{bd^2 k \left[1 - \frac{k}{3} \right]} \quad - \text{Q.E.D.} -$$

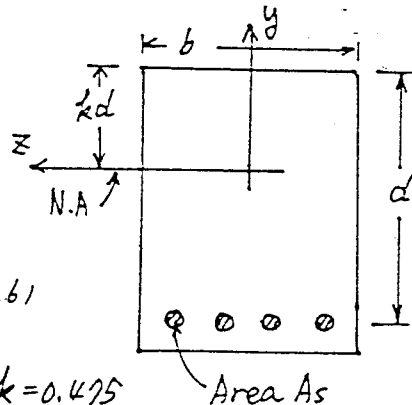
7.52 문제 (7.51) 참조.

$$E_s A_s (1-k) - \frac{1}{2} E_c b k^2 d = 0$$

$$\text{또는 } k^2 + \frac{2E_s A_s}{E_c b d} k - \frac{2E_s A_s}{E_c b d} = 0$$

$$\frac{2E_s A_s}{E_c b d} = \frac{(2)(30 \times 10^6)(2.2)}{(1.5 \times 10^6)(9)(16)} = 0.61$$

$$\therefore k^2 + 0.61k - 0.61 = 0 \quad \text{따라서 } k = 0.475$$



For $\sigma_s = 20000 \text{ psi}$ $M_b = 20000 A_s d \left(1 - \frac{k}{3}\right)$
 $= 20000 (2.2) (16) (0.842) = 592000 \text{ lb-in}$
 $(\sigma_c)_{\max} = 1350 \text{ psi}$ $M_b = 1350 \cdot \frac{1}{2} b d^2 k \left(1 - \frac{k}{3}\right)$
 $= 1350 \cdot \left(\frac{1}{2}\right) (9) (256) (0.475) (0.842) = 625000 \text{ lb-in}$
 $\therefore (M_b)_{\max} = 592000 \text{ lb-in}$
Ans. $(M_b)_{\max} = 592000 \text{ lb-in}$

7.53 그림은 문제 (7.51) 참조.

$$\sigma_s = \frac{E_s (1-k) d}{\rho} ; (\sigma_c)_{\max} = \frac{E_c k d}{\rho} \quad \therefore \frac{\sigma_s}{(\sigma_c)_{\max}} = \frac{1-k}{k} \left(\frac{E_s}{E_c} \right)$$

$$\frac{20000}{1350} = \frac{1-k}{k} \left(\frac{30 \times 10^6}{1.5 \times 10^6} \right) \quad \therefore k = 0.575$$

문제 (7.51)의 ①로부터 ; $A_s = \frac{1}{2} \frac{E_c}{E_s} (b d) \cdot \frac{k^2}{1-k}$
 $= \frac{1}{2} \left(\frac{1.5 \times 10^6}{30 \times 10^6} \right) (9) (16) \frac{(0.332)}{(0.425)} = 2.8 \text{ in}^2$

$$\therefore (5) \frac{\pi}{4} d_s^2 = 2.8 \quad (d_s ; \text{diameter of steel bars})$$

$$\therefore d_s = 0.84 \text{ in}$$

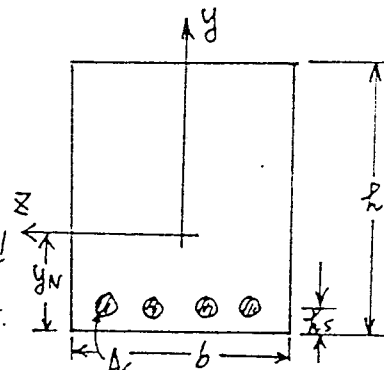
$$M_b = \sigma_s A_s d \left(1 - \frac{k}{3}\right) = 20000 (2.8) (16) \left(1 - \frac{0.575}{3}\right) = 728000 \text{ lb-in}$$

Ans. $d_s = 0.84 \text{ in}$ $M_b = 728000 \text{ lb-in}$

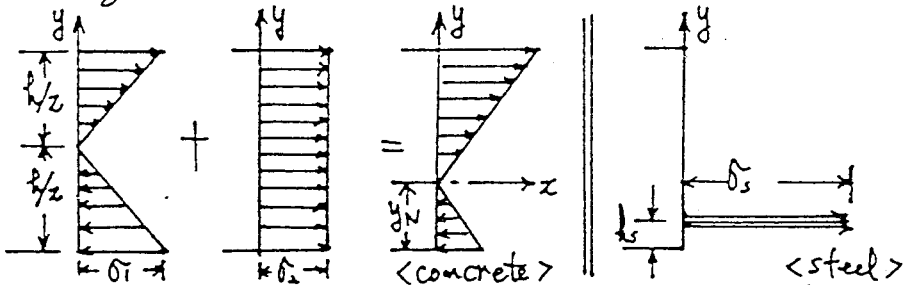
7.54 (a) Prestressing

wire가 release해 지면 beam은 곡률 반경 ρ 으로 굽게 된다.

strain과 stress의 분포는 높이가 y_N 인 중립축을 중심으로 직선적으로 될 것이다.
 concrete에 대한 응력 분포는 symmetric.



bending stress et uniform stress 로 분해될 수 있다.



centerline 주변의 bending stress et uniform stress 이 의한 moment 는 "0" 이다.

$$\text{Equilibrium; } \Sigma F_x = \sigma_2 (bh) + \sigma_s A_s = 0 \quad \text{--- ①}$$

$$\Sigma M_z = -2 \left[\left(\frac{\sigma_1}{2} \right) \left(\frac{h}{2} \right) (b) \right] \left[\frac{2}{3} \left(\frac{h}{2} \right) \right] + \sigma_s A_s \left(\frac{h}{2} - h_s \right) = 0 \quad \text{--- ②}$$

$$\text{① 식에서 } \sigma_2 = - \frac{\sigma_s A_s}{bh}$$

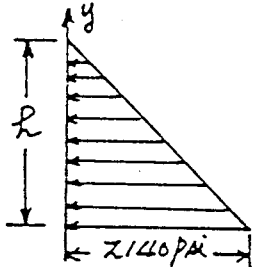
$$\text{② 식에서 } \sigma_1 = \frac{6 \sigma_s A_s}{bh^2} \left(\frac{h}{2} - h_s \right)$$

$$A_s = 76 \frac{\pi}{4} (0.15)^2 = 1.34 \text{ in}^2; b = 9"; h = 18"; h_s = 6"$$

$$\sigma_s = 130,000 \text{ psi}$$

$$\therefore \sigma_2 = \frac{-(130,000)(1.34)}{(9)(18)} = -10,700 \text{ psi}, \sigma_1 = \frac{6(130,000)(1.34)(18-6)}{(9)(18)(18)} = 10,700 \text{ psi}$$

결과적으로 concrete 이 받는 stress 의 분포는;



$\left[\frac{2h}{3} \geq h_s \geq \frac{h}{3} \right]$ 일 때는
tensile stress 는 없다.]

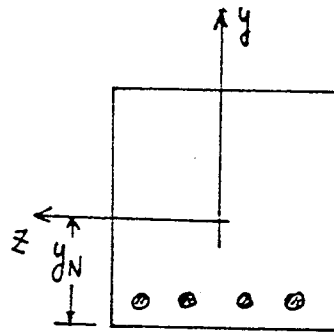
(b) Bending

Bending 이 위해에 발생되는 stress 를 증명시킨다. concrete 이 tensile stress 가 발생한다고 가정하나. 결국에는 compressive prestresses 이 위해에 상쇄된다.

문제 (7.46) 에서

$$y_N = \frac{E_c \bar{y}_c A_c + E_s \bar{y}_s A_s}{E_c A_c + E_s A_s}$$

$$\frac{1}{\rho_b} = \frac{M_b}{E_c (I_{yy})_c + E_s (I_{yy})_s}$$



($\frac{1}{\rho_b}$) ; moment M_b 에 의해 생기는 곡률의 변화량)

$$\bar{y}_c = 9'' ; A_c = (9)(18) = 162 \text{ in}^2$$

$$\bar{y}_s = 6'' ; A_s = 1.34 \text{ in}^2$$

$$E_c = 1.5 \times 10^6 \text{ psi} ; E_s = 30 \times 10^6 \text{ psi}$$

$$\therefore y_N = \frac{(1.5 \times 10^6)(9)(162) + (30 \times 10^6)(6)(1.34)}{(1.5 \times 10^6)(162) + (30 \times 10^6)(1.34)} = 8.5''$$

$$(I_{yy})_c = \left[\frac{1}{12} (9)(18)^3 + (9)(18)(9-8.5)^2 \right] = 4170 \text{ in}^4$$

$$(I_{yy})_s = [0 + (1.34)(8.5-6)^2] = 8.4 \text{ in}^4$$

$$E_c (I_{yy})_c + E_s (I_{yy})_s = 10^6 [(1.5)(4170) + (30)(8.4)] = 6512 \times 10^6 \text{ lb-in}^2$$

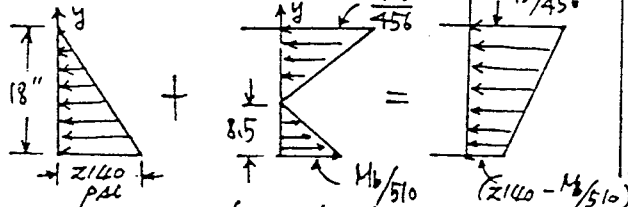
$$(\sigma_z)_{\text{concrete}} = - \frac{M_b (1.5 \times 10^6) y}{6512 \times 10^6} \quad (y: \text{중립축에서 부터 측정된 거리})$$

$$\text{Max. Compressive stress} = \frac{M_b (1.5 \times 10^6) (9.5)}{6512 \times 10^6} = \frac{M_b}{456}$$

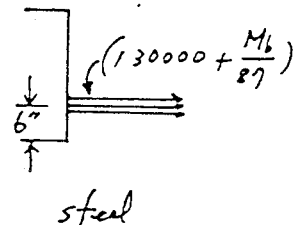
$$\text{Max. Tensile stress} = \frac{M_b (1.5 \times 10^6) (8.5)}{6512 \times 10^6} = \frac{M_b}{510}$$

$$\text{Bending Stress in Steel} = \frac{M_b (30 \times 10^6) (8.5-6)}{6512 \times 10^6} = \frac{M_b}{87}$$

(c) Resultant Stresses



concrete ; prestress + bending = resultant



Requirements

- 1) No tension in concrete ; $x(140) - \frac{M_b}{EI} > 0 \therefore M_b < 1090000 \text{ lb-in}$
- 2) Concrete compressive stress $< 2250 \text{ psi}$; $\frac{M_b}{450} < 2250 \therefore M_b < 1030000 \text{ lb-in}$
- 3) Steel tensile stress ; $130000 + \frac{M_b}{87} < 150000 \therefore M_b < 1740000 \text{ lb-in}$

$$(M_b)_{\max} = 1030000 \text{ lb-in}$$

7.55 문제 (7.46)의 경우

$$\sigma_x = \frac{-E_1 M_b y}{E_1 (I_{yy})_1 + E_2 (I_{yy})_2}$$

$$\Delta F_{yx} = \left[\int_{A_0} \sigma_x dA \right]_{x+dA} - \left[\int_{A_0} \sigma_x dA \right]_x$$

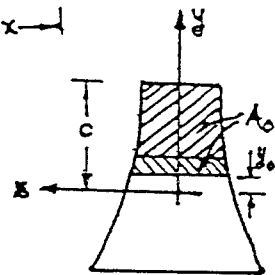
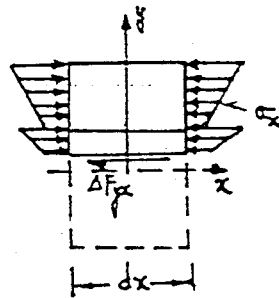
$$E_1 \frac{dF_{yx}}{dx} = \frac{d}{dx} \int_{A_0} \sigma_x dA$$

$$= \frac{d}{dx} \left[-\frac{M_b}{E_1 (I_{yy})_1 + E_2 (I_{yy})_2} \int_{A_0} E y dA \right]$$

$$= -\left(\frac{dM_b}{dx} \right) \left(\frac{1}{E_1 (I_{yy})_1 + E_2 (I_{yy})_2} \right) \left(\int_{A_0} E y dA \right)$$

$$-\frac{dM_b}{dx} = V \quad \text{and} \quad \tau_{xy} = \tau_{yx} = \frac{\partial F_{yx}}{b} = \frac{1}{b} \frac{dF_{yx}}{dx}$$

$$\therefore \tau_{xy} = \tau_{yx} = \frac{V}{b [E_1 (I_{yy})_1 + E_2 (I_{yy})_2]} \int_{A_0} E y dA \quad \text{--- a.e.d ---}$$



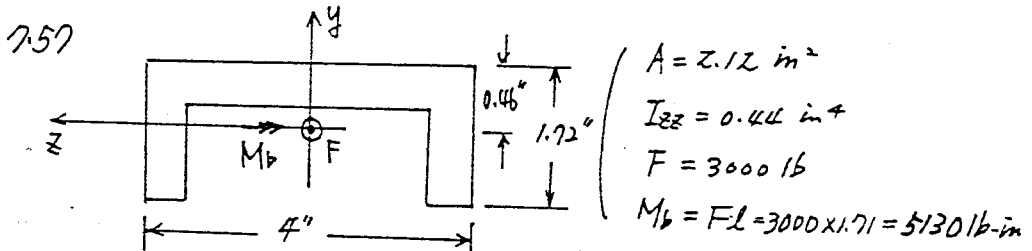
7.56 문제 (7.48)의 경우

$$\tau_{yx} = \frac{V}{E_1 I_1 + E_2 I_2} \int_{A_0} E y dA = \frac{V}{E_1 I_1 + E_2 I_2} E_1 \left[\frac{y^2}{2} \right]_{0.1}^{0.10625}$$

$$= \frac{(27000)(200 \times 10^9) \left(\frac{1}{2} \right) [(0.10625)^2 - (0.1)^2]}{(75 \times 10^9)(23.7 \times 10^{-6}) + (200 \times 10^9)(13.3 \times 10^{-6})} = 784300 \text{ (N/m)}$$

$$\therefore S = \frac{F x_z}{q_{yx}} = \frac{2 \times 22000}{784300} = 0.0561 \text{ (m)}$$

Ans. $S = 0.0561 \text{ m}$



Direct Stress; $(\sigma_x)_1 = \frac{F}{A} = \frac{3000}{2.12} = 1415 \text{ psi}$

Bending Stress; $(\sigma_x)_2 = \frac{M_b \cdot y}{I_{zz}} = \frac{5130 \times 0.46}{0.44} = 5363 \text{ psi}$

$$(\sigma_c)_2 = \frac{5130}{0.44} (0.46 - 1.72) = -14,690 \text{ psi}$$

$$\therefore (\sigma_c)_{\max} = -14,690 + 1415 = -13,275 \text{ psi}$$

$$(\sigma_t)_{\max} = 5360 + 1415 = 6775 \text{ psi} \quad \therefore \sigma_{\max} = 13275 \text{ psi}$$

Ans. $\sigma_{\max} = 13275 \text{ psi}$

7.58 문제 (7.23) 참조 $\sigma_{\max} = \frac{r \cdot L^2}{8t}$

$$2\pi r t \cdot \sigma_x = \pi r^2 p \quad \therefore \sigma_x = \frac{p \cdot r}{2t} \quad \sigma_{\max} = \sigma_x \text{ at } r$$

$$\text{따라서 } \frac{r L^2}{8t} = \frac{p \cdot r}{2t} \quad \therefore p = \frac{r L^2}{4r}$$

Ans. $\sigma_{\max} = \frac{r L^2}{8t}, \quad p = \frac{r L^2}{4r}$

7.59 cutting force를 mid-depth of cut 에 작용하는 point load로 치환하자. 그러면 collet 에 작용하는 moment or torques 는

a) due to 1600 N force ; $M_t = -(1600)(0.005) = -8 \text{ N-m}$
 $M_{bz} = (1600)(0.075) = 120 \text{ N-m}$

b) due to 605 N force ; $M_{by} = -(605)(0.005) = -3.025 \text{ N}\cdot\text{m}$

M_{by} 와 M_{bz} 는 하나의 moment $M_b = (M_{by}^2 + M_{bz}^2)^{1/2}$ 로 합성할 수 있다. y축에 대한 $\theta = \tan^{-1}(M_{bz}/M_{by})$ 각으로 합성된다.

$$\therefore M_b = \sqrt{(3.025)^2 + (120)^2} = 120.0382$$

$$\theta = \tan^{-1}(120/3.025)$$

따라서 결과적인 moment, force,

torque 는 우편의 그림과 같다.

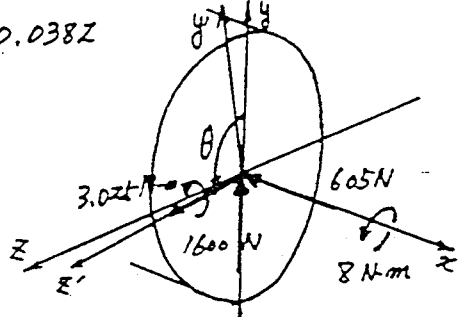
새로운 좌표축 y', z' 을 그림과 같이

잡으면 resultant bending moment 는 z' 을 중심으로 발생된다.

M_b 는 관성의 주축을 중심으로 작용한다.

$$I_{yy'} = \frac{\pi}{4} \left(\frac{0.0125}{2} \right)^4 \quad ; \quad I_x = 2(I_{yy'})$$

$$A = \pi \left(\frac{0.0125}{2} \right)^2$$



Stresses;

$$1) 605 \text{ N} ; \sigma_x = -\frac{F}{A} = -4.93 \text{ MN/m}^2$$

$$2) 1600 \text{ N} ; \tau_{xy'} = \frac{V \cdot Q}{b I_{yy'}} \quad \text{여기 } Q \text{ 는 } y' = \pm (0.0125/z) \text{ 이거나 } 0$$

$$y=0 \text{ 이거나 } Q_{\max} = \frac{2}{3} r^3$$

$$\therefore (\tau_{xy'})_{\max} = \frac{(1600) \times \left(\frac{2}{3} \right) \left(\frac{0.0125}{2} \right)^3}{(0.0125) \left(\frac{\pi}{64} \right) (0.0125)^4} = 17.38 \text{ MN/m}^2$$

$$3) M_b ; \sigma_x = \frac{M_b \cdot y'}{I_{yy'}} \quad ; \quad y' = \pm (0.0125/z) \text{ 이거나 } \max. \\ y=0 \text{ 이거나 } "0"$$

$$\therefore (\sigma_x)_{\max} = \frac{(120.0382)(0.0125/z)}{\frac{\pi}{64} (0.0125)^4} = 626 \text{ MN/m}^2$$

$$4) M_t ; \tau = \frac{M_t(r)}{I_x} \quad ; \quad r = \pm (0.0125/z) \text{ 이거나 } \max \\ r=0 \text{ 이거나 } "0"$$

$$\therefore \tau_{\max} = \frac{(8) \cdot (0.0125/z)}{2 \cdot \frac{\pi}{64} (0.0125)^4} = 20.86 \text{ MN/m}^2$$

Resulting stresses; linear superposition 을 하기 위해 문제를 단순화 시키자. 즉, bending stress 가 다른 stress 보다 훨씬 크기 때문에 max. shear stress 의 위치는 max. bending stress 의 위치와 일치하게 된다고 볼 수 있다.

따라서 그 위치는 ; $y' = (0.0125/2)$, $z' = 0$ 이다.

$$\sigma_x = -6zb - 4.93 = -630.92 \text{ MN/m}^2$$

$$\tau_{xz} = 20.86 \text{ MN/m}^2$$

$$\therefore \tau_{max} = \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + (\tau_{xz})^2} = 316.15 \text{ MN/m}^2$$

Ans. $\tau_{max} = 316.15 \text{ MN/m}^2$

7.60 one side frame 이 대해 하중 상태는 우측 그림과 같다고 생각하자.

$$\therefore \text{평형조건; } R_1 = 310 \text{ N}$$

$$R_2 = 240 \text{ N}$$

$$\begin{cases} F_1 \sin \alpha = F_2 \sin \beta \\ F_1 \cos \alpha = 550 - F_2 \cos \beta \end{cases} \begin{cases} \alpha = \tan^{-1}(500/350) = 55^\circ \\ \beta = \tan^{-1}(75/450) = 9.5^\circ \end{cases}$$

$$\therefore F_1 = 0.2 F_2 = 100 \text{ N}$$

Bending Moment

$$M_{12} = 100 \times \sqrt{(0.5)^2 + (0.35)^2} \times \frac{1}{2} = 31 \text{ (N-m)}$$

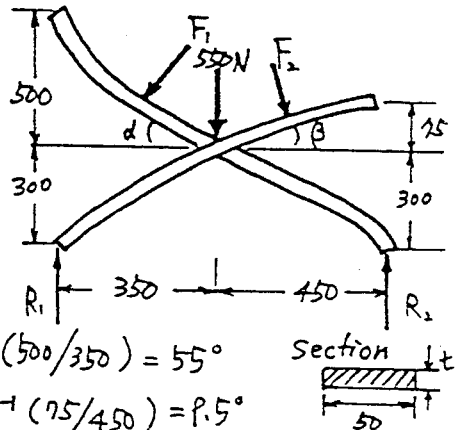
$$M_{23} = 500 \times \sqrt{(0.075)^2 + (0.45)^2} \times \frac{1}{2} = 114 \text{ (N-m)}$$

$$M_{24} = 310 \times 0.35 = 108 \text{ (N-m)}$$

$$M_{25} = 240 \times 0.45 = 108 \text{ (N-m)}$$

$$\therefore (M_b)_{max} = 114 \text{ (N-m)} \quad \sigma_y = 600 \text{ MN/m}^2$$

$$\sigma_{max} = \frac{(M_b)_{max} \cdot y_{max}}{I_{zz}} = \frac{\sigma_y}{2}$$



$$\frac{114 \times 0.5t}{\frac{0.05}{12} \times t^3} = 300 \times 10^6 \quad \therefore t = 6.75 \times 10^{-3} \text{ (m)}$$

Ams. $t = 6.75 \times 10^{-3} \text{ m}$

7.61 1) Compressive Stress due to direct load

2) Bending Stress

$$\bar{y} = \frac{\frac{1}{2}(12.5)(6.25) - \frac{1}{2}(10.9)(3.05)}{12.5 \times 6.25 - 10.9 \times 3.05} = 6.834 \text{ (mm)}$$

$$I_{yy} = 2 \left[\frac{1}{12}(1.6)(12.5)^3 + (1.6)(12.5)(6.25 - 6.834)^2 + \frac{1}{12}(3.05)(1.6)^3 + (3.05)(1.6)(12.5 - 6.834 - 0.8)^2 \right]$$

$$= 651.4 \text{ (mm}^4\text{)}$$

$$A = [12.5 + 12.5 + (6.25 - 3.2)](1.6)$$

$$= 44.88 \text{ (mm}^2\text{)}$$

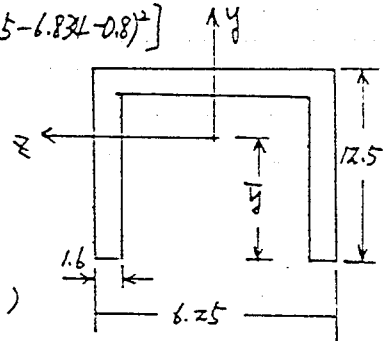
$$M_b = -300 \times 100 = -30000 \text{ (N-mm)}$$

$$\text{Compressive Stress; } \sigma_F = \frac{F}{A} = \frac{-300}{44.88} = -6.684 \text{ (N/mm}^2\text{)}$$

$$\text{Bending Stress; } \sigma_M = \frac{M_b \cdot y_{\max}}{I_{yy}} = \frac{-30000(6.843)}{651.1} = -315.3 \text{ (N/mm}^2\text{)}$$

$$\therefore \sigma_{\max} = -6.684 - 315.3 = -322 \text{ MN/mm}^2$$

이 값은 yielding stress 의 약 54% 에 해당한다. 따라서 이 값은 상당히 커서 작업중이 additional stress 가 발생한다. 따라서 300 N 은 너무 큰 하중이다.



7.62 Rod 에는 Bending moment M_b et twisting moment

$$M_t \text{ 가 동시에 작용한다. } \sigma_b = \frac{M_b \cdot d}{2I_{yy}}, \quad \tau = \frac{M_t \cdot d}{4I_{yy}} = \frac{M_t \cdot d}{2I_x}$$

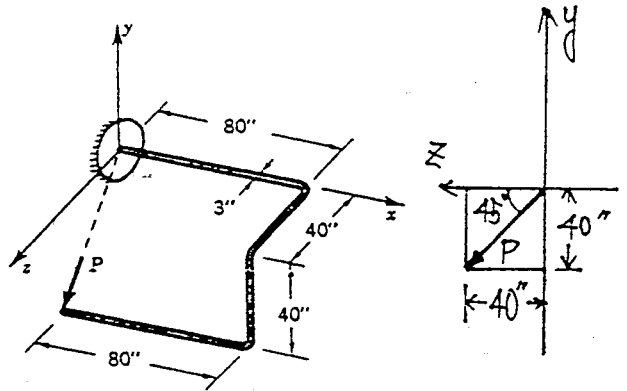
Von Mises의 yielding criterion 에 의하면
yielding 은 다음과 같은
상태에서 발생한다.

$$\bar{\sigma} = \sqrt{\frac{1}{2}(\bar{\sigma}_b^2 + \bar{\sigma}_t^2) + 3(\tau)^2} = Y$$

(Y; yield strength
in tension)

$$\bar{\sigma} = \sqrt{\left(\frac{M_b d}{2I_{yy}}\right)^2 + 3\left(\frac{M_t d}{4I_{yy}}\right)^2}$$

$$= \frac{M_b d}{2I_{yy}} \left[1 + \frac{3}{4} \left(\frac{M_t}{M_b} \right)^2 \right]^{1/2}$$



B, C, D 에 $\bar{\sigma}$ 를 계산한다. 그 중에서 maximum 이 있을 것이다.

1) just before B; $M_b = 80P, M_t = 0$ $\bar{\sigma} = \left(\frac{d}{2I_{yy}}\right) 80P$

2) just after B; $M_b = \frac{80P}{\sqrt{2}}, M_t = \frac{80P}{\sqrt{2}}$ $\bar{\sigma} = \left(\frac{d}{2I_{yy}}\right) 106P$

3) just before C; $M_b = 63.2P, M_t = 56.8P$ $\bar{\sigma} = \left(\frac{d}{2I_{yy}}\right) 87P$

4) just after C; $M_b = 63.2P, M_t = 56.8P$ $\bar{\sigma} = \left(\frac{d}{2I_{yy}}\right) 87P$

5) just before D; $M_b = \frac{80}{\sqrt{2}}P, M_t = \frac{80}{\sqrt{2}}P$ $\bar{\sigma} = \left(\frac{d}{2I_{yy}}\right) 106P$

6) just after D; $M_b = 80P, M_t = 0$ $\bar{\sigma} = \left(\frac{d}{2I_{yy}}\right) 80P$

$$(\bar{\sigma})_{\max} = 106P \left(\frac{d}{2I_{yy}} \right)$$

3" diameter rod 이 사용; $I_{yy} = \frac{\pi}{4} \left(\frac{3}{2} \right)^4 = 3.98 \text{ in}^4$

4130 HT 이 사용; $Y = 190000 \text{ psi}$

$$\therefore \frac{(106)(P)(3)}{(2)(3.98)} = 190000 \quad \therefore P = 4750 \text{ lb}$$

Ans. $P = 4750 \text{ lb}$

7.63 Bending & Torsion 의 합성에 의하여 Stress 가 발생한다.
(Direct Stress 는 무시한다.)

AB; Bending 에 의한 B점에서의 max. stress

BC; Twisting moment = 0;
Bending moment = const.

∴ Max. stress 는 minimum section 이다. 즉 B점이다.
생긴다. 이 Max. stress 는 AB 에 대한 stress 와 같다.

CD; Twisting moment = const.
Bending moment 는 D점까지 max.

∴ max. stresses at D

Von Mises Yield criterion; $\bar{\sigma} = Y$

문제 (7.62) 에서
$$\bar{\sigma} = \frac{M_b \cdot d}{2 I_{yy}} \left[1 + \frac{3}{4} \left(\frac{M_t}{M_b} \right) \right]^{1/2}$$

B; $M_b = Pa$; $M_t = 0$; $I_{yy} = \frac{\pi}{4} r^4$; $\therefore \bar{\sigma} = \sigma_b = \frac{4Pa}{\pi r^3} = 1.26 \frac{Pa}{r^3}$

D; $M_b = 2Pa$; $M_t = Pa$; $I_{yy} = \frac{\pi}{4} (1.3r)^4$

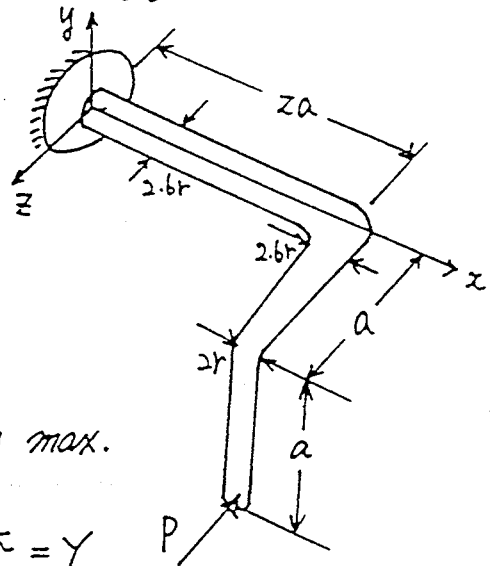
$$\therefore \bar{\sigma} = \frac{(2Pa)(2.6r)}{2(\frac{\pi}{4})(1.3r)^4} \left[1 + \frac{3}{4} \left(\frac{1}{2} \right) \right]^{1/2} = 1.36 \frac{Pa}{r^3}$$

$$\therefore (\bar{\sigma})_{\max} = 1.36 \frac{Pa}{r^3} = Y \text{ (for yielding)}$$

따라서 yielding 이 일어날 조건; $P = 0.735 \frac{r^3 Y}{a}$

만일, Tresca (max. shear stress) yield criterion 을
사용하면 $P = 0.77 \frac{r^3 Y}{a}$

Ans. Mises; $P = 0.735 r^3 Y/a$ Tresca; $P = 0.77 r^3 Y/a$



7.64 Total stress 는 Bending, Torsion, Internal, shear stress

등을 중첩함으로써 얻어진다.

Equilibrium ; 1) Piston : $F = p\pi\left(\frac{R}{\sqrt{6}}\right)^2 \approx p = \frac{6F}{\pi R^2}$

2) Tube at built in end ; $M_{bz} = -5FR$
 $M_t = -6FR$

Geometry ; $I_{yy} = \pi R^3 t$; $I_x = 2\pi R^3 t$

Stresses ; 1) Shear ; stress = 0. $\because Q = \int y dA = 0$ for N)

2) Bending ; $\sigma_x = -\frac{M_{bz} \cdot y}{I_{yy}} = -\frac{(-5FR)R}{\pi R^3 t} = \frac{5F}{\pi R t}$

3) Torsion ; $\tau_{xz} = M_t \cdot r / I_x = -(6FR)(R) / 2\pi R^3 t$
 $= -3F / \pi R t$

4) Internal Pressure ;

Circumferential Stress ; $\sigma_z = \frac{PR}{t} = \frac{6F}{\pi R t}$

Axial Stress ; $\sigma_x = \frac{PR}{2t} = \frac{3F}{\pi R t}$

Total Stresses ; $\sigma_x = \frac{5F}{\pi R t} + \frac{3F}{\pi R t} = \frac{8F}{\pi R t}$

$\sigma_z = \frac{6F}{\pi R t}$; $\tau_{xz} = \frac{-3F}{\pi R t}$

$(t/R) \ll 1$ 이므로 outer stress 는 무시한다.

Mohr's circle ; X-z plane on stress/(F/πRt) 로 그린다.

$\tau_{max} = \frac{\sigma_{max} - \sigma_{min}}{2}$

$\sigma_{max} = \sigma_x = \frac{F}{\pi R t} (7 + \sqrt{10})$

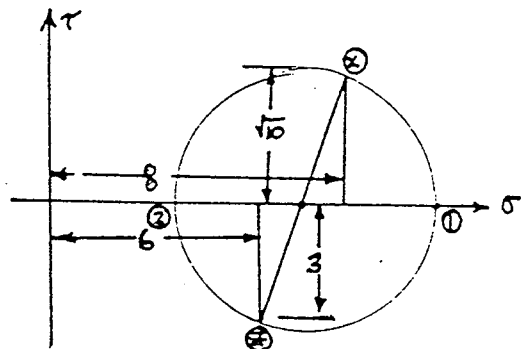
$\sigma_{min} = \sigma_y = 0$

$\therefore \tau_{max} = \frac{F}{2\pi R t} (7 + \sqrt{10})$

항복에 대해 $\tau_{max} = Y/2$

$\therefore F = \frac{\pi R t Y}{7 + \sqrt{10}} = 0.31 R t Y$

Ans. $F = 0.31 R t Y$



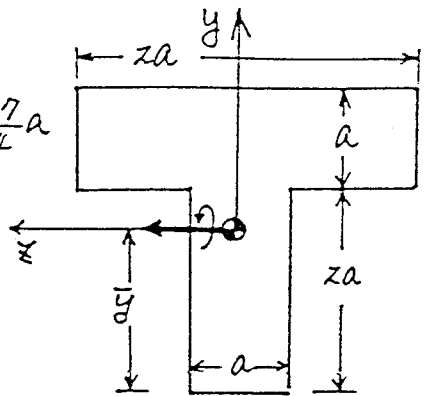
7.65 $(\sigma_x)_{\max} = Y$ 일때 $M_Y = M_b$

$$\bar{y} = \frac{(2a)(a)(a) + (2a)(a)(\frac{5a}{2})}{(2a)(a) + (2a)(a)} = \frac{7a^2}{4a^2} = \frac{7}{4}a$$

$$I_{yy} = \left[\frac{1}{12}(a)(2a)^3 + (2a)(a)\left(\frac{3a}{4}\right)^2 \right] + \left[\frac{1}{12}(2a)(a)^3 + (2a)(a)\left(\frac{3a}{4}\right)^2 \right] = \frac{37}{12}a^4$$

$$y_{\max} = \frac{7}{4}a$$

$$(\sigma_x)_{\max} = \frac{(M_b)(\frac{7a}{4})}{\frac{37}{12}a^4} = Y \quad \therefore M_b = M_Y = \left(\frac{37}{12}\right)\left(\frac{4}{7}\right)Y a^3 = \frac{37}{12}Y a^3$$



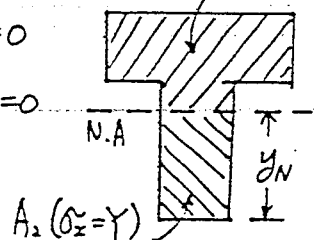
Fully Plastic ; $\sigma_x = \pm Y$ ($\sigma_x = 0$ 인 중립축은 제외) $A_1(\sigma_x = -Y)$

Equilibrium ; $\sum F_z = 0$; $-YA_1 + YA_2 = 0$

$$\sum M_z = 0 ; M_b + \int \sigma_x y dA = 0$$

첫 번째 방정식에서 $y_N = 2a$

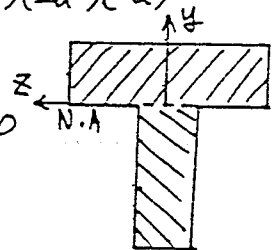
두 번째 방정식에서 $M_b = M_L$



$$\begin{aligned} \text{따라서 } M_b &= -\int \sigma_x y dA = -(-Y)(2a^2)\left(\frac{a}{2}\right) - (Y)(2a^2)(-a) \\ &= 3Ya^3 = M_L \end{aligned}$$

$$\therefore K = M_L/M_Y = (3Ya^3) / \left(\frac{37}{12}Ya^3\right) = 1.70$$

Ans. $K = 1.70$



7.66 $M_L = -\int_A \sigma_x y dA = Y_1 \int_1 y dA + Y_2 \int_2 y dA$

$$\int y dA = 2 \int_0^\pi (F d\theta)(t)(F \sin\theta) = 4F^2 t$$

$$\therefore M_L = (109.1 \times 10^6)(4)\left(\frac{0.021 + 0.015}{2}\right)^2(0.006)$$

$$+ (254.5 \times 10^6)(4)\left(\frac{0.021 + 0.021}{2}\right)^2(0.003) = 2394 \text{ (N-m)}$$

Ans. $M_L = 2394 \text{ N-m}$

7.67 2024-T4 Aluminum; $Y = 55000 \text{ psi}$

$$E_y = 55000 / 0.005 = 11 \times 10^6 \text{ psi} \quad \text{yield strain}$$

strain ϵ_x 는 중립축으로 부터의 거리에

비례한다. ④ 위치의 strain = 0.016

① 위치의 strain = -0.016, ② 위치의 strain = -0.008

③ 위치의 strain = 0.008. 이것은 모두 E_y 보다 크다 따라서, 모든 strip은 plastic 상태이다.

$\therefore$ Moment M_b 는 limit moment M_L 이 된다.

Stresses; $(\sigma_x)_1 = (\sigma_x)_2 = -55000 \text{ psi}$

$$(\sigma_x)_3 = (\sigma_x)_4 = 55000 \text{ psi}$$

$$\Sigma M_z = 0; M_b = -\int \sigma_x y dA = -\sum (\sigma_x)_i A_i \bar{y}_i$$

$$= (-55000) [(2)(\frac{1}{8})(2) + (2)(\frac{1}{8})(1)] = -82500 \text{ lb-in}$$

Unloading; Unloading 은 elastic 으로 가정한다. 그리고

residual stress 를 구하기 위해, moment $-M_b$ 이 의해서 발생하는 elastic stress 를 위에서 구한 loading stress 에 더해 준다.

$$I_{yy} = 2 \left[(2) \left(\frac{1}{8} \right) (2)^2 + (2) \left(\frac{1}{8} \right) (1)^2 \right] = 2.5 \text{ in}^4$$

Moment $-M_b$ 이 의한 Elastic stress

$$(\sigma_x)_i = -\frac{(-M_b) y_i}{I_{yy}} = \frac{M_b \cdot y_i}{I_{yy}}$$

$$\text{따라서 } (\sigma_x)_1 = (82500)(2) / (2.5) = 66000 \text{ psi} = -(\sigma_x)_4$$

$$(\sigma_x)_2 = (82500)(1) / (2.5) = 33000 \text{ psi} = -(\sigma_x)_3$$

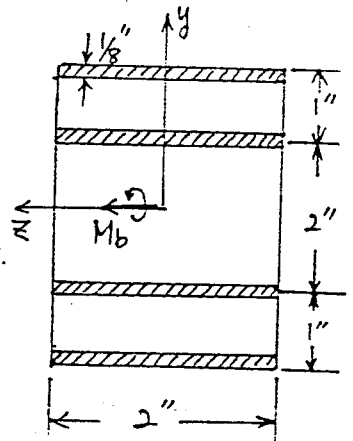
$$\therefore \text{Residual Stress } (\sigma_x)_1 = -55000 + 66000 = 11000 \text{ psi}$$

$$(\sigma_x)_2 = -55000 + 33000 = -22000 \text{ psi}$$

$$(\sigma_x)_3 = 55000 - 33000 = 22000 \text{ psi}$$

$$(\sigma_x)_4 = 55000 - 66000 = -11000 \text{ psi}$$

Ans.



7.68 Wire diameter ; $d = 0.036"$

From postage scale test ; $M_L = 0.61$ lb-in

$$\text{여기 항복에 대한 항응해 ; } M_b = M_Y = \left(\frac{\pi d^4}{64}\right) \left(\frac{\tau}{\sigma}\right) Y = \frac{\pi d^3}{32} Y$$

$$< \text{Table 7.1} > \text{에 } \frac{M_L}{M_Y} = 1.3$$

$$\therefore M_L = 1.3 M_Y = \frac{(1.3) \pi d^3 Y}{32} = 0.61 \text{ lb-in}$$

$$\therefore Y = \frac{(0.61)(32)}{(1.3)(\pi)(0.036)^3} = 102,000 \text{ psi}$$

여기에 적당한 재료는 1020 CR steel 이다.

Ans. $Y = 102,000 \text{ psi}$ material ; 1020 CR steel

7.69 (a) Butt-brazed joint

$$450 \text{ Newton 의 외하 tensile stress ; } \sigma_t = \frac{450}{0.001 \times 0.02} = 18 \text{ MN/m}^2$$

$$(\sigma_b)_p = K E y = \frac{1}{0.3} (110 \times 10^9) (0.0005) = 183.3 \text{ MN/m}^2$$

$$(\sigma_b)_s = K E y = \frac{1}{0.3} (78 \times 10^9) (0.0005) = 130 \text{ MN/m}^2$$

$$\therefore \sigma_p = \sigma_t + (\sigma_b)_p = 201.3 \text{ MN/m}^2 < \begin{matrix} \text{Fatigue limit} \\ \text{Tensile strength} \end{matrix}$$

$$\sigma_s = \sigma_t + (\sigma_b)_s = 148 \text{ MN/m}^2 > \text{Fatigue limit}$$

$\therefore$ Silver-brazing alloy 가 파괴 되었을 것이다.

$t = 1.2 \text{ mm}$ 을 정함.

$$\sigma_t = \frac{450}{0.0012 \times 0.025} = 15 \text{ MN/m}^2$$

$$(\sigma_b)_p = K E y = \frac{1}{0.3} (110 \times 10^9) (0.0006) = 220 \text{ MN/m}^2$$

$$(\sigma_b)_s = K E y = \frac{1}{0.3} (78 \times 10^9) (0.0006) = 156 \text{ MN/m}^2$$

$$\therefore \tilde{\sigma}_p = \tilde{\sigma}_t + (\tilde{\sigma}_b)_p = 235 \text{ MN/m}^2$$

$$\tilde{\sigma}_s = \tilde{\sigma}_t + (\tilde{\sigma}_b)_s = 171 \text{ MN/m}^2$$

따라서 Tensile Strength > $\tilde{\sigma}_p$ > Fatigue Limit

Tensile Strength > $\tilde{\sigma}_s$ > Fatigue Limit

$\therefore$ 불안정하다. ($\because \tilde{\sigma}_p, \tilde{\sigma}_s$ 가 Fatigue Limit 보다 크다)

(b) $t = 0.3 \text{ mm}$ 일 때;

$$\tilde{\sigma}_t = \frac{450}{0.0003 \times 0.025} = 60 \text{ MN/m}^2$$

$$(\tilde{\sigma}_b)_p = \frac{1}{0.3} (110 \times 10^9) (0.00015) = 55 \text{ MN/m}^2$$

$$(\tilde{\sigma}_b)_s = \frac{1}{0.3} (78 \times 10^9) (0.00015) = 39 \text{ MN/m}^2$$

$$\tilde{\sigma}_p = \tilde{\sigma}_t + (\tilde{\sigma}_b)_p = 115 \text{ MN/m}^2 < \text{Fatigue Limit}$$

$$\tilde{\sigma}_s = \tilde{\sigma}_t + (\tilde{\sigma}_b)_s = 99 \text{ MN/m}^2 < \text{Fatigue Limit}$$

$\therefore$ 적당하다. ($\because \tilde{\sigma}_p, \tilde{\sigma}_s$ 가 Fatigue Limit 보다 작다)

(c) Lap-brazed joint;

$$(\tilde{\sigma}_b)_p = \frac{1}{0.3} (110 \times 10^9) (0.001) = 367 \text{ MN/m}^2 > \text{Fatigue Limit}$$

$\therefore$ 불안정 하다.

(d) Modified lap-brazed joint;

(a)의 경우와 같다.

(e) Butt-brazed joint

$$\tilde{\sigma}_t = \frac{450}{0.001 \times 0.075} = 6 \text{ MN/m}^2$$

또 (a)에 따라 $(\tilde{\sigma}_b)_p = 183.3 \text{ MN/m}^2$ $(\tilde{\sigma}_b)_s = 130 \text{ MN/m}^2$

따라서 $\tilde{\sigma}_p = \tilde{\sigma}_t + (\tilde{\sigma}_b)_p = 189.3 \text{ MN/m}^2 < \text{Fatigue Limit}$

$\tilde{\sigma}_s = \tilde{\sigma}_t + (\tilde{\sigma}_b)_s = 136 \text{ MN/m}^2 > \text{Fatigue Limit}$

∴ Silver-brazing alloy가 불안정하다.

7.70 (a) ㉔의 방법이 $Q_d = Q_{max}$ 이 되어서 τ 가 가장 크기 때문에 가장 나쁘다. ($\because \tau = VQ/bI$)

(b) ㉕의 방법이 $Q_d = Q_{min}$ 이 되어서 τ 가 가장 작기 때문에 가장 좋다.

㉔의 경우; Reaction at both ends $= \frac{1}{2} \times 18 \times 10^3 (N)$

$$I = \frac{1}{12} b h^3 \Rightarrow \frac{1}{12} \times 75 \times (150^3 - 145^3) = 3.31 \times 10^6 (mm^4)$$

$$Q_{max} = 75 \cdot \left(\frac{75}{2} \right) - (72.5)(70) \left(\frac{72.5}{2} \right) = 2.697 \times 10^4 (mm^3)$$

$$\therefore \tau = \frac{VQ}{bI} = \frac{(9 \times 10^3)(2.697 \times 10^4)}{(0.005)(3.31 \times 10^6)} = 14.67 \text{ MN/mm}^2$$

$$\text{Ans } \tau = 14.67 \text{ MN/mm}^2$$

7.71 문제에서 $E_w = 0.06 E_m$.

$$(M_b)_{max} = \frac{PL}{4} \text{ (at loading point)}$$

$$I_w = \frac{1}{12} (0.15)(20t)^3 = 100t^3$$

$$I_m = 2 \left[\frac{1}{12} (0.15)t^3 + (0.15)t(10.5t)^2 \right] = 33.1t^3$$

$$k = \frac{1}{\rho} = \frac{M_b}{E_m I_m + E_w I_w} = \frac{\frac{1}{4} PL}{100t^3 \cdot E_w + 33.1t^3 E_m} = \frac{PL}{156.4 E_m t^3}$$

($\because E_w = 0.06 E_m$)

$$\sigma = k E y \text{ 이라 } \sigma_{max} = k E y_{max}$$

$$\therefore (\sigma_m)_{max} = \frac{PL}{156.4 E_m t^3} E_m \cdot 11t = 0.07033 \frac{PL}{t^2}$$

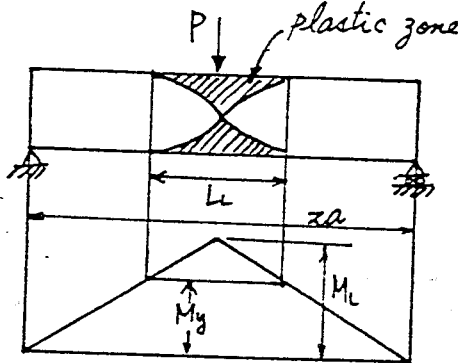
$$(\sigma_w)_{max} = \frac{PL}{156.4 E_m t^3} E_w 10t = \frac{PL}{156.4 E_m t^3} (0.06 E_m) 10t$$

$$= 0.00384 (PL/t^2)$$

$$\therefore \sigma_{max} = 0.07033 \frac{PL}{t^2}$$

$$\text{Ans. } \sigma_{max} = 0.07033 \frac{PL}{t^2}$$

7.72 (참조, text p.456~)



Elastic region 에서는 curvature 가 linear 하게 증가하다가 plastic region에 들어서면 급격히 증가하게 된다. P가 점점 더 커져서 plastic hinge를 이루게 되면 plastic 한 중간점을 중심으로 길이가

a인 두개의 beam이 hinged 된 것처럼 작용을 하게 될 것이다. 즉, moment는 전달하지 못하고, Force만을 전달한다. 따라서 plastic zone의 양 끝에서의 moment는 M_y 가 될 것이고, shear force $P/2$ 도 양쪽 hinge에 각각 $P/2$ 씩 전달하게 된다.

$$\text{평형 조건: } M_y = \frac{P}{2} \left(a - \frac{L}{2} \right), \quad P = \frac{2M_L}{a}$$

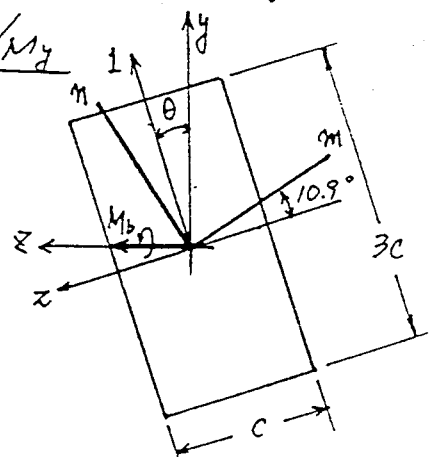
$$\therefore M_y = \frac{M_L}{a} \left(a - \frac{L}{2} \right) \quad \therefore L = 2a \left(1 - \frac{M_y}{M_L} \right) = 2a \left(1 - \frac{1}{f} \right)$$

$$\text{단 } f = M_L / M_y$$

$$\text{Ans. } L = 2a \left(1 - \frac{1}{f} \right), \quad f = M_L / M_y$$

7.73

$$\begin{cases} \frac{d\alpha}{ds_1} = 0.3849 \frac{M_{bz}}{EC^4} \\ \frac{d\beta}{ds_2} = -\alpha \frac{M_{bz}}{EC^4} \end{cases}$$



$$\frac{d\alpha}{ds_1} = \frac{d\phi}{ds} \cos\theta, \quad \frac{d\beta}{ds_2} = \frac{d\phi}{ds} \sin\theta$$

$$\therefore \left(\frac{d\phi}{ds}\right)^2 = \left(\frac{d\alpha}{ds_1}\right)^2 + \left(\frac{d\beta}{ds_2}\right)^2$$

$$\therefore \left(\frac{d\phi}{ds}\right) = \sqrt{(0.3849)^2 + (-2)^2} \cdot \frac{M_{bz}}{EC^4} = 2.037 \frac{M_{bz}}{EC^4}$$

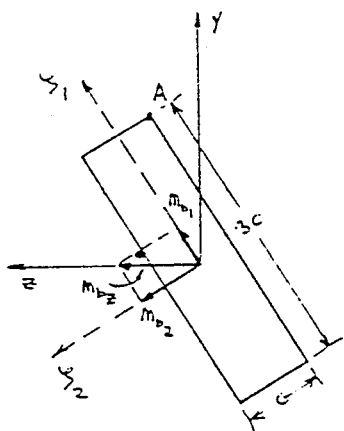
$$\alpha = \tan^{-1} \frac{(-2)}{0.3849} = -79.6^\circ$$

$$(\sigma_x)_A = -E \cdot m \left(\frac{d\phi}{ds}\right) = -E (0.7746C) (2.037 \frac{M_{bz}}{EC^4})$$

$$\approx -1.6 (M_{bz}/C^3) \quad (\because m = 0.7746C)$$

$$\text{Ans. } \frac{d\phi}{ds} = 2.037 \frac{M_{bz}}{EC^4}$$

7.74



임의의 축 ξ_1, ξ_2 에 대하여

$$\sigma_x = - \left[\frac{(\xi_1 I_{22} - \xi_2 I_{12}) M_{b2} + (\xi_1 I_{12} - \xi_2 I_{11}) M_{b1}}{I_{11} I_{22} - I_{12}^2} \right]$$

만일, ξ_1, ξ_2 가 principal axis 라면,

$I_{12} = 0$ 따라 σ_x 는 다음과 같이 된다.

$$\sigma_x = - \left[\frac{\xi_1}{I_{11}} M_{b2} - \frac{\xi_2}{I_{22}} M_{b1} \right] \quad \text{--- ①}$$

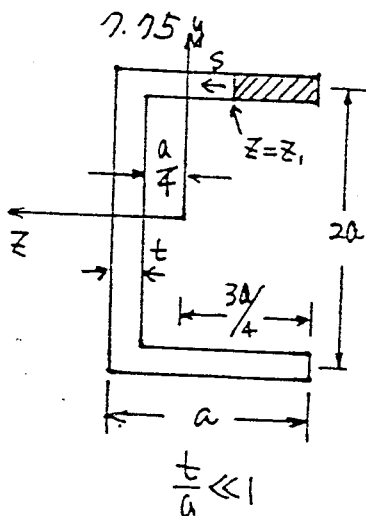
A점의 좌표; $\xi_1 = 1.5C, \xi_2 = -0.5C$

$$\therefore I_{11} = \frac{1}{12} (C) (3C)^3 = \frac{9}{4} C^4, \quad I_{22} = \frac{1}{12} (3C) (C)^3 = \frac{1}{4} C^4$$

$$M_{b1} = M_{bz} \cos 60^\circ = 0.5 M_{bz}, \quad M_{b2} = M_{bz} \sin 60^\circ = 0.866 M_{bz}$$

$$\therefore \sigma_x = -1.6 \frac{M_{bz}}{C^3}$$

$$\text{Ans. } \sigma_x = -1.6 (M_{bz}/C^3)$$



shear center는 z 축상에 있고, $I_{yz} = 0$.
 y 축 방향의 힘 V 에 의한 shear flow
 $q_{sx} = q_{xs} = q$.

i) upper leg (z 방향이 $+s$ 방향);
 first moment of area: Q

$$Q = (z_1 + \frac{3a}{4})(t)(a)$$

$$\therefore q = -\frac{VQ}{I_{yy}} = -\frac{V}{I_{yy}} at(z_1 + \frac{3a}{4})$$

$$(-\frac{3a}{4} \leq z_1 \leq \frac{a}{4} ; y=a)$$

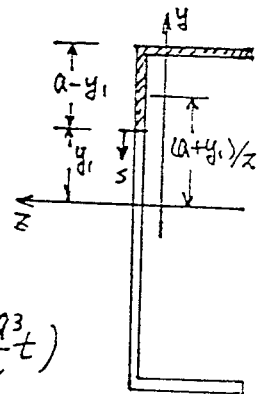
ii) lower leg ($-z$ 방향이 $+s$ 방향);

$$q = -\frac{V}{I_{yy}} at(z_1 + \frac{3a}{4}) \quad (-\frac{3a}{4} \leq z_1 \leq \frac{a}{4} ; y=-a)$$

iii) 오른쪽 그림의 빗금친 부분;

$$Q = (a^2t) + (a-y_1)(t) \frac{(a+y_1)}{2}$$

$$= \frac{3a^2t}{2} - \frac{t}{2} y_1^2 ; \quad q = -\frac{VQ}{I_{yy}}$$



Resultant z -direction force

$$\text{upper; } F_{1z} = \int_{-\frac{3a}{4}}^{\frac{a}{4}} q dz = -\frac{V}{I_{yy}} \left(\frac{a^3t}{2} \right)$$

$$\text{lower; } F_{2z} = \int_{-\frac{3a}{4}}^{\frac{a}{4}} (-q) dz = \frac{V}{I_{yy}} \left(\frac{a^3t}{2} \right)$$

Resultant y -direction force

$$F_{3y} = \int_{-a}^a (-q) dy = \frac{V}{I_{yy}} \int_{-a}^a \left(\frac{3a^2t}{2} - \frac{t}{2} y^2 \right) dy = \frac{V}{I_{yy}} \left(\frac{8}{3} a^3t \right)$$

horizontal force 에 의한 clockwise 의 moment

$$\frac{V}{I_{yy}} \left(\frac{a^3 t}{2} \right) (2a) = \frac{V}{I_{yy}} (a^4 t) \quad \text{--- ①}$$

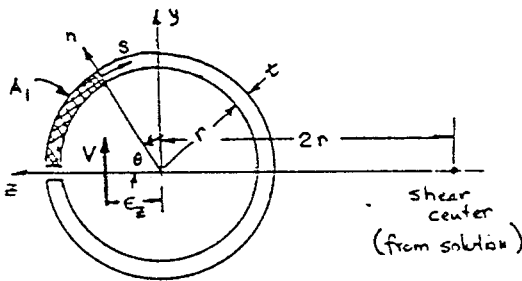
vertical force 에 의한 counter clockwise 의 moment

$$\frac{V}{I_{yy}} \left(\frac{8a^3 t}{3} \right) \cdot d \quad \text{--- ②}$$

①=② 이라 $d = \frac{3}{8}a$

Ans. 무게 중심에서 Z 방향으로 $(\frac{3}{8}a + \frac{a}{4})$

7.76



대칭성에 의해

shear center는 Z축상에
있고, $I_{yz} = 0$.

shear center가 원점
으로부터 Z의 양의
방향으로 $2r$ 만큼 떨어져
있다고 가정하면,

빋금친 부분 A_1 ; $Q = \int y dA = \int_{\phi=0}^{\phi=\theta} (r \sin \phi) (r d\phi) = r^2 (1 - \cos \theta)$

$$\therefore Q_{xs} = -\frac{V}{I_{yy}} r^4 t (1 - \cos \theta)$$

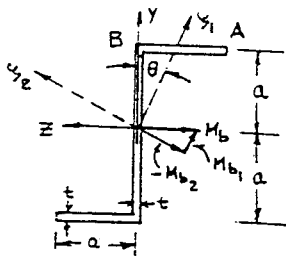
원점에 관한 Q_{xs} 의 moment; $-\int_{\theta=0}^{\theta=2\pi} Q_{xs} (r d\theta) (r)$

$$= \frac{V}{I_{yy}} (2\pi r^4 t) \quad \text{--- ①}$$

원점에 관한 shear force의 moment; $-V e_z \quad \text{--- ②}$

①=②이라 $e_z = -\frac{2\pi r^4 t}{\pi r^3 t} = -2r$ Ans. $e_z = -2r$

7.77

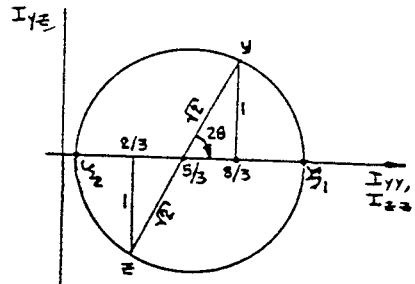


$$I_{y1} = \frac{8}{3} a^3 t$$

$$I_{z1} = \frac{2}{3} a^3 t$$

$$I_{y1z1} = -a^3 t$$

Mohr's circle : ($I/a^3 t$ is plotted)



Mohr's circle 011

$$I_1 = 3.08 a^3 t$$

$$I_2 = 0.26 a^3 t ; \theta = 22.5^\circ$$

문제 (7.74) 011

$$\tilde{\sigma}_x = \frac{M_{b1}}{I_{22}} \xi_2 - \frac{M_{b2}}{I_{11}} \xi_1$$

$$M_{b1} = M_b \sin 22.5^\circ = 0.384 M_b$$

$$M_{b2} = -M_b \cos 22.5^\circ = -0.923 M_b$$

max. stress 는 A, B 011 생긴다.

$$A; \xi_1 = 1.307 a$$

$$B; \xi_1 = 0.923 a$$

$$\xi_2 = -0.539 a$$

$$\xi_2 = 0.384 a$$

$$(\tilde{\sigma}_x)_A = \frac{(0.384 M_b)(-0.539 a)}{(0.26 a^3 t)} - \frac{(-0.923 M_b)(1.307 a)}{(3.08 a^3 t)} = -0.406 \frac{M_b}{a^2 t}$$

$$(\tilde{\sigma}_x)_B = 0.844 \frac{M_b}{a^2 t} \quad \therefore (\tilde{\sigma}_x)_{\max} = (\tilde{\sigma}_x)_B = 0.844 \frac{(M_b)_{\max}}{a^2 t}$$

$$(M_b)_{\max} = PL \quad \therefore (\tilde{\sigma}_x)_{\max} = \frac{0.844 PL}{a^2 t}$$

$$\underline{\text{Ans. } (\tilde{\sigma}_x)_{\max} = 0.844 PL / a^2 t}$$

7.78

$$I_{11} = 98.2 \text{ in}^4, \quad I_{22} = 21.4 \text{ in}^4$$

y축을 1축으로, z축을 2로

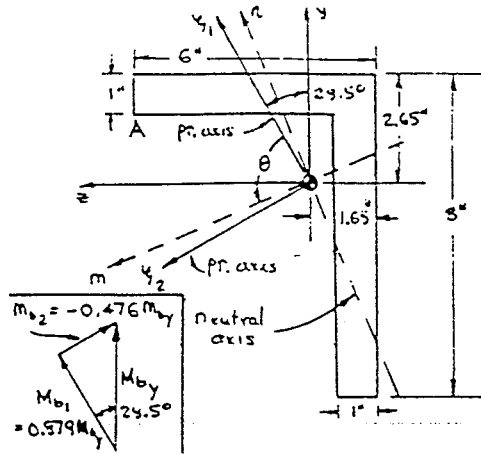
바뀌어 주면, 1, 2축은
principal axis 이므로

$$I_{12} = 0.$$

θ 를 1축으로부터 m 까지의
반시계 방향의 각도로 놓으면

$$\frac{d\phi}{ds} \cos \theta = \frac{M_{b2}}{EI_{11}}$$

$$\frac{d\phi}{ds} \sin \theta = -\frac{M_{b1}}{EI_{22}}$$



$$\therefore \tan \theta = -\frac{M_{b1} I_{11}}{M_{b2} I_{22}} = -\frac{(0.5799 M_{by})(8.2)}{(-0.476 M_{by})(21.4)} = 8.5$$

$\therefore \theta = 83.3^\circ$ 따라서 m, n 의 위치는 그림과 같다.

6개의 모서리에 대해 모두 σ_x 를 구하여 가장 큰 것이
 σ_{max} 이 된다.

max. m coordinate를 갖는 점은 A ; $\xi_1 = 3.53''$
 $\xi_2 = 3.03''$

$$\therefore (\sigma_x)_A = (\sigma_x)_{max} = \frac{M_{b1}}{I_{22}} \xi_2 - \frac{M_{b2}}{I_{11}} \xi_1 = \frac{(0.5799 M_{by})(3.03)}{21.4} - \frac{(-0.476 M_{by})(3.53)}{8.2}$$

$$= 0.141 M_{by} = 0.141 (100,000)$$

$$= 14100 \text{ (psi)} \quad \underline{\text{Ans } 14100 \text{ psi}}$$

2.79

대칭성에 의해 shear center는 z축상에 있고,

$$I_{yz} = 0.$$

vertical force V_y on the

$$Q_{xs} = Q_{sx} = -\frac{V_y}{I_{yy}} \int y dA$$

horizontal member is

$$\int y dA \approx 0 \text{ 이므로 } Q_{xs} = 0.$$

⇒ 왼쪽 vertical leg ;

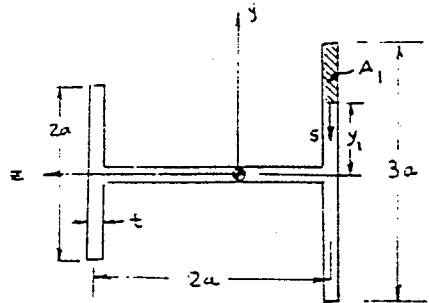
$$\int_{A_1} y dA = \int_{y_1}^{3a/2} y (\pm dy)$$

$$= \frac{\pm}{2} \left(\frac{9a^2}{4} - y_1^2 \right)$$

$$\therefore Q_{xs} = -\frac{V_y \cdot t}{I_{yy}} \left(\frac{9a^2}{4} - y_1^2 \right)$$

$$\text{total force ; } F_1 = \frac{V_y \cdot t}{2I_{yy}} \int_{-3a/2}^{3a/2} \left(\frac{9a^2}{4} - y^2 \right) dy$$

$$= \frac{V_y \cdot t}{2I_{yy}} \left(\frac{9}{2} a^3 \right)$$



ii) 오른쪽 vertical leg ;

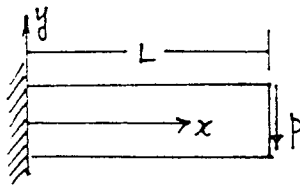
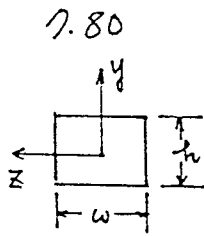
$$\int y dA = \frac{\pm}{2} (a^2 - y_1^2)$$

$$F_2 = \frac{V_y \cdot t}{2I_{yy}} \int_{-a}^a (a^2 - y^2) dy = \frac{V_y \cdot t}{2I_{yy}} \left(\frac{4}{3} a^3 \right)$$

i) ii) 에서 평형조건 $\sum M_{\text{shear center}} = 0$

$$\frac{V_y \cdot t}{2I_{yy}} \left(\frac{9}{2} a^3 \right) (\delta) = \frac{V_y \cdot t}{2I_{yy}} \left(\frac{4}{3} a^3 \right) (2a - \delta)$$

$$\therefore \delta = \frac{16}{35} a \quad \underline{\text{Ans } \delta = \frac{16}{35} a}$$



(w/h)가 작고, p 를 적당히 작음시키면 x - y plane 에 σ plane stress로 고려할 수 있다.

$$y = h/2 \text{ 에 } \sigma_x = \text{tension}; \sigma_y = 0; \tau_{xy} = 0$$

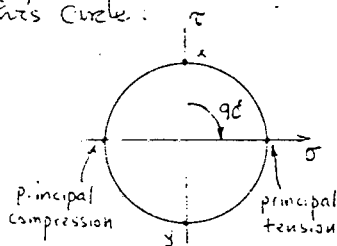
$\therefore x$ 는 principal tension direction
 y 는 " compression "

$$y = -h/2 \text{ 에 } \sigma_x = \text{compression}; \sigma_y = 0; \tau_{xy} = 0$$

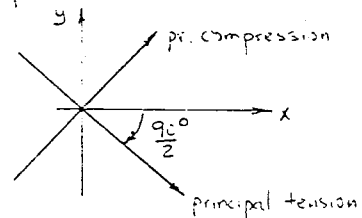
$\therefore x$ 는 principal compression direction
 y 는 " tension "

$$y = 0 \text{ 에 } \sigma_x = 0; \sigma_y = 0; \tau_{xy} = \text{negative}$$

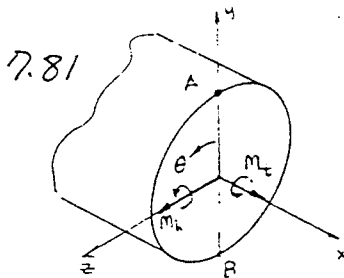
Moire's Circle:



Principal directions —



따라서 trajectories 는 옳다.



Bending 은 normal stress σ_x .
 Torsion 은 shear stress τ_{xy} 를 일으킬 것이다. (θ 는 전선 방향의 각도)

이와 같은 stress는 A, B에서 최대가 될 것이고
max. shear stress도 이와 같은 점에서 생길 것이다.

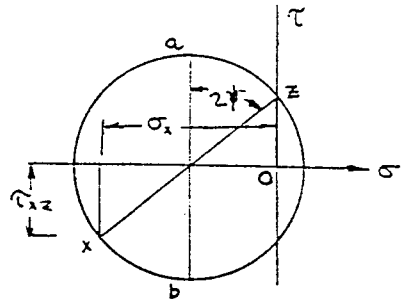
A점에서의 stress;

$$\sigma_x = -\frac{(M_b)(r)}{I_{yy}}$$

$$\tau_{xz} = \tau_{zx} = \frac{(M_t)(r)}{I_{xx}} = \frac{M_t \cdot r}{2I_{yy}}$$

$$(\because I_{xx} = \frac{\pi}{2} r^4 = 2 \left(\frac{\pi}{4} r^4 \right) = 2I_{yy})$$

다른 응력은 0이다



< Mohr's circle >

Mohr's circle에서

$$\tau_{max} = \left[\left(\frac{\sigma_x}{2} \right)^2 + (\tau_{xz})^2 \right]^{1/2} = \frac{r}{2I_{yy}} \sqrt{(M_b)^2 + (M_t)^2}$$

$$\tan 2\psi = \frac{-\sigma_x/2}{\tau_{xz}} = \frac{M_b \cdot r}{2I_{yy}} \cdot \frac{2I_{yy}}{M_t \cdot r} = \frac{M_b}{M_t}$$

$$\therefore \psi = \frac{1}{2} \tan^{-1} \left(\frac{M_b}{M_t} \right)$$

Hollow shaft에서도, solid shaft와 마찬가지로

$I_x = 2I_{yy}$ 이고, max. shear stress도 r_{max}

에서 생기므로 성립된다.

CHAPTER 8

8.1 (a) $EIy'' = -PL + px$

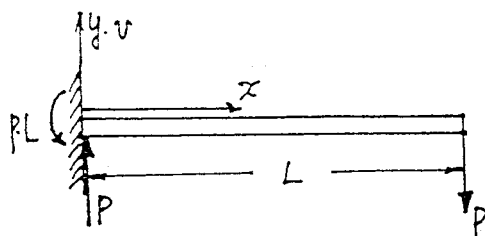
$$EIy' = -PLx + \frac{1}{2}px^2 + C_1$$

$$y'(0) = 0 \quad \therefore C_1 = 0$$

$$EIy = \frac{1}{6}px^3 - \frac{1}{2}PLx^2 + C_2$$

$$y(0) = 0 \quad \therefore C_2 = 0$$

$$\therefore y = \frac{P}{6EI} (x^3 - 3Lx^2) ; \text{Ans.}$$



(b) $EIy'' = px - pa - p\langle x-a \rangle'$

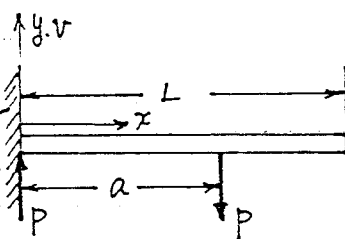
$$EIy' = \frac{1}{2}px^2 - pax - \frac{1}{2}p\langle x-a \rangle^2 + C_1$$

$$y'(0) = 0 \quad \therefore C_1 = 0$$

$$EIy = \frac{1}{6}px^3 - \frac{pa}{2}x^2 - \frac{1}{6}p\langle x-a \rangle^3 + C_2$$

$$y(0) = 0 \quad \therefore C_2 = 0$$

$$\therefore y = \frac{P}{6EI} (x^3 - 3ax^2 - \langle x-a \rangle^3) ; \text{Ans.}$$



(c) $EIy'' = \frac{3}{16}\omega_0 Lx - \frac{\sqrt{3}}{4}\omega_0 x^2 + \frac{9}{16}\omega_0 L\langle x - \frac{2}{3}L \rangle'$

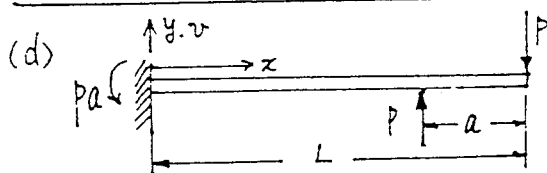
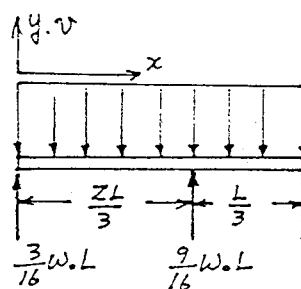
$$EIy' = \frac{3}{32}\omega_0 Lx^2 - \frac{\sqrt{3}}{12}\omega_0 x^3 + \frac{9}{32}\omega_0 L\langle x - \frac{2}{3}L \rangle^2 + C_1$$

$$EIy = \frac{1}{32}\omega_0 Lx^3 - \frac{\sqrt{3}}{48}\omega_0 x^4 + \frac{3}{32}\omega_0 L\langle x - \frac{2}{3}L \rangle^3$$

$$+ C_1 x + C_2$$

$$y(0) = 0 \quad \therefore C_2 = 0, \quad y\left(\frac{2}{3}L\right) = 0 \quad \therefore C_1 = -0.0032\omega_0 L^3$$

$$\therefore y = \frac{1}{EI} \left[\frac{1}{32}\omega_0 Lx^3 - \frac{\sqrt{3}}{48}\omega_0 x^4 + \frac{3}{32}\omega_0 L\langle x - \frac{2}{3}L \rangle^3 - 0.0032\omega_0 L^3 x \right] ; \text{Ans.}$$

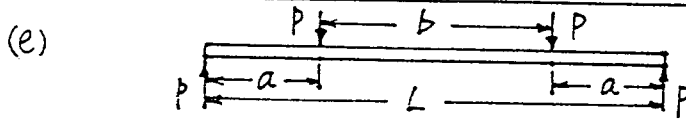


$$EIy'' = -pa + p\langle x-L+a \rangle'$$

$$EIy' = -pax + \frac{1}{2}p\langle x-L+a \rangle^2 + C_1; \quad y'(0) = 0 \quad \therefore C_1 = 0$$

$$EIy = -\frac{1}{2}pax^2 + \frac{1}{6}p\langle x-L+a \rangle^3 + C_2; \quad y(0) = 0 \quad \therefore C_2 = 0$$

$$\therefore y = \frac{p}{6EI} [-3ax^2 + \langle x-L+a \rangle^3] \quad ; \text{Ans.}$$



$$EIy'' = px - p\langle x-a \rangle' - p\langle x-a-b \rangle'$$

$$EIy' = \frac{1}{2}px^2 - \frac{1}{2}p\langle x-a \rangle^2 - \frac{1}{2}p\langle x-a-b \rangle^2 + C_1$$

$$EIy = \frac{1}{6}px^3 - \frac{1}{6}p\langle x-a \rangle^3 - \frac{1}{6}p\langle x-a-b \rangle^3 + C_1x + C_2$$

$$y(0) = 0 \quad \therefore C_2 = 0$$

$$y(2a+b) = 0; \quad \frac{1}{6}p(2a+b)^3 - \frac{1}{6}p(a+b)^3 - \frac{1}{6}pa^3 + C_1(2a+b) = 0$$

$$\therefore C_1 = -\frac{1}{2}pa(a+b)$$

$$\therefore y = \frac{p}{6EI} [x^3 - \langle x-a \rangle^3 - \langle x-a-b \rangle^3 - 3a(a+b)x] \quad ; \text{Ans.}$$

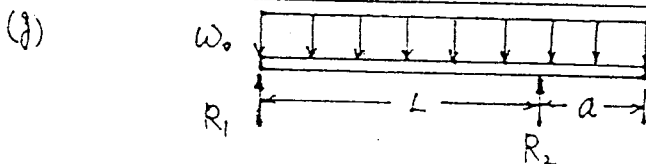


$$EIy'' = w_0 Lx - \frac{1}{2}w_0 L^2 - \frac{1}{2}w_0 x^2$$

$$EIy' = \frac{1}{2}w_0 Lx^2 - \frac{1}{2}w_0 L^2x - \frac{1}{6}w_0 x^3 + C_1; \quad y'(0) = 0 \quad \therefore C_1 = 0$$

$$EIy = \frac{1}{6}w_0 Lx^3 - \frac{1}{4}w_0 L^2x^2 - \frac{1}{24}w_0 x^4 + C_2; \quad y(0) = 0 \quad \therefore C_2 = 0$$

$$\therefore y = \frac{w_0}{24EI} (4Lx^3 - 6L^2x^2 - x^4) \quad ; \text{Ans.}$$



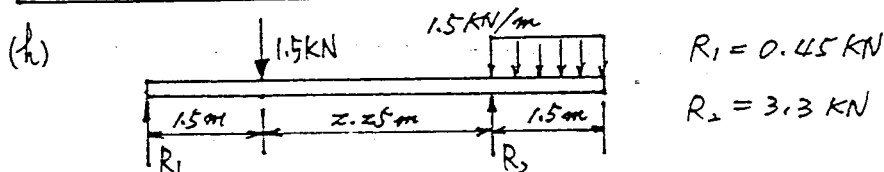
$$EIy'' = \frac{w_0(L^2 - a^2)}{2L}x - \frac{w_0(L+a)^2}{2L}\langle x-L \rangle' - \frac{1}{2}w_0x^2$$

$$EIy' = \frac{w_0(L^2 - a^2)}{4L}x^2 - \frac{w_0(L+a)^2}{4L}\langle x-L \rangle^2 - \frac{1}{6}w_0x^3 + C_1$$

$$EIy = \frac{w_0(L^2 - a^2)}{12L}x^3 - \frac{w_0(L+a)^2}{12L}\langle x-L \rangle^3 - \frac{1}{24}w_0x^4 + C_1x + C_2$$

$$y(0)=0 \therefore C_2=0, \quad y(L)=0 \therefore C_1 = \frac{1}{24}w_0L(2a^2 - L^2)$$

$$\therefore y = \frac{w_0}{EI} \left[\frac{(L^2 - a^2)}{12L}x^3 - \frac{(L+a)^2}{12L}\langle x-L \rangle^3 - \frac{1}{24}x^4 - \frac{1}{24}L(L^2 - 2a^2)x \right]$$



$$EIy'' = R_1x - P\langle x-a \rangle' + R_2\langle x-a-b \rangle' - \frac{1}{2}w_0\langle x-a-b \rangle^2$$

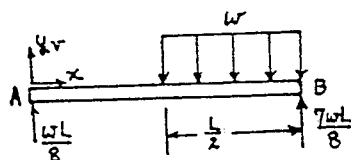
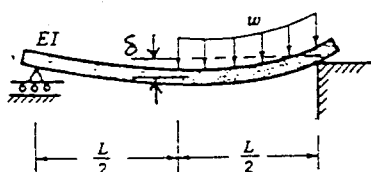
$$EIy' = \frac{1}{2}R_1x^2 - \frac{P}{2}\langle x-a \rangle^2 + \frac{R_2}{2}\langle x-a-b \rangle^2 - \frac{1}{6}w_0\langle x-a-b \rangle^3 + C_1$$

$$EIy = \frac{1}{6}R_1x^3 - \frac{P}{6}\langle x-a \rangle^3 + \frac{R_2}{6}\langle x-a-b \rangle^3 - \frac{1}{24}w_0\langle x-a-b \rangle^4 + C_1x + C_2$$

$$y(0)=0 \therefore C_2=0, \quad y(a+b)=0 \therefore C_1 = 0.2953$$

$$\therefore y = \frac{1}{EI} [0.075x^3 - 0.25\langle x-1.5 \rangle^3 + 0.55\langle x-3.75 \rangle^3 - 0.0625\langle x-3.75 \rangle^4 + 0.2953x] : \text{Ans.}$$

8.2

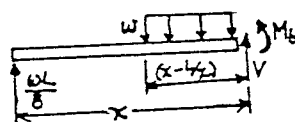


Bending Moment

$$EIy'' = M_b = \frac{wL}{8}x - \frac{w}{2}\langle x - \frac{L}{2} \rangle^2$$

$$EIy = \frac{wLx^3}{48} - \frac{w}{24}\langle x - \frac{L}{2} \rangle^4 + C_1x + C_2$$

$$y(0)=0 \therefore C_2=0, \quad y(L)=0 \therefore C_1 = -\frac{7wL^3}{384}$$



$$\therefore EIy = \frac{\omega Lx}{48} \left(x^2 - \frac{7}{8}L^2 \right) - \frac{\omega}{24} \left\langle x - \frac{L}{2} \right\rangle^4$$

$$\delta = (-y)_{x=\frac{L}{2}} = \frac{5\omega L^4}{168EI}$$

Ans. $(5\omega L^4)/(168EI)$

8.3 Bending Moment

$$EIy'' = -Px + 2P\left\langle x - \frac{L}{2} \right\rangle'$$

적분하면

$$EIy = -\frac{Px^2}{2} + \frac{P}{3}\left\langle x - \frac{L}{2} \right\rangle^3 + C_1x + C_2$$

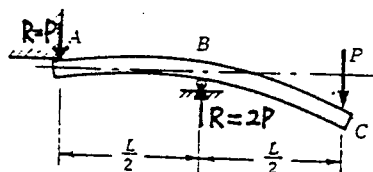
Boundary Condition $y(0) = 0, \therefore C_2 = 0$

$$y\left(\frac{L}{2}\right) = 0, \therefore C_1 = \frac{PL^2}{24}$$

$$\therefore EIy = \frac{Px}{6} \left(\frac{L^2}{4} - x^2 \right) + \frac{P}{3}\left\langle x - \frac{L}{2} \right\rangle^3$$

$$\delta = (-y)_{x=L} = \frac{PL^3}{12EI}$$

Ans. $\frac{PL^3}{12EI}$



8.4 문제 (8.1(e)) 에서 $b = L - 2a$ 를 대입하면

$$y = \frac{P}{6EI} [x^3 - \langle x-a \rangle^3 - \langle x-L+a \rangle^3 - 3a(L-a)x]$$

$$\therefore \delta = y_{x=\frac{L}{2}} = \frac{P}{6EI} \left[\left(\frac{L}{2}\right)^3 - \left(\frac{L}{2}-a\right)^3 - 3a(L-a)\frac{L}{2} \right] = \frac{Pa}{24EI} [4a^2 - 3L^2]$$

Ans. $\frac{Pa}{24EI} (4a^2 - 3L^2)$

8.5 $v_B = v_{C'}$ — ①

중첩의 원리를 이용해서 displacement를

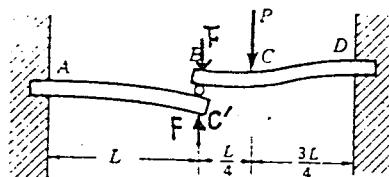
구하면,

Beam ① ; $v_{C'} = -\frac{FL^3}{3EI}$

Beam ② ; $v_B = -\left[\frac{P\left(\frac{3}{4}L\right)^3}{3EI} + \frac{L}{4} \cdot \frac{P\left(\frac{3}{4}L\right)^2}{2EI} - \frac{FL^3}{3EI} \right]$

이 두식을

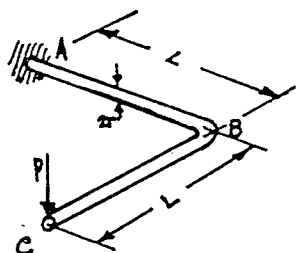
①에 대입하면.



$$F = \frac{81}{256} p$$

$$\text{Ans. } \frac{81}{256} p$$

8.6



중첩 이론.

$$\delta_c = (\delta_c)_{\text{bending}} + (\delta_c)_{\text{torsion}}$$

$$(\delta_c)_{\text{bending}} = (\delta_{BA}) + (\delta_{CB})$$

$$(\delta_c)_{\text{torsion}} = L \times (\phi_{BA})_{\text{torsion}}$$

δ_{BA} : P에 의해서 생기는 Bending Moment로 인한 A에 대한 B의 변위

δ_{CB} : " " " " B에 대한 C의 상대 변위

$$\delta_{BA} = \frac{PL^3}{3EI}; I = \frac{\pi}{4} r^4 \quad \therefore \delta_{BA} = \frac{4PL^3}{3\pi E r^4}$$

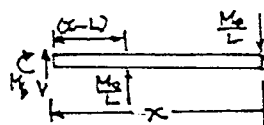
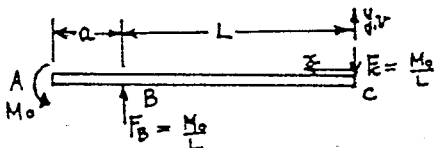
$$\delta_{CB} = \frac{PL^3}{3EI} = \frac{4PL^3}{3\pi E r^4} \quad \phi_{BA} = \frac{M \cdot L}{GI} = \frac{PL \cdot L}{G \cdot \frac{\pi}{2} r^4} \quad \text{모두를 전함하면}$$

$$\delta_c = \frac{8PL^3}{3\pi E r^4} + \frac{2PL^3}{\pi G r^4} = \frac{4PL^3}{3\pi E r^4} (5+3\nu) \quad \left(\because G = \frac{E}{2(1+\nu)} \right)$$

ν : Poisson's Ratio

$$\text{Ans. } \delta = \frac{4PL^3}{3\pi E r^4} (5+3\nu)$$

8.7



$$\text{Bending Moments; } EIy'' = -\frac{M_o}{L}x + \frac{M_o}{L}\langle x-L \rangle', \quad EIy' = -\frac{M_o}{2L}x^2 + \frac{M_o}{2L}\langle x-L \rangle^2 + C_1$$

$$\text{적분하면 } EIy = -\frac{M_o}{6L}x^3 + \frac{M_o}{6L}\langle x-L \rangle^3 + C_1x + C_2$$

$$\text{Boundary Condition } y(0)=0, \quad y(L)=0$$

$$\text{대입하면, } C_2=0, \quad C_1 = \frac{M_o L}{6}$$

$$\therefore y' = \frac{1}{EI} \left(-\frac{M_o}{2L}x^2 + \frac{M_o}{2L}\langle x-L \rangle^2 + \frac{M_o L}{6} \right)$$

$$\phi = (-y')_{x=L+a} = \frac{1}{EI} \left(\frac{M_o}{2L}(L+a)^2 - \frac{M_o}{2L}a^2 - \frac{M_o L}{6} \right) = \frac{M_o(L+3a)}{3EI}$$

$$\text{Ans. } \phi = M_o(L+3a)/(3EI)$$

8.8 Beam C.D 만을 취하자. 다른 모든 부재도 CD와 같은 형태로 변형할 것이다.

D에서의 반력은 힘 F 와 moment M_0 라 하자.

Bending Moment

$$EI\ddot{y} = Mb = Fx - M_0$$

$$EI\dot{y} = F\frac{x^2}{2} - M_0x + C_1$$

$$EIy = F\frac{x^3}{6} - M_0\frac{x^2}{2} + C_1x + C_2$$

B.C $y(0) = 0, y'(0) = 0$

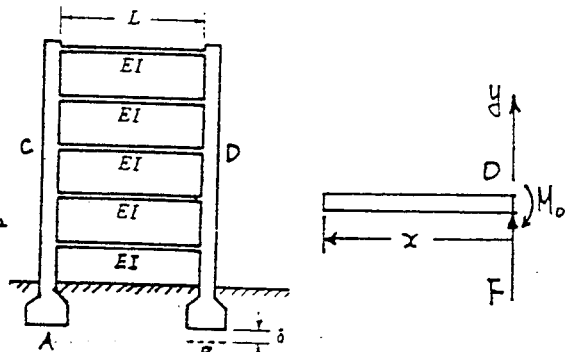
$$y(L) = \delta, y'(L) = 0$$

$$\therefore C_1 = C_2 = 0, F = \frac{12EI\delta}{L^3}, M_0 = \frac{6EI\delta}{L^2}$$

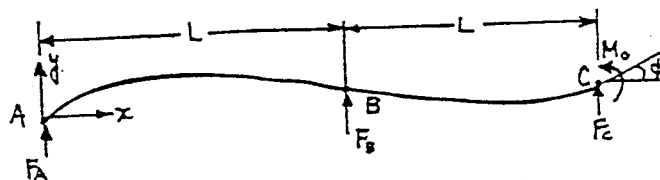
$$\therefore M_b = \frac{12EI\delta}{L^3}x - \frac{6EI\delta}{L^2}$$

$$y(L) = -\delta, \delta = -\frac{1}{EI}[-\frac{1}{6}M_0L^2] \therefore M_0 = \frac{6EI\delta}{L^2}$$

$$\text{Ans. } M_0 = \frac{6EI\delta}{L^2}$$



8.9



Equilibrium ; $\sum F_y = 0 : F_A + F_B + F_C = 0$ — ①

$$\sum M_{BZ} = 0 : F_A - F_C = \frac{M_0}{L}$$

Bending Moment ; $EIy'' = M_b = F_A x + F_B \langle x - L \rangle'$

$$EI\dot{y} = F_A \frac{x^2}{2} + \frac{F_B}{2} \langle x - L \rangle^2 + C_1$$

$$EIy = F_A \frac{x^3}{6} + \frac{F_B}{6} \langle x - L \rangle^3 + C_1x + C_2$$

B.C $y(0) = 0, y(L) = 0, y(2L) = 0$

$$\therefore C_2 = 0, C_1 = -\frac{F_A}{6}L^2$$

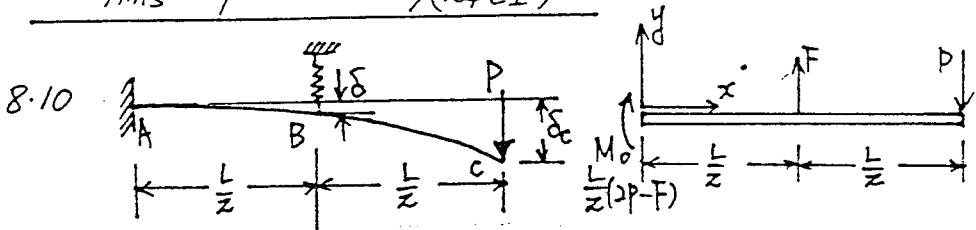
$$\frac{8F_A}{6} + \frac{F_B}{6} - \frac{2F_A}{6} = 0$$

$$\text{① ② ③ 을 ④}$$

$$F_A = -\frac{M_0}{4L} ; F_B = \frac{3M_0}{2L} ; F_C = -\frac{5M_0}{4L}$$

$$\therefore \phi \approx (y')_{x=2L} = \frac{7M_0 L}{24EI}$$

$$\text{Ans } \phi = (7M_0 L) / (24EI)$$



$$\text{Bending Moment ; } EI y'' = M_b = (P-F)x + \frac{1}{2}(2P-F)x + F(x - \frac{L}{2})'$$

$$\text{적분하면 } EI y' = (P-F)\frac{x^2}{2} + \frac{L}{2}(2P-F)x + \frac{F}{2}(x - \frac{L}{2})^2 + C_1$$

$$EI y = (P-F)\frac{x^3}{6} + \frac{L}{4}(2P-F)x^2 + \frac{F}{6}(x - \frac{L}{2})^3 + C_1 x + C_2$$

Bending Moment를 적분한 식의 Boundary Condition을 대입.

$$\text{B.C. } y(0) = 0, y'(0) = 0 \quad \therefore C_1 = C_2 = 0$$

$$\text{Spring ; } F = -k v_B = -k(y)_{x=L/2}$$

$$= -\frac{k}{EI} \left(\frac{7PL^3}{48} - \frac{FL^3}{12} \right) \quad \therefore F = \frac{7P}{4} \cdot \frac{x}{x-1} \quad \left(x \equiv \frac{kL^3}{12EI} \right)$$

$$\delta = -v_B = \frac{F}{k} = \frac{7P}{4k} \left(\frac{kL^3/12EI}{kL^3/12EI - 1} \right)$$

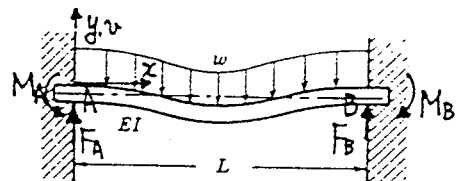
$$\delta_c = (-y)_{x=L} = \frac{PL^3}{3EI} \left[1 - \frac{25}{32 \left(\frac{24EI}{kL^3} + 1 \right)} \right]$$

$$\text{Ans. } \delta_c = \frac{PL^3}{3EI} \left[1 - \frac{25}{32 \left(\frac{24EI}{kL^3} + 1 \right)} \right]$$

$$8.11 \quad \sum F_y = 0 ; F_A + F_B = wL$$

$$\text{대칭 조건 : } M_A = M_B, F_A = F_B = \frac{wL}{2}$$

따라서 $\sum M_{A2} = 0$ 을 만족한다.



Bending Moment ; $EI y'' = M_b = \omega L \cdot \frac{x}{2} - \omega \frac{x^2}{2} - M_A$

적분하면 $EI y' = \omega L \frac{x^2}{4} - \omega \frac{x^3}{6} - M_A x + C_1$

$EI y = \omega L \frac{x^3}{12} - \omega \frac{x^4}{24} - M_A \frac{x^2}{2} + C_1 x + C_2$

Boundary Condition $y(0) = 0 \quad y'(0) = 0 \quad \text{--- ①}$

$y(L) = 0 \quad y'(L) = 0 \quad \text{--- ②}$

①에대 $C_1 = C_2 = 0$

②에대 $M_A = \frac{\omega L^2}{12} \quad \text{--- ③}$

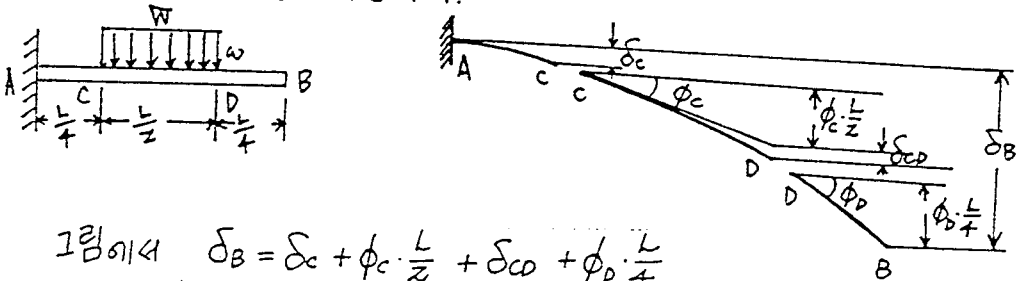
$\therefore M_b = \omega L \frac{x}{2} - \frac{\omega L^2}{12} - \frac{\omega x^2}{2} \quad \frac{dM_b}{dx} = 0 \quad \text{at } x = \frac{L}{2}$

$\therefore |M_b|_{\max} = \frac{\omega L^2}{24} \quad \text{--- ④} \quad \text{③, ④에대 } M_{\max} = \frac{\omega L^2}{12}$

$|v|_{\max} = y\left(\frac{L}{2}\right) = \frac{\omega L^4}{384EI}$

Ans. $M_{\max} = \omega L^2/12 \quad v_{\max} = \omega L^4/384EI$

8.12 중첩의 원리를 적용한다. 즉, beam 이 3개의 cantilever로 구성되어 있다고 가정하자.



1번에대 $\delta_B = \delta_C + \phi_C \cdot \frac{L}{2} + \delta_{CD} + \phi_D \cdot \frac{L}{4}$

$\delta_C = \frac{(\omega L/4)(L/4)^3}{3EI} + \frac{(\omega L^2/8)(L/4)^2}{2EI} = \frac{5\omega L^4}{768EI}$

$\phi_C = \frac{(\omega L/4)(L/4)^2}{2EI} + \frac{(\omega L^2/8)(L/4)}{EI} = \frac{3\omega L^3}{64EI}$

$\delta_{CD} = \frac{\omega(L/2)^4}{8EI} = \frac{\omega L^4}{128EI} \quad ; \quad \phi_{CD} = \frac{\omega(L/2)^3}{6EI} = \frac{\omega L^3}{48EI}$

$\therefore \phi_D = \phi_C + \phi_{CD} = (13\omega L^3)/(192EI)$

$\therefore \delta_B = \frac{5\omega L^4}{768EI} + \frac{3\omega L^3}{64EI} \cdot \frac{L}{2} + \frac{\omega L^4}{128EI} + \frac{13\omega L^3}{192EI} \cdot \frac{L}{4} = \frac{7\omega L^4}{128EI} = \frac{7}{64} \frac{\omega L^3}{EI} \quad \text{--- Q.E.D.}$

8.15 Max. bending moment은 고정단에서 생긴다. $M_{max} = \frac{1}{2} \omega L^2$

(a) $y = \frac{\omega}{EI} \left(\frac{1}{6} Lx^3 - \frac{1}{4} L^2 x^2 - \frac{1}{24} x^4 \right)$ 문제 (8.1 f) 참조

$$\therefore v_{max} = |y(L)| = \frac{\omega L^4}{8EI}$$

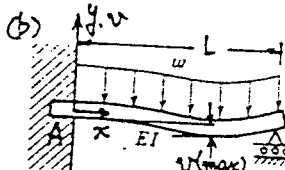
(b) $\delta_B \equiv 0$ - ①

R에 의한 B점의 처짐 = $\frac{RL^3}{3EI}$

ω 에 의한 B점의 처짐 = $-\frac{\omega L^4}{8EI}$

①에서 $\frac{RL^3}{3EI} = \frac{\omega L^4}{8EI} \therefore R = \frac{3}{8} \omega L$

Bending Moment; $M_b = EI y'' = \frac{3}{8} \omega L(L-x) - \frac{\omega}{2}(L-x)^2$



$$EI y' = -\frac{3}{16} \omega L(L-x)^2 + \frac{\omega}{6}(L-x)^3 + C_1$$

$$EI y = \frac{3}{48} \omega L(L-x)^3 - \frac{\omega}{24}(L-x)^4 + C_1 x + C_2 \quad - ②$$

B.C. $y(0) = y'(0) = 0$

C_1, C_2 를 구하여 ②에 대입하면

$$y = \frac{1}{EI} \left[\frac{1}{16} \omega L(L-x)^3 - \frac{\omega(L-x)^4}{24} + \frac{\omega L^3 x}{48} - \frac{\omega L^4}{48} \right]$$

max. deflection은 $y' = 0$ 을 만족하는 $x \approx 0.58L$ 에서 갖게 된다.

$$v_{max} = |y(0.58L)| = 0.42 \frac{\omega L^4}{48EI}$$

$\frac{dM_b}{dx} = 0$ 을 만족하는 점 $x_A = \frac{5}{8}L$

$(M_b)_{x=\frac{5}{8}L} = \frac{9}{128} \omega L^2 \quad (M_b)_{x=0} = \frac{\omega L^2}{8}$

$\therefore (M_b)_{max} = \frac{\omega L^2}{8}$

Ans. (a) $M_{max} = \omega L^2/8 \quad v_{max} = \omega L^4/8EI$

(b) $M_{max} = \omega L^2/8 \quad v_{max} = (0.42)(\omega L^4/48EI)$

8.16 (a) 문제 (8.1(g))에서 $Q=0$ 인 경우.

$$y = \frac{1}{EI} \left[\frac{\omega L x^3}{12} - \frac{\omega x^4}{24} - \frac{\omega L^3 x}{24} \right], \quad M_b = +EI y'' = \left[\frac{\omega L}{2} x - \frac{\omega}{2} x^2 \right]$$

$$v_{\max} = y\left(\frac{L}{2}\right) = \frac{5\omega L^4}{384EI}, \quad (M_b)_{\max} = |M_b|_{x=\frac{L}{2}} = \frac{\omega L^2}{8}$$

(b) $F_A = F_C$ — ①

$$F_A + F_B + F_C = 2\omega L \text{ — ②}$$

Bending Moment ;

$$EI y'' = M_b = F_A x + F_B (x-L) - \frac{\omega x^2}{2}$$

$$\therefore EI y = \frac{F_A}{6} x^3 + \frac{F_B}{6} (x-L)^3 - \frac{\omega x^4}{24} + C_1 x + C_2$$

Boundary Condition ; $y(0) = y(L) = y(2L) = 0$ 이라

$$C_2 = 0, \quad C_1 = -\frac{\omega L^3}{48}$$

$$F_A = F_C = \frac{3\omega L}{8}, \quad F_B = \frac{5}{4}\omega L$$

$$\therefore y = \frac{1}{EI} \left[\frac{3L}{48} x^3 + \frac{10L}{48} (x-L)^3 - \frac{1}{24} x^4 - \frac{L^3}{48} x \right]$$

$\frac{dy}{dx} = 0$ 을 만족하는 $x \approx 0.45L$ 이라 $v_{\max}$ 이 생긴다.

$$|v|_{\max} = \frac{\omega L^4}{200EI} = 1.91 \frac{\omega L^4}{384EI}$$

또한 AB 구간 $M_b = \frac{3\omega L x}{8} - \frac{\omega x^2}{2}$, $x = \frac{3}{8}L$ 이라 $\max$.

$$(M_b)_{x=\frac{3}{8}L} = \frac{9\omega L^2}{128}, \quad (M_b)_{x=L} = \frac{\omega L^2}{8}$$

$$\therefore |M_b|_{\max} = \frac{\omega L^2}{8}$$

Ans. (a) $R = \omega L$, $M_{\max} = \frac{1}{8}\omega L^2$, $\delta_{\max} = \frac{5\omega L^4}{384EI}$

(b) $R = \frac{3}{8}\omega L$, $M_{\max} = \frac{1}{8}\omega L^2$, $\delta_{\max} = 1.91 \frac{\omega L^4}{384EI}$

8.17 (a) $EI y'' = -M_0 + \frac{M_0}{a} (x+a-L)'$

$$EI y' = -M_0 x + \frac{M_0}{2a} (x+a-L)^2 + C_1; \quad y'(0) = 0 \quad \therefore C_1 = 0$$

$$EI y = -\frac{1}{2} M_0 x^2 + \frac{M_0}{6a} (x+a-L)^3 + C_2; \quad y(0) = 0 \quad \therefore C_2 = 0$$

$$\therefore y = \frac{M_0}{6EI} \left[-3x^2 + \frac{1}{a} \langle x-a-L \rangle^3 \right]$$

$$y(L) = \frac{M_0}{6EI} [-3L^2 + a^2] \quad \lim_{a/L \rightarrow 0} = \frac{M_0 L^2}{6EI} \left(-3 + \left(\frac{a}{L} \right)^2 \right) = -\frac{M_0 L^2}{2EI}$$

$$(b) \quad EIy'' = -M_0 \quad EIy' = -M_0 x + C_1; \quad y'(0) = 0 \quad \therefore C_1 = 0$$

$$EIy = -\frac{1}{2} M_0 x^2 + C_2; \quad y(0) = 0 \quad \therefore C_2 = 0$$

$$\therefore y = -\frac{M_0}{2EI} x^2$$

$$y(L) = -\frac{M_0 L^2}{2EI}$$

$$\text{Ans. (a) } y(L) = \frac{M_0}{6EI} [-3L^2 + a^2] \quad (b) \quad y(L) = -\frac{M_0 L^2}{2EI}$$

$(a/L) \rightarrow 0$ 일 경우에는 두 경우의 값이 서로 같다.

$$8.18 \quad g_n(x) = \langle a-x \rangle^n$$

$$\text{즉, } g_n(x) = \begin{cases} (a-x)^n & x \leq a \\ 0 & x > a \end{cases}$$

$$\int_x^\infty \langle a-x \rangle^n dx = \int_x^a \langle a-x \rangle^n dx + \int_a^\infty \langle a-x \rangle^n dx$$

$$\text{정의로부터 } \int_a^\infty \langle a-x \rangle^n dx = 0$$

$$\text{또 } \int_x^a \langle a-x \rangle^n dx = \begin{cases} \frac{(a-x)^{n+1}}{n+1} & x \leq a \\ 0 & x > a \end{cases}$$

$$\therefore \int_x^\infty \langle a-x \rangle^n dx = \frac{\langle a-x \rangle^{n+1}}{(n+1)}$$

$$\therefore \int_{x_0}^x \langle a-x \rangle^n dx = \begin{cases} \frac{(a-x_0)^{n+1}}{n+1} - \frac{(a-x)^{n+1}}{n+1} = C - \frac{(a-x)^{n+1}}{n+1} & x \leq a \\ \frac{(a-x_0)^{n+1}}{n+1} = C & x > a \end{cases}$$

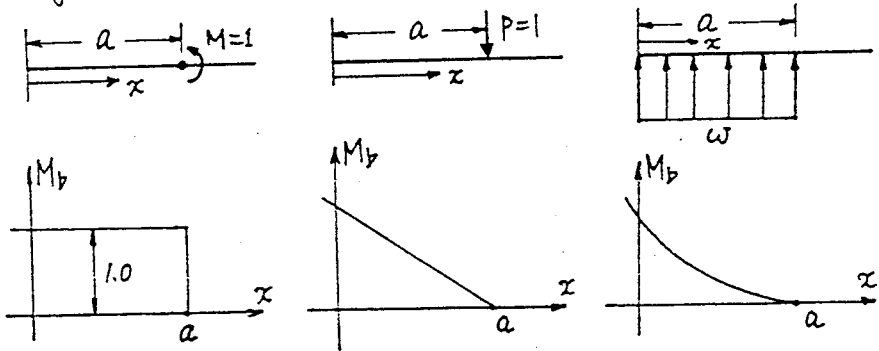
$$\therefore \int \langle a-x \rangle^n dx = \int_{x_0}^x \langle a-x \rangle^n dx = -\frac{\langle a-x \rangle^{n+1}}{n+1} + C. \quad -a.e.d.-$$

Beam에 걸리는 하중은 bending moment는 beam을 따라서
우측에서 좌측으로 나아간다.

$g_0(x)$; unit couple에 의한 moment

$g_1(x)$; unit point load에 의한 moment

$g_2(x)$; unit continuous load에 의한 moment.



$$8.19 \quad M_b = \sum_{j=1}^n P_j \langle a_j - x \rangle'$$

$n=2$ 를 취하면

$$P_1 = -P, \quad a_1 = b$$

$$P_2 = R_A, \quad a_2 = L$$

또한, $a_2 > x$ 이라면, $\langle a_2 - x \rangle' = (a_2 - x) = (L - x)$

$$\therefore M_b = R_A(L - x) - P \langle b - x \rangle' = EI y''$$

$$EI y' = -\frac{R_A}{2}(L - x)^2 + \frac{P}{2} \langle b - x \rangle^2 + C_1$$

$$EI y = \frac{R_A}{6}(L - x)^3 - \frac{P}{6} \langle b - x \rangle^3 + C_1 x + C_2$$

여기에 B.C. $y(0) = y'(0) = 0$, $y(L) = 0$ 를 대입하면

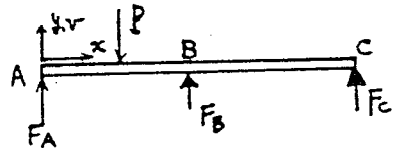
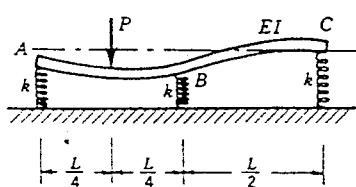
$$C_1 = \frac{1}{2}(R_A L^2 - P b^2)$$

$$C_2 = -\frac{1}{6}(R_A L^3 - P b^3)$$

$$\therefore R_A = \frac{P}{2} \cdot \frac{b^2}{L^3} (3L - b)$$

$$\text{Ans. } R_A = \frac{P b^2}{2 L^3} (3L - b)$$

8.20



변형이 일어난 후에 A점을 기준으로 잡으면

δ = displacement of A, $\delta - v_B$ = displacement of B

$\delta - v_C$ = displacement of C

Force - deflection; $F_A = k\delta$

$$F_B = k(\delta - v_B) = F_A - kv_B$$

$$F_C = k(\delta - v_C) = F_A - kv_C$$

Equilibrium; $\sum F_y = 0$; $F_A + F_B + F_C = P$ 이고, $3F_A - k(v_B + v_C) = P$ — ①

$$\sum M_A = 0; \frac{PL}{4} = F_B \cdot \frac{L}{2} + F_C \cdot L = L \left[\frac{3}{2}F_A - k \left(\frac{v_B}{2} + v_C \right) \right]$$

$$\text{이므로 } 6F_A - 2k(v_B + v_C) = P \quad \text{--- ②}$$

Bending Moment;

$$EIv'' = M_b = F_A \cdot x - P \langle x - \frac{L}{4} \rangle' + F_B \langle x - \frac{L}{2} \rangle'$$

$$= F_A \cdot x - P \langle x - \frac{L}{4} \rangle' + (F_A - kv_B) \langle x - \frac{L}{2} \rangle'$$

$$\text{적분하면; } EIv = \frac{F_A}{6}x^3 - \frac{P}{6} \langle x - \frac{L}{4} \rangle^3 + \frac{(F_A - kv_B)}{6} \langle x - \frac{L}{2} \rangle^3 + C_1x + C_2$$

Boundary Condition;

$$v(0) = 0, \quad v(\frac{L}{2}) = v_B, \quad v(L) = v_C$$

$$\therefore C_2 = 0$$

$$EIv_B = \frac{F_A \cdot L^3}{48} - \frac{PL^3}{384} + C_1 \frac{L}{2} \quad \text{--- ③}$$

$$EIv_C = \frac{F_A \cdot L^3}{6} - \frac{27PL^3}{384} + \frac{(F_A - kv_B) \cdot L^3}{48} + C_1 L \quad \text{--- ④}$$

① ② ③ ④를 풀면

$$F_A = \frac{P(\frac{7}{2} + \frac{13}{384}N)}{(6 + \frac{N}{12})}, \quad F_B = \frac{P(2 + \frac{22}{384}N)}{(6 + \frac{N}{12})}, \quad F_C = \frac{P(\frac{1}{2} - \frac{3}{384}N)}{(6 + \frac{N}{12})}, \quad N \equiv \frac{kL^3}{EI}$$

$$\text{Ans. } F_A = P(\frac{7}{2} + \frac{13}{384}N) / (6 + \frac{N}{12}), \quad F_B = P(2 + \frac{22}{384}N) / (6 + \frac{N}{12})$$

$$F_C = P(\frac{1}{2} - \frac{3}{384}N) / (6 + \frac{N}{12}) \quad \text{where } N \equiv \frac{kL^3}{EI}$$

8.21 (a) Bending Moment ;

$$EI y'' = -M_0 + F_1 x - p \langle x - 0.5255L \rangle'$$

$$EI y' = -M_0 x + \frac{F_1}{2} x^2 - \frac{1}{2} p \langle x - 0.5255L \rangle^2 + C_1$$

$$EI y = -\frac{1}{2} M_0 x^2 + \frac{F_1}{6} x^3 - \frac{1}{6} p \langle x - 0.5255L \rangle^3 + C_1 x + C_2$$

Boundary Condition ; $y'(0) = 0$, $y(0) = 0$

$$\therefore C_1 = C_2 = 0$$

$$y(L) = y''(L) = 0$$

$$\therefore -\frac{1}{2} M_0 L^2 + \frac{1}{6} F_1 L^3 + \frac{1}{6} p (0.4745L)^3 = 0 \quad \text{--- ①}$$

$$-M_0 + F_1 L - 0.4745 p L = 0 \quad \text{--- ②}$$

$$\text{①} \text{ ②} \Rightarrow F_1 = 0.7385 p ; M_0 = 0.2640 p L$$

$$\text{i) } \beta < 0.5255 \text{ 인 경우 } y'' = 0 \text{ 이라 } \therefore x = \frac{M_0}{F_1} = 0.3575 L$$

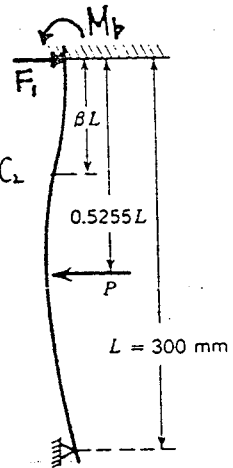
$$\text{ii) } \beta \geq 0.5255 \text{ 인 경우 } y'' = 0 \text{ 이라 } \therefore x = L$$

$$(b) \phi_1 = y'(0.3575L) = -0.04719 \frac{pL^2}{EI}$$

$$\phi_2 = y'(L) = 0.1053 \frac{pL^2}{EI} \quad \therefore \beta = 1 \quad \phi_{max} = 0.1053 \frac{pL^2}{EI}$$

$$(c) k = \frac{S}{L} = \frac{2 \phi_{max} \times 1.5}{L} = 0.3159 \frac{L^2}{EI}$$

$$\text{Ans. (a)(b)(c) 이라 } \beta = 1. \quad \phi_{max} = 0.1053 \frac{pL^2}{EI} \quad k = 0.3159 \frac{L^2}{EI}$$



8.22 Castigliano's theorem ; $\delta = \frac{\partial U}{\partial P}$ (277.93 뉴턴의 생각)

$$U = \frac{1}{2} \int \frac{M_b^2}{EI} dx = \frac{1}{4EI} \int_0^{\frac{L}{2}} p^2 (L-x)^2 dx + \frac{1}{2EI} \int_{\frac{L}{2}}^L p^2 (L-x)^2 dx$$

$$= \frac{p^2}{4EI} \left[-\frac{(L-x)^3}{3} \right]_0^{\frac{L}{2}} + \frac{p^2}{2EI} \left[-\frac{(L-x)^3}{3} \right]_{\frac{L}{2}}^L = \frac{3p^2 L^3}{32EI}$$

$$\therefore \delta = \frac{\partial U}{\partial p} = \frac{3pL^3}{16EI}$$

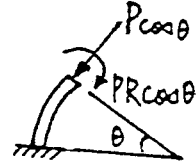
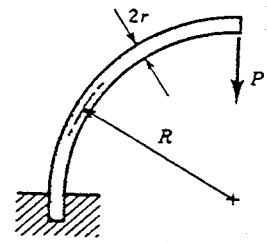
$$\text{Ans. } \delta = 3pL^3 / 16EI$$

8.23 Castigliano's theorem 을 적용하면.

$$\begin{aligned} U &= \int \frac{F^2}{2EA} ds + \int \frac{M^2}{2EI} ds \\ &= \int_0^{\pi} \frac{P^2 \cos^2 \theta}{2EA} (R d\theta) + \int_0^{\pi} \frac{P^2 R^2 \cos^2 \theta}{2EI} (R d\theta) \\ &= \frac{P^2 R}{8r^2 E} + \frac{P^2 R^3}{2r^2 E} \end{aligned}$$

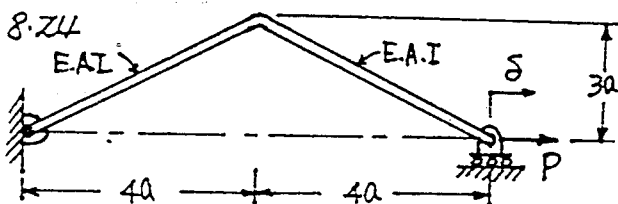
$$\therefore \delta = \frac{\partial U}{\partial P} = \frac{PR}{4r^2 E} + \frac{PR^3}{r^2 E} \quad \therefore \frac{\delta_a}{\delta_b} = \frac{r^2}{4R^2}$$

Ans. $\delta = \frac{P \cdot R}{4r^2 E} + \frac{PR^3}{r^2 E}$



-Q.E.D-

8.24

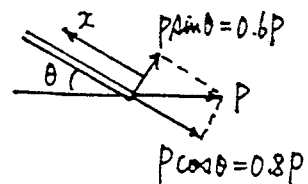


Castigliano's theorem 을 적용하면.

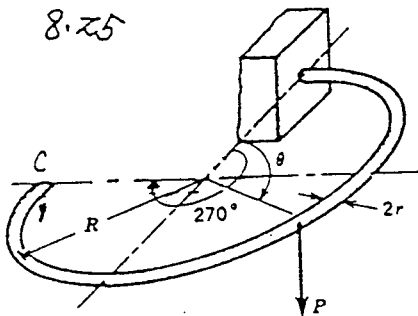
$$\begin{aligned} U &= \int \frac{F^2}{2EA} dx + \int \frac{M^2}{2EI} dx \\ &= 2 \left[\int_0^{5a} \frac{(0.8P)^2}{2EA} dx + \int_0^{5a} \frac{(0.6P \cdot x)^2}{2EI} dx \right] = \frac{16P^2 a}{5EA} + 5 \frac{P^2 a^3}{EI} \end{aligned}$$

$$\therefore \delta = \frac{\partial U}{\partial P} = \frac{32Pa}{5EA} + \frac{10Pa^3}{EI}$$

Ans. $\delta = \frac{32Pa}{5EA} + \frac{10Pa^3}{EI}$



8.25



(1) Vertical deflection (임의의 각; α)

$$(M_b)_\alpha = -PR \sin(\theta - \alpha)$$

$$M_b = -R \sin(\alpha - \frac{\pi}{2}) = -R \cos \alpha$$

$$(M_t)_\alpha = PR(1 - \cos(\theta - \alpha))$$

$$M_t = R(1 + \cos(\alpha - \frac{\pi}{2})) = R(1 + \sin \alpha)$$

$$\begin{aligned}
 \therefore \delta v &= \int \frac{M_b \cdot m_b}{EI} ds + \int \frac{M_t \cdot m_t}{G \cdot J} ds \quad \left(\alpha > 0 \text{ 일 때 } M_b = M_t = 0 \text{ 이므로 } \right. \\
 &\quad \left. \alpha = 0 \text{ 까지만 적분해 주면 된다. } \right) \\
 &= \int_0^\theta \frac{PR^2 \sin(\theta - \alpha) \cos \alpha}{EI} R d\theta + \int_0^\theta \frac{PR^2 \{1 - \cos(\theta - \alpha)\} (1 + \sin \alpha)}{GJ} R d\theta \\
 &= \frac{PR^3}{EI} \left[\sin \theta \cdot \left(\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right) - \cos \theta \cdot \frac{\sin^2 \theta}{2} \right] + \frac{PR^3}{GJ} \left[\theta - \cos \theta \sin \theta \right. \\
 &\quad \left. + \sin \theta \cos \theta - \sin \theta - \cos \theta + 1 - \cos \theta \cdot \frac{\sin^2 \theta}{2} + \frac{1}{4} \sin \theta \sin 2\theta - \frac{1}{2} \theta \sin \theta \right] \\
 \therefore \delta v &= \frac{PR^3 \theta}{2EI} \sin \theta + \frac{PR^3}{GJ} \left(1 + \theta - \frac{1}{2} \theta \sin \theta - \sin \theta - \cos \theta \right)
 \end{aligned}$$

(2) Horizontal deflection

C점에 Horizontal load

H가 있다고 가정을 하면,

임의점에서의 접선력은

F_H 라 하면,

$$F_A = \sin\left(\frac{3}{2}\pi - \alpha\right) \times H = -H \cos \alpha$$

$$(M_b)_H = HR \sin\left(\frac{3}{2}\pi - \alpha\right) = -HR \cos \alpha$$

$$M_t = PR(1 - \cos(\theta - R))$$

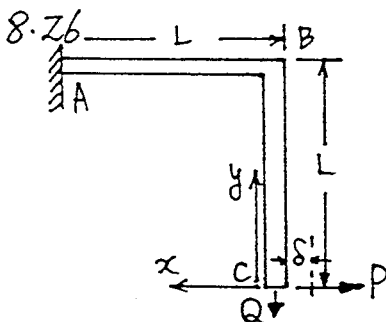
$$(M_b)_v = PR \sin(\theta - R)$$

Castigliano's theorem 이용.

$$U = \int_0^{\frac{3}{2}\pi} \frac{H^2 R}{2EA} \cos^2 \alpha d\alpha + \int_0^{\frac{3}{2}\pi} \frac{R^3}{2EI} [H^2 \cos^2 \alpha + p^2 \sin^2(\theta - \alpha)] d\alpha$$

$$\int_0^{\frac{3}{2}\pi} \frac{R^3}{2GJ} \cdot p^2 \cdot (1 - \cos(\theta - \alpha))^2 d\alpha, \quad \left(\frac{\partial U}{\partial H} \right)_{H=0} = 0 \quad \therefore \delta_H = 0$$

$$\text{Ans. } \delta_v = \frac{PR^3 \theta}{2EI} \sin \theta + \frac{PR^3}{GJ} \left(1 + \theta - \frac{1}{2} \theta \sin \theta - \sin \theta - \cos \theta \right), \quad \delta_H = 0$$



Castigliano's theorem을 적용.

$$\text{7ㄷ AB ; } \begin{cases} M_b = PL - Qx \\ F_A = P \end{cases}$$

$$\text{7ㄷ BC ; } \begin{cases} M_b = P \cdot y \\ F = Q \end{cases}$$

$$\text{Strain energy; } U = \int_0^L \frac{(PL-Qx)^2}{2EI} dx + \int_0^L \frac{p^2}{2EA} dx + \int_0^L \frac{p^2 y^2}{2EI} dy + \int_0^L \frac{Q^2}{2EA} dy$$

$$\frac{\partial U}{\partial Q} = \int_0^L \frac{2(PL-Qx)}{2EI} (-x) dx + \int_0^L \frac{2Q}{2EA} dy$$

$$= \frac{QL^3}{3EI} - \frac{PL^3}{2EI} + \frac{QL}{EA} = 0 \therefore Q = \frac{3L^2 A}{2L^2 A + 6EI} P \quad \text{--- (1)}$$

$$\begin{aligned} \delta = \frac{\partial U}{\partial P} &= \int_0^L \frac{(PL-Qx)}{EI} L dx + \frac{PL}{EA} + \frac{PL^3}{3EI} \\ &= \frac{PL^3}{EI} - \frac{QL^3}{2EI} + \frac{PL}{EA} + \frac{PL^3}{3EI} = \left[\frac{4L^3}{3EI} + \frac{L}{EA} - \frac{L^3}{2EI} \left(\frac{3L^2 A}{2L^2 A + 6EI} \right) \right] P \\ &= 0.025 \end{aligned}$$

$$I = \frac{\pi}{64} (0.013)^4, \quad A = \frac{\pi}{4} (0.013)^2, \quad L = 1.25 \text{ m}$$

$$E = 2 \times 10^{11} \text{ N/m}^2 \text{ 을 대입하면} \quad P = 6.15 \text{ N} \text{ 같이 대입}$$

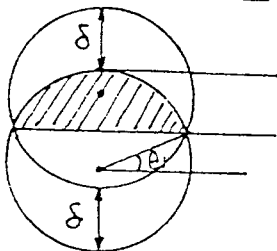
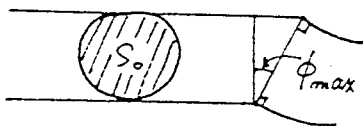
$$\therefore Q = 9.23 \text{ N}$$

$$\text{i) 구간 AB} \quad \sigma_{\max} = \frac{P}{A} + \frac{M_b}{I} \cdot \frac{d}{2} = 35.70 \text{ MN/m}^2$$

$$\text{ii) 구간 BC} \quad \sigma_{\max} = \frac{Q}{A} + \frac{M_b}{I} \cdot \frac{d}{2} = 35.7 \text{ MN/m}^2$$

$$\text{Ans. } \sigma_{\max} = 35.7 \text{ MN/m}^2$$

$$8.27) \text{ 문제 (8.1 (g)) 같이 } \delta_{\max} = \frac{5\omega L^4}{384EI}, \quad \phi_{\max} = \frac{\omega L^3}{24EI}$$



$$S_0 = \pi (0.072)^2 \cos \phi_{\max} \approx \pi R_i^2 \quad (\because \phi_{\max} \approx 0)$$

$$\text{그럼 같이 } \sin \theta_1 = \frac{\delta}{2R_i}$$

θ_1 이 infinitesimal 이라고 가정하면

$$\sin \theta_1 \approx \theta_1 = \frac{\delta}{2R_i}$$

$$S_1 = 2 \int_{\theta_1}^{\frac{\pi}{2}} 2R_i \cos \theta (R_i \cos \theta) d\theta = 2R_i^2 \left(\frac{\pi}{2} - \theta_1 - \sin \theta_1 \cos \theta_1 \right)$$

$$= \pi R_i^2 - 2R_i \frac{\delta}{R_i} = \frac{1}{2} \pi R_i^2 = \frac{1}{2} S_0 = 8.486 \times 10^{-3} \text{ (m}^2\text{)}$$

$$\therefore \frac{\delta}{R_i} = \frac{1}{4} \cdot \frac{1}{\pi} \quad \text{따라서 위의 가정이 맞다.}$$

$$\left(\omega = (1.79 \times 10^3)(9.8) \times \frac{\pi}{4} [0.15^2 - 0.144^2] = 105.8 \text{ N/m} \right.$$

$$\left. I = \pi R^3 t = 3.744 \times 10^{-6} \text{ m}^4 \quad \text{\textcircled{\scriptsize 2}} \text{에 대입하면} \right.$$

$$\therefore \frac{5 \cdot (105.8) \cdot L^4}{384 \cdot (200 \times 10^9) (3.744 \times 10^{-6})} = \frac{1}{4\pi} (0.072)$$

$$L = 7.47 \text{ m}$$

$$(2) \omega = (1.79 \times 10^3) \times \frac{\pi}{4} (0.15^2 - 0.126^2) = 397.2 \text{ N/m}$$

$$I = \frac{\pi}{64} (0.15^4 - 0.126^4) = 1.248 \times 10^{-5} \text{ m}^4$$

$$\delta_{\max} = \frac{5 \times (397.2) (7.47)^4}{384 \times (200 \times 10^9) \times 1.248 \times 10^{-5}} = 6.452 \times 10^{-3}$$

$$\text{if } \theta_1 \ll 1, \sin \theta_1 = \theta_1 = \frac{\delta}{2R_i} = \frac{6.452 \times 10^{-3}}{0.126} = 0.0512 \text{ (rad)}$$

$$\therefore S_1 = 2 \times (0.063)^2 \left[\frac{\pi}{2} - 0.0512 - 0.0512 \times 1 \right] = 0.01166 \text{ (m}^2\text{)}$$

$$\therefore \Delta S = 0.01166 - 8.486 \times 10^{-3} = 3.17 \times 10^{-3} \text{ (m}^2\text{)}$$

$$\text{따라서 } \frac{\Delta S}{S_1} \times 100 = \frac{3.17 \times 10^{-3}}{8.486 \times 10^{-3}} \times 100 = 37.36 (\%) \text{ 증가}$$

$$\underline{\text{Ans. } L = 7.47 \text{ m, } 37.36\%}$$

8.28 문제 (8.1 (g))에 대해

$$\delta_{\max} = \frac{5\omega L^4}{384EI} = \frac{1}{360} L$$

$$\therefore I_{\min} = \frac{5 \times 360 \omega L^3}{384 E} = \frac{5 \times 360 \times 15000 \times 3.6^3}{384 \times 7 \times 10^9} = 4.686 \times 10^{-4} \text{ (m}^4\text{)}$$

$$\underline{\text{Ans. } I_{\min} = 4.686 \times 10^{-4} \text{ m}^4}$$

8.29 Bending Moment;

$$EIv'' = M_b = \frac{M_0}{L}x$$

정답하면 $EIV = \frac{M_0}{6L}x^3 + C_1x + C_2$

Boundary Condition;

$$v(0) = v(L) = 0$$

$$\therefore C_2 = 0, \quad C_1 = -\frac{M_0 L}{6}$$

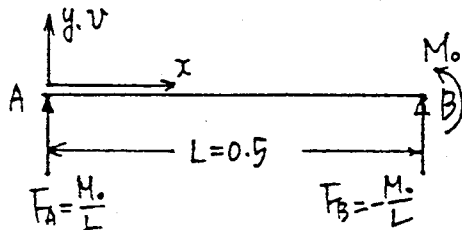
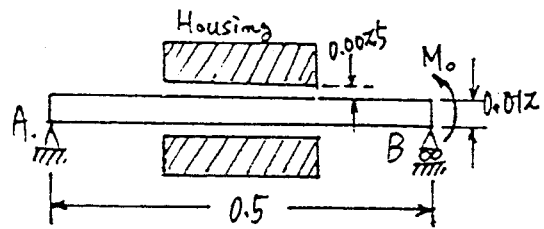
$$\therefore v = \frac{1}{EI} \left\{ \frac{M_0}{6} \left(\frac{x^3}{L} - xL \right) \right\}$$

$|v|_{\max}$ 은 $\frac{dv}{dx} = 0$ 을 만족하는

$$x = \frac{L}{\sqrt{3}} \text{ 이다. } |v|_{\max} = \frac{M_0}{6EI} L^2 \left(\frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{3} \right) = -\frac{\sqrt{3} M_0 L^2}{27EI}$$

$$\therefore \frac{\sqrt{3} M_0 (0.5)^2}{27 \times (200 \times 10^9) \left(\frac{\pi}{64} \times 0.02^4 \right)} = 1.25 \times 10^{-3} \quad \therefore M_0 = 15.87 \text{ N-m}$$

Ans. $M_0 = 15.87 \text{ N-m}$



8.30 Beam AB

Bending Moment

$$EIv'' = -M_0 - F_2x + F_1 \left\langle x - \frac{L}{2} \right\rangle$$

$$EIv' = -M_0x - \frac{1}{2}F_2x^2 + \frac{1}{2}F_1 \left\langle x - \frac{L}{2} \right\rangle^2 + C_1$$

$$EIv = -\frac{1}{2}M_0x^2 - \frac{1}{6}F_2x^3 + \frac{1}{6}F_1 \left\langle x - \frac{L}{2} \right\rangle^3 + C_1x + C_2$$

Boundary Condition $v'(0) = v(0) = 0$. $\therefore C_1 = C_2 = 0$

$$\therefore EIv = -\frac{1}{2}M_0x^2 - \frac{1}{6}F_2x^3 + \frac{1}{6}F_1 \left\langle x - \frac{L}{2} \right\rangle^3 \quad \text{--- ①}$$

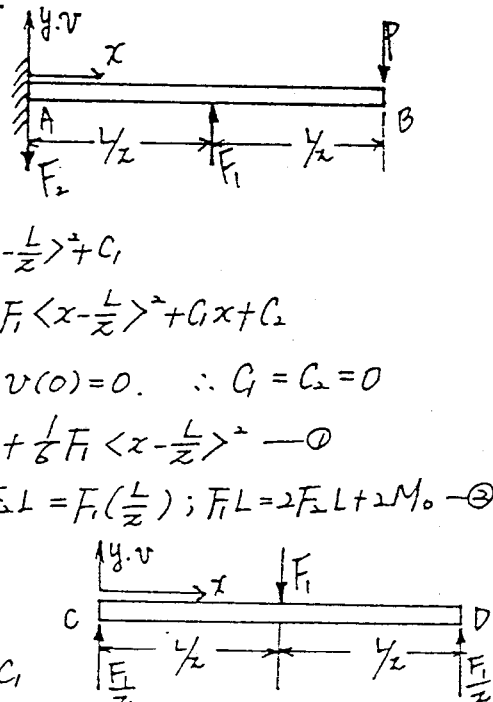
Equilibrium; $\sum M_0 = 0$. $M_0 + F_2L = F_1 \left(\frac{L}{2} \right)$; $F_1L = 2F_2L + 2M_0$ --- ②

Beam CD

$$EIv'' = \frac{1}{2}F_1x + (-F_1) \left\langle x - \frac{L}{2} \right\rangle$$

$$EIv' = \frac{1}{4}F_1x^2 - \frac{1}{2}F_1 \left\langle x - \frac{L}{2} \right\rangle^2 + C_1$$

$$EIv = \frac{1}{12}F_1x^3 - \frac{1}{6}F_1 \left\langle x - \frac{L}{2} \right\rangle^3 + C_1x + C_2$$



$$v(0) = v(L) = 0 \quad \therefore C_2 = 0. \quad C_1 = -\frac{1}{16} F_1 L^2$$

$$\therefore v = \frac{F_1}{EI} \left[\frac{1}{12} x^3 - \frac{1}{6} \left\langle x - \frac{L}{2} \right\rangle^3 - \frac{1}{16} L^2 x \right] - (3)$$

①③ 사이의 Geometry를 따름. $v_{AB}(\frac{L}{2}) = v_{CD}(\frac{L}{2})$

$$-\frac{1}{8} M_0 L - \frac{F_2}{48} L^3 = -\frac{1}{48} F_1 L^3$$

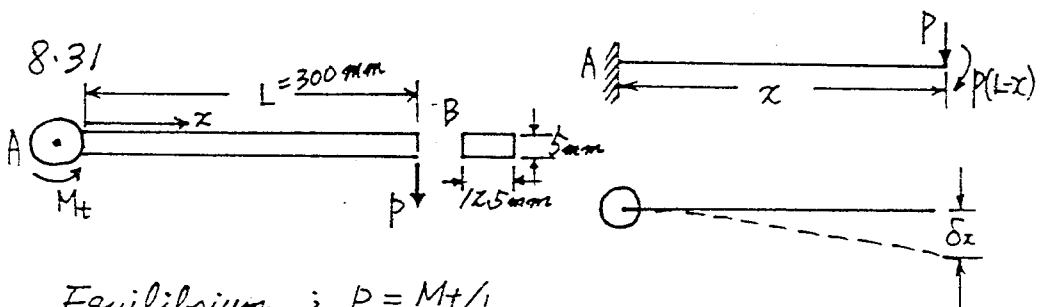
$$\therefore F_1 L = F_2 L + 6 M_0. \quad \text{--- ④} \quad \Sigma F_y = 0; F_1 = F_2 + P \quad \text{--- ⑤}$$

②④⑤에서, $F_1 = \frac{5}{3} P, F_2 = \frac{2}{3} P, M_0 = \frac{1}{6} PL$

따라서 $v(\frac{L}{2}) = -\frac{F_1 L^3}{48 EI} = -\frac{5 PL^3}{144 EI} \quad \therefore \delta_E = \frac{3 PL^3}{144 EI}$

$$\begin{aligned} v(L) &= \frac{1}{EI} \left(-2 M_0 L^2 - \frac{1}{6} F_2 L^3 + \frac{F_1}{48} L^3 \right) \\ &= \frac{1}{EI} \left(-\frac{PL^3}{3} - \frac{PL^3}{9} + \frac{PL^3}{144} \right) = -\frac{63 PL^3}{144 EI} \end{aligned}$$

Ans. $\delta_P = \frac{63 PL^3}{144 EI}$



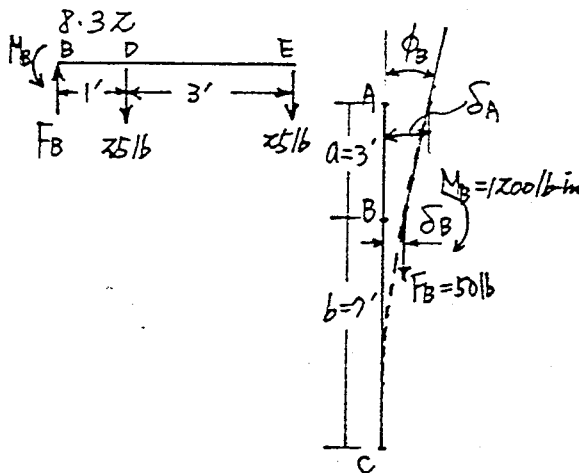
Equilibrium ; $P = M_t / L$

중첩의 원리에 의한 x점에서의 deflection;

$$\delta_x = \frac{Px^3}{3EI} + \frac{P(L-x)x^2}{2EI} = \frac{P}{EI} \left[\frac{x^3}{3} + \frac{x^2(L-x)}{2} \right]$$

$$\begin{aligned} x=0.3 \text{ m 일 때 } \delta_x &= \frac{P}{EI} \left[\frac{(0.3)^3}{3} + \frac{(0.3)^2(0.38-0.3)}{2} \right] \\ &= \frac{M_t}{(0.38)(200 \times 10^9)(1.302 \times 10^{-10})} \left[\frac{(0.3)^3}{3} + \frac{(0.3)^2(0.08)}{2} \right] \\ &= 1.2733 \times 10^{-3} M_t \end{aligned}$$

Ans. $\delta = 1.2733 \times 10^{-3} M_t \cdot m$ M_t 는 $N \cdot m$.



평형 조건; $F_B = 50 \text{ lb}$

$M_B = 1200 \text{ lb-in}$

Signpost 의 자중과 Axial force F_B 는 Signpost 에 bending-moment 를 야기시키지만, 작은 값이므로 무시한다.

중첩의 원리; $\delta_A = \delta_B + AB \times \phi_B$
 $= \delta_B + a \phi_B$

$\delta_B = \frac{M_B \cdot b^2}{2EI}$, $\phi_B = \frac{M_B \cdot b}{EI}$ $a = 36"$
 $b = 84"$

따라서 $\delta_A = \frac{M_B \cdot b}{EI} \left(\frac{b}{2} + a \right) = \frac{84 \times 1200}{\frac{\pi}{64} (256 - 150) \times 30 \times 10^6} \times 78 = 0.0503"$

Ans. $\delta_A = 0.0503"$

8.33 Pontoon 이 δ_p 만큼 갈아 앉는다면, 상방으로 작용하는 힘은;

$F_p = \gamma A \delta_p$

대칭성에 의해서, 양단에서의 반력은 서로 같다.

그리고, bridge 는 중양점 (P) 에서의 기울기는

"0" 이다. 따라서 bridge 의 반은 한쪽 끝이

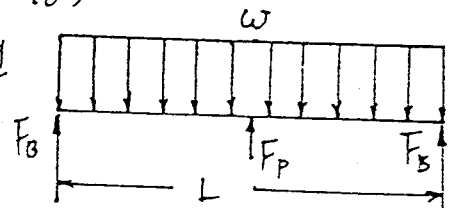
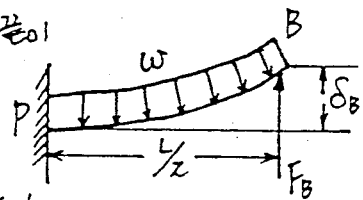
고정된 cantilever 로 생각할 수 있다.

그런후에 중첩의 원리를 적용하자.

$\delta_B = \frac{F_B \cdot (\frac{L}{2})^3}{3EI} - \frac{w \cdot (\frac{L}{2})^4}{8EI} = \frac{L^3}{8EI} \left(\frac{F_B}{3} - \frac{wL}{16} \right)$

이것은 B 점이 고정되어 있다고 생각하면 δ_p 도 될 것이다.

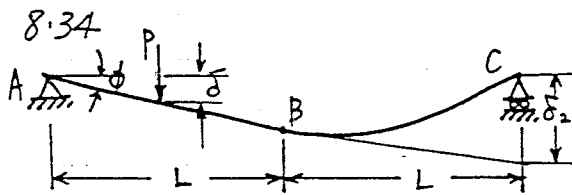
$\therefore F_p = \gamma A \delta_p = \frac{\gamma A L^3}{8EI} \left(\frac{F_B}{3} - \frac{wL}{16} \right)$



전체 bridge의 평형 조건에 따라 $\sum F_B + F_P = \omega L$

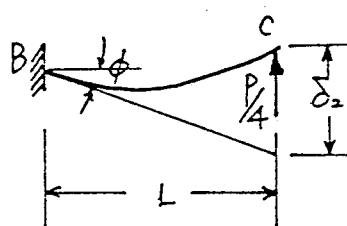
$$\therefore F_B = \frac{\left(1 + \frac{\gamma AL^3}{128EI}\right)}{\left(2 + \frac{\gamma AL^3}{128EI}\right)} \quad \therefore \delta_P = \frac{5\omega L^4}{384EI\left(1 + \frac{\gamma AL^3}{48EI}\right)}$$

Ans. $\delta = \frac{5\omega L^4}{384EI\left(1 + \frac{\gamma AL^3}{48EI}\right)}$



중첩의 원리.

Rigid beam의 기울기를 ϕ 로 놓으면, 그림에서 $\delta \approx \phi L$
 $\delta_2 \approx 2\phi L$



Slope ϕ 로, B점이 고정된 Cantilever beam으로 BC를 고려하면.

$$\delta_2 = \frac{P}{4} \cdot \frac{(L)^3}{3EI} = 2\phi L$$

$$\therefore \phi = \frac{PL^2}{24EI}$$

따라서 $(\delta)_{1/2} = \phi \cdot \frac{L}{2} = \frac{PL^3}{48EI}$; 하중 상태에서의 처짐.

Ans. $\delta = \frac{PL^3}{48EI}$

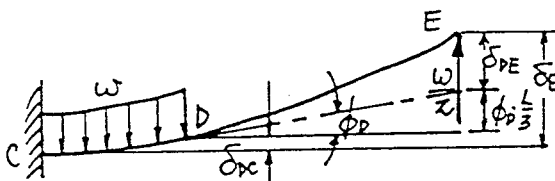
8.35 분포 하중 $\omega = 3W/L$, 대칭에 의하여 양단에서의 반력은

$W/2$ 이다. 또 대칭성에 의하여

중양점의 기울기는 "0"이다.

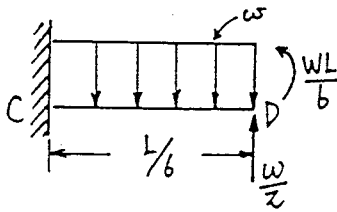
따라서 중양점이 "0" 기울기로

고정되어 있는 cantilever로 고려할



수 있다. $\delta_E = \delta_{DC} + \phi_D \cdot \frac{L}{3} + \delta_{DE}$

Beam CD;



$$\begin{aligned}\delta_{DC} &= -\frac{w(\frac{L}{6})^4}{8EI} + \frac{w}{2} \frac{(\frac{L}{6})^3}{3EI} + \frac{wL}{6} \cdot \frac{(\frac{L}{6})^2}{2EI} \\ &= \frac{29wL^3}{10368EI}\end{aligned}$$

$$\begin{aligned}\phi_D &= -\frac{w(\frac{L}{6})^3}{6EI} + \frac{w}{2} \frac{(\frac{L}{6})^2}{2EI} + \frac{wL}{6} \frac{(\frac{L}{6})}{EI} \\ &= \frac{42wL^2}{1296EI}\end{aligned}$$

$$\delta_{DE} = \frac{\frac{w}{2}(\frac{L}{3})^3}{3EI} = \frac{wL^3}{162EI}$$

$$\therefore \delta_E = \frac{wL^3}{EI} \left(\frac{1}{162} + \frac{29}{10368} + \frac{14}{1296} \right) = \frac{205wL^3}{10368EI}$$

이것은 고정점을 E로 보면 $\delta_C = \frac{205wL^3}{10368EI}$

exact solution ; $\delta_{exact} = \frac{wL^3}{48EI}$

$$\therefore \frac{\delta_{approx}}{\delta_{exact}} = \frac{10368}{48 \times 205} \approx 1.05 \quad (\text{약 } 5\% \text{의 error})$$

Ans. $\delta = (205wL^3)/(10368EI)$ $\delta_{approx}/\delta_{exact} = 1.05$

8.36 i) CD에 대해 $EIV'' = M_b = 0$

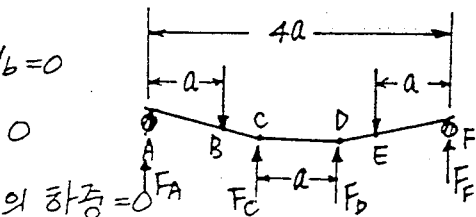
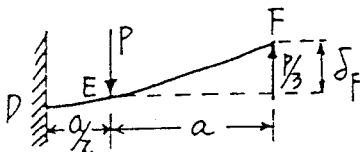
ii) CD에 대해 $\frac{dM_b}{dx} = V_y = 0$

iii) CD에 대한 임의점에서의 하중 = 0

iv) C·D점에서만 point force가 작용한다. 그렇지 않으면 $M_b \neq 0$.

평형조건 ; $F_A + F_C = F_D + F_F = P$

$M_b = 0$ at C and D $\therefore F_A = F_F = P/3, F_D = F_C = 2P/3$



DF를 D점이 고정된 cantilever로 생각하자. D점에서의 기울기는 "0". 그리고, 중첩의 원리를 사용하면

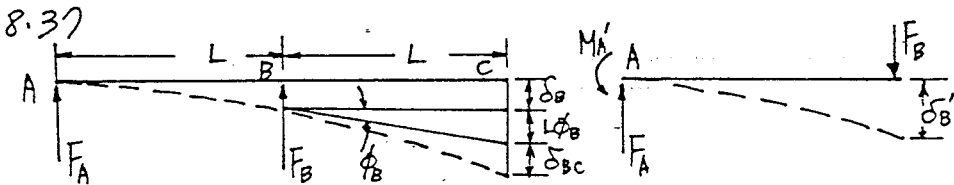
$$\text{Deflection at E} = \frac{Pa}{3} \left(\frac{a}{2}\right)^2 - \frac{\left(\frac{2P}{3}\right)\left(\frac{a}{2}\right)^3}{2EI} \quad \text{Deflection of EF} = \frac{\left(\frac{P}{3}\right)(a)^3}{3EI}$$

$$\text{Slope at E} = \frac{\left(\frac{Pa}{3}\right)\left(\frac{a}{2}\right)}{EI} - \frac{\left(\frac{2P}{3}\right)\left(\frac{a}{2}\right)^2}{2EI}$$

$$\therefore \delta_F = (\text{deflection at E}) + a \times (\text{slope at E}) + (\text{deflection of EF})$$

$$= \frac{5Pa^3}{24EI} \quad \therefore P = \frac{24EId}{5a^3}$$

Ans. $P = (24EId) / (5a^3)$



$$\text{Upper; } \delta_B = \frac{(P-F_B)L^3}{3EI} + \frac{PL \cdot L^2}{2EI} \quad \delta_B' = \frac{F_B L^3}{3EI}$$

$$= \frac{L^3}{6EI} (5P - 2F_B)$$

가하학적 적합조건; $\delta_B = \delta_B' \quad \therefore \frac{F_B L^3}{3EI} = \frac{L^3}{6EI} (5P - 2F_B)$

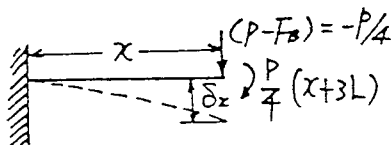
$$\therefore F_B = \frac{5}{4}P$$

따라서, $\delta_C = \delta_B + L\phi_B + \delta_{BC}$

$$= \frac{5}{12} \frac{PL^3}{EI} + L \cdot \left[\frac{PL}{EI} + \frac{\left(-\frac{P}{4}\right)L^2}{2EI} \right] + \frac{PL^3}{EI} = \frac{13PL^3}{8EI}$$

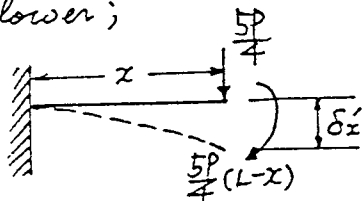
Deflection curves; Method of superposition

upper; $x \leq L \quad \delta_x = \frac{P}{4} \cdot \frac{(x+3L)x^2}{2EI} - \frac{P}{4} \cdot \frac{x^3}{3EI}$



$$= \frac{P}{4EI} \left\{ \frac{x^3}{6} + \frac{3}{2}x^2L \right\} \quad x \leq L$$

lower;



$$\delta'_x = \frac{(\frac{5P}{4}) \cdot x^3}{3EI} + \frac{(\frac{5P}{4})(L-x)x^2}{2EI}$$

$$= \frac{P}{4EI} \left\{ \frac{5}{2} Lx^2 - \frac{5}{6} x^3 \right\} \quad 0 \leq x \leq L$$

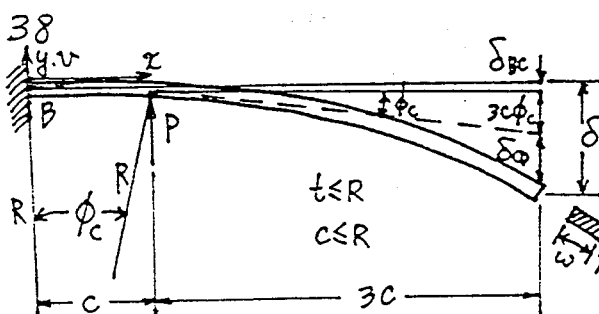
$$\delta'_x - \delta_x = \frac{P}{4EI} (Lx^2 - x^3) > 0 \quad \text{for } 0 < x < L$$

$$= 0 \quad \text{for } x = L$$

$\therefore$ 가정은 정확하다.

Ams. $\delta = (13PL^3)/(8EI)$, The assumption is correct.

8.38



i) $V'' = -\frac{1}{R} = \text{const.}$
 $= \frac{M_b}{EI}$

ii) $\frac{dM_b}{dx} = V_y = 0$

iii) Intensity of loading은
 0이다.

iv) C점에서 위로 힘 P가 작용한다. 따라서 $M_b = \text{const.}$

$x \leq C$ 인 경우 $M_b = -P(4C-x) + P(C-x) = -3PC$

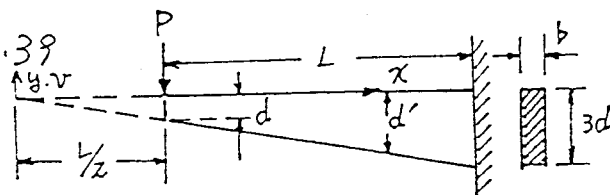
$$\therefore P = -\frac{M_b}{3C} = \frac{EI}{3CR}$$

중첩의 원리; $\delta_D = \delta_{BC} + 3C \cdot \phi_C + \delta_{CD}$

$$= \frac{1}{2} \frac{C^2}{R} + 3C \cdot \frac{C}{R} + \frac{P(3C)^3}{3EI} = \frac{13C^2}{2R}$$

Ams. $\delta_D = (13C^2)/(2R)$

8.39



임의의 점 x에서

$$d'(\text{depth}) = 2dx/L$$

$$\therefore I_x = \frac{bd^3}{12} = \frac{2bd^3x^2}{3L^3}$$

Bending Moments $EI_x \cdot v'' = M_b = -p(x - \frac{L}{2}), \frac{L}{2} \leq x \leq \frac{3L}{2}$

I_x 에 대해 적분; $\frac{2Ebd^3}{3L^3} v'' = -\frac{P}{x^3} (x - \frac{L}{2})$

적분하면; $\frac{2Ebd^3}{3L^2} v' = \frac{P}{x} - \frac{PL}{4x^2} + C_1$

$\frac{2Ebd^3}{3L^2} v = p \ln x + \frac{PL}{4x} + C_1 x + C_2$

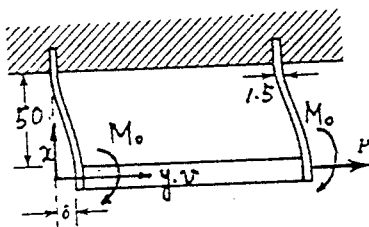
Boundary Condition ; $v(\frac{3}{2}L) = v'(\frac{3}{2}L) = 0$

$\therefore v = \frac{3PL^3}{2Ebd^3} \left\{ \ln \frac{2x}{3L} + \frac{2}{3} + \frac{L}{4x} - \frac{5}{9} \frac{x}{L} \right\}$

$x = \frac{L}{2}$ 이라 $v = \frac{3PL^3}{2Ebd^3} \left(\frac{8}{9} - \log_e 3 \right)$

$\delta \equiv |v|_{y/2} = \frac{3PL^3}{2Ebd^3} \left(\log_e 3 - \frac{8}{9} \right) ; \text{Ans.}$

8.40



(대칭이므로 한쪽의 vertical rod 만을 고려한다.)

(a) Bending moment

$EIV'' = M_0 - \frac{1}{2}Px$

$EIV' = M_0 x - \frac{1}{4}Px^2 + C_1$

$EIV = \frac{1}{2}M_0 x^2 - \frac{1}{12}Px^3 + C_1 x + C_2$

Boundary Condition

$v'(0) = v'(L) = 0, v(L) = 0$

따라서 $C_1 = 0, M_0 = \frac{1}{4}PL$

$C_2 = \frac{PL^3}{12} - \frac{1}{2}M_0 L^2 = -\frac{1}{24}PL^3$

$\therefore EIV = \frac{1}{8}PLx^2 - \frac{1}{12}Px^3 - \frac{1}{24}PL^3$

$\delta = v(0) = \frac{PL^3}{24EI}$

$I = \frac{1}{12} (6 \times 10^{-3}) (1.5 \times 10^{-3})^3 = 1.6875 \times 10^{-12} \text{ (m}^4\text{)}$

$E = 100 \text{ GN/m}^2$ 값 위에 대입하면

$$\delta = \frac{(50 \times 10^{-3})^3 \cdot P}{24 \cdot (100 \times 10^9) (1.6875 \times 10^{-12})} \quad \therefore P = 32400 \delta \text{ (N)}$$

δ 의 단위는 m.

$$(b) EI v'' = M_0 - \frac{P}{2}x = \frac{P}{4}L - \frac{P}{2}x$$

$$\text{따라서 } (M_b)_{\max} = \frac{1}{4}PL$$

$$\sigma_y = \frac{(M_b)_{\max}}{I} \left(\frac{t}{2} \right) = \frac{\frac{1}{4}(32400\delta)}{(1.6875 \times 10^{-12})} (50 \times 10^{-3}) \cdot \frac{(1.5 \times 10^{-3})}{2}$$

$$= 1.8 \times 10^8 \delta = 350 \times 10^6 \text{ (N/m}^2 \text{)} (\because \text{yielding loader})$$

$$\therefore \delta = 1.94 \text{ mm}$$

달아야 하기 때문

Ans. $P = 32400 \delta \text{ (N)}$. $\delta = 1.94 \text{ (mm)}$

$$8.41 \quad \frac{dV_s}{dx} = \frac{(\tau_{xy})_{\max}}{G} = \frac{3P}{2Gb^h}$$

$$(\tau_{xy})_{\max} = \frac{3P}{2bh} \text{ for rectangular sections}$$

$$V_s(x) = \int_{x=0}^x \frac{3P}{2Gb^h} dx + V_s(0) = \frac{3Px}{2Gb^h} + 0$$

$$V_b(x) = \frac{Px^3}{3EI} + \frac{P(L-x)x^2}{2EI}; \text{ 중첩의 원리 적용.}$$

$$\frac{V_s(L)}{V_b(L)} = \frac{3PL/2Gb^h}{PL^3/3EI} = \frac{9EI}{2Gb^hL^2}; I = \frac{1}{12}bh^3; \frac{E}{G} = 2(1+\nu)$$

$$\therefore \left(\frac{V_s}{V_b} \right)_{x=L} = \frac{3(1+\nu)}{4} \frac{h^2}{L^2}; \nu \leq 0.5 \text{ (Thermodynamics 0141)}$$

$$\frac{h}{L} \leq 10^{-1} \text{ 이면 } \left(\frac{V_s}{V_b} \right)_{x=L} \leq \frac{3 \times 1.5}{4} \times 10^{-2} = 1.025 \times 10^{-2}$$

$\therefore$ Shear deflection 은 약 1% 정도

$$\left(\frac{V_s}{V_b} \right)_{x=L} = \frac{3(1+\nu)}{4} \cdot \frac{h^2}{L^2} \quad - Q.E.D. -$$

8.42 Castigliano's Theorem을 사용하자.

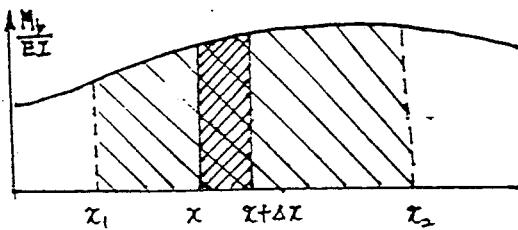
$$\tau_{xy} = \frac{V}{2I} \left[\left(\frac{h}{z} \right)^2 - y^2 \right]$$

$$\begin{aligned} U_s &= \int \frac{\tau_{xy}^2}{2G} dx dy dz = \frac{1}{2G} \cdot \frac{V^2}{4I^2} \int \left(\frac{h^2}{z^2} - y^2 \right) dx dy dz \\ &= \frac{V^2}{8GI^2} \cdot b \cdot L \left[\frac{h^4}{16} \cdot \frac{h}{z} - \frac{h^2}{z} \cdot \frac{1}{3} \left(\frac{h}{z} \right)^3 + \frac{1}{5} \left(\frac{h}{z} \right)^5 \right] \times z \\ &= \frac{2V^2 L h^2}{480GI^2} \cdot b h^3 = \frac{2V^2 h^2 L}{40GI} = \frac{(1+\nu)}{10EI} \cdot V^2 h^2 L \end{aligned}$$

Castigliano's theorem; $\frac{\partial U_s}{\partial V} = \frac{(1+\nu) V h^2 L}{5EI} = \delta_s$

$$\therefore \left(\frac{\delta_s}{\delta_b} \right)_{x=L} = \frac{3(1+\nu)}{5} \cdot \frac{h^2}{L^2} \quad \text{--- Q. E. D. ---}$$

8.43



(a) 곡률이 매우 크다면, Δx 는
중립축을 따라서 측정되어야
한다. 구간 dx 이서 곡률 반경이
일정 하다고 가정하자.

그러면 구간 dx 의 양끝에서의 접선 사이의 각도 $d\phi$ 는;

$$d\phi = \Delta x / R$$

$$\frac{1}{R} \approx \frac{d^2 v}{dx^2} = \frac{M_b}{EI} \quad \therefore d\phi = \frac{M_b \cdot \Delta x}{EI} = \frac{M_b}{EI} \text{ 구간에서 빛금친 부분}$$

적분하면, $\phi_{12} = \int_{x_1}^{x_2} \frac{M_b}{EI} dx$; $\frac{M_b}{EI}$ 구간에서 x_1 과 x_2 사이의 면적

(b) 기하학적인 형태에서 Δx 에 의한 ϵ 의 증가량 $d\epsilon$ 은

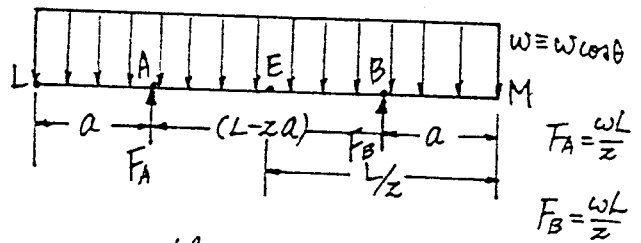
$$d\epsilon = (x_2 - x) d\phi = (x_2 - x) \cdot \frac{M_b}{EI} \cdot \Delta x; \quad x_2 \text{에 대한 빛금 부분의 1차 moment}$$

적분하면 $\epsilon_{12} = \int_{x_1}^{x_2} d\epsilon = \int_{x_1}^{x_2} (x_2 - x) \cdot \frac{M_b}{EI} dx$; x_2 에 대한 x_1 과 x_2 사이의 M_b/EI 구간의 1차 moment

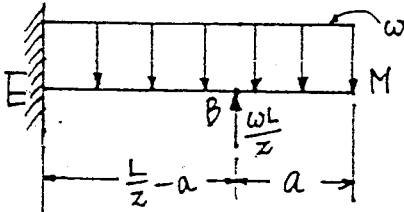
8.44 임의점에서의 하중

i) axial load ; $\frac{W}{L} \sin \theta$

ii) normal load ; $\frac{W}{L} \cos \theta$



EM을 E점의 기둥기 "O"으로 고정된 cantilever로 생각한다.



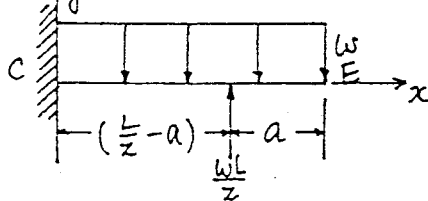
Load M이 작용하는 단면들은 E점에서의 단면에 평행하여야 한다. 왜냐하면 EL과 EM의 처점은 E점에 대하여 대칭이기 때문이다.

중첩 이론에 의해서 M점에서의 기둥기 ϕ_M

$$\phi_M = -\frac{w(\frac{L}{2})^3}{6EI} + \frac{wL}{z} \cdot \frac{(\frac{L}{2} - a)^2}{2EI} = 0$$

$$\therefore a = L \left(\frac{\sqrt{3}-1}{2\sqrt{3}} \right) = 0.211L \quad \text{Ans. } a = 0.211L$$

8.45 y, v



8.44와 같은 방법으로 E점이 기둥기 "O"으로 고정되어 있는 cantilever로 CE를 생각하자.

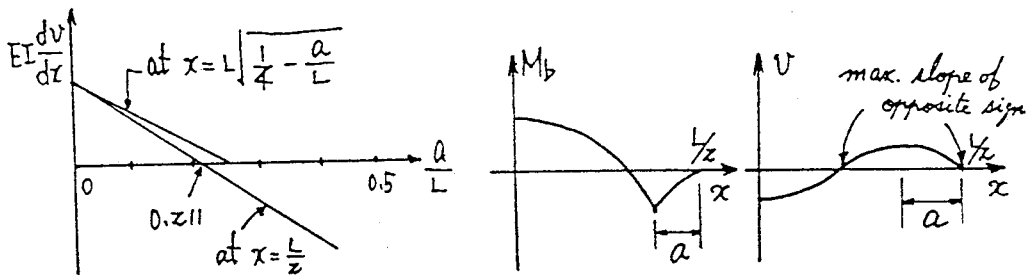
$$EI v'' = M_b = \frac{wL}{z} \left\langle \frac{L}{2} - a - x \right\rangle' - \frac{w}{z} \left(\frac{L}{2} - x \right)^2$$

Boundary Condition $v(0) = v'(0) = 0$

$$v' = \frac{1}{EI} \left\{ -\frac{wL}{4} \left\langle \frac{L}{2} - a - x \right\rangle^2 + \frac{w}{6} \left(\frac{L}{2} - x \right)^3 + \frac{3(L-2a)^3 - L^2}{48} \omega L \right\}$$

$$(v')_{\max} \text{ 는 } M_b = 0 \text{ or } \begin{cases} x = \sqrt{\frac{1}{4} - \frac{a}{L}} \cdot L & \text{for } \frac{a}{L} < \frac{1}{4} \\ x = \frac{L}{2} & \text{for } \frac{a}{L} > \frac{1}{4} \end{cases}$$

$\frac{a}{L}$ 과 v' 과의 관계를 그래프로 그리면 ;



$\frac{dv}{dx}$ 가 절대값은 같고, 부호가 반대인 2개의 극값을 갖도록 (a/L) 의 값을 취하면 $0.211 < \frac{a}{L} < 0.25$

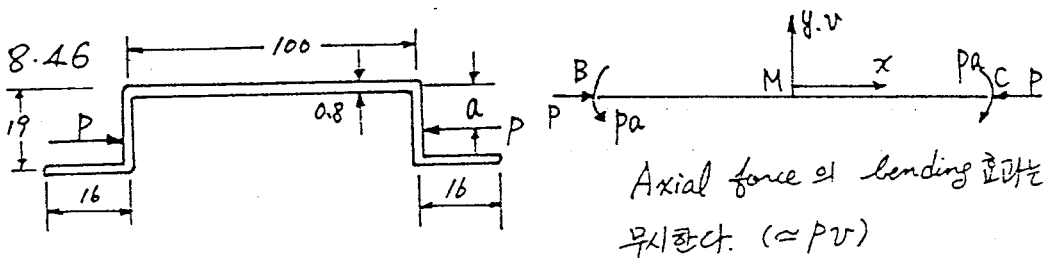
(a/L) 값이 이 값의 범위에서 변하면, $|\frac{dv}{dx}|$ 의 값이 증가한다.

따라서 (a/L) 의 선택은 $\min \cdot [|\frac{dv}{dx}|_{\max}]$ 을 만족해야 한다.

Trial-Error method에 의해서 $(a/L) \approx 0.22$

$$\text{이때 } \left(\frac{dv}{dx}\right)_{x=\sqrt{\frac{1}{4}-\frac{a}{L}} \cdot L} = -\left(\frac{dv}{dx}\right)_{x=\frac{L}{2}}$$

$$\text{Ans. } (a/L) = 0.22$$



$$\text{따라서 } M_b = EI v'' = -pa = -\frac{3}{8}p$$

$$\text{Boundary Condition ; } v(0) = v'(0) = 0$$

$$\therefore v = -\frac{P}{EI} \cdot \frac{3}{16} x^2 ; M \text{ 점이 극치점인 포물선이 된다.}$$

$$E = 200 \times 10^9 \text{ N/m}^2, I = \frac{bh^3}{12} = \frac{1}{12} (0.013)(0.0008)^3 = 5.547 \times 10^{-13} \text{ m}^4$$

$$\therefore v \approx -0.005 P x^2 \quad |v|_{\max} = 0.02 p$$

$$|v|_{\max} = 1.25 \text{ mm 라 두면 } p = 11.68 \text{ N}$$

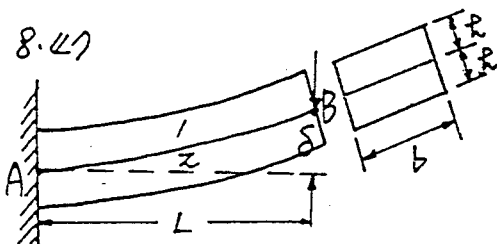
$$\text{Stress ; } (\sigma_x)_{\max} = \frac{M \cdot y}{I_{yy}} = \frac{(11.68)(0.0095)}{5.547 \times 10^{-13}} \times \frac{0.0008}{2} = 80.03 \text{ MN/m}^2$$

$$\text{Steel Data ; } \left(\begin{array}{l} 1020 \text{ HR ; } Y \approx 255 \text{ MN/m}^2 \text{ 안정계수 ; } 6 \\ 1020 \text{ CR ; } Y \approx 600 \text{ MN/m}^2 \text{ 안정계수 ; } 7.5 \\ 4130 \text{ HT ; } Y = 1880 \text{ MN/m}^2 \text{ 안정계수 ; } 17.3 \end{array} \right)$$

$\therefore$ 1020 HR 이 가장 적당하며, plastic yielding 은 $P = 37.38 \text{ N}$ 이에서 생긴다.

Ans. $v = -\frac{Pax^2}{2EI}$; parabola , 1020 HR , $P_Y = 37.38 \text{ N}$

8.47

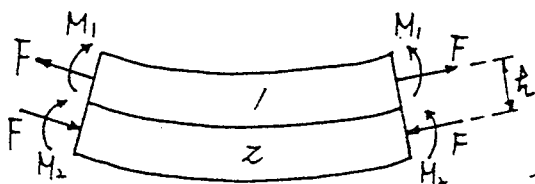


A, B are areas of stress concentration

i) 일반 beam theory 에서 plane section 은 변형후에도 plane 을 유지한다.

ii) 어떤 net force 나 moment 는 어떤 단면을 통해서도 전달될 수 있다.

따라서, stress 와 strain 은 beam 종에서 변화가 없다. beam 과 beam 사이에는 크기는 같고 방향이 반대인 힘이 전달된다.



평형조건 ; 단면종에서 tensile stress 가 일정하다고 가정하면,
 $M_1 + M_2 = F \cdot h$

기하학적 적합조건 ; 접촉면에서의

strain 과 곡률반경은 두 beam 에서 서로 같아야 한다.

$$\text{strain ; } \epsilon_1 = \frac{F}{E_1 b h} + \frac{M_1 \cdot \frac{h}{2}}{E_1 I} + \alpha_1 T \quad \text{--- ①}$$

$$\epsilon_2 = \frac{F}{E_2 b h} + \frac{M_2 \cdot \frac{h}{2}}{E_2 I} + \alpha_2 T \quad \text{--- ②}$$

$$\text{곡률 ; } \frac{1}{R_1} \approx \frac{M_1}{E_1 I}, \quad \frac{1}{R_2} \approx \frac{M_2}{E_2 I} \quad \text{where } I = \frac{b h^3}{12}$$

$$R_1 = R_2, \quad M_1 = \frac{E_1}{E_2} M_2 \quad \text{또 } M_1 + M_2 = M_2 \left(1 + \frac{E_1}{E_2}\right) = F \cdot h$$

$$\therefore M_2 = \frac{F \cdot R}{1 + \frac{E_1}{E_2}}, \quad M_1 = \frac{F \cdot R}{1 + \frac{E_2}{E_1}}$$

① ② 에 대입하면, $E_1 = E_2$ 로 놓으면

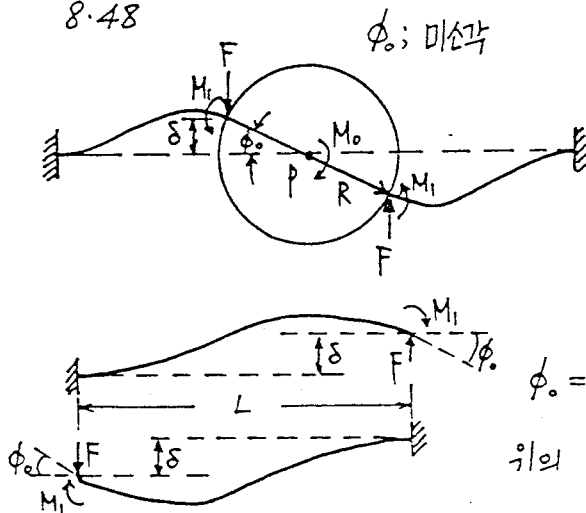
$$M_2 = \frac{(\alpha_2 - \alpha_1) T}{\frac{1 + \frac{E_1}{E_2}}{E_1 b R^2} + \frac{1 + \frac{E_1}{E_2}}{E_2 b R^2} + \frac{6}{b R^2} \left(\frac{1}{E_1} + \frac{1}{E_2} \right)}$$

원의 기하학에서; $\delta = \frac{L^2}{2R} = \frac{M_2 L^2}{2 E_2 I} = \frac{6 M_2 L^2}{E_2 b R^3}$

따라서 $\delta = \frac{(\alpha_2 - \alpha_1) T L^2}{R} \cdot \frac{6}{14 + \frac{E_1}{E_2} + \frac{E_2}{E_1}}$

Ans. $\delta = 6(\alpha_2 - \alpha_1) T L^2 E_1 E_2 / R (14 E_1 E_2 + E_1^2 + E_2^2)$

8.48



기하학적 적합조건; $\delta = R \phi_0$

Strip과 disc에 작용하는 힘;

Disc의 평형조건;

$$\sum M_p = 0; M_0 = 2FR + 2M_1 \quad \text{--- ①}$$

Strip의 변형;

$$\delta = \frac{FL^3}{3EI} - \frac{M_1 L^2}{2EI}$$

$$\phi_0 = \frac{M_1 L}{EI} - \frac{FL^2}{2EI} \quad \delta = R \phi_0$$

위의 3식과 ①식에서

$$F = \frac{M_0}{2R + \frac{2}{3}L \times \left(\frac{2L+3R}{L+2R} \right)}$$

$$\phi_0 = -\frac{FL^2}{2EI} + \frac{FL^2}{3EI} \cdot \frac{2L+3R}{L+2R}$$

$$\therefore K \equiv \frac{M_0}{\phi_0} = \frac{EI}{L^2} \cdot \frac{2R + \frac{2}{3}L \times \left(\frac{2L+3R}{L+2R} \right)}{-\frac{1}{2} + \frac{1}{3} \cdot \frac{2L+3R}{L+2R}}$$

$$K = \frac{8EI}{L^3} (3R^2 + 3RL + L^2)$$

Ans. $K = \frac{8EI}{L^3} (3R^2 + 3RL + L^2)$

8.49 문제는 오른쪽 그림과 같이
치환할 수 있다.

A점에서의 bending deflection은
B점에서의 bending deflection과 같다.

$$(\delta_A)_{\text{bending}} = \frac{P(10a)^3}{3E \cdot \frac{1}{2} \cdot 2a \cdot a^3} = \frac{2000P}{Ea}$$

Torsion에 의한 deflection $(\delta)_{\text{torsion}} = a\phi$

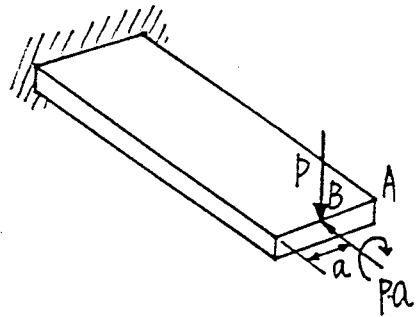
$$\frac{M_t}{\phi} = C_2 \cdot \frac{G a' b'^3}{L} \quad \text{where } (C_2 = 0.229, \text{ (by table (6.1))})$$

$$a' = 2a, b' = a, M_t = Pa$$

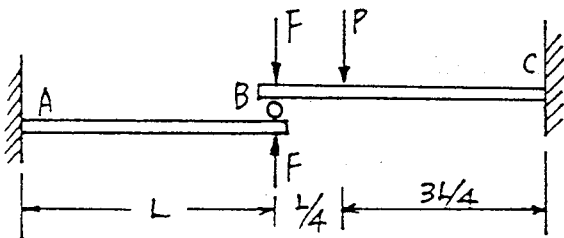
$$\therefore \phi = \frac{Pa \times 10a}{0.229 \times 2a \times a^3} = \frac{5P}{0.229Ga^2}$$

$$\therefore \delta_A = (\delta)_{\text{bending}} + (\delta)_{\text{torsion}} = \frac{P}{a} \left\{ \frac{2000}{E} + \frac{5}{0.229G} \right\} = \frac{P}{Ea} \{ 2000 + 56.2(1+\nu) \}$$

$$\text{Ans. } \delta = \frac{P}{Ea} \{ 2000 + 56.2(1+\nu) \}$$



8.50



$$\text{Ans. } P = \frac{8}{3L} M_L$$

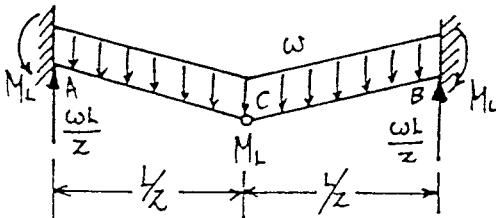
Limiting value P를 구하는
것이므로, $M_A = M_C = M_L$.

$$FL = M_L,$$

$$-FL + \frac{3}{4}PL = M_L$$

$$\therefore P = \frac{8}{3L} M_L$$

8.51 문제 (8.11)에서 $M_A = M_B = M_L$ 평형조건 $\sum M_A = 0$;



$$M_L = -M_L + \frac{1}{2}wL\left(\frac{L}{2}\right) - \frac{1}{2}w\left(\frac{L}{2}\right)^2$$

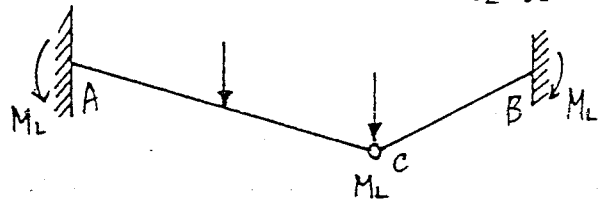
$$\therefore w = \frac{16M_L}{L^2}$$

$$\text{Ans. } w = 16M_L/L^2$$

8.52

i) $\frac{a}{L} \leq \frac{1}{2} \quad \sum M_B = \sum M_C = 0;$
 $-M_L + F_1 L = M_L$
 $-M_L + 3F_1 L - 2(1 - \frac{a}{L})PL - Pa = -M_L$
 $\therefore 6M_L = (2L - a)P \quad \therefore P = \frac{6M_L}{2L - a}$

ii) $\frac{a}{L} \geq \frac{1}{2} \quad \sum M_B = \sum M_C = 0;$
 $-M_L + F_1 2L - P(L - a) = M_L$
 $-M_L + 3F_1 L - 2P(L - a) - Pa = -M_L$
 $\therefore P = \frac{6M_L}{L + a}$



Ans. $a \geq \frac{L}{2}; P = \frac{6M_L}{L + a} \quad a \leq \frac{L}{2}; P = \frac{6M_L}{2L - a}$

8.53 material 1; $(\sigma_y)_1 = Y$, material 2; $(\sigma_y)_2 = 0.5Y$

i) material 1 이 더 붕괴가 발생되면,

$$M_L = \frac{3}{4}P \cdot \frac{1}{4}L = \frac{3}{16}PL$$

$$M_L = \frac{3}{16}PL = (\sigma_y)_1 \cdot \frac{b^2}{2} \cdot \frac{b}{4} \times L \quad \therefore P = \frac{4}{3} \frac{b^3 Y}{L}$$

ii) material 2 이 더 붕괴가 발생되면,

$$M_L = \frac{1}{4}P \cdot \frac{L}{2} = \frac{1}{8}P \cdot L$$

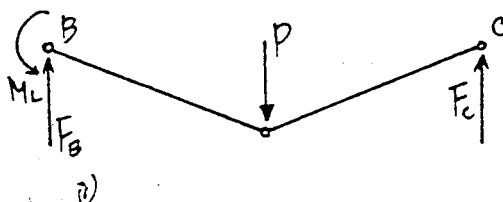
$$M_L = \frac{1}{8}P \cdot L = \frac{1}{2}Y \cdot \frac{b^2}{2} \cdot \frac{b}{4} \cdot L \quad \therefore P = \frac{b^3 Y}{L}$$

i) ii) 이 때 ii)의 $P = \frac{b^3 Y}{L}$ 가 더 작은 값이다. 따라서 $P = \frac{b^3 Y}{L}$ 이 때 material 2가 먼저 붕괴한다.

Ans. $P = \frac{b^3 Y}{L}$

8.54

i) $-M_L + F_A(\frac{L}{4}) = M_L$
 $-M_L + \frac{1}{2}F_A L - 2P(\frac{L}{4}) = -M_L$
 $\therefore P = \frac{8}{L} M_L$



$$\begin{aligned} \text{ii)} \quad & -M_L + F_C \left(\frac{L}{2}\right) = M_L \\ & -M_L + F_C \left(\frac{L}{2}\right) - P \left(\frac{L}{4}\right) = -M_L \\ & \therefore P = \frac{16}{L} M_L \end{aligned}$$

i) ii) 에서 작은 값 $P = \frac{8M_L}{L}$ 에서 i) 의 형태로 분리된다.

Ans. $P = 8M_L/L$

8.55 Bending moment;

$$EI v'' = -M_A + F_A x - P \left\langle x - \frac{L}{3} \right\rangle'$$

$$EI v' = -M_A \cdot x + \frac{1}{2} F_A \cdot x^2 - \frac{1}{2} P \left\langle x - \frac{L}{3} \right\rangle^2 + C_1$$

$$EI v = -\frac{1}{2} M_A \cdot x^2 + \frac{1}{6} F_A x^3 - \frac{1}{6} P \left\langle x - \frac{L}{3} \right\rangle^3 + C_1 x + C_2$$

Boundary Condition;

$$v'(0) = 0 \therefore C_1 = 0, \quad v(0) = 0 \therefore C_2 = 0$$

$$v'(L) = 0; -M_A \cdot L + \frac{1}{2} F_A \cdot L^2 - \frac{2}{9} P L^2 = 0 \quad \text{--- ①}$$

$$v(L) = 0; -\frac{1}{2} M_A \cdot L^2 + \frac{1}{6} F_A L^3 - \frac{4}{81} P L^3 = 0 \quad \text{--- ②}$$

①② 에서 $F_A = \frac{20}{27} P, \quad M_A = \frac{4}{27} PL$

따라서, B점에서의 moment; $M_B = EI v''\left(\frac{L}{3}\right) = -\frac{4}{27} PL + \frac{20}{27} P \left(\frac{L}{3}\right) = \frac{8}{81} PL$

C점에서의 moment; $M_C = EI v''(L) = -\frac{4}{27} PL + \frac{20}{27} PL - \frac{2}{3} PL = -\frac{2}{27} PL$

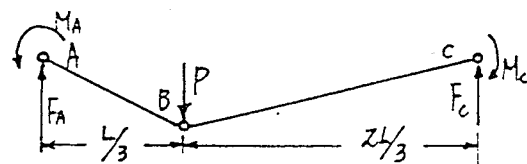
$\therefore M_{\max} = M_A = \frac{4}{27} PL \therefore P_y = \frac{27}{4} \frac{M_y}{L} (\because M_{\max} = M_{\text{yielding}})$

$$-M_L + F_A \cdot \frac{L}{3} = M_L; F_A = \frac{6M_L}{L}$$

$$-M_L + F_A \cdot L - \frac{2}{3} PL = -M_L; P = \frac{3}{2} F_A$$

$$\therefore P_L = \frac{9M_L}{L}$$

$$\therefore \frac{P_L}{P_y} = \frac{\frac{9}{L} \cdot M_L}{\frac{27}{4} \cdot \frac{M_y}{L}} = \frac{4}{3} \frac{M_L}{M_y} = \frac{4}{3} K \quad \text{--- Q. E. D. ---}$$



8.56 (8.55)과 같은 경우로.

$$EIv'' = -M_0 + Fx - P\langle x - \frac{L}{2} \rangle', \quad EIv' = -M_0x + \frac{1}{2}Fx^2 - \frac{P}{2}\langle x - \frac{L}{2} \rangle^2 + C_1$$

$$EIv = -\frac{M_0}{2}x^2 + \frac{1}{6}Fx^3 - \frac{P}{8}\langle x - \frac{L}{2} \rangle^3 + C_1x + C_2$$

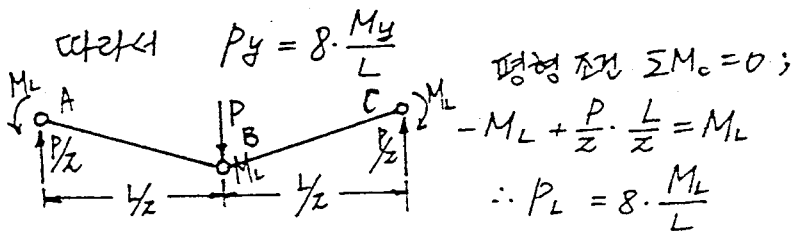
B.C $v'(0) = 0 \therefore C_1 = 0$, $v(0) = 0 \therefore C_2 = 0$

$$v'(L) = 0; -M_0L + \frac{1}{2}FL^2 - \frac{1}{8}PL^2 = 0 \quad -\text{①}$$

$$v(L) = 0; -\frac{1}{2}M_0L + \frac{1}{6}FL^3 - \frac{P}{48}L^3 = 0 \quad -\text{②}$$

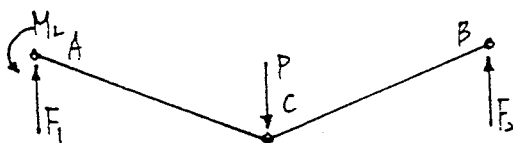
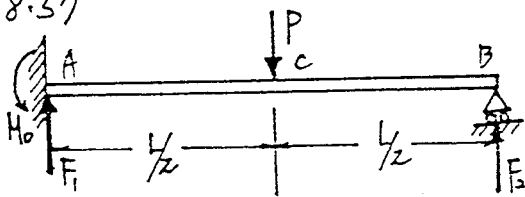
①②에서 $F_A = \frac{P}{2}$, $M_0 = \frac{1}{8}PL$

또, $M_B = EIv''(\frac{L}{2}) = -\frac{1}{8}PL + \frac{1}{4}PL = \frac{1}{8}PL \therefore M_{\max} = \frac{1}{8}PL$



$$\therefore P_L/P_y = (8 \cdot M_L/L) / (8 \cdot \frac{M_y}{L}) = \frac{M_L}{M_y} = k \quad - Q. E. D. -$$

8.57



평형 조건 $\Sigma M_A = \Sigma M_C = 0;$

$$-M_L + F_2 \cdot \frac{L}{2} = M_L$$

$$-M_L + F_2 \cdot L - \frac{1}{2}PL = 0 \quad \therefore P = \frac{6M_L}{L}$$

문제(8.19)에서

$$F_1 = \frac{11}{16}P, \quad F_2 = \frac{1}{2}P \cdot \frac{1}{3} \times \frac{L^2}{4} (3L - \frac{L}{2}) = \frac{5}{16}P$$

$$\therefore M_0 = \frac{1}{2}PL - \frac{5}{16}PL = \frac{3}{16}PL$$

$$M_B = \frac{1}{2}F_2L = \frac{5}{32}PL$$

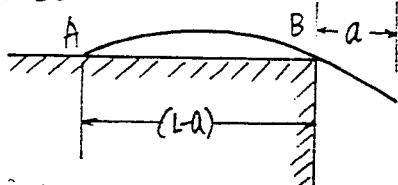
따라서 $M_{\max} = \frac{3}{16}PL$

$$\therefore P_y = \frac{16}{3} \cdot \frac{M_y}{L}$$

$$\therefore P_L = 6 \cdot \frac{M_L}{L}$$

$$\therefore \frac{P_L}{P_y} = \frac{\frac{6}{L} M_L}{\frac{16}{3L} M_y} = \frac{9}{8} \frac{M_L}{M_y} = \frac{9}{8} K \quad - \text{G. E. D. -}$$

8.58



$$F_B(L-a) = \frac{1}{2} \omega L^2 \quad \therefore F_B = \frac{\omega L^2}{2(L-a)}$$

$$F_A = \omega L - F_B = \frac{\omega L(L-2a)}{2(L-a)}$$

Bending Moment;

$$EI v'' = \frac{\omega L(L-2a)}{2(L-a)} x - \frac{1}{2} \omega x^2 + \frac{\omega L^2}{2(L-a)} \langle x+a-L \rangle'$$

$$EI v' = \frac{\omega L(L-2a)}{4(L-a)} x^2 - \frac{1}{6} \omega x^3 + \frac{\omega L^2}{4(L-a)} \langle x+a-L \rangle^2 + C_1$$

$$EI v = \frac{\omega L(L-2a)}{12(L-a)} x^3 - \frac{1}{24} \omega x^4 + \frac{\omega L^2}{12(L-a)} \langle x+a-L \rangle^3 + C_1 x + C_2$$

B.C. $v(0) = v(L-a) = 0 \quad \therefore C_2 = 0$

$$C_1 = -\frac{\omega}{24} (L^3 - 3aL + a^2L + a^3)$$

따라서
$$v = \frac{1}{EI} \left[\frac{\omega L(L-2a)}{12(L-a)} x^3 + \frac{\omega L^2}{12(L-a)} \langle x+a-L \rangle^3 - \frac{1}{24} \omega x^4 - \frac{\omega}{24} (L^3 - 3aL + a^2L + a^3) x \right]$$

8.59 문제에 $\frac{d^2 v}{dx^2} = \frac{1}{E} \frac{I_{yz} M_{by} + I_{yy} M_{bz}}{I_{yy} I_{zz} - I_{yz}^2} \quad \frac{d^2 w}{dx^2} = -\frac{1}{E} \frac{I_{zz} M_{by} + I_{yz} M_{bz}}{I_{yy} I_{zz} - I_{yz}^2}$

$$I_{zz} = \frac{8}{3} ta^3, \quad I_{yy} = \frac{2}{3} ta^3, \quad I_{yz} = -ta^3$$

$$M_{bz} = -P(L-x), \quad M_{by} = 0$$

$$\therefore \frac{d^2 v}{dx^2} = \frac{-\frac{2}{3} P(L-x)}{Ea^3 t \left(\frac{2}{3} \cdot \frac{8}{3} - 1 \right)} = -\frac{6P}{7E} (L-x) \cdot \frac{1}{a^3 t}$$

적분하여 B.C $v'(0) = v(0) = 0$ 를 대입하면,

$$v = -\frac{8P}{7EA^3t} \left(\frac{1}{2}x^2 - \frac{1}{6}x^3 \right) \therefore v(L) = -\frac{2PL^3}{7EA^3t}$$

$$\frac{d^2w}{dx^2} = \frac{1}{EA^3t} \cdot \frac{-P(L-x)}{\left(\frac{2}{3} \cdot \frac{8}{3} - 1\right)} = \frac{-9P}{7EA^3t} (L-x)$$

적분하여, B.C $w'(0) = w(0) = 0$ 을 대입하면

$$w = -\frac{9P}{7EA^3t} \left(\frac{L}{2}x^2 - \frac{1}{6}x^3 \right) \therefore w(L) = -\frac{3PL^3}{7EA^3t}$$

Ans. $v(L) = (-2PL^3)/(7EA^3t)$, $w(L) = (-3PL^3)/(7EA^3t)$

8.60 문제 (7.7) (7.78) 보이기

$$\left(\begin{array}{l} I_{yz} = 32.3 \text{ in}^4 \quad I_{yy} = 38.8 \text{ in}^4 \quad I_{zz} = 80.8 \text{ in}^4 \\ E = 30 \times 10^6 \text{ psi} \quad M_{bz} = 100,000 \text{ lb-in} \end{array} \right)$$

문제 (8.59) 식을 이용하면,

$$\frac{d^2v}{dx^2} = \frac{1}{E} \cdot \frac{I_{yz}M_{bz} + I_{yy}M_{bz}}{I_{yy}I_{zz} - I_{yz}^2} = \frac{(32.3)(10^5)}{(30 \times 10^6)\{(38.8)(80.8) - (32.3)^2\}} \\ = 5.147 \times 10^{-5}$$

B.C $v'(0) = v(0) = 0$ 을 이용하면

$$v = 2.574 \times 10^{-5} x^2 \therefore v(120) = 0.3706 \text{ in}$$

$$\frac{d^2w}{dx^2} = -\frac{(80.8)(10^5)}{(30 \times 10^6)\{(38.8)(80.8) - (32.3)^2\}} = -1.284 \times 10^{-4}$$

B.C $w'(0) = w(0) = 0$ 을 이용하면

$$w = (-6.418 \times 10^{-5}) x^2 \therefore w(120) = -0.9242 \text{ in}$$

Ans vertical = 0.3706 in, horizontal = -0.9242 in

$$8.61 \quad \delta = \frac{L^2}{2R} = \frac{L^2}{2\rho}, \quad M_o = \frac{1}{\rho} EI \quad \therefore \delta = \frac{M_o L^2}{2EI}$$

$$\phi \text{ 를 미소각으로 가정: } \phi \approx \frac{L}{\rho} = \frac{M_o L}{EI} \quad - \text{ Q.E.D. -}$$

$$8.62 \quad (a) \quad \phi_x = \frac{M_x \cdot L}{2GJ} = \frac{2(1+\nu)}{2E} \cdot \frac{M_x}{\frac{\pi}{2} r^4} \cdot R = \frac{2(1+\nu)}{\pi r^4 E} R M_x$$

$$(\because G = \frac{E}{2(1+\nu)})$$

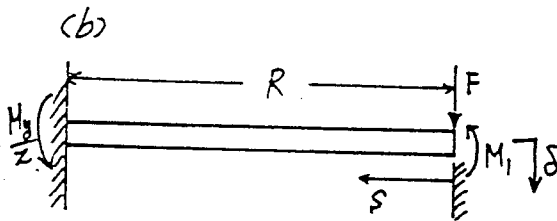
$$\therefore k_x = \frac{M_x}{\phi_x} = \frac{\pi r^4 E}{2(1+\nu) \cdot R}$$

$$\text{평형 조건 } \Sigma M = 0;$$

$$\frac{1}{2} M_y = F \cdot R - M_1$$

$$\delta = \frac{FR^3}{3EI} - \frac{M_1 R^2}{2EI}$$

$$\phi_y = \frac{FR^2}{2EI} - \frac{M_1 R}{EI}$$



가하학적 적합 조건;

$$\delta = R \phi_y; \quad \frac{1}{3} FR - \frac{1}{2} M_1 = \frac{1}{2} F - M_1$$

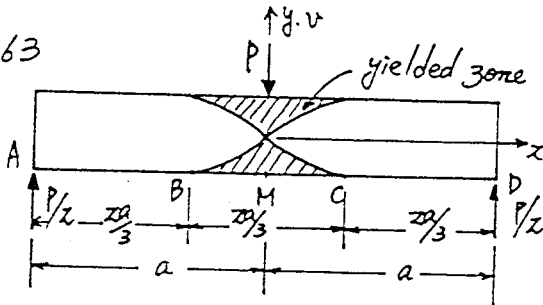
$$\therefore M_1 = \frac{1}{3} FR = \frac{2EI \phi_y}{R} \therefore F = \frac{3M_y}{4R}$$

$$\text{따라서 } \phi_y = \frac{FR^2}{2EI} - \frac{FR^2}{3EI} = \frac{FR^2}{6EI} = \frac{M_y \cdot R}{8EI}$$

$$\therefore k_y = \frac{M_y}{\phi_y} = \frac{8EI}{R} = \frac{2\pi r^4 E}{R} \quad \left(\frac{k_y}{k_x} = 4(1+\nu) \leq 6 \right)$$

$$\text{Ans. } k_x = \pi r^4 E / 2(1+\nu) \cdot R \quad k_y = 2\pi r^4 E / R$$

8.63



moment-curvature;

$$M_b = \frac{3}{2} M_y \left\{ 1 - \frac{1}{3} \left[\frac{(y/\rho)}{(1/\rho)} \right]^2 \right\} \quad \text{--- ①}$$

$$|x| \leq \frac{a}{3}$$

$$M_b = \frac{E I_{yy}}{\rho}, \quad \frac{a}{3} < |x| \leq a \quad \text{--- ②}$$

ρ : radius of curvature

$$\text{또, } M_y = \frac{b h^2}{6} \gamma, \quad \frac{1}{\rho} \approx \frac{d^2 y}{dx^2}, \quad \left(\frac{1}{\rho} \right)_y = \frac{\epsilon_y}{R/x} = \frac{2\gamma}{E h}$$

$$\text{그럼에 } M_b = \frac{P(a-x)}{2} \quad 0 < x < a$$

① ② 식에 대입하면,

$$\frac{P(a-x)}{z} = \frac{3}{z} M_y \left\{ 1 - \frac{1}{3} \left[\frac{2Y}{ER} / \frac{d^2v}{dx^2} \right]^2 \right\}, |x| \leq \frac{a}{3} \quad \text{--- ③}$$

$$\frac{P(a-x)}{z} = EI \frac{d^2v}{dx^2} \quad \frac{a}{3} \leq |x| \leq a \quad \text{--- ④}$$

또, table 8.1에 $(M_b)_{x=0} = 1.5 M_y = M_L$

$$\therefore \frac{Pa}{z} = 1.5 M_y \quad \text{or} \quad p = \frac{3 M_y}{a}$$

따라서 ③ ④ 식은 다음과 같이 된다.

$$\frac{1}{3} \left[\left(\frac{2Y}{ER} \right) / \left(\frac{d^2v}{dx^2} \right) \right] = \left(\frac{x}{a} \right)^{1/2} \quad |x| \leq \frac{a}{3} \quad \text{--- ⑤}$$

$$EI \frac{d^2v}{dx^2} = \frac{3 M_y}{2a} (a-x) \quad \frac{a}{3} < |x| < a \quad \text{--- ⑥}$$

B.C $v'(0) = 0, \quad v(a) = 0,$

$$\text{⑤ 식에서의 } v(\pm \frac{a}{3}) = \text{⑥ 식에서의 } v(\pm \frac{a}{3})$$

$$\text{⑤ 식에서의 } v'(\pm \frac{a}{3}) = \text{⑥ 식에서의 } v'(\pm \frac{a}{3})$$

적분하여 B.C를 이용하면

$$v = -\frac{20}{27} \cdot \frac{a^2 M_y}{EI} + \frac{2\sqrt{a} Y}{3\sqrt{3} ER} \cdot \frac{4}{3} x^{3/2} \quad |x| \leq \frac{a}{3}$$

$$v = \frac{1}{EI} \left\{ (a-x) M_y \cdot a \left[\frac{1}{4} \left(1 - \frac{x}{a} \right)^2 - 1 \right] \right\} \quad \frac{a}{3} \leq |x| \leq a$$

$$Y = \frac{6 M_y}{b h^2}, \quad I = \frac{b h^3}{12} \quad \text{을 대입하면}$$

$$v = -\frac{a^2 M_y}{EI} \left[\frac{20}{27} - \frac{4}{3\sqrt{3}} \left(\frac{x}{a} \right)^{3/2} \right] \quad |x| \leq \frac{a}{3}$$

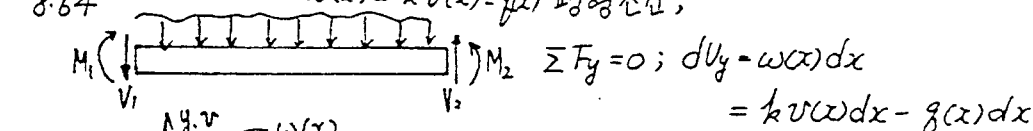
$$v = -\frac{M_y \cdot a^2}{4EI} \left(3 - \frac{x}{a} \right) \left[1 - \left(\frac{x}{a} \right)^2 \right] \quad \frac{a}{3} \leq |x| \leq a \quad \text{--- Q.E.D. ---}$$

$$x=0 \text{ 일 때 } v_p = -\frac{20 M_y \cdot a^2}{27 EI}$$

$$\text{문제 (8.4)에 } v\left(\frac{L}{2}\right) = -\frac{PL^3}{48EI} \Rightarrow v_c = -\frac{M_y \cdot a^2}{3EI} \quad \left(\because L=2a, M_y = \frac{1}{2} Pa \right)$$

$$\therefore \frac{v_p}{v_c} = \frac{\left(-\frac{20}{27} \right)}{\left(-\frac{1}{3} \right)} = \frac{20}{9} \quad \text{--- Q.E.D. ---}$$

8.64 $w(x) = k v(x) - g(x)$ 평형조건;



$$\sum F_y = 0; \quad dV_y - w(x) dx = k v(x) dx - g(x) dx$$

$$\sum M_z = 0; \quad V_y dx = -dM_b$$

$$\frac{dM_b}{dx} = -V_y, \quad \frac{d^2 M_b}{dx^2} = -\frac{dV_y}{dx} = g(x) - k v(x)$$

Beam bending 이력 $M_b = EI \frac{d^2 v}{dx^2}$

$$\therefore EI \frac{d^4 v}{dx^4} + k v(x) = g(x) \quad \text{--- ①}$$

$g(x) = 0$ 인 경우에는 $v = e^{sx}$ 가 ①식의 해가 된다.

S 를 가하기 위해 $v = e^{sx}$ 를 ①식에 대입하면

$$(EIS^4 + k)v = 0 \quad \therefore S^4 = -\frac{k}{EI}, \quad \beta^4 = -\frac{S^4}{4} = \frac{k}{4EI} \text{로 놓으면}$$

$$S = \pm \beta \pm i\beta$$

$$\therefore v = e^{(\beta+i\beta)x}, \quad v = e^{(\beta-i\beta)x}, \quad v = e^{(-\beta+i\beta)x}, \quad v = e^{(-\beta-i\beta)x}$$

①식에서 $g(x) = 0$ 이면, linear 이므로, 위의 4개의 해의 combination도 해가 된다.

$$v = A e^{(\beta+i\beta)x} + B e^{(\beta-i\beta)x} + C e^{(-\beta+i\beta)x} + D e^{(-\beta-i\beta)x}$$

$$= e^{\beta x} [(A+B) \cos \beta x + (A-C) \sin \beta x]$$

$$+ e^{-\beta x} [(C+D) \cos \beta x + (C-D) \sin \beta x]$$

$$= e^{-\beta x} (C_1 \cos \beta x + C_2 \sin \beta x) + e^{\beta x} (C_3 \cos \beta x + C_4 \sin \beta x)$$

$x=0$ 이 point load가 작용하고, beam이 매우 길다면

$$x \rightarrow \infty \text{ 이면 } v = v' = 0$$

$$\therefore x > 0 \text{ 이면 } C_3 = C_4 = 0 \quad \therefore v = e^{-\beta x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

또, Boundary Condition ; $v'(0) = 0$ --- ②

또 다른 B.C.가 $x=0$ 근처의 점에서의 shear stress에 의해 생긴다.

$$2V_y + kv dx = P$$

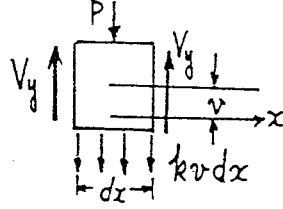
$$dx \rightarrow 0 \text{ 일때 } V_y = \frac{P}{2} = -EI v''' \quad \text{--- ㉓}$$

① ㉓을 $v = e^{-\beta x} (C_1 \cos \beta x + C_2 \sin \beta x)$ 이 대입하여,

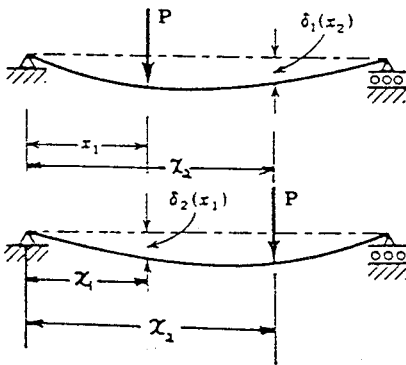
$$C_1 = C_2 = -\frac{P\beta}{2k}$$

$$\delta \equiv -v(0) = -C_1$$

$$= \frac{P\beta}{2k} = \frac{P}{2k} \left(\frac{k}{4EI} \right)^{1/4} = \frac{P}{(64EI k^3)^{1/4}} \quad \text{--- Q. E. D. ---}$$



8.65 Table (8.1) 을 4.



$$(a) \delta(x_2) = \frac{P(L-x_1)}{6LEI} \left[\frac{L}{(L-x_1)} (x_2-x_1)^3 \right. \\ \left. - x_2^3 + \{L^2 - (L-x_1)^2\} x_2^2 \right]$$

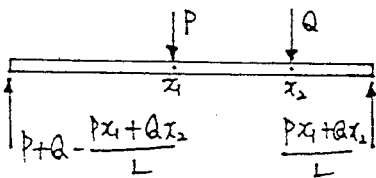
$$= \frac{P}{6EIL} x_1 (L-x_2) (2Lx_2 - x_1^2 - x_2^2) \quad \text{--- ㉔}$$

$$\delta(x_1) = \frac{P(L-x_2)}{6LEI} \left[\frac{L}{(L-x_2)} (x_1-x_2)^3 \right. \\ \left. - x_1^3 + \{L^2 - (L-x_2)^2\} x_1^2 \right]$$

$$= \frac{P}{6EIL} x_1 (L-x_2) (2Lx_2 - x_1^2 - x_2^2) \quad \text{--- ㉕}$$

$\therefore$ ㉔ ㉕ 을 4 $\delta(x_1) = \delta(x_2) \quad \text{--- Q. E. D. ---}$

(b) Castigliano's theorem.



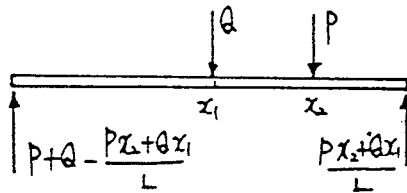
$$M_b = \left(P+Q - \frac{Px_1+Qx_2}{L} \right) x$$

$$- P \langle x-x_1 \rangle' - Q \langle x-x_2 \rangle'$$

$$U = \int \frac{M_b^2}{2EI} dx$$

$$\therefore \delta(x_2) = \left(\frac{\partial U}{\partial Q} \right)_{Q=0} = \int \frac{M_b \cdot m}{EI} dx = \frac{1}{EI} \int M_b \left[\left(1 - \frac{x_2}{L} \right) x - \langle x-x_2 \rangle' \right] dx$$

$$= \frac{P}{EI} \int \left[\left(1 - \frac{x_1}{L} \right) x - \langle x-x_1 \rangle' \right] \left[\left(1 - \frac{x_2}{L} \right) x - \langle x-x_2 \rangle' \right] dx \quad \text{--- ㉖}$$



같은 요령으로

$$M_b = \left(p+Q - \frac{Px_2+Qx_1}{L} \right) x - Q \langle x-x_1 \rangle' - p \langle x-x_2 \rangle'$$

$$\therefore \delta(x_1) = \left(\frac{\partial U}{\partial Q} \right)_{Q=0} = \int \frac{M_b \cdot m}{EI} dx$$

$$= \frac{P}{EI} \int \left[\left(1 - \frac{x_2}{L} \right) x - \langle x-x_2 \rangle' \right] \left[\left(1 - \frac{x_1}{L} \right) x - \langle x-x_1 \rangle' \right] dx \quad \text{--- ②}$$

$$\text{① ② 같기} \quad \delta(x_1) = \delta(x_2) \quad - Q.E.D. -$$

8.66 material property (copper) $\left(\begin{array}{l} \rho = 8.95 \times 10^3 \text{ kg/m}^3 \\ E = 117 \text{ GN/m}^2 \end{array} \right.$

$$I = \frac{\pi}{64} d^4 = \frac{\pi}{64} (0.025)^4 = 1.917 \times 10^{-8} (\text{m}^4)$$

mirror 의 위치는 rod 의 중앙 이므로.

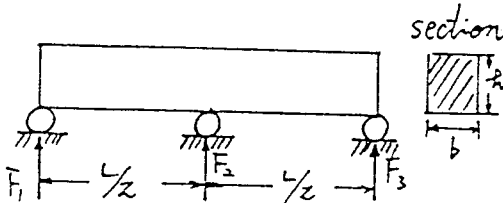
$$M_b = \frac{1}{8} \omega L^2$$

또 mirror 는 rod 가 처져 생기는 곡률과 일치하는 곡률을 갖게 될 것이다.

$$\begin{aligned} \frac{d\phi}{ds} &= \frac{M_b}{EI} = \frac{\frac{1}{8} (8.95 \times 10^3) \left(\frac{\pi}{4} \right) (0.025)^2 (9.8) (2.5)^2}{(117 \times 10^9) (1.917 \times 10^{-8})} \\ &= \frac{1}{0.015} (\text{m}) \quad \therefore \rho = 66.68 \text{ m} \end{aligned}$$

Ans. $\rho = 66.68 \text{ m}$

8.67



Bending moment ;

$$EI v'' = F_1 x - \frac{1}{2} \omega x^2 + F_2 \left\langle x - \frac{L}{2} \right\rangle'$$

$$\begin{aligned} EI v' &= \frac{1}{2} F_1 x^2 - \frac{1}{6} \omega x^3 \\ &\quad + \frac{1}{2} F_2 \left\langle x - \frac{L}{2} \right\rangle^2 + C_1 \end{aligned}$$

$$EIV = \frac{1}{6}F_1x^3 - \frac{1}{24}\omega x^4 + \frac{1}{6}F_2\left(x - \frac{L}{2}\right)^3 + C_1x + C_2$$

Boundary Condition ;

$$v(0) = 0 \quad \therefore C_2 = 0$$

$$v'\left(\frac{L}{2}\right) = 0 ; \frac{1}{8}F_1L^2 - \frac{1}{48}\omega L^3 + C_1 = 0 \quad \therefore C_1 = -\frac{1}{8}F_1L^2 + \frac{1}{48}\omega L^3$$

$$v\left(\frac{L}{2}\right) = 0 ; \frac{1}{128}\omega L^4 - \frac{1}{24}F_1L^3 = 0$$

$$\therefore F_1 = \frac{3}{16}\omega L, \quad F_2 = \frac{5}{8}\omega L$$

max. bending moment 는 $v''' = 0$ 을 만족하는 곳에서 발생.

$$i) x = \frac{3}{16}L ; (M_b)_{\frac{3}{16}L} = \frac{9}{512}\omega L^2$$

$$ii) x = \frac{1}{2}L ; (M_b)_{\frac{1}{2}L} = -\frac{1}{32}\omega L^2 \quad \therefore (M_b)_{\max} = \frac{\omega}{32}L^2$$

따라서 가운데 바퀴에서 파리가 떨어진다.

8.68 문제 (8.1 (g)) 에서

$$(a) (M_b)_{\max} = \frac{1}{8}\omega L^2, \quad \sigma_{\max} = \frac{(M_b)_{\max}}{I} \cdot \frac{h}{2} = \frac{3\omega L^2}{4bh^2}$$

$$\omega L = W \text{ 이므로 } W = \frac{4bh^2}{3L}(\sigma_{\max}), \quad W_B = \gamma b h L$$

$$\therefore \frac{W}{W_B} = \frac{4h}{3L^2} \left(\frac{\sigma_{\max}}{\gamma} \right)$$

$$(b) v = \frac{\omega}{EI} \left[\frac{L}{12}x^3 - \frac{1}{24}x^4 - \frac{1}{24}L^3x \right]$$

$$\therefore \delta_{\max} = \frac{\omega}{EI} \cdot \frac{5L^4}{384} = \frac{5WL^3}{384EI} \quad (\because W = \omega L)$$

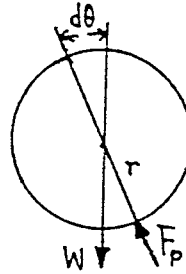
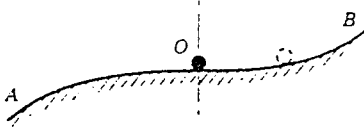
$$\therefore W = \frac{32EIb^2h^3\delta_{\max}}{5L^4} \quad \left. \vphantom{\frac{32EIb^2h^3\delta_{\max}}{5L^4}} \right) \therefore \frac{W}{W_B} = \frac{32h^2\delta_{\max}}{5L^4} \left(\frac{E}{\gamma} \right)$$

$$W_B = \gamma b h L$$

— Q. E. D. —

CHAPTER 9

9.1



$\sum F = 0$ 이 아니다.

따라서 평형은 있을 수 없다. 또, 불균형 moment

$$M_p = W r d\theta \quad (G)$$

구를 O점에서 부러 구르게 할 것이다. 따라서 평형은 불안정하다.

9.2 A점에서의 tension = P

$$\sum M_{Bz} = 0;$$

$$(kx + P \sin \theta) \sqrt{L^2 + x^2} = P \cos \theta \cdot x$$

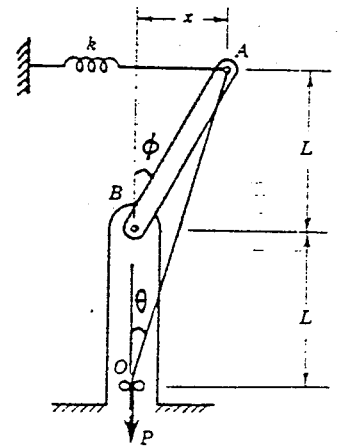
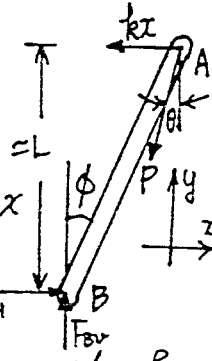
$$\theta \ll 1 \quad \sin \theta \approx x/L$$

$$\sqrt{L^2 + x^2} \approx L$$

$$\cos \theta = 1 \quad \therefore \frac{Px}{L} = x \left(k + \frac{P}{L} \right)$$

$$\therefore P = 2kL$$

$$\text{Ans. } P = 2kL$$



9.3(a) B, C는 아무런 moment도 전달할 수 없기 때문에, A, D점의 반력은 AB, DC에만 전해진다.

$$\sum M_D = 0; (F_{AV} + 2F_{AH})L = M + kxL - 0$$

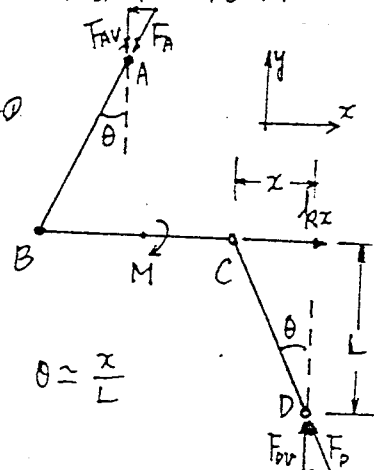
$$\sum F_y = 0; F_{AV} = F_{DV} - 0$$

$$\sum F_x = 0; F_{AH} + F_{DH} = kx - 0$$

$$\text{Geometry; } F_{AH} \approx F_{AV} \cdot \frac{x}{L} - 0$$

$$F_{DH} \approx F_{DV} \cdot \frac{x}{L} - 0$$

$$0 \sim 0 \text{ 에서 } M = \frac{kL^2}{2}; \text{ Ans.}$$



(b) M이 반대 방향으로 작용하면, stability 문제는 없다. AD, CD가 tension으로 분리되거나, BC가 bending에 의해 분리 될 때까지 M이 작용.

9.4 평형 조건 $\sum M_{BZ} = 0$; $P(L \sin \theta + \epsilon) = k\theta$

θ 가 미소각이면, $\sin \theta \approx \theta$

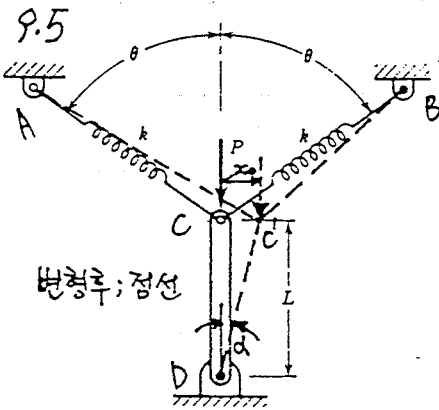
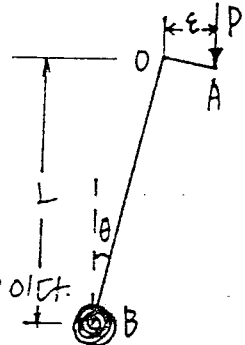
$$\therefore \theta = \frac{P\epsilon}{k - PL}$$

$P \rightarrow \frac{k}{L}$ 이면 $\theta \rightarrow \infty$ 이다.

따라서, $P = \frac{k}{L}$ 는 한계 하중 (limiting load) 이다.

$\therefore P < \frac{k}{L}$ 이면 평형은 안정이다.

Ans. $\theta = (P\epsilon)/(k - PL)$, $P < k/L$



변형률; 점선

그림에서 α 를 미소각으로 가정하면

$$\delta_{AC} \approx x \cos \theta, \delta_{BC} \approx -x \sin \theta$$

평형 조건; $\sum M_{DZ} = 0$;

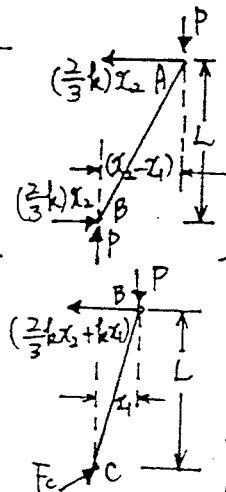
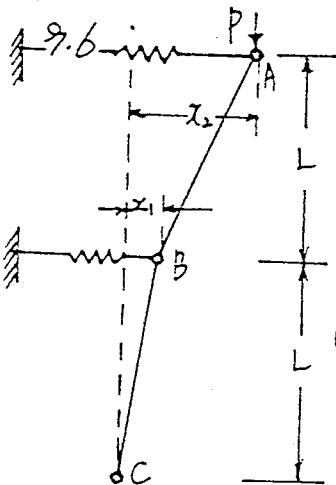
$$Px = F_{AC} L \sin \theta - F_{BC} L \sin \theta$$

$$= (kx \cos \theta + kx \sin \theta) L \sin \theta$$

$$= kx L \sin \theta (\sin \theta + \cos \theta)$$

$$\therefore P_{cr} = k L \sin \theta (\sin \theta + \cos \theta)$$

Ans. $P_{cr} = k L \sin \theta (\sin \theta + \cos \theta)$



구간 AB의 평형 조건;

$$\sum M_B = 0; P(x_2 - x_1) = \frac{2}{3} k x_2 L \quad \text{--- ①}$$

구간 BC의 평형 조건;

$$\sum M_C = 0;$$

$$(P)(x_1) = k(x_1 + \frac{2}{3} x_2) L \quad \text{--- ②}$$

식 ①②가 해를 갖아야만 평형을 만족하게 된다. ②에서 x_1 을 제거하면

$$\left[\left(2 - \frac{kL}{P} \right) \left(1 - \frac{2}{3} \frac{kL}{P} \right) - 1 \right] x_2 = 0.$$

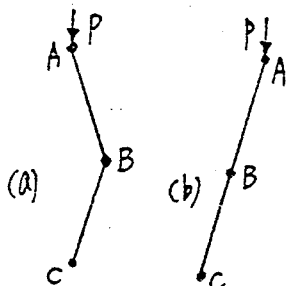
만약 $x_2 = 0$ 이면 $x_1 = 0$ 인 *trivial solution*을 얻게된다. 따라서 *non-trivial solution*을 얻으려면,

$$\left[\left(2 - \frac{kL}{P} \right) \left(1 - \frac{2}{3} \frac{kL}{P} \right) - 1 \right] = 0 \quad \therefore P = \frac{kL}{3} \text{ or } P = 2kL$$

Stability

$x_2 = -x_1$ 이면 $P = \frac{kL}{3}$ 로 *neutral equilibrium*을 유지한다. (a)

$x_2 = \frac{3}{2}x_1$ 이면 $P = 2kL$ 로 *neutral equilibrium*을 유지한다. (b)



(a) 는 $P < \frac{kL}{3}$ 일 경우 안정

$P > \frac{kL}{3}$ 일 경우 불안정

(b) 는 $P < 2kL$ 일 경우 안정

$P > 2kL$ 일 경우 불안정

구조물은 모든 가능한 상황에서 안정되어야 하므로

$P < \frac{kL}{3}$ 을 만족해야 한다. Ans. $P < \frac{kL}{3}$

9.7 그림에서

$$2k(AM - AB) \sin \theta = P$$

$$AM \cos \phi = AB \cos \theta$$

θ 와 ϕ 를 미소각으로 보면

$$\sin \theta \approx \theta$$

$$\cos \phi \approx 1 - \frac{\phi^2}{2}, \quad \cos \theta \approx 1 - \frac{\theta^2}{2} \quad \therefore P = kL\theta(\phi^2 - \theta^2) \quad \text{--- ①}$$

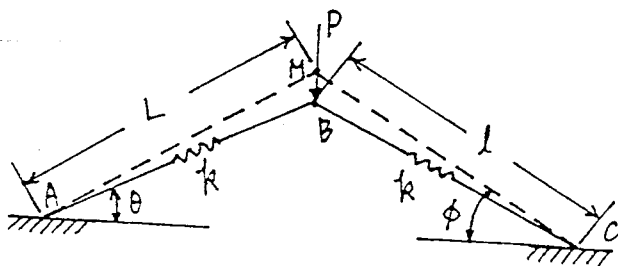
snap through 는 $\frac{dP}{d\theta} = kL(-\phi^2 + 3\theta^2) = 0$ 이여야 한다.

$$\therefore \theta = \pm \frac{\phi}{\sqrt{3}} \quad \text{①에 대입하면} \quad P = \pm \frac{2kL\phi^3}{3\sqrt{3}} \quad \text{--- snap through}$$

stability ; $d\theta = \epsilon$ 이고 $P = \text{const.}$ 이면

$$\text{restoring force ; } P_{\text{rest.}} = kL(\theta + \epsilon)[-\phi^2 + (\theta + \epsilon)^2] - P = kL\epsilon(-\phi^2 + 3\theta^2)$$

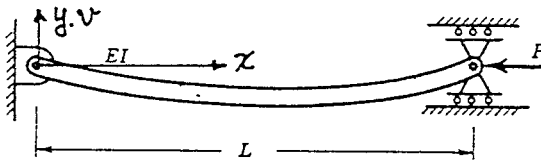
즉, $\phi^2 < 3\theta^2$ 일때 안정. $|\theta| > \frac{\phi}{\sqrt{3}}$; 안정 $|\theta| \leq \frac{\phi}{\sqrt{3}}$; 불안정.



Ans. $P = kL\theta(\phi^2 - \theta^2)$, snap through: $P = \pm \frac{2kL\phi^3}{3\sqrt{3}}$

$|\theta| \leq \frac{\phi}{\sqrt{3}}$ 일때 불안정, $|\theta| > \frac{\phi}{\sqrt{3}}$ 일때 안정.

9.8



$$EI v'' = M_b = -Pv \quad \text{--- ①}$$

$$\therefore EI v'' + Pv = 0$$

$$v'' + \frac{P}{EI} v = v'' + k^2 v = 0$$

$$(k^2 = \frac{P}{EI})$$

이 방정식의 일반해는: $v = A \cos kx + B \sin kx$

Boundary Condition: $v(0) = v(L) = 0$

$$\therefore A = 0, \quad B \sin kL = 0$$

Non-trivial sol. 은 $\sin kL = 0 \quad \therefore kL = n\pi = \sqrt{\frac{P}{EI}} L$

따라서 $P_{cr} = \frac{\pi^2 EI}{L^2}$ 여기서 좌굴하중은 $n=1$ (최소) 일때만 의미가 있으므로.

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$

Ans. $P_{cr} = \frac{\pi^2 EI}{L^2}$

9.9 Eq. (9.10) 식으로 부터

$$v = C_1 + C_2 x + C_3 \sin \sqrt{\frac{P}{EI}} x + C_4 \cos \sqrt{\frac{P}{EI}} x \quad \text{--- ①}$$

Boundary Condition $v(0) = v(L) = v'(0) = v'(L) = 0$ ①에 대입.

$$\therefore C_1 + C_4 = 0, \quad C_2 + C_3 \sqrt{\frac{P}{EI}} = 0$$

$$C_1 + C_2 L + C_3 \sin \sqrt{\frac{P}{EI}} L + C_4 \cos \sqrt{\frac{P}{EI}} L = 0$$

$$C_2 + C_3 \sqrt{\frac{P}{EI}} \cos L \sqrt{\frac{P}{EI}} - C_4 \sqrt{\frac{P}{EI}} \sin \sqrt{\frac{P}{EI}} L = 0$$

위의 4식에서 C_2, C_4 를 소거하면

$$C_1 (1 - \cos \sqrt{\frac{P}{EI}} L) + C_3 (\sin \sqrt{\frac{P}{EI}} L - \sqrt{\frac{P}{EI}} L) = 0 \quad \text{--- ②}$$

$$C_1 \sqrt{\frac{P}{EI}} \sin \sqrt{\frac{P}{EI}} L + C_3 \sqrt{\frac{P}{EI}} (\cos \sqrt{\frac{P}{EI}} L - 1) = 0 \quad \text{--- ③}$$

②③ 에서 non-trivial sol. 은 계수 C_1, C_3 이 어떠한 determinant = 0

이어야 한다.

$$\therefore (1 - \cos \beta L)(\cos \beta L - 1) - \sin^2 \beta L + \beta L \sin \beta L = 0, \quad \beta \equiv \sqrt{\frac{P}{EI}}$$

$$\sin \frac{\beta L}{2} \left(\frac{\beta L}{2} \cos \frac{\beta L}{2} - 4 \sin \frac{\beta L}{2} \right) = 0$$

$$i) \sin \frac{\beta L}{2} = 0 \text{ 일 경우 } \frac{\beta L}{2} = n\pi \quad \therefore p_{cr} = \frac{(2n^2)\pi^2 EI}{L^2}$$

$$v_m = C_1 \left(1 - \cos \frac{2n\pi x}{L} \right)$$

좌굴하중은 첫번째 ($n=1$) 일 때만 의미를 갖는다.

$$\therefore p_{cr} = (4\pi^2 EI) / L^2, \quad v = C_1 \left(1 - \cos \frac{2\pi x}{L} \right)$$

ii) $\tan \frac{\beta L}{2} = \frac{\beta L}{2}$ 인 경우 $\Rightarrow$ 보다 큰 좌굴 하중이 계산된다.

$$ii) \text{에서 } p_{cr} = (4\pi^2 EI) / L^2 : \text{Ans.}$$

9.10 Eg (9.9) 에서

$$EI \frac{d^4 v}{dx^4} + P \frac{d^2 v}{dx^2} = 0$$



방정식의 해는 ; $v = C_1 + C_2 x + C_3 \sin \sqrt{\frac{P}{EI}} x + C_4 \cos \sqrt{\frac{P}{EI}} x$

Boundary Condition ; $v(0) = v'(0) = 0$.

또한 shear force $= EI \frac{d^3 v}{dx^3} = 0$, moment $= M_b = EI \frac{d^2 v}{dx^2} = 0$

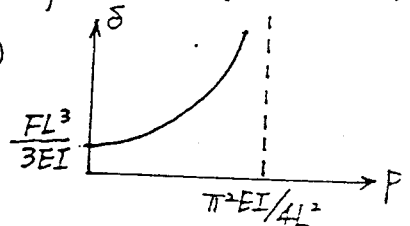
따라서 $C_1 = -C_4 = \delta + \frac{F}{P} L$

$$C_3 = -C_2 \sqrt{\frac{EI}{P}} = \left(\delta + \frac{F}{P} L \right) \cot kL \quad k \equiv \sqrt{\frac{P}{EI}}$$

$$\therefore v = \left(\delta + \frac{F}{P} L \right) (1 - \cos kx) - \left(\delta + \frac{F}{P} L \right) \cot kL (kx - \sin kx)$$

$$\delta = (v)_{x=L} = \left(\delta + \frac{F}{P} L \right) (1 - kL \cot kL)$$

$$\therefore \delta = \frac{FL}{P} \frac{(\sin kL - kL \cos kL)}{\cos kL}$$



Ans. $\delta = \left(\frac{F}{P} L \right) (\sin kL - kL \cos kL) / (\cos kL)$

9.11 Eq. (9.21)은 다음과 같이 쓸 수 있다.

$$\frac{P}{AE} \geq \frac{Y}{E} \quad \text{for plastic flow} \quad \text{--- (1)}$$

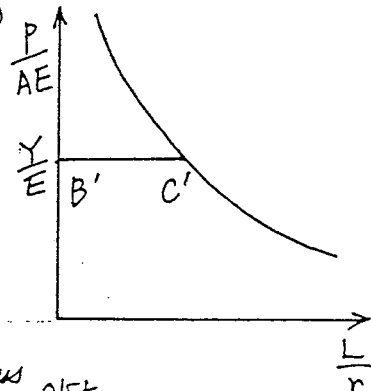
Eq. (9.22)은 다음과 같이 쓸 수 있다.

$$\frac{P}{AE} < \frac{\pi^2}{4} \cdot \frac{I}{AL^2} = \frac{\pi^2}{4} \cdot \frac{r^2}{L^2} \quad \text{for stability}$$

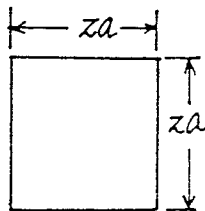
$$\text{또 } \frac{P}{AE} \geq \frac{\pi^2}{4} \left(\frac{r}{L}\right)^2 \quad \text{for instability} \quad \text{--- (2)}$$

③식은 cantilever column에 대한 instability locus를 나타낸다. column의 재료나 기하학적인 형상에 무관하게 $\left(\frac{P}{AE}\right)$ 처 $\left(\frac{L}{r}\right)^2$ 의 graph는 고정된 곡선을 형성한다.

BC에 대응되는 좌표는 $\frac{Y}{E} = \frac{\text{yield stress}}{\text{modulus}}$ 이다.



9.12



i) Yielding ; $P = AY = 4a^2 Y$

1020 HR steel ; $Y = 255 \text{ MN/m}^2$

문제에서 주어진 하중 ; $P = 50 \text{ kN}$

$$\therefore a = \sqrt{\frac{50 \times 10^3}{4 \times 255 \times 10^6}} = 0.007 \text{ (m)}$$

ii) Buckling ; $P = \frac{\pi^2}{4} \cdot \frac{EI}{L^2}$; $I = \frac{bh^3}{12} = \frac{4}{3}a^4$

$$\therefore P = \frac{\pi^2}{3} \cdot \frac{Ea^4}{L^2} \quad \therefore a^4 = \frac{3L^2 P}{\pi^2 E}$$

$$\therefore a = \sqrt[4]{\frac{3 \times 2^2 \times 50 \times 10^3}{\pi^2 \times 200 \times 10^9}} = 0.02348 \text{ (m)}$$

i) ii)의 값은 0.02348 (m) ; 한 변의 길이 $2a = 0.04696$

Ans 0.04696 (m) ; 한 변의 길이

9.13 i) A점에 관한 moment 이다

$$P_c = 2kL = 2 \times 80 \times 10^3 = 160,000 \text{ N}$$

ii) Yielding

$$P_{cr} = YA = \frac{\pi}{4} d^2 Y = \frac{\pi}{4} (25 \times 10^{-3})^2 (400 \times 10^6) = 196,350 \text{ N}$$

iii) B점이 옆으로 이동하지 않는다면,

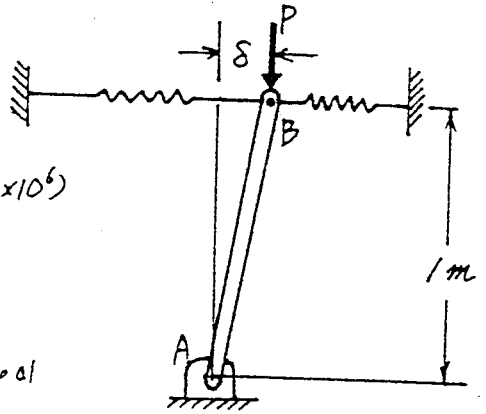
양단이 hinged 된 beam 으로 buckling 이

가능하다.

$$P_{cr} = \frac{\pi^2 EI}{L^2} = \pi^2 \times (76 \times 10^9) \cdot \frac{\pi}{64} (0.025)^4 = 14,383 \text{ N}$$

i) ii) iii) 에서 buckling 이 가장 먼저 일어나며, $P_{cr} = 14,383 \text{ N}$

Ans. Buckling 이 가장 먼저 일어난다. $P_{cr} = 14,383 \text{ N}$



9.14 (Eg. 9.17) 에서 $\delta = \frac{M_0}{P} \left(\frac{1 - \cos L \sqrt{\frac{P}{EI}}}{\cos L \sqrt{\frac{P}{EI}}} \right)$

미소각 θ 에 대해서 $\theta \approx L \sqrt{\frac{P}{EI}}$, $1 - \cos \theta \approx \frac{\theta^2}{2}$, $\cos \theta \approx 1.0$

$$\therefore \delta \approx \frac{M_0}{P} \cdot \frac{\theta^2}{2} = \frac{M_0 L^2}{2EI} \quad - \text{Q.E.D.} -$$

9.15 (a) Yielding ; $(P_{cr})_Y = YA$ (Y; Yielding Stress)

$$\text{Buckling ; } (P_{cr})_B = \frac{\pi^2 EI}{L^2}$$

위의 그식에서 작은 P_{cr} 값이 critical value 이다.

(b) 평형조건 ; $(L \cos \theta)(kL \sin \theta) = PL \sin \theta \therefore p = kL \cos \theta$

(c) Stable 이 되려면 restoring force 가 (+) 이여야 한다.

$$P_{res.} = kL \cos(\theta + \epsilon) - kL \cos \theta = -kL \epsilon \sin \theta > 0$$

$$(\because \epsilon \ll 1 \quad \sin \epsilon \approx \epsilon, \quad \cos \epsilon \approx 1)$$

$$\therefore \sin \theta < 0 \text{ 에서 stable 하다 } \therefore \pi < \theta < 2\pi$$

(d) $\epsilon = \theta$ 를 미소각으로 가정하면.

(b)에서 $P = kL \cos \theta = kL (1 - \frac{1}{2}\theta^2) \therefore P = \frac{1}{2}kL(2 - \theta^2)$

(e) stable 하려면 restoring force는 (+)이어야 한다.

$$P_{res.} = \frac{1}{2}kL \{ 2 - (\theta + \epsilon')^2 \} - \frac{1}{2}kL(2 - \theta^2)$$

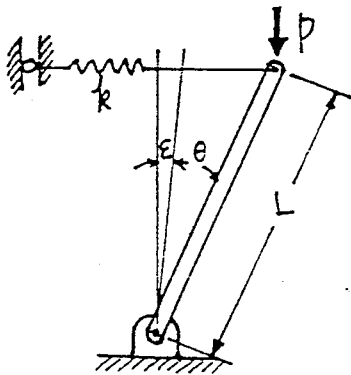
$$= \frac{1}{2}kL(-\epsilon')(2\theta + \epsilon') < 0$$

$P_{res.} < 0$ 이므로 θ 의 관례값이 항상 unstable 하다.

(f) $\frac{dP}{d\theta} = -kL\theta = 0$ 인 점에서 P 가 maximum value를 갖는다.

$\therefore P_{max} = kL$ at $\theta = 0$.

(g) 평형조건 $\sum M_A = 0$; $P L \sin \theta = kL \cos \theta \cdot (L \sin \theta - L \sin \epsilon)$



$$\therefore P = kL \cdot \cos \theta \cdot (1 - \frac{\sin \epsilon}{\sin \theta})$$

$$\frac{dP}{d\theta} = kL \left(\frac{\sin \epsilon \cdot \cos \theta}{\sin^2 \theta} \right) \cos \theta$$

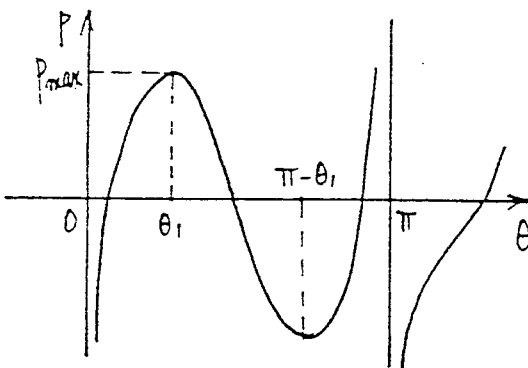
$$- kL \sin \theta \left(1 - \frac{\sin \epsilon}{\sin \theta} \right)$$

$$= kL \frac{\sin \epsilon \cdot \cos^2 \theta - \sin^3 \theta + \sin \epsilon \sin^2 \theta}{\sin^2 \theta}$$

$$= kL \frac{\sin \epsilon - \sin^3 \theta}{\sin^2 \theta}$$

$$\frac{dP}{d\theta} = 0 ; \sin^3 \theta_1 = \sin \epsilon \text{ 인 } (0 < \theta_1 < \frac{\pi}{2}) \theta_1, \pi - \theta_1 \text{ 에서}$$

P 의 극값을 갖는다. $P - \theta$ 의 graph를 그리면



$0 < \theta < \theta_1$; stable

$\theta_1 < \theta < \pi - \theta_1$; unstable

$\pi - \theta_1 < \theta < \pi$; stable

θ 는 안정된 상태를 유지하려 한다. $\therefore \theta = \theta_1$ 에서

$\theta = \pi - \theta_1$ 으로 jump가 발생하게 된다.

$$(k) \quad (d) \text{에러} \quad p = kL \left(1 - \frac{\theta^2}{z}\right) = kL \left(1 + \beta \frac{x^2}{L^2}\right) \quad \therefore \underline{\beta = -\frac{1}{z}}$$

$$9.16 \quad \bar{\sigma} = \left\{ \frac{1}{z} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] \right\}^{1/2}$$

Biaxial tension 을 받는 spherical shell 에서는 $\sigma_1 = \sigma_2$, $\sigma_3 = 0$

$$\text{평형 조건; } p = \frac{z \sigma_1 t_0}{r_0} \quad (p = \text{pressure}) \quad \text{--- ①}$$

$$\text{위에서 } \bar{\sigma} = \sigma_1 = \sigma_2 \quad \text{이므로} \quad p = \frac{z \bar{\sigma} t_0}{r_0} \quad \text{--- ②}$$

$$\bar{\epsilon} = \left\{ \frac{z}{3} [(d\epsilon_I - d\epsilon_{II})^2 + (d\epsilon_I - d\epsilon_{III})^2 + (d\epsilon_{II} - d\epsilon_{III})^2] \right\}^{1/2}$$

체적의 미소 변화 윗에서의 체적 증가; $d(\Delta V)$

$$d(\Delta V) = (d\epsilon_I + d\epsilon_{II} + d\epsilon_{III}) \cdot \Delta V$$

체적의 증가는 없으므로 $d\epsilon_I + d\epsilon_{II} + d\epsilon_{III} = 0$.

spherical shell 에서, $d\epsilon_I = d\epsilon_{II} = \frac{dr_0}{r_0} \quad \therefore d\epsilon_{III} = -2 \frac{dr_0}{r_0}$

$$\text{즉 } \frac{dt_0}{t_0} = -2 \frac{dr_0}{r_0} \quad \therefore \bar{\epsilon} = \int \frac{2}{3} \frac{dr_0}{r_0} \quad \text{or } d\bar{\epsilon} = \frac{2}{3} \frac{dr_0}{r_0}$$

②식을 log 를 취하여 미분하면

$$\frac{dp}{p} = \frac{d\bar{\sigma}}{\bar{\sigma}} + \frac{dt_0}{t_0} - \frac{dr_0}{r_0} = \frac{d\bar{\sigma}}{\bar{\sigma}} - \frac{3}{2} d\bar{\epsilon} \quad \text{--- ③}$$

$$\text{초기에 } \frac{d\bar{\sigma}}{d\bar{\epsilon}} = E \text{ 이므로, } \frac{d\bar{\sigma}}{\bar{\sigma}} - \frac{3}{2} d\bar{\epsilon} > 0 \quad \text{즉 } \frac{d\bar{\sigma}}{d\bar{\epsilon}} > \frac{3}{2} \bar{\sigma}$$

따라서 초기에는 $dp > 0$ 또는 p 가 증가한다.

$$\text{또, } \frac{d\bar{\sigma}}{d\bar{\epsilon}} = \frac{3}{2} \bar{\sigma} \text{ 일때 } dp = 0 \quad \frac{d\bar{\sigma}}{d\bar{\epsilon}} < \frac{3}{2} \bar{\sigma} \text{ 일때 } dp < 0$$

p 는 감소

$\therefore$ Max. pressure 는 $\frac{d\bar{\sigma}}{d\bar{\epsilon}} = \frac{3}{2} \bar{\sigma}$ 일때 생긴다. --- Q.E.D. ---

$$\text{③식을 적분; } \ln p = \ln \bar{\sigma} - \frac{3}{2} \bar{\epsilon} + \ln C_1$$

$$\text{또는 } p = C_1 \bar{\sigma} \exp\left(-\frac{3}{2} \bar{\epsilon}\right)$$

$$\bar{\epsilon} \ll 1 \text{ 이면 } \exp\left(-\frac{3}{2}\bar{\epsilon}\right) \approx 1$$

$$\therefore p \approx C_1 \bar{\sigma} = \frac{2\bar{\sigma}t_0}{r_0} \quad \therefore C_1 = \frac{2t_0}{r_0}$$

$$\text{따라서, } p = \frac{2t_0\bar{\sigma}}{r_0} \exp\left(-\frac{3}{2}\bar{\epsilon}\right)$$

Max. pressure 는 $\bar{\sigma} = \bar{\sigma}_{\max}$, $\bar{\epsilon} = \bar{\epsilon}_{\max}$ 이에서 생긴다.

$$p_{\max} = \frac{2t_0\bar{\sigma}_{\max}}{r_0} \exp\left(-\frac{3}{2}\bar{\epsilon}_{\max}\right)$$

$\bar{\sigma}_{\max}$ 과 $\bar{\epsilon}_{\max}$ 의 결정.

임의의 점 A에서의 접선의 기울기라

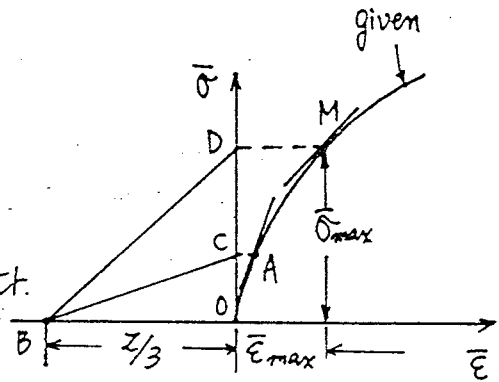
BC의 기울기가 같아지는 점에서

$\bar{\sigma}_{\max}$ 과 $\bar{\epsilon}_{\max}$ 을 나타낸다.

M에서 근사적으로 다음 식이 성립한다.

$$\frac{d\bar{\sigma}}{d\bar{\epsilon}} = \frac{3}{2}\bar{\sigma} \left(= \frac{OD}{OB} \right)$$

$$\text{Ans. } p_{\max} = \frac{2t_0\bar{\sigma}_{\max}}{r_0} \exp\left(-\frac{3}{2}\bar{\epsilon}_{\max}\right)$$



$$9.17 \quad F = A\bar{\sigma}$$

$$A = \frac{A_0 \cdot L_0}{L_0 + \bar{\sigma}} = \frac{A_0}{1 + \frac{\bar{\sigma}}{L_0}} = \frac{A_0}{1 + \bar{\epsilon}} \quad \therefore F = \frac{A_0 \bar{\sigma}}{1 + \bar{\epsilon}}, \quad \frac{F}{A_0} = \frac{\bar{\sigma}}{1 + \bar{\epsilon}} \quad - \text{A.E.D.} -$$

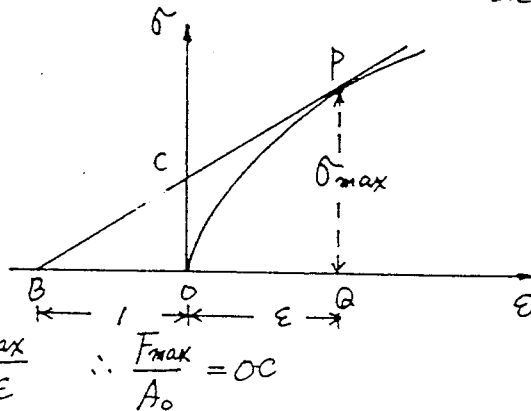
이런 graph 에서

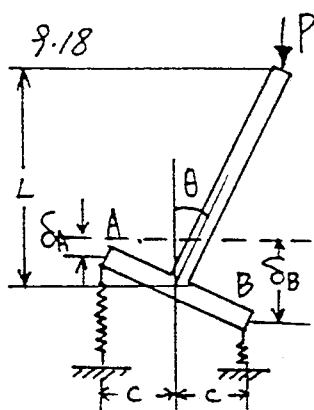
$$\bar{\sigma}_{\max} = PQ$$

$$\frac{F_{\max}}{A_0} = \frac{\bar{\sigma}_{\max}}{1 + \bar{\epsilon}}$$

$$\text{또, } \frac{OC}{QP} = \frac{1}{1 + \bar{\epsilon}}$$

$$\therefore OC = \frac{QP}{1 + \bar{\epsilon}} = \frac{\bar{\sigma}_{\max}}{1 + \bar{\epsilon}} \quad \therefore \frac{F_{\max}}{A_0} = OC$$





$$F_A = \frac{P}{2} \left(1 - \frac{L\theta}{c}\right) \quad - ①$$

$$F_B = \frac{P}{2} \left(1 + \frac{L\theta}{c}\right) \quad - ②$$

$$\delta_B - \delta_A = 2c\theta \quad - ③$$

force-deflection relation;

$$F_A = k_c \delta_A, \quad F_B = k_c \delta_B \quad - ④$$

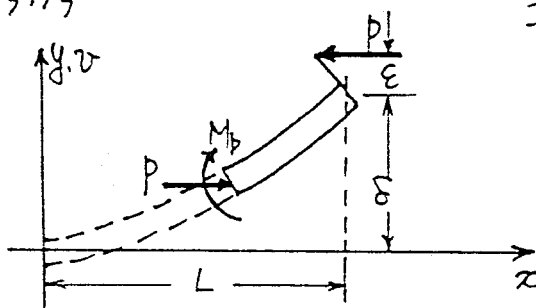
$$③④ \text{ 이기 } F_B - F_A = 2k_c c \theta \quad - ⑤$$

$$①② \text{ 를 } ⑤ \text{ 에 대입하면 } \theta \left(2k_c c - \frac{PL}{c}\right) = 0$$

$$\therefore \text{non-trivial sol. } P_{cr} = \frac{2k_c c^2}{L} \quad (\text{임의의 } \theta \text{ 에 대해})$$

$$\underline{\text{Ans } P_{cr} = 2k_c c^2 / L}$$

9.19



그림에서

$$M_b = p(\delta + \epsilon - v) = EI v''$$

$$\therefore EI v'' = p(\delta + \epsilon - v) \quad - ①$$

$$\phi = -(\delta + \epsilon - v) \text{ 라 놓으면 } \quad - ②$$

①식은 다음과 같이 된다.

$$EI \frac{d^2 \phi}{dx^2} + \frac{P}{EI} \phi = 0 \quad - ③$$

③식의 일반해;

$$\phi = C_1 \sin \sqrt{\frac{P}{EI}} x + C_2 \cos \sqrt{\frac{P}{EI}} x$$

$$②식에서 \quad v = \delta + \epsilon + C_1 \sin \sqrt{\frac{P}{EI}} x + C_2 \cos \sqrt{\frac{P}{EI}} x$$

Boundary Condition;

$$v(0) = v'(0) = 0, \quad v(L) = \delta \text{ 를 대입하면}$$

$$C_1 = 0 \quad C_2 = -(\delta + \epsilon) \quad \therefore v = (\delta + \epsilon) [1 - \cos \sqrt{\frac{P}{EI}} x]$$

$$\therefore v(L) = \delta = (\delta + \epsilon) [1 - \cos \sqrt{\frac{P}{EI}} L]$$

$$\text{따라서 } \delta = \frac{\epsilon (1 - \cos \sqrt{\frac{P}{EI}} L)}{\cos \sqrt{\frac{P}{EI}} L} = \epsilon (\sec \sqrt{\frac{P}{EI}} L - 1) \quad - \text{Q.E.D.} -$$

9.20 평형조건: $p(x+\epsilon) = 2kLx(1 + \beta \frac{x^2}{L^2})$ - ①

$\frac{dp}{dx} = 0$ 일 때 max. value ; $\frac{dp}{dx} = (x+\epsilon)(1 + 3\beta \frac{x^2}{L^2}) - x(1 + \beta \frac{x^2}{L^2}) = 0$ - ②

$$\therefore x + \epsilon = \frac{x(1 + \beta \frac{x^2}{L^2})}{1 + 3\beta \frac{x^2}{L^2}}$$

위의 $(x+\epsilon)$ 을 ①에 대입하면,

$$p_{max} = \frac{2kLx(1 + \beta \frac{x^2}{L^2})}{x + \epsilon} = 2kL(1 + 3\beta \frac{x^2}{L^2})$$

$p_{cr} = 2kL \therefore \frac{p_{cr} - p_{max}}{p_{cr}} = 1 - \frac{p_{max}}{p_{cr}} = -3\beta \frac{x^2}{L^2}$ - Q.E.D. -

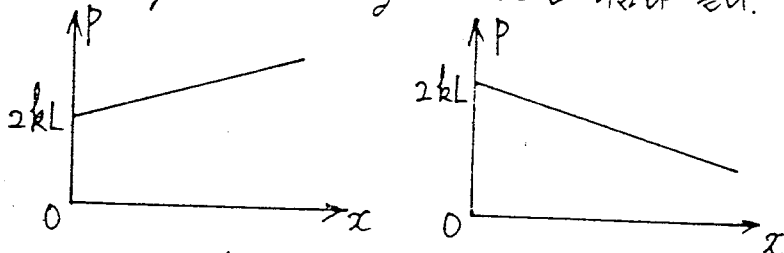
②식에서 $\beta \frac{x^2}{L^2} = -\frac{1}{3}(\epsilon + 3\beta \epsilon \frac{x^2}{L^2}) = -\frac{1}{3}\epsilon \frac{p_{max}}{p_{cr}}$

$$\therefore (1 - \frac{p_{max}}{p_{cr}})^{\frac{3}{2}} = \frac{3\sqrt{3}\beta}{2} \cdot \frac{\epsilon}{L} \cdot \frac{p_{max}}{p_{cr}} \quad - \text{Q.E.D.} -$$

9.21 $f = kx(1 + \alpha x/L)$, $px = 2kx(1 + \alpha x/L)L$

$$\therefore p = 2kL(1 + \alpha x/L)$$

따라서 postbuckling curve는 다음과 같다.



$p(x+\epsilon) = 2kx(1 + \alpha x/L)L$ - ①

$\frac{dp}{dx} = 0$ 일 때 max. value ;

$\frac{dp}{dx} = (1 + 2\alpha x/L)(x+\epsilon) - x(1 + \alpha x/L) = 0$ - ②

②식에서 $p_{max} = \frac{2kx(1 + \alpha x/L)L}{x + \epsilon} = 2kL(1 + 2\alpha x/L)$

따라서 $\frac{P_{cr} - P_{max}}{P_{cr}} = 1 - \frac{P_{max}}{P_{cr}} = -2\alpha x/L$ - Q.E.D. -

또, $(1 - \frac{P_{max}}{P_{cr}})^2 = -4\alpha \cdot (-\alpha) \frac{x^2}{L^2}$

③식에서 $\epsilon + \alpha \frac{x^2}{L} + 2\alpha \epsilon \frac{x}{L} = 0$ 이므로 $-\alpha \frac{x^2}{L^2} = \frac{\epsilon}{L} (1 + 2\alpha x/L)$

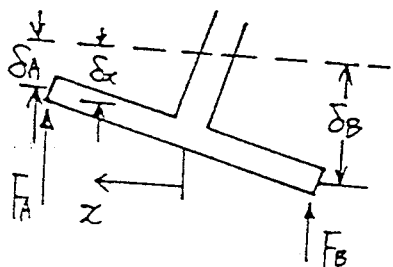
따라서 $(1 - \frac{P_{max}}{P_{cr}})^2 = -4\alpha \cdot \frac{\epsilon}{L} (1 + 2\alpha x/L) = -4\alpha \cdot \frac{\epsilon}{L} \frac{P_{max}}{P_{cr}}$ - Q.E.D. -

9.22 $P_e - 2k_e c\theta = P_t + 2k_t c\theta$ - ①

$P_t = \frac{2c^2}{L} k_t$, $P_e = \frac{2c^2}{L} k_e$ 이므로 ①식에 대입하면

$\theta = \frac{P_e - P_t}{2(k_e + k_t)} \cdot \frac{1}{c} = \left(\frac{k_e - k_t}{k_e + k_t} \right) \left(\frac{c}{L} \right)$ - Q.E.D. -

9.23



평형 조건; $P(L\theta + \epsilon) = (F_B - F_A)c$ - ①

$P = F_A + F_B$

원편 1람에서

$\delta_A - \delta_x = -(c-x)\theta$

$\delta_B - \delta_x = (c+x)\theta$

$\therefore F_A = \frac{1}{2}P_0 - k_e(c-x)\theta$, $F_B = \frac{1}{2}P_0 + k_t(c+x)\theta$ - ②

②를 ①에 대입; $P(L\theta + \epsilon) = [(k_e + k_t)c - (k_e - k_t)x]\theta \cdot c(1 + \beta\theta^2)$

$P_A = P_{x=c}$ 이므로 $P(L\theta + \epsilon) = 2k_t c^2 \theta (1 + \beta\theta^2)$ - ③

$\frac{dP}{d\theta} = 0$ 일 때 max. value; $\frac{dP}{d\theta} = (1 + 3\beta\theta^2)(\epsilon + L\theta) - L\theta(1 + \beta\theta^2) = 0$

$\therefore \frac{1 + \beta\theta^2}{L\theta + \epsilon} = \frac{1 + 3\beta\theta^2}{L\theta}$ - ④

③④에서 $(P_A)_{max} = \frac{2k_t c^2}{L} (1 + 3\beta^2)$

이제 $\beta < 0$ 이므로,

$(P_A)_{\max}$ 은 $P_x (= \frac{2k_x C^2}{L})$ 보다 작다. $\therefore P_x > P_r$

— Q.E.D. —